

# The second row of a random Latin square: towards a precise form of Cameron’s conjecture

Working notes

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## Abstract

Let  $L$  be a uniformly random Latin square of order  $n$  whose first row is the identity, and let  $\sigma$  be its second row, a uniformly “Latin-tilted” random derangement. Cameron conjectured [*A niggling problem*, blog post, 24 Jan 2015] that the law of  $\sigma$  tends *very rapidly* to the uniform distribution on derangements. These notes make the conjecture quantitative and assemble exact, structural, and numerical evidence for the following refined picture. Write  $N_n(\sigma)$  for the number of completions of the  $2 \times n$  rectangle  $\begin{pmatrix} \text{id} \\ \sigma \end{pmatrix}$ , which depends only on the cycle type  $\lambda$  of  $\sigma$ ; set  $C_n(\lambda) = N_n(\sigma)/(n-2)!$ . (i) **Refined law (conjectural)**:  $\log C_n(\lambda)$  is, up to a constant, *additive* over the cycles of  $\lambda$ :  $\log C_n(\lambda) = c_0(n) + \sum_{\ell \geq 2} h_n(\ell) \lambda_\ell + (\text{small})$ , with  $h_n(2) = c_2 n^{-3}(1 + o(1))$ ,  $c_2 \approx 1.3$ , and  $h_n(\ell) \asymp n^{1-2\ell}$ ; consequently  $\max_{\sigma, \tau} N_n(\sigma)/N_n(\tau) = 1 + \Theta(n^{-2})$  and  $d_{\text{TV}} \asymp n^{-3}$  — polynomially, not exponentially, fast. (ii) **Exact layer mechanism**: for the first extension row the analogue of the refined law is a theorem. With  $M_n$  the ménage numbers, the number of third rows compatible with  $\begin{pmatrix} \text{id} \\ \sigma \end{pmatrix}$ ,  $\sigma$  of type  $(\ell, n-\ell)$ , equals  $M_n + M_{n-2\ell}$  exactly — an identity which turns out to be an instance of the classical theory of discordant permutations of Touchard and Riordan — giving a per- $\ell$ -cycle effect  $(1 + o(1)) n^{-2\ell}$ ; exact computations of the fourth row show the same effect again, supporting the mechanism “ $n$  layers  $\times n^{-4}$  per layer =  $n^{-3}$ ”. A *pair inclusion–exclusion identity* reduces the fourth layer entirely to punctured Touchard–Riordan counts and extends the layer table to  $n = 18$ ; an *exact switch calculus* (comparing adjacent cycle-type representatives whose boards differ by one  $2 \times 2$  switch, via Wronskian identities for path matching polynomials) proves the fourth-layer law:  $r_4 = r_3(1 + O(1/n))$ ; the calculus transfers layer by layer — a triple inclusion–exclusion identity gives  $r_5 = r_3(1 + O(1/n))$  as well, with exact confirmation at  $n \leq 9$  — so that each of the first three extension layers contributes the same  $(1 + o(1))n^{-4}$  per 2-cycle. The same toolkit yields as by-products exact joint intercalate statistics of  $3 \times n$  rectangles: the three pair counts are asymptotically independent Poisson(1/2), with  $\mathbb{E}[X_{12}] - \frac{1}{2} = \frac{M_{n-4}}{2M_n}(1 + O(n^{-2}))$  (the third row tilts the second by exactly the layer coefficient) and pairwise covariances  $(\frac{1}{2} + o(1))n^{-2}$ . (iii) **An exact identity and a reduction**: the switching multigraph of Cavenagh–Greenhill–Wanless yields an *exact* identity expressing adjacent completion ratios through a single structural probability: for a split of a part  $\alpha + \beta$  into  $\alpha \neq \beta$ ,  $C_n(\lambda)/C_n(\mu) = \frac{1}{2}(1 + \bar{q})$  where  $\bar{q} \in [0, 2]$  is twice the probability that an explicit “post-flip” column pair is a  $B$ -pair. A row-swap involution forces the two constituent events of  $\bar{q}$  to have exactly equal probability. A *marked-pair identity* then eliminates the joining measure and the post-flip events altogether:  $2C_n(\lambda)/C_n(\mu) - 1$  equals the exact  $B:A$  odds of a single marked column pair  $\{j, \sigma^\alpha(j)\}$  in a single uniform square of  $S(\lambda)$  — an identity with a short self-contained proof, valid uniformly in  $\alpha, \beta$  (including the previously anomalous  $\alpha = \beta$  case) and verified exhaustively for  $n \leq 7$ . Modulo an equidistribution hypothesis for this single bit — verified exhaustively for  $n \leq 6$ , exactly tabulated for  $n \leq 11$ , and tested

by Monte Carlo beyond — the weak (total-variation) form of Cameron’s conjecture follows, and with it the statement that almost all loops have multiplication group  $S_n$ . A transfer programme for the Kwan–Petrova–Sawhney re-randomisation machinery, whose quantifiers match the required  $o(1/\log n)$  budget, is laid out for this bit.

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# 1 The conjecture and its context

## 1.1 Setup and notation

A *Latin square* of order  $n$  is an  $n \times n$  array over the symbol set  $[n] = \{1, \dots, n\}$  in which every symbol occurs exactly once in each row and each column. Rows are permutations of  $[n]$  (row  $r$  maps  $j \mapsto L(r, j)$  after identifying columns with symbols). For two rows  $r, s$  the permutation  $\sigma_{r,s}$  defined by  $\sigma_{r,s}(L(r, j)) = L(s, j)$  is a *derangement* of  $[n]$ . We write  $\mathcal{D}_n$  for the set of derangements,  $D_n = |\mathcal{D}_n|$ .

Throughout,  $\sigma$  denotes the second row of a uniformly random Latin square whose first row is the identity; equivalently  $\sigma = \sigma_{1,2}$  under the convention row 1 = id. For a derangement  $\sigma$  let

$$N_n(\sigma) = \#\{\text{Latin squares with rows } 1, 2 = (\text{id}, \sigma)\}, \quad \mathbb{P}(\text{second row} = \sigma) = \frac{N_n(\sigma)}{\sum_{\tau \in \mathcal{D}_n} N_n(\tau)}.$$

Cycle types of derangements are partitions  $\lambda = (2^{\lambda_2} 3^{\lambda_3} \dots) \vdash n$  with no part 1; we write  $\kappa(\lambda) = \sum_{\ell} \lambda_{\ell}$  for the number of parts and  $\gamma(\lambda) = n! / \prod_{\ell} \lambda_{\ell}! \ell^{\lambda_{\ell}}$  for the number of derangements of type  $\lambda$ .

**Lemma 1.1** (type dependence and within-type uniformity).  *$N_n(\sigma)$  depends only on the cycle type of  $\sigma$ . Consequently, conditionally on its cycle type, the second row is uniform on the derangements of that type.*

*Proof.* For  $\pi \in S_n$ , relabelling symbols by  $\pi$  and then columns by  $\pi^{-1}$  is a bijection of Latin squares sending rows  $(\text{id}, \sigma) \mapsto (\text{id}, \pi \sigma \pi^{-1})$ .  $\square$

Following [3] we set, for  $\lambda \vdash n$  with no part 1,

$$C_n(\lambda) = \frac{N_n(\sigma)}{(n-2)!} \quad (\sigma \text{ of type } \lambda),$$

(the completions of a  $2 \times n$  rectangle fall into classes of size  $(n-2)!$  under permutation of the remaining rows), and define two probability measures on cycle types:

$$P_n(\lambda) = \frac{\gamma(\lambda) C_n(\lambda)}{\sum_{\mu} \gamma(\mu) C_n(\mu)}, \quad Q_n(\lambda) = \frac{\gamma(\lambda)}{D_n}.$$

$P_n$  is the law of the cycle type of  $\sigma$ ;  $Q_n$  is the law of the cycle type of a uniform derangement. By Lemma 1.1, for any set  $S \subseteq \mathcal{D}_n$  of derangements,

$$|\mathbb{P}(\sigma \in S) - |S|/D_n| \leq d_{\text{TV}}(P_n, Q_n), \quad (1)$$

so total variation at the level of cycle types controls total variation at the level of derangements; in fact the two distances are *equal*, because the two laws of  $\sigma$  share the same conditional (uniform) distribution within each type.

## 1.2 The conjecture

Cameron's blog post [1] states:

*The probability distribution on derangements defined as above (where the probability of a derangement is the probability that a random Latin square with first row the identity has that derangement as its second row) tends very rapidly to the uniform distribution as  $n$  tends to infinity.*

The weak form was formalised by Cavenagh, Greenhill and Wanless as

**Conjecture 1.2** ([3, Conj. 6.1]).  $d_{\text{TV}}(P_n, Q_n) = o(1)$  as  $n \rightarrow \infty$ .

Even Conjecture 1.2 is open. The purpose of these notes is (a) to propose a sharp quantitative form of the full conjecture, based on new exact and numerical analysis; (b) to isolate exact structure (a “layer” theorem for the third row, and an exact switching identity) that explains the observed rates; and (c) to give a reduction of Conjecture 1.2 to a single equidistribution hypothesis about an explicit local event, together with extensive verification.

### 1.3 Known results

We use the following throughout;  $[n]$ -asymptotics are as  $n \rightarrow \infty$ .

- **CGW switching bounds** [3]. For partitions  $\lambda, \mu$  adjacent in the split/join graph (i.e.  $\mu$  refines  $\lambda$  by splitting one part in two),  $\frac{1}{2}\gamma(\lambda)/\gamma(\mu) \leq |S(\lambda)|/|S(\mu)| \leq \frac{3}{2}\gamma(\lambda)/\gamma(\mu)$ , where  $S(\lambda)$  is the set of Latin squares whose first two rows have row-cycle type  $\lambda$ . Iterating:  $P_n(\lambda)$  agrees with  $Q_n(\lambda)$  within a factor exponential in  $\kappa(\lambda)$ , whence, e.g., almost all cycle types have  $P_n/Q_n \in [n^{-3/2}, n^{3/2}]$ , the single-cycle probability is  $2n^{-2} \leq P_n((n)) \leq 2n^{-2/3}$  (the upper bound under CGW(3/2), Remark 19.10), and  $\mathbb{P}(\sigma \text{ even}) \in [\frac{1}{4} - o(1), \frac{3}{4} + o(1)]$ . They also computed  $C_n$  *exactly* for  $n \leq 11$  [3, Tables 1–2]; those tables are the data source for §2.
- **Intercalates.** An *intercalate* is a  $2 \times 2$  Latin subsquare; the number of intercalates inside rows 1, 2 equals  $\lambda_2(\sigma)$  exactly. McKay–Wanless [15] and, much more precisely, Kwan–Sawhney–Simkin [12] determined the global intercalate count: whp  $\mathbf{N} = (1 + o(1))n^2/4$ , with large-deviation rates  $\exp(-\Theta(n^2))$  (lower tail) and  $\exp(-\Theta(n^{4/3} \log n))$  (upper tail).
- **Parities.** Kwan–Petrova–Sawhney [9] proved that the vector of row parities of a random Latin square converges in total variation to  $n$  independent fair coins (resolving a different conjecture of Cameron), introducing *stable intercalate switchings* and a quantitative comparison with the triangle-removal process. Their results do not imply Conjecture 1.2, but their machinery is discussed in §14.
- **Loops motivation.** Cameron [2] proved that almost all quasigroups have multiplication group  $S_n$  or  $A_n$ ; Häggkvist–Janssen [6] ruled out  $A_n$  (the proportion of all-even Latin squares is  $O(c^n)$ ,  $c < 1$ ); Luczak–Pyber [14] proved that almost no permutation lies in a transitive subgroup other than  $S_n, A_n$ . Conjecture 1.2 would transfer this to *loops* (reduced Latin squares) via (1): almost all loops would have multiplication group  $S_n$ , hence trivial character theory [3, Lemma 6.2].

**Remark 1.3** (reduced squares: the conditioning is invisible). A loop on  $[n]$  with identity 1 is a *reduced* Latin square (first row *and* first column the identity), so its second row satisfies the extra constraint  $\sigma(1) = 2$ , whereas  $P_n$  is defined from squares normalised in the first row only. This

costs nothing: sorting the rows of a first-row-normalised square by their first-column entries is an  $(n-1)!$ -to-one map onto reduced squares which is exchangeable in the rows  $2, \dots, n$ , so the second row of a uniform loop is distributed as  $\sigma$  *conditioned on*  $\{\sigma(1) = 2\}$ . By Lemma 1.1 both  $P_n$  and  $Q_n$  are uniform on each derangement class, and on any fixed class  $\sigma(1)$  is uniform on  $\{2, \dots, n\}$  (conjugate by permutations fixing 1); hence  $\mathbb{P}(\sigma(1) = 2 \mid \lambda) = \frac{1}{n-1}$  independently of the type, the cycle-type law of the loop's second row is *exactly*  $P_n$ , and the total-variation distance between the conditioned laws equals  $d_{\text{TV}}(P_n, Q_n)$  exactly. (Verified by complete enumeration at  $n = 5, 6$ : the three laws — second row, loop second row, conditioned second row — coincide as exact rationals.) All statements in these notes therefore apply verbatim to loops.

## 2 The refined picture: exact data analysis for $n \leq 11$

All statements in this section are *exact* computations from the CGW tables of  $C_n(\lambda)$ ,  $4 \leq n \leq 11$ , re-verified in two independent ways: the published fractions  $P_n(\lambda)$  are reproduced exactly, and  $\sum_{\lambda} \gamma(\lambda) C_n(\lambda) (n-2)! n! = L_n$  reproduces the known numbers of Latin squares up to  $n = 11$  [16].

### 2.1 Additivity over cycles

The completion counts are astonishingly rigid. Along the chain of types  $(n) \rightarrow (2, n-2) \rightarrow (2, 2, n-4) \rightarrow \dots$ , each added 2-cycle multiplies  $C_n$  by almost exactly the same factor:

| $n$ | step                            | $C_n(\mu')/C_n(\mu) - 1$   | $\times n^3$ |
|-----|---------------------------------|----------------------------|--------------|
| 10  | $(10) \rightarrow (2, 8)$       | $+1.320265 \times 10^{-3}$ | 1.3203       |
| 10  | $(2, 8) \rightarrow (2^2, 6)$   | $+1.321183 \times 10^{-3}$ | 1.3212       |
| 10  | $(2^2, 6) \rightarrow (2^3, 4)$ | $+1.323079 \times 10^{-3}$ | 1.3231       |
| 10  | $(2^3, 4) \rightarrow (2^5)$    | $+2.647253 \times 10^{-3}$ | 2.6473       |
| 11  | $(11) \rightarrow (2, 9)$       | $+9.609945 \times 10^{-4}$ | 1.2791       |
| 11  | $(2, 9) \rightarrow (2^2, 7)$   | $+9.620287 \times 10^{-4}$ | 1.2805       |
| 11  | $(2^2, 7) \rightarrow (2^3, 5)$ | $+9.610360 \times 10^{-4}$ | 1.2791       |
| 11  | $(2^3, 5) \rightarrow (2^4, 3)$ | $+9.750180 \times 10^{-4}$ | 1.2977       |

The one apparent exception, the last step at  $n = 10$ , changes the type by *two* 2-cycles ( $\lambda_2: 3 \rightarrow 5$ , removing a 4-cycle), and its increment is exactly twice the others; the last step at  $n = 11$  adds a 3-cycle as well, and the excess  $+1.4 \times 10^{-5}$  matches the fitted 3-cycle coefficient below. This motivates:

**Observation 2.1** (additive fit). Fit  $\log C_n(\lambda) = c_0(n) + \sum_{\ell \geq 2} h_n(\ell) \lambda_{\ell}$  by least squares (the gauge freedom  $h(\ell) \mapsto h(\ell) + t\ell$  is fixed by  $h_n(n) = 0$ ). Then, using all types at each order:

| $n$ | $h_n(2)$               | $h_n(3)$               | $h_n(4)$              | $\max_{\lambda}  \text{resid} $ |
|-----|------------------------|------------------------|-----------------------|---------------------------------|
| 10  | $1.323 \times 10^{-3}$ | $5.12 \times 10^{-5}$  | $-6.9 \times 10^{-6}$ | $3.2 \times 10^{-5}$            |
| 11  | $9.605 \times 10^{-4}$ | $1.544 \times 10^{-5}$ | $1.2 \times 10^{-6}$  | $1.8 \times 10^{-6}$            |

At  $n = 11$  the additive model reproduces all fourteen  $\log C_{11}(\lambda)$  values to within  $1.8 \times 10^{-6}$ , while the leading coefficient is  $9.6 \times 10^{-4}$ : additivity holds to relative precision  $\approx 0.2\%$ , and improves rapidly with  $n$ .

There is *no* detectable dependence on the sign of  $\sigma$ : adding a  $\pm \text{sgn}(\sigma)$  term to the fit yields a coefficient ( $4.5 \times 10^{-7}$  at  $n = 11$ ) below the residual noise. Consistently, the parity of  $\sigma$  is explained entirely by the derangement count disparity (see §3.1).

## 2.2 Scaling of the coefficients

The fitted  $h_n(2)$ , and two independent estimators of it — the chain increments above, and the tilt of the expected number of rows-(1, 2) intercalates,  $2n^3(\mathbb{E}_{P_n}[\lambda_2] - \frac{1}{2})$  — agree and scale as  $n^{-3}$ :

| $n$                                              | 6    | 7    | 8    | 9    | 10   | 11   |
|--------------------------------------------------|------|------|------|------|------|------|
| $n^3 h_n(2)$ -proxy (chain)                      | 10.3 | 2.01 | 1.71 | 1.16 | 1.32 | 1.28 |
| $2n^3(\mathbb{E}_{P_n} \lambda_2 - \frac{1}{2})$ | 22.0 | 0.93 | 2.55 | 1.06 | 1.33 | 1.28 |

Odd and even  $n$  approach the limit from different sides (as CGW already noticed for their coefficient of variation), and  $n \leq 11$  cannot pin the limit precisely; the two largest orders give

$$c_2 := \lim_{n \rightarrow \infty} n^3 h_n(2) \approx 1.30 \pm 0.05 \quad (\text{numerically; existence of the limit is conjectural}). \quad (2)$$

The 3-cycle coefficient is much smaller ( $n^4 h_n(3) = 0.69, 0.51, 0.23$  at  $n = 9, 10, 11$ ), consistent with  $h_n(3) = O(n^{-4})$  and, in view of the layer analysis of §5, conjecturally  $\asymp n^{-5}$ ; coefficients for  $\ell \geq 4$  are below the resolution of the  $n \leq 11$  data.

## 3 Precise forms of the conjecture

We can now state the proposed quantitative hierarchy. Recall  $c_2 \approx 1.30$  from (2).

**Conjecture 3.1** (refined law). *There are quantities  $h_n(\ell)$ ,  $2 \leq \ell \leq n$ , with*

$$\log C_n(\lambda) = c_0(n) + \sum_{\ell \geq 2} h_n(\ell) \lambda_\ell + O(n^{-4}) \quad \text{uniformly in } \lambda,$$

*such that  $h_n(2) = c_2 n^{-3}(1 + o(1))$  for a constant  $c_2 > 0$  (numerically  $c_2 \approx 1.30$ ), and, for each fixed  $\ell \geq 3$ ,  $h_n(\ell) = O(n^{-4})$  (conjecturally  $h_n(\ell) = c_\ell n^{1-2\ell}(1 + o(1))$ ).*

**Conjecture 3.2** (rates). *As  $n \rightarrow \infty$ :*

$$(a) \max_{\substack{\sigma, \tau \in \mathcal{D}_n \\ \lambda = (n)}} \frac{N_n(\sigma)}{N_n(\tau)} = 1 + \frac{c_2}{2} n^{-2}(1 + o(1)), \text{ the extremes being attained at } \lambda = (2^{\lfloor n/2 \rfloor} \dots) \text{ and } \lambda = (n);$$

$$(b) d_{\text{TV}}(P_n, Q_n) = \frac{c_2 e^{-1/2}}{2} n^{-3}(1 + o(1)) \approx 0.394 n^{-3}, \text{ and the same for the total variation distance between the law of the second row and the uniform distribution on } \mathcal{D}_n.$$

Part (b) follows from Conjecture 3.1 by a direct computation: to leading order  $P_n(\lambda) \propto Q_n(\lambda) e^{h_n(2)\lambda_2}$ ,  $\lambda_2$  is asymptotically Poisson(1/2) under  $Q_n$ , and  $\mathbb{E}|X - \mathbb{E}X| = e^{-1/2}$  for  $X \sim \text{Poisson}(1/2)$ . The exact  $n \leq 11$  data match both predictions strikingly well:

| $n$                           | 8    | 9    | 10   | 11   | predicted limits:                             |
|-------------------------------|------|------|------|------|-----------------------------------------------|
| $n^3 d_{\text{TV}}(P_n, Q_n)$ | 0.62 | 0.34 | 0.40 | 0.39 | 0.394 and $c_2/2 \approx 0.65$ ( $n$ even),   |
| $n^2 (\max / \min - 1)$       | 1.65 | 0.37 | 0.66 | 0.47 | smaller along odd $n$ (no type $(2^{n/2})$ ). |

**Remark 3.3** (“very rapidly”). The convergence is thus *polynomial*, with small constants — not exponential. At  $n = 7$  the extreme completion counts already differ by only 1.4% (Cameron’s observation), yet the ratio  $1 + \Theta(n^{-2})$  decays slowly thereafter: uniformity to 0.1% in sup-ratio requires  $n \gtrsim 36$ . For every *fixed* set  $S \subseteq \mathcal{D}_n$  of derangements, however,  $\mathbb{P}(\sigma \in S) = |S|/D_n + O(n^{-3})$  — rapid indeed.

### 3.1 Consequences and cross-checks

**Parity.** The number of even minus odd derangements of  $[n]$  is  $(-1)^{n-1}(n-1)$ , so under  $Q_n$ ,  $\mathbb{P}(\text{even}) - \frac{1}{2} = (-1)^{n-1}(n-1)/(2D_n)$ , of order  $n^{-1}e/(n-1)!$ . If Conjecture 3.2(b) holds *at the level of the sign functional* with its natural precision, then the same asymptotics should hold under  $P_n$ . The exact data agree:

| $n$                                           | 7                      | 8                      | 9                      | 10                     | 11                     |
|-----------------------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| $\mathbb{P}_{P_n}(\text{even}) - \frac{1}{2}$ | $+1.813 \cdot 10^{-3}$ | $-1.998 \cdot 10^{-4}$ | $+2.952 \cdot 10^{-5}$ | $-3.168 \cdot 10^{-6}$ | $+3.250 \cdot 10^{-7}$ |
| $(-1)^{n-1} \frac{n-1}{2D_n}$                 | $+1.618 \cdot 10^{-3}$ | $-2.360 \cdot 10^{-4}$ | $+2.996 \cdot 10^{-5}$ | $-3.371 \cdot 10^{-6}$ | $+3.405 \cdot 10^{-7}$ |
| ratio                                         | 1.12                   | 0.85                   | 0.985                  | 0.94                   | 0.95                   |

This suggests the sharp parity conjecture  $\mathbb{P}(\sigma \text{ even}) - \frac{1}{2} \sim (-1)^{n-1}(n-1)/(2D_n)$  — a statement finer, for this particular functional, than Conjecture 3.2(b), since the parity imbalance is super-exponentially small while  $d_{\text{TV}} \asymp n^{-3}$ . (It is the  $\sigma_{1,2}$ -analogue of the row-parity phenomena studied in [9].)

**Intercalates.** Since the intercalates inside rows  $(1,2)$  number exactly  $\lambda_2$ , symmetry over row pairs gives  $\mathbb{E}[\mathbf{N}] = \binom{n}{2} \mathbb{E}_{P_n}[\lambda_2]$  for the total intercalate count  $\mathbf{N}$ . Under  $Q_n$ ,  $\mathbb{E}[\lambda_2] = \binom{n}{2} D_{n-2}/D_n = \frac{1}{2} + O(1/(n-2)!)$ . Conjecture 3.1 thus predicts a third-order term in the intercalate expectation:

$$\mathbb{E}[\mathbf{N}] = \frac{n^2}{4} - \frac{n}{4} + \frac{c_2 + o(1)}{4n}.$$

The known asymptotics  $\mathbb{E}[\mathbf{N}] \sim n^2/4$  [12] match the leading term; the finer terms appear to be new and give an independent interpretation of  $c_2$ .

**Loops.** By (1) and §1.3, even the weak Conjecture 1.2 implies that almost all loops have multiplication group  $S_n$  and trivial character theory; Conjecture 3.2(b) would give an  $O(n^{-3})$  (plus Łuczak–Pyber) error term.

## 4 The third-row layer: exact results

Why  $n^{-3}$ ? The mechanism becomes visible if one builds the Latin square row by row. Write

$$N_n(\sigma) = \prod_{k=2}^{n-1} \frac{N_n^{(k+1)}(\sigma)}{N_n^{(k)}(\sigma)}, \quad N_n^{(k)}(\sigma) = \#\{k \times n \text{ Latin rectangles with rows } 1, 2 = (\text{id}, \sigma)\},$$

so that  $\log N_n(\sigma)$  is a sum of  $n-2$  *layer contributions*. The first layer is exactly solvable.

For  $k=2$ ,  $N_n^{(3)}(\sigma) = \text{per}(J - I - P_\sigma)$  counts the permutations *discordant* with both  $\text{id}$  and  $\sigma$ . The forbidden bipartite graph  $I \cup P_\sigma$  is a disjoint union of even cycles, one cycle  $C_{2\ell}$  per  $\ell$ -cycle of  $\sigma$ ; by inclusion–exclusion

$$\text{per}(J - B) = \sum_{k \geq 0} (-1)^k m_k(B) (n-k)!, \quad (3)$$

where  $m_k(B)$  is the number of  $k$ -matchings of  $B$ , and the matching generating polynomial  $M_B(x) = \sum_k m_k(B)x^k$  factorises over the cycles, with

$$M_{C_m}(x) = \alpha_+^m + \alpha_-^m, \quad \alpha_{\pm} = \frac{1}{2}(1 \pm \sqrt{1+4x}), \quad \alpha_+ \alpha_- = -x.$$

**Lemma 4.1** (fusion; Touchard–Riordan). *For  $a \geq b \geq 1$ ,  $M_{C_{2a}}(x) M_{C_{2b}}(x) = M_{C_{2(a+b)}}(x) + x^{2b} M_{C_{2(a-b)}}(x)$ , with the convention  $M_{C_0} \equiv 2$ .*

*Proof.* Expand both products in  $\alpha_{\pm}$ :  $(\alpha_+^{2a} + \alpha_-^{2a})(\alpha_+^{2b} + \alpha_-^{2b}) = \alpha_+^{2(a+b)} + \alpha_-^{2(a+b)} + (\alpha_+ \alpha_-)^{2b} (\alpha_+^{2(a-b)} + \alpha_-^{2(a-b)})$ , and  $(\alpha_+ \alpha_-)^{2b} = x^{2b}$ .  $\square$

This is classical: in Riordan’s normalisation of *associated* ménage polynomials  $m_k(x) = x^k M_{C_{2k}}(-x^{-1})$  one has the Chebyshev form  $m_k(x) = 2 T_{2k}(\sqrt{x}/2)$  and the factor-free product rule  $m_j m_k = m_{j+k} + m_{j-k}$  [19, Ch. 8, eqs. (24)–(25)].

Let  $M_n$  denote the ménage numbers  $0, 1, 2, 13, 80, 579, 4738, 43387, 439792, 4890741, \dots$  ( $n = 2, 3, \dots$ ), i.e.  $M_n = \text{per}(J - I - P_c)$  for an  $n$ -cycle  $c$ ; classical asymptotics (e.g. via (3)) give  $M_n = n! e^{-2} (1 - \frac{1}{n} + O(n^{-2}))$ .

**Theorem 4.2** (third row, exact; essentially Touchard–Riordan). *Let  $2 \leq \ell \leq n/2$  and let  $\sigma$  have cycle type  $(\ell, n - \ell)$ . Then*

$$N_n^{(3)}(\sigma) = M_n + M_{n-2\ell} \quad (\text{with } M_0 := 2 \text{ when } \ell = n/2).$$

*More generally, for any type  $\lambda \cup \{a, b\}$  with  $a \geq b$ ,  $a - b \geq 2$ ,*

$$N_n^{(3)}(\lambda \cup \{a, b\}) = N_n^{(3)}(\lambda \cup \{a + b\}) + N_{n-2b}^{(3)}(\lambda \cup \{a - b\}),$$

*and when  $a = b$  the last term is replaced by  $2 N_{n-2b}^{(3)}(\lambda)$ .*

*Proof.* Insert Lemma 4.1 into (3): the factor  $x^{2b}$  shifts the matching index by  $2b$ , turning  $(n - k)!$  into  $((n - 2b) - k')!$ , which reproduces the inclusion–exclusion sum (3) on the ground set of size  $n - 2b$  for the reduced type. (Verified by exhaustive computation for all cycle types for  $n \leq 25$ .)  $\square$

**Corollary 4.3** (per-cycle effect at the third row). *For fixed  $\ell \geq 2$ ,*

$$\rho_\ell(n) := \frac{N_n^{(3)}((\ell, n - \ell))}{N_n^{(3)}((n))} - 1 = \frac{M_{n-2\ell}}{M_n} = \frac{1}{(n)_{2\ell}} \left( 1 - \frac{2\ell}{n^2} + O(n^{-3}) \right) = n^{-2\ell} \left( 1 + \binom{2\ell}{2} \frac{1}{n} + O(n^{-2}) \right).$$

Numerically (exact rational computations up to  $n = 400$ , Richardson extrapolation):  $n^4 \rho_2 = 1 + 6/n + 21/n^2 + 42/n^3 + \dots$  and  $n^6 \rho_3 = 1 + 15/n + 134/n^2 + \dots$ , matching the corollary ( $\binom{4}{2} = 6$ ,  $\binom{6}{2} = 15$ ; the third coefficients pick up the  $-2\ell/n^2$  correction from the ménage asymptotics).

Moreover the *additivity* of §2.1 is already present, and provable, at this layer: iterating Theorem 4.2 shows that

$$\log N_n^{(3)}(\lambda) = \log M_n + \sum_{\ell \geq 2} \lambda_\ell \frac{M_{n-2\ell}}{M_n} + (\text{pair interactions of smaller order}),$$

and an exact least-squares check at  $n = 30$  (all 1039 cycle types) gives a maximal additivity residual  $1.1 \times 10^{-10}$  against a leading coefficient  $h^{(3)}(2) = M_{26}/M_{30} = 1.5 \times 10^{-6}$ .

**Remark 4.4** (attribution). Permutations discordant with two given permutations are a classical topic going back to Touchard [20] and Riordan [18], and Chapter 8, §3 of Riordan’s book [19] contains everything needed for Theorem 4.2: the rook polynomial of the cycle class  $(k) =$

$(1^{k_1}2^{k_2}\dots)$  factorises over cycles (his eq. (22)), the associated polynomials satisfy the Chebyshev product rule (eqs. (24)–(25)), and his Theorem 2 reduces the hit polynomial of any class to a linear combination  $\sum_j A_j(1-t)^j U_{n-j}(t)$  of ménage hit polynomials. Our identity  $N_n^{(3)}((\ell, n-\ell)) = M_n + M_{n-2\ell}$  is the evaluation at  $t = 0$  of the case  $m_\ell m_{n-\ell} = m_n + m_{n-2\ell}$ , and the general recursion of Theorem 4.2 follows from eq. (26) there; Riordan attributes the substance to Touchard [20]. What we add in this section is only the quantitative reading relevant to Cameron’s conjecture: the per- $\ell$ -cycle effect  $\rho_\ell(n) = M_{n-2\ell}/M_n \sim n^{-2\ell}$  with its full expansion, and the additivity of  $\log N^{(3)}$  over cycles. Section 8.4 of [19] carries the classical theory one layer further in aggregate form — the three-line Latin rectangle count  $K(3, n) = \sum_k \binom{n}{k} D_{n-k} D_k U_{n-2k}$  — and the “ $k$  discordant permutations” literature (Moser’s very reduced  $4 \times n$  rectangles [17]; Whitehead [21]) is the natural classical toolkit for the layer-4 analogue raised in §20.

## 5 Layer accumulation: the $n^{-3}$ mechanism

Corollary 4.3 gives a per-2-cycle effect  $\asymp n^{-4}$  at the first layer. The full effect is  $h_n(2) \asymp n^{-3}$  — precisely  $n$  times larger. This suggests that *every* layer contributes  $\asymp n^{-4}$ , the influence of the fixed first two rows persisting (neither amplified nor washed out) as the rectangle grows. The fourth layer can still be computed exactly (summing  $\text{per}(J - I - P_\sigma - P_\tau)$  over all third rows  $\tau$ ; Ryser–Gray-code permanents):

| $n$ | 2-cycle effect              |                        | 3-cycle effect              |                        |
|-----|-----------------------------|------------------------|-----------------------------|------------------------|
|     | layer 3 ( $= M_{n-4}/M_n$ ) | layer 4                | layer 3 ( $= M_{n-6}/M_n$ ) | layer 4                |
| 8   | $4.221 \times 10^{-4}$      | $5.015 \times 10^{-4}$ | 0                           | $-7.0 \times 10^{-5}$  |
| 9   | $2.996 \times 10^{-4}$      | $3.134 \times 10^{-4}$ | $2.305 \times 10^{-5}$      | $1.974 \times 10^{-5}$ |
| 10  | $1.819 \times 10^{-4}$      | $1.872 \times 10^{-4}$ | $4.548 \times 10^{-6}$      | $4.093 \times 10^{-6}$ |
| 11  | $1.184 \times 10^{-4}$      | $1.205 \times 10^{-4}$ | $2.658 \times 10^{-6}$      | $2.469 \times 10^{-6}$ |

(The layer- $k$  effect is  $\log[L_k((\ell, n-\ell))/L_k((n))]$  up to lower order, with  $L_k(\lambda) = N^{(k+1)}/N^{(k)}$ ; at  $n = 9$  the exact 3-cycle layer-3 value is  $1/M_9 = 1/43387$ , a pleasing instance of Theorem 4.2.)

The layer-4 effect equals the layer-3 effect within 1–5% at  $n = 10, 11$ ; the layer-5 effect (now exactly computed for  $n \leq 9$ , §8) equals the layer-4 effect within 3% at  $n = 9$ . Summing the exactly known layers 3, 4 and comparing with the exactly known total at  $n = 11$ : layers 3 + 4 contribute  $2.39 \times 10^{-4}$  of  $h_{11}(2) = 9.61 \times 10^{-4}$ ; the remaining seven layers average  $1.03 \times 10^{-4} \approx (1.51) n^{-4}$  — the same order, declining towards the final (forced) layers. Writing the conjectural layer profile as  $t_k(n) = \gamma(k/n) n^{-4}(1 + o(1))$  with  $\gamma(0) = 1$  (by Corollary 4.3), the constant in (2) is

$$c_2 = \int_0^1 \gamma, \quad \gamma(0) = 1.$$

Determining  $\gamma$  — e.g. via permanent asymptotics for complements of random  $k \times n$  rectangles containing a planted intercalate — is, we believe, the most promising route to the sharp constant. The same mechanism predicts  $h_n(\ell) \asymp n \cdot n^{-2\ell} = n^{1-2\ell}$  for each fixed  $\ell \geq 3$ , which is consistent with (though not yet demanded by) the  $n \leq 11$  data.

**Remark 5.1** (a finite-size caveat on  $c_2$ ). The estimate  $c_2 \approx 1.30$  in (2) is the value of  $n^3 h_n(2)$  at  $n = 10, 11$ . The layer analysis shows that this quantity carries a substantial finite-size inflation: the early-layer weights track  $1/(n)_4 = n^{-4}(1 + 6/n + \dots)$ , whose  $n \leq 11$  values exceed  $n^{-4}$  by

50–70%. If, say,  $\gamma \equiv 1$ , then  $n^3 h_n(2) \approx (n-2)n^3/(n)_4$  would still be  $\approx 1.5$  at  $n = 11$  while converging to  $c_2 = 1$ . The exact totals at  $n = 11$  (layer sum  $14.1 n^{-4}$  against  $9 \times 1.73 n^{-4} = 15.6 n^{-4}$  for nine constant-weight layers) show the late layers contributing  $\approx 0.9$  of the early-layer weight on average at that size. We therefore regard the *limit*  $c_2$  as substantially uncertain — plausibly anywhere in  $[0.8, 1.5]$  — while  $n^3 h_n(2) \approx 1.3$  remains an accurate description of the exact data at  $n = 10, 11$  (and, through Conjecture 3.2(b), of  $d_{TV}$  at those orders). Determining  $\gamma$  in the bulk is the key open computation; §6 settles its behaviour at the fourth layer.

## 6 A pair calculus for the fourth row

The fourth layer is governed by  $N_n^{(4)}(\sigma) = \sum_{\tau} \text{per}(J - B_2 - P_{\tau})$ , a sum over third rows of permanents of complements of *three*-regular boards, for which no product formula is available. The following identity nevertheless reduces everything to the Touchard–Riordan calculus, extended to punctured (path-and-cycle) boards.

For a partial matching  $M$  of the complement of  $B_2 = I \cup P_{\sigma}$  (a set of cells, no two in a row or column, all avoiding  $B_2$ ), let

$$R(M; \sigma) = \#\{\tau \in S_n : \tau \text{ avoids } B_2, \tau \supseteq M\}.$$

Deleting the rows and columns of  $M$  cuts the cycles of  $B_2$  into disjoint paths and cycles; with  $m_k$  the matching numbers of the resulting board and  $N = n - |M|$ ,

$$R(M; \sigma) = \sum_k (-1)^k m_k (N - k)!, \quad (4)$$

computable from the matching polynomials of paths ( $m_k(P_e) = \binom{e-k+1}{k}$ , the Chebyshev partner of (3)) and cycles. We verified (4) against exhaustive enumeration for all tested puncture sets at  $n = 6, 7$ .

**Proposition 6.1** (pair inclusion–exclusion).

$$N_n^{(4)}(\sigma) = \sum_{j \geq 0} (-1)^j \sum_{M \in \mathcal{M}_j} R(M; \sigma)^2,$$

where  $\mathcal{M}_j$  is the set of  $j$ -cell partial matchings of the complement of  $B_2$ .

*Proof.*  $N^{(4)}$  counts ordered pairs  $(\tau, \eta)$  of  $B_2$ -avoiding permutations with no common cell. Inclusion–exclusion over the common-cell set (which is automatically a partial matching) gives the sum; for fixed  $M$ , the pairs with  $\tau, \eta \supseteq M$  number  $R(M; \sigma)^2$  by independence. (Verified exactly at  $n = 6, 7$ : e.g.  $N_7^{(4)}((7)) = 83600$  both ways.)  $\square$

Probabilistically: if  $\tau, \eta$  are *independent* uniform  $B_2$ -avoiding rows and  $X = |\tau \cap \eta|$  their number of common cells, then with  $t_j = T_j/T_0$ ,  $T_j = \sum_{M \in \mathcal{M}_j} R(M)^2$ :

$$\Phi(\sigma) := \frac{N_n^{(4)}(\sigma)}{N_n^{(3)}(\sigma)^2} = \mathbb{P}(X = 0), \quad t_j = \mathbb{E} \binom{X}{j}, \quad \underbrace{r_4 - r_3}_{\text{layer-4 minus layer-3 effect}} = \Delta_{\lambda} \log \mathbb{P}(X = 0),$$

where  $\Delta_{\lambda}$  denotes the difference between  $\sigma$  of types  $(2, n-2)$  and  $(n)$ . The layer-4 effect thus equals the layer-3 effect *plus the cycle-type sensitivity of a pairwise-clash probability* — and the latter is very small:

**Observation 6.2** (near-uniformity of cell probabilities). Writing  $p_c = R(\{c\}; \sigma)/N^{(3)}(\sigma)$  for the probability that a random  $B_2$ -avoiding row passes through the admissible cell  $c$  (so  $\sum_c p_c = n$  over the  $n(n-2)$  admissible cells),

$$\mu := \mathbb{E}X = t_1 = \sum_c p_c^2 = \frac{n}{n-2} + n(n-2) \text{Var}_c(p_c) \geq \frac{n}{n-2},$$

and exact computation shows the variance term to be astonishingly small:  $t_1 - \frac{n}{n-2} \approx 2.1 \times 10^{-7}$  at  $n = 12$  (type (12)), decaying like  $n^{-6\pm}$  and depending on the cycle type only in its last digits. Equivalently, the cell probabilities are uniform to relative  $O(n^{-3})$ : at  $n = 20$  (type (20)) the exact profile is flat to  $10^{-8}$  across all cells except at  $\omega$ -distance 2 (and its mirror  $n-1$ ), where a single-cell deficit  $\approx -1.4 n^{-3}$  appears: the value  $\tau(i) = \sigma^2(i)$  is slightly disfavoured. (Both facts are provable by path fusion; we record them as computed.) This is another appearance of the  $n^{-3}$  scale of the master conjecture.

## 6.1 The fourth layer, extended

Truncations  $S_J = \sum_{j \leq J} (-1)^j T_j$  of Proposition 6.1 give layer-4 effects with rapidly cancelling errors: against the exact values (available for  $n \leq 11$  from §5), the  $J = 2$  and  $J = 3$  estimates bracket the truth within a few percent and their average is accurate to  $\approx 0.4\%$ . This extends the layer-4 table:

| $n$ | $r_3 = M_{n-4}/M_n$     | $r_4$                    | $r_4 \cdot (n)_4$ | $r_4/r_3$ |
|-----|-------------------------|--------------------------|-------------------|-----------|
| 8   | $4.2212 \times 10^{-4}$ | $5.0153 \times 10^{-4}$  | 0.843             | 1.188     |
| 9   | $2.9963 \times 10^{-4}$ | $3.1340 \times 10^{-4}$  | 0.948             | 1.046     |
| 10  | $1.8190 \times 10^{-4}$ | $1.8715 \times 10^{-4}$  | 0.943             | 1.029     |
| 11  | $1.1839 \times 10^{-4}$ | $1.2049 \times 10^{-4}$  | 0.954             | 1.018     |
| 12  | $8.0011 \times 10^{-5}$ | $8.074 \times 10^{-5} *$ | 0.959             | 1.009     |
| 13  | $5.5940 \times 10^{-5}$ | $5.629 \times 10^{-5} *$ | 0.966             | 1.006     |
| 14  | $4.0247 \times 10^{-5}$ | $4.043 \times 10^{-5} *$ | 0.971             | 1.005     |
| 15  | $2.9676 \times 10^{-5}$ | $2.978 \times 10^{-5} *$ | 0.976             | 1.003     |
| 16  | $2.2351 \times 10^{-5}$ | $2.241 \times 10^{-5} *$ | 0.979             | 1.003     |
| 17  | $1.7149 \times 10^{-5}$ | $1.718 \times 10^{-5} *$ | 0.981             | 1.002     |
| 18  | $1.3374 \times 10^{-5}$ | $1.340 \times 10^{-5} *$ | 0.984             | 1.002     |

(\*: series estimate  $\frac{1}{2}(\text{eff}(2) + \text{eff}(3))$ , with a validated systematic of  $\approx -0.4\%$ ;  $n \leq 11$  rows are exact.) The evidence for

$$r_4(n) = \frac{1 + o(1)}{(n)_4} = r_3(n) (1 + O(n^{-1})) \quad (5)$$

is now overwhelming; the exact  $n \leq 11$  values show the relative excess  $r_4/r_3 - 1$  decaying from 4.6% ( $n = 9$ ) to 1.8% ( $n = 11$ ), already at the boundary of the series systematics beyond that. The decomposition of  $r_4 - r_3 = \Delta_\lambda \log \mathbb{P}(X = 0)$  through the truncations shows the difference is *not* dominated by  $\Delta\mu$  (which contributes only  $\approx 5\%$  of it at  $n = 11$ ) but spread across the higher coincidence statistics — consistent with its smallness and rapid decay. Section 7 turns this evidence into an exact “switch calculus” and a proof of (5) (Theorem 7.15).

## 7 An exact switch calculus for the fourth layer

This section develops the machinery for TODO 1 of the programme: a proof that the fourth-layer effect equals the third-layer effect to relative  $O(1/n)$ ,

$$r_4(n) = r_3(n) (1 + O(n^{-1})), \quad r_3(n) = \frac{M_{n-4}}{M_n}. \quad (6)$$

The idea is to compare the two cycle types through *adjacent representatives*: derangements  $\sigma$  (type  $(n)$ ) and  $\sigma'$  (type  $(2, n-2)$ ) whose boards differ by a single  $2 \times 2$  switch. All  $\sigma$ -dependent quantities then differ by four-term brackets of punctured counts, and these brackets are governed *exactly* by Wronskian-type identities for path matching polynomials — the punctured analogues of the fusion identity (Lemma 4.1). The programme is carried out in full: every step is an exact identity, an exhaustively verified finite case analysis, or a uniform Bonferroni estimate, and each is cross-checked against exact computation.

### 7.1 Adjacent representatives and the switch decomposition

Since  $N_n^{(k)}$  depends only on the cycle type (Lemma 1.1), we may choose

$$\sigma = (1\,2\,3\cdots n), \quad \sigma' = (1\,2)(3\,4\cdots n).$$

Then  $\sigma(i) = \sigma'(i)$  except at  $i = 2, n$ :

$$\begin{aligned} B_2(\sigma) \setminus B_2(\sigma') &= \{d_1, d_2\}, & B_2(\sigma') \setminus B_2(\sigma) &= \{e_1, e_2\}, \\ d_1 &= (2, 3), \, d_2 = (n, 1), & e_1 &= (2, 1), \, e_2 = (n, 3). \end{aligned}$$

Let  $B_0 = B_2(\sigma) \cap B_2(\sigma')$ , a board of  $2n-2$  cells whose bipartite graph is a disjoint union of two paths,

$$P^{(1)} : c_1 - r_1 - c_2 - r_2 \quad (3 \text{ edges}), \quad P^{(2)} : c_3 - r_3 - c_4 - \cdots - c_n - r_n \quad (2n-5 \text{ edges}),$$

where  $r_i, c_j$  denote row and column vertices. The four cells  $d_1, d_2, e_1, e_2$  avoid  $B_0$  and connect the four path-ends pairwise:  $d_1 = r_2 c_3$  and  $d_2 = r_n c_1$  join the two paths (*crosswise*), while  $e_1 = r_2 c_1$  and  $e_2 = r_n c_3$  close each path onto itself. Adding  $\{d_1, d_2\}$  to  $B_0$  yields the single cycle  $C_{2n}$  of  $B_2(\sigma)$ ; adding  $\{e_1, e_2\}$  yields the  $C_4 \cup C_{2n-4}$  of  $B_2(\sigma')$ .

Let  $A_0$  be the set of permutations avoiding  $B_0$ ,  $N_0 = |A_0|$ , and for cells  $z_1, \dots$  let  $R_0(M)$  denote the number of  $\tau \in A_0$  with  $\tau \supseteq M$  (a punctured count, computable by (4); all components of the punctured board are *paths*). Write

$$N_0^{(4)} = \#\{(\tau, \eta) \in A_0^2 : \tau \cap \eta = \emptyset\}, \quad G(z) = \#\{(\tau, \eta) \in A_0^2 : \tau \cap \eta = \emptyset, z \in \tau\},$$

and analogously  $G_2(z, z')$  ( $z, z' \in \tau$ ) and  $H(z, z')$  ( $z \in \tau, z' \in \eta$ ).

**Lemma 7.1** (switch decomposition; exact). *With  $\Delta N^{(4)} = N_n^{(4)}(\sigma') - N_n^{(4)}(\sigma)$ ,*

$$\begin{aligned} \Delta N^{(4)} &= 2 \text{Br}_G + 2[G_2(e_1, e_2) - G_2(d_1, d_2)] + 2[H(e_1, e_2) - H(d_1, d_2)], \\ \text{Br}_G &:= G(d_1) + G(d_2) - G(e_1) - G(e_2). \end{aligned} \quad (7)$$

*Proof.* A pair of disjoint rows avoiding  $B_2(\sigma)$  is a pair of disjoint members of  $A_0$  avoiding  $d_1, d_2$ ; inclusion–exclusion over the events  $d_i \in \tau \cup \eta$  gives  $N^{(4)}(\sigma) = N_0^{(4)} - U(d_1) - U(d_2) + U(d_1, d_2)$  with  $U(z) = 2G(z)$  (a cell lies in at most one row of a disjoint pair) and  $U(z, z') = 2G_2(z, z') + 2H(z, z')$  (using  $H(z, z') = H(z', z)$ , by swapping  $\tau, \eta$ ). Similarly for  $\sigma'$  with  $e_1, e_2$ ; subtract. (Verified by complete enumeration at  $n = 6$ :  $\Delta N^{(4)} = 10 = 2 \cdot 6 + 2 \cdot (-1) + 2 \cdot 0$ ; and  $n = 7$ :  $400 = 2 \cdot 224 + 2 \cdot 4 - 2 \cdot 28$ ; `switch.py`.)  $\square$

**Remark 7.2** (a hidden symmetry). The map  $\Psi(i, c) = (3 - c, 3 - i) \bmod n$  is an involution of the cell grid that preserves  $B_0$ , swaps  $d_1 \leftrightarrow d_2$ , and fixes  $e_1$  and  $e_2$ . It induces the involution  $\tau \mapsto \Psi(\tau)$ ,  $\Psi(\tau)(i) = 3 - \tau^{-1}(3 - i)$ , of  $A_0$ , which preserves disjointness of pairs. Hence  $G(d_1) = G(d_2)$  *exactly* (confirmed by exact computation at  $n = 8, \dots, 11$ ), while  $G(e_1)$  and  $G(e_2)$  are merely each  $\Psi$ -invariant.

## 7.2 The path toolkit

Let  $p_e(x) = \sum_k m_k(P_e)x^k$  be the matching polynomial of the path with  $e$  edges ( $p_0 = 1$ ,  $p_1 = 1 + x$ ,  $p_e = p_{e-1} + xp_{e-2}$ ), so that with  $\alpha_{\pm}$  as in §4,  $p_e = (\alpha_+^{e+2} - \alpha_-^{e+2})/(\alpha_+ - \alpha_-)$ . Extend to negative indices by this formula:

$$p_{-1} = 1, \quad p_{-2} = 0, \quad p_{-e} = -(-x)^{2-e} p_{e-4} \quad (e \geq 3).$$

**Lemma 7.3** (path Wronskian). *For all integers  $a, b$ ,*

$$p_a p_b - p_{a-1} p_{b+1} = -(-x)^{b+2} p_{a-b-3}. \quad (8)$$

*In particular  $p_2 p_{C-1} - p_1 p_C = -x^3 p_{C-4}$  and  $p_2 p_C - p_3 p_{C-1} = -x^4 p_{C-5}$ .*

*Proof.* Put  $d_m = (\alpha_+^m - \alpha_-^m)/(\alpha_+ - \alpha_-)$ , so  $p_e = d_{e+2}$ . In the product  $(\alpha_+ - \alpha_-)^2 (p_a p_b - p_{a-1} p_{b+1})$  the “diagonal” terms  $\alpha_{\pm}^{a+b+4}$  cancel and the cross terms give

$$-(\alpha_+ \alpha_-)^{b+2} \left[ \alpha_+^{a-b} + \alpha_-^{a-b} - \alpha_+^{a-b-1} \alpha_- - \alpha_-^{a-b-1} \alpha_+ \right] = -(-x)^{b+2} (\alpha_+ - \alpha_-) (\alpha_+^{a-b-1} - \alpha_-^{a-b-1}),$$

using  $\alpha_+ \alpha_- = -x$ ; divide by  $(\alpha_+ - \alpha_-)^2$ . (Verified symbolically for  $a, b \in [0, 11]$ ; `fusion_lemmas.py`.)  $\square$

The cycle polynomial decomposes over paths:  $c_m := \alpha_+^m + \alpha_-^m = d_{m+1} + x d_{m-1}$ , i.e.

$$M_{C_{2m}}(x) = p_{2m-1}(x) + x p_{2m-3}(x). \quad (9)$$

For a polynomial  $f$  and integer  $m \geq 0$  define the *IE evaluation*  $\langle f \rangle_m = \sum_k (-1)^k [x^k] f \cdot (m - k)!$ , so that the count of permutations of an  $m$ -set avoiding a board with matching polynomial  $f$  is  $\langle f \rangle_m$ , and

$$\langle x^s f \rangle_m = (-1)^s \langle f \rangle_{m-s}. \quad (10)$$

## 7.3 The empty bracket: an exact identity

Define, for a partial matching  $M$  avoiding  $B_0$  and compatible with a pin  $z$  (i.e.  $M \cup \{z\}$  is again a matching avoiding  $B_0$ ), the *bracket*

$$\text{Br}(M) = R_0(M \cup d_1) + R_0(M \cup d_2) - R_0(M \cup e_1) - R_0(M \cup e_2).$$

**Proposition 7.4** (empty bracket).  $\text{Br}(\emptyset) = M_{n-4}$ .

*Proof.* Pinning a cell  $(i, c)$  deletes the vertices  $r_i, c_c$  from  $B_0$ . Each of the four special vertices  $r_2, c_1, c_3, r_n$  has degree 1 in  $B_0$ , so each pin removes exactly two edges; the resulting boards are unions of two or three paths. Grouping the bracket as  $(R_0(d_1) - R_0(e_1)) + (R_0(d_2) - R_0(e_2))$ , both differences compare end-deletions of two paths within a common board, and Lemma 7.3 gives, as polynomial identities on the boards,

$$f_{d_1} - f_{e_1} = -x^3 p_{2n-9}, \quad f_{d_2} - f_{e_2} = -x^4 p_{2n-11}$$

(the computation is the generic case of Lemma 7.6 below with  $A = 2$ ,  $C = 2n - 5$ , resp.  $A'' = 3$ ,  $C'' = 2n - 6$ ). Hence by (10),

$$\text{Br}(\emptyset) = \langle p_{2n-9} \rangle_{n-4} - \langle p_{2n-11} \rangle_{n-5} = \langle p_{2(n-4)-1} + x p_{2(n-4)-3} \rangle_{n-4} = \langle M_{C_{2(n-4)}} \rangle_{n-4} = M_{n-4},$$

using (9). (Verified exactly for  $6 \leq n \leq 14$ ; `switch.py`, check 1.)  $\square$

Applying the same identity to single rows instead of pairs gives  $\Delta N^{(3)} = g(d_1) + g(d_2) - g(e_1) - g(e_2) = M_{n-4}$  with  $g(z) = R_0(\{z\})$  — an alternative, “local” proof of the  $\ell = 2$  case of Theorem 4.2: the two double-pin terms  $R_0(\{d_1, d_2\})$  and  $R_0(\{e_1, e_2\})$  coincide because both pin pairs delete the *same* four vertices  $r_2, r_n, c_1, c_3$ .

## 7.4 Structure of the bracket for general punctures

Fix a matching  $M$  avoiding  $B_0$  with  $\text{rows}(M) \cap \{2, n\} = \text{cols}(M) \cap \{1, 3\} = \emptyset$  (so that all four pins are compatible; the remaining cases are the *incompatible classes* treated in §7.7). Let  $G_M$  denote  $B_0$  punctured by  $M$  and let  $f_{M \cup z}$  be the matching polynomial of  $G_M$  minus the two vertices of  $z$ , so  $R_0(M \cup z) = \langle f_{M \cup z} \rangle_{n-|M|-1}$  and the bracket polynomial is

$$B_M = f_{M \cup d_1} + f_{M \cup d_2} - f_{M \cup e_1} - f_{M \cup e_2}.$$

**Lemma 7.5** (valuation).  $[x^0]B_M = [x^1]B_M = 0$ , and

$$[x^2]B_M = (\mathbf{1}_{c_2\text{-column} \notin M} - \mathbf{1}_{c_n\text{-column} \notin M})(\mathbf{1}_{r_3\text{-row} \notin M} - \mathbf{1}_{r_1\text{-row} \notin M}).$$

*In particular the valuation of  $B_M$  is at least 3 unless  $M$  meets exactly one of the columns  $\{2, n\}$  and exactly one of the rows  $\{1, 3\}$ , and it is always at least 2.*

*Proof.*  $[x^0]$ : each  $f$  has constant term 1.  $[x^1]$ :  $m_1$  of a vertex-deleted graph is  $E - \deg u - \deg v$  for nonadjacent  $u, v$ ; the four deletions use each of  $r_2, r_n, c_1, c_3$  twice with opposite signs, so both the  $E$ - and the degree-terms cancel.  $[x^2]$ :  $m_2(G \setminus \{u, v\}) = \binom{E - \deg u - \deg v}{2} - A(G \setminus \{u, v\})$  where  $A$  counts adjacent edge-pairs. Writing  $a = \deg r_2$ ,  $b = \deg c_3$ ,  $c = \deg r_n$ ,  $d = \deg c_1$  (degrees in  $G_M$ ), the binomial terms contribute  $(a - c)(b - d)$  by the quadratic identity  $f(a+b) + f(c+d) - f(a+d) - f(c+b) = (a - c)(b - d)$  for  $f(t) = \binom{E-t}{2}$ . Since the graph is bipartite, a row-vertex and a column-vertex have no common neighbour, so  $A(G \setminus \{u, v\}) = A(G) - \binom{\deg u}{2} - \binom{\deg v}{2} - \sum_{w \sim u} (\deg w - 1) - \sum_{w \sim v} (\deg w - 1)$ ; every one of these terms appears twice with opposite signs in the bracket, so the  $A$ -terms cancel identically. Finally  $\deg r_2 = \mathbf{1}[\text{column } 2 \notin \text{cols}(M)]$  (its only  $B_0$ -edge is the cell  $(2, 2)$ ), and similarly  $\deg r_n = \mathbf{1}[c_n \text{ free}]$ ,  $\deg c_3 = \mathbf{1}[r_3 \text{ free}]$ ,  $\deg c_1 = \mathbf{1}[r_1 \text{ free}]$ . (Exhaustive verification for all  $|M| \leq 2$  at  $n = 9, 10$ : minimal valuation 2, and valuation 2 occurs exactly in the predicted classes; `valuation_scan.py`.)  $\square$

**Lemma 7.6** (master formula). *Let  $M$  be as above. In  $G_M \setminus r_2$ , let  $A$  be the edge-length of the path-piece containing  $c_1$  and  $C$  the edge-length of the piece containing  $c_3$ ; in  $G_M \setminus r_n$ , let  $A''$  and  $C''$  be the edge-lengths of the pieces containing  $c_1$  and  $c_3$ . Then, as an identity of polynomials,*

$$B_M = (-x)^{A+1} p_{C-A-2} \cdot f_1^{\text{rest}} - (-x)^{A''+1} p_{C''-A''-2} \cdot f_2^{\text{rest}}, \quad (11)$$

where  $f_i^{\text{rest}}$  are the matching polynomials of the remaining pieces. Moreover  $A \leq 2$  and  $A'' \leq 3$  always, and in the window-avoiding case —  $M$  meets none of rows 1, 3, columns 2,  $n$ , and no piece-end lies within distance 8 of  $c_3$  or  $r_n$  — one has  $A = 2$ ,  $A'' = 3$ ,  $C'' \in \{C, C-1\}$ , and

$$B_M = -x^3 p_{C-4} p_D \prod_{\text{mid}} -x^4 p_{C-5} p_{D-1} \prod_{\text{mid}}, \quad (12)$$

with  $D$  the edge-length of the piece containing  $r_n$  and  $\prod_{\text{mid}}$  the product over the interior pieces.

*Proof.* Group  $B_M = (f_{M \cup d_1} - f_{M \cup e_1}) + (f_{M \cup d_2} - f_{M \cup e_2})$ . The first difference deletes  $r_2$  from  $G_M$  and then compares deletion of the end-vertices  $c_3$  (in its piece of length  $C$ ) and  $c_1$  (in its piece of length  $A$ ); these lie in distinct components (they descend from the distinct components  $P^{(2)}, P^{(1)}$  of  $B_0$ ), so the difference is  $[p_{C-1} p_A - p_{C P A-1}] \cdot f_1^{\text{rest}}$ , which is  $(-x)^{A+1} p_{C-A-2} f_1^{\text{rest}}$  by (8). The second difference is handled identically after deleting  $r_n$  (with the roles of  $c_1, c_3$  reversed, whence the minus sign).  $A \leq 2$ ,  $A'' \leq 3$  because every piece of  $P^{(1)}$  has at most 3 edges. In the window-avoiding case  $P^{(1)}$  is intact, so  $A = 2$  ( $P^{(1)} \setminus r_2 = c_1 r_1 c_2$ ) and  $A'' = 3$ ; deleting  $r_n$  shortens only the  $r_n$ -piece, so  $C'' = C$  if the  $c_3$ - and  $r_n$ -pieces are distinct ( $C'' = C - 1$  if  $M$  leaves  $P^{(2)}$  uncut). Substituting gives (12). (Verified as exact integer identities on random generic  $M$  with  $1 \leq |M| \leq 4$  at  $n = 9, 11, 13$ ; `fusion_lemmas.py`.)  $\square$

## 7.5 A uniform estimate for punctured path boards

**Lemma 7.7** (uniform punctured estimate). *There are absolute constants  $m_0, C_0$  with the following property. Let  $F$  be a disjoint union of paths and cycles with  $E$  edges, considered as a board in an  $m \times m$  grid,  $E \leq 2m$ ,  $m \geq m_0$ . Then*

$$\langle M_F \rangle_m = m! e^{-E/m} (1 + \theta), \quad |\theta| \leq C_0/m.$$

*Proof.* Write  $S = \langle M_F \rangle_m / m! = \sum_k (-1)^k m_k / (m)_k$ . By Bonferroni (the summands are the inclusion–exclusion of the events “the permutation hits a fixed cell of  $F$ ”), the partial sums alternate around  $S$ , so for every  $K$ ,  $|S - \sum_{k < K} (-1)^k m_k / (m)_k| \leq m_K / (m)_K$ . For  $k \leq K := \lceil 2 \log m \rceil$  we use the two-sided bounds  $\frac{1}{k!} \prod_{i < k} (E - 3i) \leq m_k \leq E^k / k!$  (each edge of a max-degree-2 graph is adjacent to at most two others) and  $(m)_k = m^k e^{-k^2/2m + O(k^3/m^2)}$ , giving  $m_k / (m)_k = \frac{(E/m)^k}{k!} (1 + \delta_k)$  with  $|\delta_k| \leq C k^2 (1/E + 1/m)$  when  $3K^2 \leq E$ , and  $m_k / (m)_k \leq \frac{(E/m)^k}{k!} e^{k^2/m}$  always. Summing,  $\sum_{k < K} (-1)^k \frac{(E/m)^k}{k!} (1 + \delta_k) = e^{-E/m} + O(1/m)$  (the weighted error  $\sum_k \frac{(E/m)^k}{k!} k^2 (1/E + 1/m) = O(1/m)$  uniformly in  $E \leq 2m$ , including small  $E$ ), the truncation tails of the exponential series and the Bonferroni tail being  $O((2e/K)^K) = O(m^{-2})$ . If  $E < 3K^2$  the same computation with the crude bound  $|\delta_k| \leq C k^2 / E$  for  $k \leq \sqrt{E/3}$  and Bonferroni truncation there gives the claim directly.  $\square$

**Corollary 7.8** (weights and moments). *For  $n \geq n_0$  and any matching  $M$  avoiding  $B_0$  with  $|M| = j \leq n/2$ :  $q(M) := R_0(M)/N_0 \leq 2e^{6j/n}/(n)_j$ , and consequently*

$$\sum_{j \geq 0} (j+1)^4 \sum_{M \in \mathcal{M}_j} q(M)^2 \leq C_1 \quad (\text{absolute constant}).$$

Moreover  $N_0^{(4)} = \sum_M (-1)^{|M|} R_0(M)^2$  (pair inclusion–exclusion, Proposition 6.1 on  $B_0$ ), and  $c_2^{-1} N_0^2 \leq N_0^{(4)} \leq N_0^2$  for an absolute  $c_2$  and  $n \geq n_0$ .

*Proof.* The weight bound follows from Lemma 7.7 applied to numerator and denominator ( $E$  changes by at most  $4j$  under puncturing). The moment bound follows from  $|\mathcal{M}_j| \leq (n(n-2))^j/j!$ . The lower bound on  $N_0^{(4)}$  follows from Bonferroni in the pair inclusion–exclusion together with  $t_j := \sum_{\mathcal{M}_j} q(M)^2 = \mu^j/j! (1 + O((j+1)/n))$  for  $j \leq 5$ ,  $\mu = t_1 \in [1, 1.5]$ , which is a direct consequence of Lemma 7.7.  $\square$

## 7.6 The generic stratum

**Proposition 7.9** (generic per-puncture estimate). *There are  $n_0, C$  such that for  $n \geq n_0$ , every window-avoiding  $M$  with  $|M| = j \leq n/2$  satisfies*

$$\langle B_M \rangle_{n-1-j} = \frac{M_{n-4}}{M_n} R_0(M) (1 + \delta_M), \quad |\delta_M| \leq C \frac{j+1}{n}.$$

*Proof.* By (12) and (10),  $\langle B_M \rangle_{n-1-j} = \langle p_{C-4} p_D \prod_{\text{mid}} \rangle_{n-4-j} - \langle p_{C-5} p_{D-1} \prod_{\text{mid}} \rangle_{n-5-j}$ . Apply Lemma 7.7 to both terms and to  $R_0(M) = \langle p_3 p_C p_D \prod_{\text{mid}} \rangle_{n-j}$ ; the edge counts of the two bracket boards are  $E(G_M) - 7$  and  $E(G_M) - 9$ , so the exponential factors agree with that of  $R_0(M)$  up to  $e^{O((j+1)/n)}$ , while the factorial prefactors give

$$\frac{(n-4-j)! - (n-5-j)! e^{O(1/n)}}{(n-j)!} = \frac{1}{(n-j)_4} \left(1 + O\left(\frac{1}{n-j}\right)\right).$$

Since  $M_{n-4}/M_n = (n)_4^{-1} (1 + O(n^{-2}))$  and  $(n)_4/(n-j)_4 = 1 + O(j/n)$  for  $j \leq n/2$ , the claim follows.  $\square$

## 7.7 Boundary and incompatible classes; the remaining lemma

Expanding each  $G$ -term of  $\text{Br}_G$  by pair inclusion–exclusion over common cells  $M \subseteq \tau \cap \eta$  (which automatically avoid the pinned row and column) gives the exact stratified form

$$\text{Br}_G = \sum_{j \geq 0} (-1)^j S_j, \quad S_j = \sum_{M \in \mathcal{M}_j} \left[ \sum_{z \in \{d_1, d_2\}} R_0^\dagger(M \cup z) - \sum_{z \in \{e_1, e_2\}} R_0^\dagger(M \cup z) \right] R_0(M), \quad (13)$$

where  $R_0^\dagger(M \cup z) = R_0(M \cup z)$  if  $M \cup \{z\}$  is a matching,  $= R_0(M)$  if  $z \in M$ , and  $= 0$  otherwise. ( $S_0 = M_{n-4} N_0$  by Proposition 7.4; verified together with  $S_1, S_2, S_3$  against direct enumeration at  $n = 8, 9$ ; `switch_j1.py`.) The matchings  $M$  fall into three classes:

- (a) *window-avoiding*  $M$ : controlled by Proposition 7.9;
- (b) *compatible boundary*  $M$  (all pins compatible but  $M$  meets the window): write  $B_M = (f_{M \cup d_1} - f_{M \cup e_1}) + (f_{M \cup d_2} - f_{M \cup e_2})$ ; by Lemma 7.6 each group is a single  $x^{s_i}$ -shifted path product with  $s_1, s_2 \geq 2$ . If both  $s_i \geq 3$ , Lemma 7.7 gives  $|\langle B_M \rangle_{n-1-j}| \leq C(n-j)^{-4} R_0(M)$ . If some  $s_i = 2$  then, unless  $M$  makes the two-hit configuration of Lemma 7.5 (extra measure  $O((j+1)^2/n^2)$ , crude bound  $C(n-j)^{-3} R_0(M)$ ), the  $x^2$  coefficients of the two groups cancel ( $[x^2] B_M = 0$ ), the two shifted products have edge counts differing by  $O(1)$ , and the difference of their Lemma 7.7 evaluations is again  $O((n-j)^{-4}) R_0(M)$ . With Lemma 7.10, the class-(b) total is  $O((j+1)^2/n) \cdot (M_{n-4}/M_n) \sum_{\mathcal{M}_j} R_0(M)^2$  per stratum;

- (c) *incompatible classes* ( $M$  meets row 2, row  $n$ , column 1, or column 3, or contains a pin cell): here the bracket degenerates to two-term or one-term combinations whose individual sizes are as large as  $(n-j)^{-2}R_0(M)$  (two-term,  $x^1$ -level) or  $(n-j)^{-1}R_0(M)$  (one-term,  $x^0$ -level).

The class-(c) terms are *not* small individually, but they cancel in structured pairs. The key device is a family of *line-swap pairings*: bijections  $M \mapsto \varphi(M)$  between two classes which leave the deleted row- and column-sets of the *pinned* counts unchanged — so that the pinned factors agree exactly — while the free factors  $R_0(M)$ ,  $R_0(\varphi M)$  differ by a Wronskian (Lemma 7.3), hence by a relative  $O(n^{-3})$ . We first record the measure bound used throughout.

**Lemma 7.10** (line measure). *For any fixed line (row or column)  $\ell$  and  $j \leq n/2$ ,  $\sum_{M \in \mathcal{M}_j, M \text{ meets } \ell} R_0(M)^2 \leq C \frac{j}{n} \tilde{C}^j N_0^2 / j!$  with absolute  $C, \tilde{C}$ .*

*Proof.* By Lemma 7.7,  $R_0(M) \leq C' R_0(M \setminus z) / (n-j)$  for  $z \in M$ ; sum over the  $\leq n$  cells  $z$  of  $\ell$  and over  $M' = M \setminus z \in \mathcal{M}_{j-1}$ , and use Corollary 7.8.  $\square$

**Lemma 7.11** (pairing). *The total contribution to  $\sum_j (-1)^j S_j$  of the classes (c-2) ( $M$  meets one pinned row and one pinned column) and (c-3) ( $M$  contains a pin cell), and the difference  $G_2(e_1, e_2) - G_2(d_1, d_2)$ , are all  $O(n^{-1}(M_{n-4}/M_n)N_0^2)$ .*

*Proof.* (c-2). Consider the class {row 2, column 3}: all pins except  $d_2$  are incompatible, so the stratum term is  $+R_0(M \cup d_2)R_0(M)$ . Map  $\varphi$ : replace the column-3 cell  $(\rho, 3) \in M$  by  $(\rho, 1)$ ; then  $\varphi M$  lies in the class {row 2, column 1}, whose only compatible pin is  $e_2$ , with term  $-R_0(\varphi M \cup e_2)R_0(\varphi M)$ . The deletions of  $M \cup d_2$  and  $\varphi M \cup e_2$  are the *same* rows and the same columns ( $\{3\} \cup \{1\}$  in both), so  $R_0(M \cup d_2) = R_0(\varphi M \cup e_2)$  *exactly* (verified on random instances; `switch.py` suite). The free factors differ by the deletion of  $c_3$  versus  $c_1$  from the same board with  $r_2$  already removed, which is the master-formula comparison:  $R_0(M) - R_0(\varphi M) = \langle -x^3 p_{C-4} \cdot f^{\text{rest}} \rangle$ , a relative  $O(n^{-3})$ . Since the class carries measure  $O((j+1)^2/n^2)$  (Lemma 7.10 twice) and the pinned factor a further  $(n-j)^{-1}$ , the paired total is  $O(n^{-2} \cdot n^{-1} \cdot n^{-3} \cdot n N_0^2)$  summed over strata, i.e.  $O(n^{-5})N_0^2 = O(n^{-1}(M_{n-4}/M_n)N_0^2)$ . The pairs {row  $n$ , col 1} / {row  $n$ , col 3} (surviving pins  $d_1/e_1$ ) are handled by the same map;  $\varphi$ -exceptional cases ( $\rho = 1$ , where  $(\rho, 1) \in B_0$ ) are paired instead by the row swap  $(2, \gamma) \mapsto (n, \gamma)$  against the class {row  $n$ , col 3}, with the same exact pinned-factor equality and an  $x^2$  Wronskian, and measure smaller by a further  $n^{-1}$ ; deeper exceptional cells carry measure  $O(n^{-4})$  and the crude one-term bound suffices. Exact totals confirm the mechanism: the two class totals (all strata  $j \leq 4$ ) are +4.277 vs -4.276 in units of  $M_{n-4}N_0$  at  $n = 8$ , and +3.9012 vs -3.9025 at  $n = 9$  — cancellation to relative  $O(n^{-2})$ .

(c-3). For  $M \ni d_1$  the surviving terms are  $+R_0(M)^2 + R_0(M \cup d_2)R_0(M)$  (pins  $e_1, e_2$  are incompatible with  $d_1$ 's row and column); for  $M \ni e_1$  they are  $-R_0(M)^2 - R_0(M \cup e_2)R_0(M)$  ( $d_1, d_2$  incompatible). Pair  $M \leftrightarrow M' = M \setminus d_1 \cup e_1$  (same rows; columns 3  $\leftrightarrow$  1). Again the double-deletion sets of  $M \cup d_2$  and  $M' \cup e_2$  coincide, so the cross terms cancel up to the free-factor Wronskian; and  $R_0(M)^2 - R_0(M')^2 = (R_0(M) - R_0(M'))(R_0(M) + R_0(M'))$  with  $R_0(M) - R_0(M')$  the *same*  $c_3$ -vs- $c_1$  comparison with  $r_2$  deleted, again  $-x^3 p_{C-4}$ -suppressed. With class measure  $O(n^{-2})$  this gives  $O(n^{-2} \cdot n^{-3})N_0^2$  per stratum,  $O(n^{-5})N_0^2$  in total; same for the  $d_2/e_2$  pairing. (Exact check at  $n = 9$ , strata  $j \leq 3$ : the four pin-cell class totals are -33.15, +33.00, -33.15, +33.18 in units of  $M_5 N_0$ , pairwise sums -0.15 and +0.03.)

*$G_2$ -difference.* In the pair inclusion-exclusion for  $G_2(d_1, d_2)$  and  $G_2(e_1, e_2)$  the compatible- $M$  ranges coincide (both pin pairs delete rows 2,  $n$  and columns 1, 3), and the pinned factors

$R_0(M \cup \{d_1, d_2\}) = R_0(M \cup \{e_1, e_2\})$  agree exactly; so all terms cancel except those with  $M$  containing a pin cell, which are handled exactly as in (c-3).  $\square$

**Lemma 7.12** (*H-difference*).  $H(e_1, e_2) - H(d_1, d_2) = O(n^{-1}(M_{n-4}/M_n)N_0^2)$ .

*Proof.* Both  $H$ 's expand over the *same*  $M$ -range ( $M$  must avoid rows  $2, n$  and columns  $1, 3$ , and cannot contain a pin cell since the pins lie in  $\tau \setminus \eta$  resp.  $\eta \setminus \tau$ ), with terms  $R_0(M \cup e_1)R_0(M \cup e_2) - R_0(M \cup d_1)R_0(M \cup d_2)$ . Writing the difference as  $R_0(M \cup e_1)[R_0(M \cup e_2) - R_0(M \cup d_2)] + R_0(M \cup d_2)[R_0(M \cup e_1) - R_0(M \cup d_1)]$ , each bracket is a single pin-difference, i.e. an  $x^s$ -shifted single path product by Lemma 7.6 with  $s \geq 3$  off the window and  $s \geq 2$  on it (Lemma 7.5 logic); with the two pinned factors each  $O(R_0(M)/(n-j))$  this gives  $O(R_0(M)^2 n^{-5})$  per generic  $M$  and  $O(R_0(M)^2 n^{-4})$  on the window (measure  $O((j+1)/n)$ ); summing with Corollary 7.8 gives the claim.  $\square$

It remains to control class (c-1):  $M$  meets exactly one pinned line, say row 2 (the cases row  $n$ , column 1, column 3 being its images under  $\Psi$  and under transposition of the roles of  $d$ - and  $e$ -cells), so that the bracket is the two-term  $R_0(M \cup d_2) - R_0(M \cup e_2)$ . The dangerous sub-case is a *doubly degenerate* window hit, where this bracket is only  $x^1$ -suppressed; it is handled by an exact *column-exchange identity* and one more line-swap pairing.

**Lemma 7.13** (column exchange). *Let  $M$  avoid rows  $n, 3$  and columns  $1, 3$  and meet both rows 1 and 2. Then, exactly,*

$$R_0(M \cup d_2) - R_0(M \cup e_2) = -R_0(M \cup \{(3, 1), (n, 3)\}).$$

*Dually, if  $M$  avoids rows  $n, 1$  and columns  $1, 3$  and meets rows 2 and 3, then  $R_0(M \cup d_2) - R_0(M \cup e_2) = +R_0(M \cup \{(1, 3), (n, 1)\})$ . And if  $M$  meets all three rows  $1, 2, 3$ , the bracket vanishes identically.*

*Proof.* Since rows  $1, 2$  meet  $M$ , the cells  $(1, 1), (1, 2), (2, 2)$  of  $B_0$  are punctured and column 1 carries no constraint in the reduced system; both counts live on the rows  $\neq n$ , and  $R_0(M \cup d_2)$  (resp.  $R_0(M \cup e_2)$ ) counts the assignments leaving column 1 (resp. column 3) unused. Given  $\tau$  with column 1 unused, move the row  $\tau^{-1}(3)$  from column 3 to column 1: the new cell is always admissible (column 1 is free of constraints), so this is an injection into the column-3-unused family; its image misses exactly the  $\tau'$  with  $\tau'(3) = 1$ , whose count is  $R_0(M \cup \{(3, 1), (n, 3)\})$  — note  $(3, 1)$  avoids  $B_0$  and the deleted lines, and leaving column 3 unused while deleting row  $n$  is the deletion set of  $\{(n, 3)\}$ . The dual statement is the same exchange with the roles of columns 1 and 3 reversed (column 3 is constraint-free exactly when its unique  $B_0$ -cell  $(3, 3)$  is punctured, i.e. row 3 meets  $M$ ). If rows  $1, 2, 3$  all meet  $M$ , both columns are constraint-free and the two counts coincide. (All three statements verified as exact integer identities on random class- $M$  at  $n = 9, 11, 13$ ; `switch.py` suite.)  $\square$

**Lemma 7.14** (boundary). *The total contribution of class (c-1) to  $\sum_j (-1)^j S_j$  is  $O(n^{-1}(M_{n-4}/M_n)N_0^2)$ .*

*Proof.* Sub-classify by the window hits accompanying the row-2 hit. (i) *No other special hit:*  $A'' = 2$  in Lemma 7.6, an  $x^3$ -shifted single product:  $|\text{bracket}| \leq C(n-j)^{-4}R_0(M) \cdot (n-j)$ , and Lemma 7.10 gives a class total  $O((j+1)/n)(M_{n-4}/M_n)$  per stratum, as in class (b). (ii) *One additional hit making  $A'' = 1$  (column 2) or shortening the  $c_3$ -side pieces:* valuation  $\geq 2$  with measure  $O((j+1)^2/n^2)$ ; crude bounds suffice. (iii) *Row 1 also hit ( $A'' = 0$ , the  $x^1$  case):* by Lemma 7.13 the stratum term is exactly  $-R_0(M)R_0(M \cup \{(3, 1), (n, 3)\})$ . Pair this class with the

class “rows 2, 3 hit” via the row swap  $\varphi : (1, \gamma) \mapsto (3, \gamma)$  on the row-1 cell:  $\varphi$  maps one class onto the other (for the  $O(1)$  exceptional columns  $\gamma$  where  $(3, \gamma) \in B_0$  or collisions occur, see below), and by the dual identity the partner’s term is  $+R_0(\varphi M)R_0(\varphi M \cup \{(1, 3), (n, 1)\})$ . The two pinned factors agree *exactly*: the deletion sets of  $M \cup \{(3, 1), (n, 3)\}$  and  $\varphi M \cup \{(1, 3), (n, 1)\}$  are the same rows and the same columns. Hence the paired contribution is  $[R_0(M) - R_0(\varphi M)]R_0(M \cup \{(3, 1), (n, 3)\})$ , and  $R_0(M) - R_0(\varphi M)$  compares the deletion of  $r_1$  versus  $r_3$  from a common board — a Wronskian difference, of relative size  $O(1/n)$  by Lemmas 7.3 and 7.7 (measured:  $\approx -2/n^2$ ). Since the paired class carries measure  $O((j+1)^2/n^2)$  and the pinned factor two powers, the total is  $O(n^{-2} \cdot n^{-2} \cdot n^{-1})N_0^2 = O(n^{-5})N_0^2$  per stratum family, summable by Corollary 7.8. The  $\varphi$ -exceptional column slices ( $\gamma$  with  $(3, \gamma) \in B_0$ , i.e.  $\gamma \in \{3, 4\}$ , or  $\gamma$  colliding with  $M$ ’s columns) are single-column sub-families of a class of size  $O(n^{-4})N_0^2$ , hence  $O(n^{-5})N_0^2$  by the crude bounds, as are the deeper multi-hit sub-classes; and the three-row class “rows 1, 2, 3” contributes 0 identically by Lemma 7.13. The mirror families (row  $n$ ; columns 1, 3) are identical under  $\Psi$  and under exchanging the roles of the  $d$ - and  $e$ -pins.  $\square$

Exact class computations confirm the mechanism: the rows- $\{1, 2\}$  family totals  $-0.0611 M_4 N_0$  at  $n = 8$  and  $-0.0528 M_5 N_0$  at  $n = 9$  ( $\approx -0.5/n$  in main-term units) while its individual strata are  $\Theta(n)$  larger; the pinned-factor equalities and both column-exchange identities are integer-exact on all tested instances (`switch.py` suite).

The full quantity is measured directly by exact permanent sums: with  $E^\pm[A]$  the mean number of disjoint partners over  $\tau \ni d$ -cells resp.  $e$ -cells,

$$\frac{\text{Br}_G N_0}{M_{n-4} N_0^{(4)}} = 0.8919, 0.8685, 0.8818, 0.8929 \quad (n = 8, 9, 10, 11),$$

and the deviation decomposes as  $m^+(E^+[A] - E^-[A])/(M_{n-4}E[A]) = -1.17/n \pm 2\%$  ( $n = 9, 10, 11$ ) plus  $(E^-[A] - E[A])/E[A] = O(n^{-2})$  (`anatomy.c`), exactly the  $1 - O(1/n)$  shape required by Theorem 7.15.

**Theorem 7.15** (fourth layer). *For  $n \geq n_0$ ,*

$$\log \frac{N_n^{(4)}((2, n-2))}{N_n^{(4)}((n))} = 2 \frac{M_{n-4}}{M_n} + O(n^{-5}), \quad \text{hence} \quad r_4(n) = r_3(n)(1 + O(n^{-1})).$$

*Proof.* Combine (7) and (13). Strata with  $j > n/2$  are bounded crudely by  $|\mathcal{M}_j| (n-j)! (n-j-1)!$ , which is superexponentially small against  $N_0^2$ ; assume  $j \leq n/2$  throughout. Classes (a) and (b): by Proposition 7.9 and the class-(b) bound, using Corollary 7.8 to sum unsigned errors,

$$\sum_j (-1)^j S_j^{(a)+(b)} = \frac{M_{n-4}}{M_n} \left[ \sum_M (-1)^{|M|} R_0(M)^2 \right] (1 + O(n^{-1})) = \frac{M_{n-4}}{M_n} N_0^{(4)} (1 + O(n^{-1})),$$

where re-completing the class-(c) matchings into the full signed sum  $\sum_M (-1)^{|M|} R_0(M)^2 = N_0^{(4)}$  costs another  $O(n^{-1})$ -relative term ( $O(1/n)$  of the pair measure lies in class (c)). Class (c) and the  $G_2$ -,  $H$ -differences are  $O(n^{-1}(M_{n-4}/M_n)N_0^2)$  by Lemmas 7.11, 7.12 and 7.14, and  $N_0^2 \leq e^{\mu+o(1)}N_0^{(4)}$  by Corollary 7.8. Hence  $\Delta N^{(4)} = 2(M_{n-4}/M_n)N_0^{(4)}(1 + O(1/n))$ . Finally  $N_0^{(4)} = N_n^{(4)}((n))(1 + O(1/n))$  (the switch decomposition for  $N^{(4)}(\sigma)$  against  $N_0^{(4)}$ , with  $G(z) \leq e^\mu N_0^{(4)}/(n-2)$ ), so  $\Delta \log N^{(4)} = 2M_{n-4}/M_n + O(n^{-5})$ , and  $r_4 - r_3 = \Delta \log N^{(4)} - 2 \Delta \log N^{(3)} = O(n^{-5}) = r_3 \cdot O(n^{-1})$ .  $\square$

**Remark 7.16** (general  $\ell$ ; the layer- $k$  programme). Choosing  $\sigma = (1 \cdots n)$  and  $\sigma' = (1 \cdots \ell)(\ell+1 \cdots n)$  gives boards differing by one switch at distance  $\ell$ ; the same calculus applies with  $P^{(1)}$  of length  $2\ell - 1$ , and the empty bracket is  $M_{n-2\ell}$  by the same Wronskian computation, predicting  $r_4^{(\ell)} = \rho_\ell(1 + O(1/n))$  for the  $\ell$ -cycle layer-4 effect. More significantly, nothing in the preceding subsections is specific to *two* added rows: the switch decomposition and the bracket calculus apply verbatim to  $N^{(k)}$  for any fixed  $k$ , with pair counts replaced by  $(k-2)$ -tuple counts and Lemma 7.7 applied to the correspondingly punctured boards. Section 8 carries this out in full for  $k = 5$ . The obstruction to  $r_k = r_3(1 + o(1))$  for growing  $k$  is that the boards seen by the last rows are no longer unions of two paths — this is where the cluster expansion of TODO 4 takes over.

## 8 The fifth layer: the switch calculus transfers

The remark closing Section 7 promised that nothing in the switch calculus is specific to *two* added rows. This section makes that promise concrete for the fifth layer: we derive the triple inclusion–exclusion identity, transfer the switch decomposition and the bracket calculus, and obtain

$$r_5(n) = r_3(n)(1 + O(n^{-1})), \quad (14)$$

where  $r_5$  is the fifth-layer effect  $\Delta_\lambda \log(N^{(5)}/N^{(4)})$  between types  $(2, n-2)$  and  $(n)$ . Exact computations (independent of the calculus) confirm the result at  $n = 8, 9$  with striking precision. Everything is written at the same working-note rigor as Section 7: exact identities and finite verifications are labelled as such, and the two genuinely new bookkeeping steps (the spectator factors in the strata analysis) are flagged explicitly.

### 8.1 A triple inclusion–exclusion identity

$N^{(5)}(\sigma)$  is the number of ordered triples  $(\tau, \eta, \theta)$  of  $B_2$ -avoiding permutations that are pairwise cell-disjoint. Expanding each of the three pairwise-disjointness indicators by inclusion–exclusion over subsets of that pair’s coincidence set — the three expansions are independent — gives the analogue of Proposition 6.1.

**Proposition 8.1** (triple inclusion–exclusion). *Let  $M_1, M_2, M_3$  range over matchings of the complement of  $B_2$  (the coincidence sets of  $(\tau, \eta)$ ,  $(\tau, \theta)$ ,  $(\eta, \theta)$  respectively), subject to compatibility: each pairwise union  $M_i \cup M_j$  is again a matching (shared cells are allowed; two distinct cells of  $M_i, M_j$  may not share a line). Then*

$$N^{(5)}(\sigma) = \sum_{(M_1, M_2, M_3)} (-1)^{|M_1|+|M_2|+|M_3|} R(M_1 \cup M_2) R(M_1 \cup M_3) R(M_2 \cup M_3), \quad (15)$$

with  $R(\cdot)$  the punctured counts of (4). Equivalently, in resummed form: compatibility of the triple is equivalent to  $U = M_1 \cup M_2 \cup M_3$  being a matching, and collecting the label multiplicities cell by cell,

$$N^{(5)}(\sigma) = \sum_U \sum_{U=U_{ab} \sqcup U_{ac} \sqcup U_{bc} \sqcup U_{abc}} (-1)^{|U_{ab}|+|U_{ac}|+|U_{bc}|} 2^{|U_{abc}|} R(A) R(B) R(C), \quad (16)$$

where  $A = U_{ab} \cup U_{ac} \cup U_{abc}$ ,  $B = U_{ab} \cup U_{bc} \cup U_{abc}$ ,  $C = U_{ac} \cup U_{bc} \cup U_{abc}$  are the constraint sets of  $\tau, \eta, \theta$ : every coincidence cell lies in at least two of the three constraint sets.

*Proof.* For fixed  $\tau, \eta, \theta$  avoiding  $B_2$ , the product of the three indicators  $\prod_{\text{pairs}} \mathbf{1}[\text{disjoint}]$  expands as  $\sum_{M_1 \subseteq \tau \cap \eta} (-1)^{|M_1|} \dots$ , independently in the three pairs; exchanging sums gives (15), the count of triples with  $\tau \supseteq M_1 \cup M_2$  etc. factorizing since the three rows are constrained separately (and vanishing unless each union is a matching). For (16): if two cells of the  $M_i$  share a line, compatibility of the relevant pair forces them equal, so pairwise compatibility is the same as  $U$  being a matching; each  $u \in U$  carries the nonempty label set  $T(u) \subseteq \{1, 2, 3\}$  of the  $M_i$  containing it, the constraint-set membership of  $u$  depends only on  $T(u)$  ( $|T| = 1$  gives exactly two memberships,  $|T| \geq 2$  gives all three), and summing  $(-1)^{|T|}$  over the label sets with a given membership pattern gives weight  $-1$  for each two-membership pattern and  $+3 - 1 = +2$  for the full pattern. (Both forms verified against brute-force enumeration of all triples at  $n = 5$ , and (16) additionally at  $n = 6, 7$ , both cycle types:  $N^{(5)}$  values 36; 4032, 4224; 2185152, 2198160. `triple_ie.py`.)  $\square$

**Remark 8.2** (the truncated series is slowly convergent). Truncating (16) at total order  $q = |M_1| + |M_2| + |M_3| \leq Q$  gives an alternating series whose natural parameter is the *total* coincidence intensity of the three pairs,  $3\mu \approx 6$  — three times that of the pair series of Section 6. Convergence in  $Q$  is correspondingly slower: at  $n = 8$  the  $Q \leq 3$  truncation underestimates the fifth-layer effect by 24% (exact anchor below), at  $n = 9$  by 6%; the truncation error shrinks rapidly with  $n$  ( $Q = 3$  values 0.922, 0.942, 0.964, 0.970, 0.974 at  $n = 10, \dots, 14$  in the  $r_5 \cdot (n)_4$  normalisation of the table below, each presumably a few percent low, approaching 1 monotonically as the layer-4 sequence did). We therefore use the truncated series only as a consistency check at larger  $n$  and rely on exact enumeration (`n5_direct2.c`) at  $n \leq 9$ ; the switch calculus below is what replaces series estimation as the source of the theorem. (`layer5_production.py`.)

## 8.2 The switch decomposition for triples

Adopt the setting of Section 7: adjacent representatives  $\sigma, \sigma'$ , common board  $B_0$  (two paths), pins  $d_1 = (2, 3)$ ,  $d_2 = (n, 1)$ ,  $e_1 = (2, 1)$ ,  $e_2 = (n, 3)$ , and  $A_0$  the set of  $B_0$ -avoiding permutations. Write

$$N_0^{(5)} = \#\{(\tau, \eta, \theta) \in A_0^3 \text{ pairwise disjoint}\}, \quad G^{(5)}(z) = \#\{\text{such triples with } z \in \tau\},$$

and analogously  $G_2^{(5)}(z, z')$  ( $z, z' \in \tau$ ) and  $H^{(5)}(z, z')$  ( $z \in \tau, z' \in \eta$ ).

**Lemma 8.3** (switch decomposition, fifth layer; exact). *With  $\Delta N^{(5)} = N^{(5)}(\sigma') - N^{(5)}(\sigma)$ ,*

$$\begin{aligned} \Delta N^{(5)} &= 3 \text{Br}_G^{(5)} + 3[G_2^{(5)}(e_1, e_2) - G_2^{(5)}(d_1, d_2)] + 6[H^{(5)}(e_1, e_2) - H^{(5)}(d_1, d_2)], \\ \text{Br}_G^{(5)} &:= G^{(5)}(d_1) + G^{(5)}(d_2) - G^{(5)}(e_1) - G^{(5)}(e_2). \end{aligned} \quad (17)$$

*Proof.* As for Lemma 7.1: inclusion–exclusion over the events  $d_i \in \tau \cup \eta \cup \theta$ , with  $U(z) = 3G^{(5)}(z)$  (choice of the row through  $z$ ) and  $U(z, z') = 3G_2^{(5)}(z, z') + 6H^{(5)}(z, z')$  (one row through both, or an ordered pair of distinct rows). Verified by complete enumeration at  $n = 6$  ( $\Delta N^{(5)} = 192 = 288 - 24 - 72$ ) and  $n = 7$  ( $13008 = 15384 + 2496 - 4872$ ); and at  $n = 8$  the pieces computed on  $B_0$  (`anatomy5.c`:  $3\text{Br}_G^{(5)} = 2649312$ ,  $3\Delta G_2^{(5)} = 159840$ ,  $6\Delta H^{(5)} = -596544$ ) sum to 2212608, exactly the value of  $\Delta N^{(5)}$  from the independent direct enumeration of  $N^{(5)}$  (`n5_direct2.c`); likewise at  $n = 9$ :  $1632687552 + 69975336 - 375847080 = 1326815808 = \Delta N^{(5)}$ .  $\square$

**Remark 8.4** ( $\Psi$  acts diagonally). The involution  $\Psi$  of Remark 7.2 preserves  $A_0$  and disjointness, hence acts on triples coordinatewise; consequently  $G^{(5)}(d_1) = G^{(5)}(d_2)$  *exactly* (confirmed at  $n = 7, 8, 9$ : e.g.  $G^{(5)}(d_1) = G^{(5)}(d_2) = 357845442448$  at  $n = 9$ ).

### 8.3 The stratified bracket; spectator factors

Expand  $G^{(5)}(z)$  by Proposition 8.1 on the board  $B_0$  with the pin  $z$  carried by  $\tau$ 's factor. In the notation of (16) ( $A = \tau$ 's constraint set,  $B, C$  the other two),

$$\text{Br}_G^{(5)} = \sum_A B_A w(A), \quad w(A) := \sum_{\substack{(M_1, M_2, M_3) \\ M_1 \cup M_2 = A}} (-1)^{|M_1|+|M_2|+|M_3|} R_0(B) R_0(C), \quad (18)$$

where  $B_A = \sum_{z \in \{d_1, d_2\}} R_0^\dagger(A \cup z) - \sum_{z \in \{e_1, e_2\}} R_0^\dagger(A \cup z)$  is *the same* four-term bracket as in (13), now indexed by  $A$ . Two exact identities anchor (18):

- (i)  $\sum_A R_0(A) w(A) = N_0^{(5)}$  (the regrouped (16) on  $B_0$ );
- (ii)  $w(\emptyset) = \sum_{M_3} (-1)^{|M_3|} R_0(M_3)^2 = N_0^{(4)}$  (the pair inclusion–exclusion on  $B_0$ ), so the  $A = \emptyset$  stratum equals  $M_{n-4} N_0^{(4)}$  *exactly* by Proposition 7.4.

(Both identities verified exactly on  $B_0$  at  $n = 6, 7$ ; `switch5.py, check3`.) Since  $B_A$  is unchanged, the entire bracket toolkit of Section 7 — Lemma 7.3, Lemma 7.5, Lemma 7.6, Proposition 7.9 — applies verbatim in the  $A$ -slot. What is new is the weight:  $R_0(M)$  is replaced by the spectator weight  $w(A)$ . The transfer rests on two observations.

*First observation.* Every cell of  $A$  lies in  $B$  or  $C$  (cells of  $M_1$  lie in  $B$ , cells of  $M_2$  in  $C$ ). Hence a line-swap pairing  $\varphi$  moving a cell of  $A$  (as in Lemma 7.11 or Lemma 7.13) moves the *same* cell in exactly one spectator factor when the cell has a single label, and in both when it is shared; the third factor is untouched in the first case. The pinned-factor equalities of the pairing lemmas — which involve only  $B_A$ , through the deleted row- and column-sets of the pinned counts — are therefore untouched, while the free-factor difference becomes, by the product rule,

$$R_0(B)R_0(C) - R_0(\varphi B)R_0(\varphi C) = [R_0(B) - R_0(\varphi B)]R_0(C) + R_0(\varphi B)[R_0(C) - R_0(\varphi C)],$$

each bracket a single line-swap difference controlled by Lemma 7.3 exactly as at the fourth layer, i.e. a relative  $O(n^{-3})$  per factor.

*Second observation (spectator stability).* The pairing must also re-index the spectator sum over  $M_3$ : an  $M_3$  compatible with  $(M_1, M_2)$  maps to one compatible with  $\varphi(M_1, M_2)$  by the identity map except on the  $O(1/n)$ -measure set of  $M_3$  meeting the two swapped lines, which is absorbed by (the  $B_0$ -analogue of) Lemma 7.10 applied inside  $w$ . Thus  $|w(A) - w(\varphi A)| = O(n^{-1}) \bar{w}(A)$  with  $\bar{w}$  the unsigned weight, and the paired-class cancellations of Lemmas 7.11, 7.12, 7.13 and 7.14 survive with the same  $O(n^{-1})$  headroom. Finally, Lemma 7.7 and Corollary 7.8 apply to  $R_0(B)$ ,  $R_0(C)$  individually (any max-degree-2 board), so the unsigned sums  $\sum_A |B_A| \bar{w}(A)$  obey the same stratum bounds as at the fourth layer with  $N_0^2$  replaced by an  $N_0^{(5)}$ -scale; the  $G_2$ - and  $H$ -differences of (17) are handled by the same pairings with one or two pins carried by spectator slots.

These two observations are the only new content; we record the outcome.

**Theorem 8.5** (fifth layer). *For  $n \geq n_0$ ,*

$$\log \frac{N_n^{(5)}((2, n-2))}{N_n^{(5)}((n))} = 3 \frac{M_{n-4}}{M_n} + O(n^{-5}), \quad \text{hence} \quad r_5(n) = r_4(n)(1 + O(n^{-1})) = r_3(n)(1 + O(n^{-1})).$$

*Proof.* Run the proof of Theorem 7.15 on (17) and (18), with Proposition 7.9 in the  $A$ -slot, weights  $w(A)$  in place of  $R_0(M)$ , and the two observations above supplying the paired-class and unsigned-sum bounds; the generic strata resum, via identity (i), to  $\text{Br}_G^{(5)} = (M_{n-4}/M_n) N_0^{(5)} (1 + O(1/n))$ , and the pin-pin differences of (17) are  $O(n^{-1}(M_{n-4}/M_n)N_0^{(5)})$ . Then  $N_0^{(5)} = N^{(5)}((n))(1 + O(1/n))$  as before, giving  $\Delta \log N^{(5)} = 3M_{n-4}/M_n + O(n^{-5})$ , and  $r_5 - r_3 = \Delta \log N^{(5)} - \Delta \log N^{(4)} - \Delta \log N^{(3)} = O(n^{-5})$ .  $\square$

## 8.4 Exact fifth-layer data

Direct enumeration (64/128-bit cell masks; `n5_direct.c`, `n5_direct2.c`) gives  $N^{(5)}$  exactly at  $n \leq 9$ ; the  $N^{(3)}$  and  $N^{(4)}$  values reproduce the known exact tables, certifying the enumeration.

| $n$ | $N^{(5)}((n))$ | $N^{(5)}((2, n-2))$ | $r_3(n)_4$ | $r_4(n)_4$ | $r_5(n)_4$ | $\frac{\Delta \log N^{(4)}}{2M_{n-4}/M_n}$ | $\frac{\Delta \log N^{(5)}}{3M_{n-4}/M_n}$ |
|-----|----------------|---------------------|------------|------------|------------|--------------------------------------------|--------------------------------------------|
| 6   | 4032           | 4224                | 0          | 2.454      | 14.29      | —                                          | —                                          |
| 7   | 2185152        | 2198160             | 1.450      | 2.560      | 0.976      | 1.382                                      | 1.146                                      |
| 8   | 1539331584     | 1541544192          | 0.709      | 0.842      | 0.862      | 1.094                                      | 1.134                                      |
| 9   | 1417404643008  | 1418731458816       | 0.906      | 0.948      | 0.976      | 1.023                                      | 1.041                                      |

Here  $r_k(n)_4$  abbreviates  $r_k \cdot n(n-1)(n-2)(n-3)$ , and  $r_k = \Delta_\lambda \log(N^{(k)}/N^{(k-1)})$ . At  $n = 8, 9$ :  $r_5/r_4 = 1.023, 1.030$  — the fifth-layer effect tracks the fourth to 3% already at  $n = 9$  (small  $n = 6, 7$  are off-trend, as for every other layer statistic). On the anatomy side,  $\text{Br}_G^{(5)} = 5128, 883104, 544229184$  at  $n = 7, 8, 9$  with  $\text{Br}_G^{(5)} N_0 / (M_{n-4} N_0^{(5)}) = 0.612, 0.696, 0.722$ , consistent with a  $1 - c/n$  approach with  $c \approx 2.5$  — about twice the fourth-layer constant  $c \approx 1.17$ , as the two spectator rows suggest.

**Remark 8.6** (all types at once). The direct enumeration is cheap enough at  $n \leq 8$  to cover *every* derangement type (`n5_types.c`). Writing  $\Delta_\lambda = \log N^{(k)}(\lambda) - \log N^{(k)}((n))$ , the exact  $n = 8$  data are:

| $\lambda$    | $\Delta_\lambda \log N^{(3)}$ | $\Delta_\lambda \log N^{(4)}$ | $\Delta_\lambda \log N^{(5)}$ | $\frac{\Delta \log N^{(4)}}{2\Delta \log N^{(3)}}$ | $\frac{\Delta \log N^{(5)}}{3\Delta \log N^{(3)}}$ |
|--------------|-------------------------------|-------------------------------|-------------------------------|----------------------------------------------------|----------------------------------------------------|
| (2, 6)       | $+4.220 \cdot 10^{-4}$        | $+9.234 \cdot 10^{-4}$        | $+1.436 \cdot 10^{-3}$        | 1.094                                              | 1.134                                              |
| (3, 5)       | 0                             | $-6.95 \cdot 10^{-5}$         | $-2.14 \cdot 10^{-4}$         | —                                                  | —                                                  |
| (4, 4)       | $+4.220 \cdot 10^{-4}$        | $+8.570 \cdot 10^{-4}$        | $+1.314 \cdot 10^{-3}$        | 1.015                                              | 1.038                                              |
| (2, 2, 4)    | $+1.266 \cdot 10^{-3}$        | $+2.714 \cdot 10^{-3}$        | $+4.213 \cdot 10^{-3}$        | 1.072                                              | 1.110                                              |
| (2, 3, 3)    | $+4.220 \cdot 10^{-4}$        | $+7.865 \cdot 10^{-4}$        | $+1.022 \cdot 10^{-3}$        | 0.932                                              | 0.807                                              |
| (2, 2, 2, 2) | $+2.951 \cdot 10^{-3}$        | $+6.384 \cdot 10^{-3}$        | $+1.006 \cdot 10^{-2}$        | 1.082                                              | 1.136                                              |

Three reads. (i) *Uniformity over types*: every 2-cycle-bearing type shows the same finite- $n$  factor  $\approx 1.09$  (layer 4) and  $\approx 1.11$ – $1.14$  (layer 5) as the adjacent pair (2, 6) vs. (8), with an  $O(1/n)$ -size spread — the per-type version of Theorems 7.15–8.5. (ii) *The  $M_{n-2\ell}$  pattern*: a 3-cycle has  $M_{n-6} = M_2 = 0$ , and indeed its layer-4,5 effects are not zero but an order smaller (and negative) than the 2-cycle effects. (iii) *The  $M_0 = 2$  convention*: for  $\lambda = (4, 4)$  the leading bracket is  $M_{n-2\ell} = M_0 = 2$ , the same value as the 2-cycle's  $M_4 = 2$  at  $n = 8$  — and indeed (4, 4) tracks (2, 6) closely at layers 3, 4 and 5, a three-layer confirmation of the convention entering Lemma 9.4. The deviation of (2, 3, 3) is itself structured: its layer-4 effect equals  $\Delta \log N^{(4)}(2, 6) + 2 \Delta \log N^{(4)}(3, 5)$  to  $2 \cdot 10^{-6}$  (0.3% of its value;  $1.3 \cdot 10^{-5}$ , i.e. 1.3%, at layer 5) — additivity over cycles at the fourth and fifth layers, within the accuracy that the neglected  $\ell \geq 5$  cycle terms allow one to test at a single  $n$ .

**Remark 8.7** (general  $k$ ; the profile at the origin). By induction on  $k$ , the identical transfer (a  $(k-2)$ -tuple inclusion–exclusion; one more spectator factor per layer) gives, for each *fixed*  $k$ ,

$$\Delta_\lambda \log N^{(k)} = (k-2) \frac{M_{n-4}}{M_n} + O_k(n^{-5}), \quad r_k(n) = r_3(n)(1 + O_k(1/n)),$$

with constants growing with  $k$  (each spectator contributes its own  $O(1/n)$  bookkeeping). In the language of the layer profile  $\gamma$  of Section 5,  $\gamma$  is flat at the origin: every fixed layer contributes the same  $(1 + o(1)) M_{n-4}/M_n$  per 2-cycle. The uniformity in  $k$  needed for the bulk of the profile (layers  $k \asymp tn$ ) remains the province of the cluster expansion (TODO 4): there the boards are no longer two paths and the calculus here does not apply.

## 9 Three-row rectangles: exact moments and a joint Poisson limit

The same calculus resolves the “cycle-type-refined  $K(3, n)$ ” question at the level of 2-cycles. Let  $R$  be a uniform  $3 \times n$  Latin rectangle and let  $X_{12}, X_{13}, X_{23}$  be the numbers of intercalates between the three pairs of rows (equivalently, the 2-cycle counts of the three row permutations  $\sigma_{ab}$ ).

**Lemma 9.1** (exchangeability).  *$(X_{12}, X_{13}, X_{23})$  is exchangeable under the full symmetric group on the three pair labels.*

*Proof.* Permuting the rows of a rectangle is a bijection of rectangles and induces the corresponding permutation of the three unordered pairs.  $\square$

**Theorem 9.2** (joint Poisson limit). *For each fixed  $(a, b, c)$ ,*

$$\mathbb{E}[(X_{12})_a (X_{13})_b (X_{23})_c] = \left(\frac{1}{2}\right)^{a+b+c} (1 + O_{a,b,c}(n^{-1})),$$

*so  $(X_{12}, X_{13}, X_{23})$  converges jointly to three independent Poisson(1/2) variables.*

The proof rests on two uniform estimates, both immediate from Lemma 7.7, whose Bonferroni proof applies verbatim to any board of maximum degree 2 (unions of paths and cycles).

**Lemma 9.3** (uniform third-row estimates). *Uniformly over derangement types,  $N^{(3)}(\sigma) = n! e^{-2}(1 + O(1/n))$ ; and for every partial matching  $M$  avoiding  $B_2(\sigma)$  with  $|M| = j \leq n/2$ ,*

$$R(M; \sigma) = (n-j)! e^{-E_M/(n-j)} (1 + O(\frac{1}{n-j})) = (n-j)! e^{-2} (1 + O(\frac{j+1}{n})),$$

*where  $E_M \in [2n - 4j, 2n - 2j]$  is the number of cells of the punctured board. In particular  $q(M) := R(M; \sigma)/N^{(3)}(\sigma) \leq C e^{6j/n}/(n)_j$ .*

*Proof of Theorem 9.2.* Normalise the first row to id; the uniform measure is then uniform over pairs  $(\sigma, \tau)$  with  $\sigma \in \mathcal{D}_n$  and  $\tau$  avoiding  $B_2(\sigma)$ , and  $X_{12} = \lambda_2(\sigma)$ ,  $X_{13} = \lambda_2(\tau)$ ,  $X_{23} = \lambda_2(\tau\sigma^{-1})$ . Since distinct 2-cycles of a permutation are disjoint,

$$\mathbb{E}[(X_{12})_a (X_{13})_b (X_{23})_c] = \sum_{(A,B,C)} \mathbb{P}(A \subseteq \text{cyc}_2(\sigma), B \subseteq \text{cyc}_2(\tau), C \subseteq \text{cyc}_2(\tau\sigma^{-1})),$$

the sum over ordered tuples  $A$  of  $a$  pairwise disjoint 2-sets, and similarly  $B$  ( $b$  sets) and  $C$  ( $c$  sets). A 2-cycle  $\{u, v\}$  of  $\tau\sigma^{-1}$  prescribes the  $\tau$ -cells  $(\sigma^{-1}(u), v), (\sigma^{-1}(v), u)$ ; a 2-cycle  $\{i, j\}$  of  $\tau$  prescribes  $(i, j), (j, i)$ .

*Generic configurations.* Call  $(A, B, C)$  generic if the  $2(a+b+c)$  indices appearing in  $A \cup B \cup C$  are pairwise distinct. There are  $\binom{n}{2}^{a+b+c} (1 + O_{a,b,c}(1/n))$  generic tuples. Fix one. First,  $\mathbb{P}(A \subseteq \text{cyc}_2(\sigma)) = D_{n-2a}/D_n = (n)_{2a}^{-1} (1 + O(1/(n-2a)!))$ . Conditionally on  $\{A \subseteq \text{cyc}_2(\sigma)\}$ , the derangement  $\sigma$  restricted to the complement of the support of  $A$  is uniform; the  $B$ - and  $C$ -prescriptions form a matching  $M(\sigma)$  of exactly  $2(b+c)$   $\tau$ -cells avoiding  $B_2(\sigma)$  *unless*  $\sigma$  satisfies one of  $O_{b,c}(1)$  degeneracy events, each of conditional probability  $O(1/n)$ : a prescribed cell may land on  $B_2(\sigma)$  (e.g.  $\sigma(i) = j$  for a  $B$ -pair  $\{i, j\}$ , or  $v = \sigma^{-1}(u)$  for a  $C$ -pair), or two prescriptions may conflict in a row or column (e.g.  $\sigma^{-1}(u)$  in the support of  $B$ ); for a generic tuple all index slots are distinct, so no two prescribed cells can *coincide*, and on every degeneracy event the prescriptions are inconsistent,  $q = 0$ . Off these events, Lemma 9.3 gives  $\mathbb{P}(\tau \supseteq M(\sigma) \mid \sigma) = q(M(\sigma)) = (n)_{2(b+c)}^{-1} (1 + O(1/n))$ . Hence each generic tuple contributes  $(n)_{2a}^{-1} (n)_{2(b+c)}^{-1} (1 + O_{a,b,c}(1/n))$ , and the generic sum is  $\binom{n}{2}^{a+b+c} (n)_{2a}^{-1} (n)_{2(b+c)}^{-1} (1 + O(1/n)) = 2^{-(a+b+c)} (1 + O_{a,b,c}(1/n))$ .

*Collision configurations.* The remaining tuples are classified by their collision pattern: which equalities hold among the  $2(a+b+c)$  index slots — finitely many patterns for fixed  $(a, b, c)$ . A pattern with  $s \geq 1$  independent equalities has  $O(n^{2(a+b+c)-s})$  tuples, so it suffices to show that the per-tuple contribution is  $O(n^{-2(a+b+c)} \cdot n^{s-1})$ . Bound it as follows:  $\mathbb{P}(A \subseteq \text{cyc}_2(\sigma)) \leq Cn^{-2a}$ ; conditionally on  $\sigma$ , if the prescriptions are consistent they determine a matching of  $2(b+c) - t$  cells, where  $t$  counts coincidences among prescribed cells, and Lemma 9.3 gives  $q \leq Cn^{-2(b+c)+t}$ . Two  $B$ -cells, or two  $C$ -cells, never coincide (the pairs within  $B$ , within  $C$ , are disjoint); a coincidence of the  $B$ -cell  $(i, j)$  with the  $C$ -cell  $(\sigma^{-1}(u), v)$  requires the index equality  $v = j$  (counted in  $s$ ) *and* the  $\sigma$ -equation  $u = \sigma(i)$ , of conditional probability  $O(1/n)$ ; and  $t$  such coincidences involve  $t$  distinct such  $\sigma$ -equations, of joint probability  $O(n^{-t})$ . (A full coincidence of a  $B$ -pair with a  $C$ -pair is impossible: it would force  $\sigma(i) = i$ .) Hence the per-tuple bound is  $Cn^{-2a} \cdot n^{-t} \cdot n^{-2(b+c)+t} = Cn^{-2(a+b+c)}$ , which beats the required bound by  $n^{s-1} \geq 1$  — with room to spare unless  $s = 1$ , where it is exactly what is needed.

Summing,  $\mathbb{E}[(X_{12})_a (X_{13})_b (X_{23})_c] = 2^{-(a+b+c)} (1 + O_{a,b,c}(1/n))$ , and the joint Poisson limit follows by the method of moments.  $\square$

The toolkit computes the low joint moments *exactly* (certified against brute-force enumeration at  $n = 7$ ):

| $n$ | $\mathbb{E}X_{12} - \frac{1}{2}$ | ratio to $\frac{M_{n-4}}{2M_n}$ | $\mathbb{E}[X_{12}X_{13}]$ | $n^2 \text{Cov}(X_{12}, X_{13})$ | $\mathbb{E}\binom{X_{13}}{2}$ |
|-----|----------------------------------|---------------------------------|----------------------------|----------------------------------|-------------------------------|
| 8   | $+5.162 \times 10^{-4}$          | 2.45                            | 0.26230                    | 0.754                            | 0.12761                       |
| 9   | $+1.143 \times 10^{-4}$          | 0.76                            | 0.25873                    | 0.698                            | 0.12466                       |
| 10  | $+9.467 \times 10^{-5}$          | 1.041                           | 0.25689                    | 0.679                            | 0.12511                       |
| 11  | $+5.886 \times 10^{-5}$          | 0.9944                          | 0.25552                    | 0.661                            | 0.12502                       |
| 12  | $+4.005 \times 10^{-5}$          | 1.0010                          | 0.25453                    | 0.646                            | 0.12502                       |
| 13  | $+2.797 \times 10^{-5}$          | 1.0000                          | 0.25378                    | 0.633                            | 0.12501                       |

(All three  $\mathbb{E}X_{ab}$  agree exactly, as do the three mixed products — Lemma 9.1 at the level of exact rationals.) Two structural features deserve note.

The next two statements need a quantitative form of the fusion expansion of  $N^{(3)}$ , of independent interest: it proves the additivity of  $\log N^{(3)}$  over cycles (§4) with an explicit second-order remainder.

**Lemma 9.4** (fusion expansion with remainder). *There are  $n_0, C$  such that for all  $n \geq n_0$  and every derangement type  $\lambda \vdash n$  with  $\kappa = \kappa(\lambda)$  parts,*

$$N^{(3)}(\lambda) = M_n + \sum_{\ell \geq 2} \lambda_\ell M_{n-2\ell} + E(\lambda), \quad |E(\lambda)| \leq C \kappa^2 (n-8)!,$$

with the conventions  $M_0 = 2$ ,  $M_1 = -1$  (the IE values  $\langle M_{C_{2m}} \rangle_m$ ) and  $M_m = 0$  for  $m < 0$ . Exhaustive computation over all types gives  $\max_\lambda |E(\lambda)|/(\kappa^2(n-8)!) \approx 0.06$  at  $n = 12, 16, 20$ , and  $E(\lambda) \geq 0$  except for the single type  $(\frac{n}{2}, \frac{n}{2})$ , where  $E = -2$ .

*Proof.* Induction on  $\kappa$ ; the case  $\kappa = 1$  is trivial ( $E = 0$ ). For  $\kappa \geq 2$  pick two parts as follows: two equal parts  $b = a$  if any exist, otherwise the smallest and the largest,  $b < a$  (then  $a - b \geq 2$  unless  $\kappa = 2$  and  $\lambda = (m, m+1)$ , a single boundary case computed directly from  $M_{C_2} = 1 + 2x$ : there  $N^{(3)} = M_n + \langle x^{2m}(1+2x) \rangle_n = M_n - 1$  and  $E = 0$  under the stated conventions). Theorem 4.2 gives  $N^{(3)}(\lambda) = N^{(3)}(\mu) + N_{n-2b}^{(3)}(\nu)$ , where  $\mu$  replaces  $\{a, b\}$  by  $a+b$  and  $\nu$  (on  $n-2b$  points) replaces them by  $a-b$  (for  $a = b$ :  $2N_{n-2b}^{(3)}(\lambda \setminus \{a, b\})$ , same bookkeeping). Apply the induction hypothesis to both terms. In the sum, the first-order terms reassemble as  $\sum_\ell \lambda_\ell M_{n-2\ell}$  plus the correction terms

$$M_{n-2(a+b)} + \sum_{\ell\text{-parts of } \nu \setminus \{a-b\}} M_{(n-2b)-2\ell} \leq M_{n-8} + \kappa M_{n-8}$$

(each index is at most  $n-8$  because  $a, b, \ell \geq 2$ ), the term  $M_{(n-2b)-2(a-b)} = M_{n-2a}$  cancelling the deficit of  $\mu$ 's first-order sum. Hence  $|E(\lambda)| \leq |E(\mu)| + |E(\nu)| + C'(\kappa+1)(n-8)! \leq C(\kappa-1)^2(n-8)! + C\kappa^2(n-2b-8)! + C'(\kappa+1)(n-8)!$ , and since  $(n-2b-8)! \leq (n-12)! \leq (n-8)!/n^3$ , the right side is at most  $C\kappa^2(n-8)!$  once  $C \geq 2C'$  and  $n \geq n_0$ .  $\square$

We also record the standard moment computation for derangement cycle counts: for  $s_\ell \geq 0$  with  $m = \sum_\ell \ell s_\ell \leq n/2$ ,

$$\mathbb{E}_{Q_n} \left[ \prod_\ell (\lambda_\ell)_{s_\ell} \right] = \prod_\ell \ell^{-s_\ell} \cdot \frac{(n)_m D_{n-m}}{D_n} = \prod_\ell \ell^{-s_\ell} \left( 1 + O\left(\frac{1}{(n-m)!}\right) \right), \quad (19)$$

by counting derangements with an ordered tuple of marked cycles.

**Proposition 9.5** (the third row tilts the second).  $\mathbb{E}[X_{12}] - \frac{1}{2} = \frac{M_{n-4}}{2M_n} (1 + O(n^{-2}))$ .

*Proof.*  $X_{12} = \lambda_2(\sigma)$  where  $\mathbb{P}(\sigma) \propto N^{(3)}(\sigma)$  on  $\mathcal{D}_n$ , so with  $w(\lambda) = N^{(3)}(\lambda)/M_n$  and  $\mathbb{E}_Q$  the uniform-derangement mean,

$$\mathbb{E}[X_{12}] - \mathbb{E}_Q[\lambda_2] = \frac{\text{Cov}_Q(\lambda_2, w)}{\mathbb{E}_Q[w]}.$$

By Lemma 9.4,  $w = 1 + \sum_\ell \rho_\ell \lambda_\ell + E(\lambda)/M_n$  with  $\rho_\ell = M_{n-2\ell}/M_n$ . By (19),  $\text{Var}_Q(\lambda_2) = \frac{1}{2} + O(1/(n-4)!)$  and  $\text{Cov}_Q(\lambda_2, \lambda_\ell) = O(1/(n-2-\ell)!)$  for  $\ell \geq 3$  (both are differences of products

of factorial moments, and the Poisson values cancel exactly). For the remainder,  $\kappa \leq n/2$  gives the crude bound  $|\text{Cov}_Q(\lambda_2, E/M_n)| \leq Cn^2(n-8)!/M_n = O(n^{-6})$ . Hence

$$\text{Cov}_Q(\lambda_2, w) = \rho_2 \cdot \frac{1}{2} + O(n^{-6}), \quad \mathbb{E}_Q[w] = 1 + O(n^{-4}),$$

and  $\mathbb{E}_Q[\lambda_2] = \frac{1}{2} + O(1/(n-2)!)$  by (19); assembling,  $\mathbb{E}[X_{12}] - \frac{1}{2} = \frac{1}{2}\rho_2(1 + O(n^{-2}))$ , as claimed (the  $O(n^{-2})$  is dominated by  $O(n^{-6})/\rho_2$ ). The exact table above confirms the formula to  $10^{-4}$  relative by  $n = 13$  — a sharp quantitative cross-check of the layer framework: *conditioning on one extra row tilts the second row's 2-cycle count by exactly the layer-3 coefficient times the variance.*  $\square$

**Proposition 9.6** (pair-pair covariance).  $\text{Cov}(X_{12}, X_{13}) = \frac{1/2 + O(n^{-1} \log n)}{n^2} > 0$ .

*Proof.* Write  $h(\sigma) = \mathbb{E}[X_{13} \mid \sigma] = \sum_{\{i,j\} \text{ adm}} q_{ij}(\sigma)$ , where  $q_{ij} = R(M_{ij}; \sigma)/N^{(3)}(\sigma)$ ,  $M_{ij} = \{(i, j), (j, i)\}$ , and the sum runs over the *admissible* pairs (both cells off  $B_2(\sigma)$ ); their number is  $\binom{n}{2} - n + \lambda_2(\sigma)$  exactly (the excluded pairs are the  $\{i, \sigma(i)\}$ , of which  $\lambda_2$  coincide in pairs). A 2-cycle  $\{i, j\}$  of  $\sigma$  thus makes  $\{i, j\}$  doubly forbidden but *frees* the two pairs that  $i$  and  $j$  would otherwise each exclude. Split

$$h(\sigma) = \left[\binom{n}{2} - n + \lambda_2(\sigma)\right] (n)_2^{-1} + \Delta(\sigma), \quad \Delta(\sigma) = \sum_{\text{adm}} [q_{ij}(\sigma) - (n)_2^{-1}],$$

so that, under the  $N^{(3)}$ -tilted measure of the three-row ensemble,

$$\text{Cov}(X_{12}, X_{13}) = (n)_2^{-1} \text{Var}(\lambda_2) + \text{Cov}(\lambda_2, \Delta).$$

As in Proposition 9.5 (second factorial moments via Lemma 9.4 and (19)),  $\text{Var}(\lambda_2) = \frac{1}{2} + O(n^{-2})$ .

$\Delta(\sigma)$  depends only on the cycle type, and we claim the *join-stability* bound: for types  $\lambda, \mu$  one join apart,  $|\Delta(\lambda) - \Delta(\mu)| \leq Cn^{-3}$ . Choose adjacent representatives as in §7 (boards differing in one  $2 \times 2$  switch). For each admissible pair,  $R(M_{ij}; \sigma') - R(M_{ij}; \sigma)$  equals the four-term bracket  $\text{Br}(M_{ij})$  of §7 (the double-pin terms cancel as identical punctured boards, and if  $M_{ij}$  meets a pinned line the bracket degenerates to the compatible terms). By Lemmas 7.5 and 7.6: for the  $O(n^2)$  pairs avoiding the switch window the bracket is  $x^3$ -shifted and pinned, giving relative differences  $O(n^{-4})$ ; for the  $O(n)$  pairs meeting one window line, valuation  $\geq 2$  gives  $O(n^{-3})$ ; for the  $O(1)$  pairs meeting two pinned lines the crude one-term bound gives  $O(n^{-1})$ . Together with the  $N^{(3)}$ -normalisation ( $\Delta_\lambda \log N^{(3)} = O(n^{-4})$ ) and the  $O(1)$  pairs entering or leaving the admissible set (each contributing  $q - (n)_2^{-1} = O(n^{-3})$ ), this sums to  $|\Delta(\lambda) - \Delta(\mu)| \leq C(n^2 \cdot n^{-4} + n \cdot n^{-3} + 1 \cdot n^{-1}) \cdot (n)_2^{-1} + O(n^{-4}) \leq Cn^{-3}$ .

Any type with  $\kappa$  parts is  $\kappa - 1$  joins from  $(n)$ , so  $|\Delta(\lambda) - \Delta((n))| \leq C\kappa n^{-3}$ , and since the tilted density is bounded,  $\mathbb{E}[\kappa^2] \leq C \log^2 n$ ; Cauchy-Schwarz gives  $|\text{Cov}(\lambda_2, \Delta)| \leq Cn^{-3} \log n$ . Assembling,  $\text{Cov}(X_{12}, X_{13}) = \frac{1}{2}(n)_2^{-1}(1 + O(n^{-1} \log n))$ . The exact values  $n^2 \text{Cov} = 0.661, 0.646, 0.633$  at  $n = 11, 12, 13$  extrapolate to  $\approx 0.48 + 2.0/n$ , consistent with the constant  $\frac{1}{2}$  at these small orders.  $\square$

**Remark 9.7.** Theorem 9.2 is the  $k = 3$  instance of the expectation that intercalate structure across any  $O(1)$  rows is asymptotically that of an independent model; it may be seen as the (easy) rectangle-regime companion of the full-square intercalate results of [12]. We have not found it stated in the literature; the classical aggregate  $K(3, n)$  theory ([19, Ch. 8 §4]) does not resolve joint pair statistics.

## 10 An exact switching identity and the B-pair reformulation

The switchings of [3] do more than bound ratios: their multigraph is exactly balanced on one side, and this yields an *identity*. We briefly recall the notions (see [3, §2–3] for details).

Fix a Latin square  $L$  (of order  $n$ ; we take the CGW parameters  $m = n$ ,  $F = \emptyset$ ). The first two rows decompose into *row cycles* (corresponding to the cycles of  $\sigma_{1,2}$ ). For two columns  $j \neq j'$  with  $j' \notin \{\omega(j), \omega^{-1}(j)\}$  — where  $\omega$  is the column map defined by  $L(1, \omega(j)) = L(2, j)$  — the pair  $\{j, j'\}$  is an *A-pair* if rows 1, 2 lie in different column cycles of the column pair, and a *B-pair* otherwise. The *switch* of  $L$  at  $\{j, j'\}$  trades the row cycle (among the two through  $j, j'$ ) containing the minimal symbol; the *flip* at an A-pair  $\{j, j'\}$  whose columns lie in different row cycles of lengths  $\alpha, \beta$  trades the column cycle of  $\{j, j'\}$  through row 2, and joins the two row cycles into one of length  $\alpha + \beta$ .

Let  $\mu \vdash n$  contain parts  $\alpha \neq \beta$  ( $\alpha, \beta \geq 2$ ) and let  $\lambda$  be  $\mu$  with  $\alpha, \beta$  merged into  $\alpha + \beta$ . Let  $L$  be uniform on  $S(\mu) = \{L : \sigma_{1,2}(L) \text{ has type } \mu\}$ , pick an  $\alpha$ -cycle and a  $\beta$ -cycle of  $L$  uniformly, and columns  $j, j'$  uniformly in them. Perform the CGW joining step: if  $\{j, j'\}$  is a B-pair, switch at  $\{j, j'\}$  (making it an A-pair); then flip at  $\{j, j'\}$ , obtaining  $L' \in S(\lambda)$  in which  $\{j, j'\}$  lies in a single  $(\alpha + \beta)$ -cycle with  $(\omega')^\beta(j) = j'$ , where  $\omega' = \omega_{L'}$ . Define the two *post-flip events*

$$B_1 = \{\{\omega'^{-1}(j), \omega'^{-1}(j')\} \text{ is a B-pair of } L'\}, \quad B_2 = \{\{\omega'(j), \omega'(j')\} \text{ is a B-pair of } L'\},$$

(the shifted pairs are automatically admissible when  $\alpha \neq \beta$ ), and put  $\bar{q} = \bar{q}(\mu \rightarrow \lambda) = \mathbb{P}(B_1) + \mathbb{P}(B_2) \in [0, 2]$ .

**Theorem 10.1** (exact identity;  $\alpha \neq \beta$ ). *With the notation above,*

$$\frac{|S(\lambda)|}{|S(\mu)|} = \frac{\gamma(\lambda)}{\gamma(\mu)} \cdot \frac{1 + \bar{q}}{2}, \quad \text{equivalently} \quad \boxed{\bar{q} = 2 \frac{C_n(\lambda)}{C_n(\mu)} - 1.}$$

*Proof (source and verification).* This is the exact content of the CGW splitting/joining multigraph  $G_S = G_J$  [3, §3]: the splitting side is *regular* — every  $L' \in S(\lambda)$  produces exactly two edges for each of its  $\lambda_{\alpha+\beta}$  cycles of length  $\alpha + \beta$  and each of the  $\alpha + \beta$  column pairs at  $\omega$ -distance  $\beta$  within the cycle ([3, Eq. (2)]) — while the joining side produces, for each of the  $\mu_\alpha \mu_\beta \alpha \beta$  instances,  $1 + \mathbf{1}_{B_1} + \mathbf{1}_{B_2}$  edges. Equating the edge counts and using  $\gamma(\lambda)/\gamma(\mu) = \mu_\alpha \mu_\beta \alpha \beta / ((\mu_{\alpha+\beta} + 1)(\alpha + \beta))$  gives the identity. Because the bookkeeping is delicate, we certified the exact statement independently: by *complete enumeration* of all Latin squares of orders  $n = 5$  ( $\mu = (2, 3)$ : both sides equal 2) and  $n = 6$  ( $\mu = (2, 4)$ : both sides equal 10/11, from 3,041,280 instances), and statistically at  $n = 8$  (§13). The min-symbol tie-break in the switch is averaged over symbol relabellings, which is the invariant formulation of the CGW rule.  $\square$

**Remark 10.2** (the case  $\alpha = \beta$ ). For  $\alpha = \beta = 2$  the two shifted pairs coincide and, with the event counted twice, the identity again holds exactly (verified at  $n = 6$ ,  $\mu = (2, 2, 2)$ : both sides 4/7). For  $\alpha = \beta \geq 3$  our reading of the joining bookkeeping does *not* reproduce the splitting count (at  $n = 6$ ,  $\mu = (3, 3)$ : joining average 5/6 against the required 3/4), so the exact identity in that case needs a corrected accounting; we do not use it — see the routing argument in §12. (The CGW *inequalities* are unaffected.)

**Lemma 10.3** (symmetry of the two events).  $\mathbb{P}(B_1) = \mathbb{P}(B_2)$  *exactly, for every  $n, \mu, \alpha, \beta$ . Hence*  $\mathbb{P}(B_i) = \bar{q}/2 = C_n(\lambda)/C_n(\mu) - \frac{1}{2}$ .

*Proof sketch.* The involution  $\iota$  that transposes rows 1 and 2 preserves  $S(\mu)$  and the instance measure (it fixes the row-cycle column sets, the  $A/B$  status of every pair, and the min-symbol rule, and conjugates  $\omega \mapsto \omega^{-1}$ ), and it exchanges the two shifted pairs, mapping the event  $B_1$  for the instance of  $L$  onto the event  $B_2$  for the corresponding instance of  $\iota(L)$ . We verified the exchange *instance-by-instance* over 7,000 instances at  $n = 6$  (both  $A$ - and  $B$ -instances, both switch variants); the exact equalities  $\mathbb{P}(B_1) = \mathbb{P}(B_2) = 5/11, 5/12, 2/7$  in the three  $n = 6$  enumerations confirm it in aggregate. A self-contained algebraic proof (chasing  $\iota$  through the flip, whose defining column cycle is the one through row 2) is routine but fiddly; we record it as such.  $\square$

Theorem 10.1 converts the CGW exact tables into exact values of a structural probability of random Latin squares. For the join  $(2, n-2) \rightarrow (n)$ :

| $n$                                     | 7        | 8        | 9        | 10       | 11       |
|-----------------------------------------|----------|----------|----------|----------|----------|
| $\bar{q} = 2C_n((n))/C_n((2, n-2)) - 1$ | 0.988372 | 0.993345 | 0.996822 | 0.997363 | 0.998080 |
| $\frac{1}{2}(1 - \bar{q})n^3$           | 1.99     | 1.70     | 1.16     | 1.32     | 1.28     |

Thus each post-flip event has probability  $\frac{1}{2} - \Theta(n^{-3})$ : the deficit is precisely the refined-law coefficient. By contrast a *generic* admissible column pair is a  $B$ -pair with probability  $\frac{1}{2} + \Theta(n^{-2})$  (§13), and two marked points of a uniform random derangement lie in a common cycle with probability  $\frac{1}{2} + (1 + o(1))/n$ . The post-flip pairs are extraordinarily well balanced, and this balance *is* the conjecture:

**Hypothesis 10.4** (B-pair equidistribution, EQ( $\delta$ )). *For every pair  $\mu \rightarrow \lambda$  as above with  $\alpha \neq \beta$ ,  $|\bar{q}(\mu \rightarrow \lambda) - 1| \leq \delta(n)$ .*

By Theorem 10.1, EQ( $\delta$ ) with  $\delta(n) = O(n^{-3+\varepsilon})$  is *equivalent* to the corresponding precision in Conjecture 3.1 across  $\alpha \neq \beta$  joins; EQ( $o(1/\log n)$ ) already implies the weak conjecture (§12). In §11 we prove an identity that removes the joining measure and the events  $B_1, B_2$  from this picture entirely.

## 11 The marked-pair identity: eliminating the post-flip events

The identity of Theorem 10.1 expresses the completion ratio through the post-flip events  $B_1, B_2$  of the *joining* measure. In this section we prove a cleaner exact identity that eliminates the joining measure, the switch tie-break, and the events  $B_1, B_2$  altogether: the ratio  $C_n(\lambda)/C_n(\mu)$  is the *A-versus-B odds of a single marked column pair in a single uniform square of  $S(\lambda)$* . The proof is self-contained (it does not use the multigraph of [3]), covers  $\alpha = \beta$  without any modification — resolving the bookkeeping discrepancy of Remark 10.2 — and yields the lower bound  $C_n(\lambda)/C_n(\mu) > \frac{1}{2}$  for free.

Throughout, squares are normalised to have first row id, so the column map is  $\omega = \sigma$  ( $1 \circ \omega(j) = 2 \circ j$  with the first row the identity gives  $\omega(j) = \sigma(j)$ ). For the proofs it is convenient to work with *unnormalised* squares (equivalently, symbol-relabelling classes): every instance count below then acquires a common factor  $n!$ , which cancels from every ratio, and the maps we use (flip, switch) act on unnormalised squares without any renormalisation step. Statuses, cycle types, and column positions are relabelling-invariant, so we freely state results in the normalised picture.

**Definition 11.1** (marked instances). Let  $\lambda \vdash n$  contain a part  $m = \alpha + \beta$  ( $\alpha, \beta \geq 2$ ), and let  $\mu$  be  $\lambda$  with one part  $m$  split into  $\alpha, \beta$ . The *splitting instance space* is

$$X = X(n, \lambda; \alpha, \beta) = \{(L', P) : L' \in S(\lambda), P = \{j, \sigma^\alpha(j)\} \text{ with } j \text{ in an } m\text{-cycle of } \sigma(L')\},$$

unordered pairs counted once. Thus every  $L'$  carries exactly  $\lambda_m m$  marked pairs when  $\alpha \neq \beta$  and  $\lambda_m \alpha$  when  $\alpha = \beta$ . Since  $\alpha, \beta \geq 2$  every marked pair is admissible, and  $X = X_A \sqcup X_B$  according to the  $A/B$  status of  $P$  in  $L'$ . The *joining instance space* is

$$J = J(n, \mu; \alpha, \beta) = \{(L, P) : L \in S(\mu), P = \{j, j'\}, j \text{ in an } \alpha\text{-cycle, } j' \text{ in a different } \beta\text{-cycle}\},$$

again with  $J = J_A \sqcup J_B$  by the status of  $P$  in  $L$ .

The counting arithmetic is immediate: with  $D_\nu$  the number of derangements of cycle type  $\nu$  and  $N(\nu) = C_n(\nu)(n-2)!$ ,

$$\frac{2|X|}{|J|} = \frac{2 D_\lambda N(\lambda) \lambda_m m}{D_\mu N(\mu) \mu_\alpha \mu_\beta \alpha \beta} = \frac{2 C_n(\lambda)}{C_n(\mu)} \quad (\alpha \neq \beta), \quad (20)$$

using  $D_\lambda/D_\mu = \alpha \mu_\alpha \beta \mu_\beta / (m \lambda_m)$ ; for  $\alpha = \beta$  one has  $D_\lambda/D_\mu = \alpha^2 \mu_\alpha (\mu_\alpha - 1) / (2m \lambda_m)$ ,  $|J| = |S(\mu)| \binom{\mu_\alpha}{2} \alpha^2$  and  $|X| = |S(\lambda)| \lambda_m \alpha$ , and (20) holds verbatim. The content of the identity is therefore the following count.

**Lemma 11.2** (cycle trades preserve pair status). *Let  $P = \{j, j'\}$  be an admissible column pair of a Latin square  $L$  and let  $C$  be one column cycle of  $P$ . Trading  $C$  (swapping the entries of columns  $j, j'$  in the rows of  $C$ ) inverts the pair permutation  $\pi_P$  on  $C$  and fixes it elsewhere; the partition of rows into column cycles of  $P$  is unchanged. In particular the  $A/B$  status of  $P$  itself is invariant under the CGW flip at  $P$ , which trades the full column cycle through row 2.*

*Proof.* Write  $\pi = \pi_P$ , so  $\pi(r)$  is the row of column  $j'$  holding the symbol  $L(r, j)$ ;  $C$  is  $\pi$ -closed and the trade swaps  $L(r, j) \leftrightarrow L(r, j')$  for  $r \in C$ . Fix  $r \in C$  and let  $\pi'$  be the new pair permutation. The new  $L(r, j)$  is the old  $L(r, j')$ ; the row of the new column  $j'$  holding that symbol is found by solving (new  $L(s, j')$ ) = (old  $L(r, j')$ ): for  $s \in C$  the new  $L(s, j')$  is the old  $L(s, j)$ , and old  $L(s, j) =$  old  $L(r, j')$  exactly when  $\pi(s) = r$ . Hence  $\pi' = \pi^{-1}$  on  $C$ ; off  $C$  nothing changes. A cycle inverted in place has the same support, so the partition into cycles is unchanged.  $\square$

**Lemma 11.3** (switch toggle). *Let  $(L, P) \in J$  and let  $Q$  be the row cycle through one column of  $P$  (the two row cycles are distinct). Trading  $Q$  (the CGW switch) changes  $\pi_P$  by precomposition with the transposition of rows 1, 2: the cycles of  $\pi_P$  through rows 1, 2 are split if rows 1, 2 are in one cycle, and joined if in two. Hence the switch toggles the status of  $P$ , preserves  $S(\mu)$  and the instance data (it inverts one  $\sigma$ -cycle in place), and is an involution: the trade preserves the traded cycle's symbol set (the two rows carry the same symbols on its columns), so the min-symbol rule re-selects the same cycle. Consequently  $|J_A| = |J_B|$ , exactly.*

*Proof.* Say  $Q$  is the row cycle through  $j$ ; its columns include  $j$  but not  $j'$ . The trade swaps the row-1 and row-2 entries in the columns of  $Q$ ; for  $\pi_P$  only column  $j$  matters, where the entries of rows 1, 2 are exchanged. Thus  $\pi'_P = \pi_P \circ \tau_{12}$ , and multiplying by a transposition splits or joins the cycles through its two points. Rows 1, 2 lie in one cycle of  $\pi_P$  exactly when  $P$  is a  $B$ -pair. The trade swaps the two rows' entries along the columns of  $Q$ , which inverts the corresponding  $\sigma$ -cycle in place: the cycle type and the column sets of all row cycles are preserved, so  $(L, P) \mapsto (\text{sw}(L), P)$  maps  $J$  to  $J$ .  $\square$

**Theorem 11.4** (marked-pair identity). *For every  $n$ ,  $\lambda$ , and split  $m = \alpha + \beta$  with  $\alpha, \beta \geq 2$  (including  $\alpha = \beta$ ),*

$$|J_A| = |J_B| = |X_A|, \quad \text{hence} \quad \boxed{\frac{C_n(\lambda)}{C_n(\mu)} = \frac{|X|}{2|X_A|}, \quad 2 \frac{C_n(\lambda)}{C_n(\mu)} - 1 = \frac{|X_B|}{|X_A|}.}$$

*In particular  $C_n(\lambda) > \frac{1}{2} C_n(\mu)$  unconditionally, and Hypothesis 10.4 (EQ( $\delta$ )) is equivalent to*

$$\left| \mathbb{P}(P \text{ is a } B\text{-pair}) - \frac{1}{2} \right| \leq \frac{\delta}{4}(1 + o(1)),$$

*for  $(L', P)$  uniform on  $X$ : the law of a single bit — whether rows 1, 2 lie in one column cycle of the marked pair  $\{j, \sigma^\alpha(j)\}$  — of a single uniform square of  $S(\lambda)$ .*

*Proof.* By (20) it suffices to prove  $|J| = 2|X_A|$ . The flip at an  $A$ -pair  $P$  whose columns lie in one row cycle at  $\sigma$ -distances  $(\alpha, \beta)$  splits that cycle into an  $\alpha$ - and a  $\beta$ -cycle and maps  $S(\lambda) \rightarrow S(\mu)$ ; by Lemma 11.2 the pair  $P$  is still an  $A$ -pair afterwards, so  $x \mapsto \text{unflip}(x)$  maps  $X_A$  into  $J_A$ . The flip at a fixed pair is an involution (the traded cycle again contains row 2, with the same row support), so  $\text{join} = \text{flip}$  inverts it:  $X_A \leftrightarrow J_A$  is a bijection. Composing with the switch of Lemma 11.3 gives a bijection  $X_A \leftrightarrow J_B$ , namely  $x \mapsto \text{sw}(\text{unflip}(x))$  with inverse  $(L, P) \mapsto \text{flip}(\text{sw}(L))$  — the map “switch-then-flip”, which is exactly the CGW joining step at a  $B$ -pair. Hence  $|J| = |J_A| + |J_B| = 2|X_A|$ .  $\square$

**Remark 11.5** (what became of the events  $B_1, B_2$ ). Theorem 11.4 recovers Theorem 10.1 and sharpens it: the joining measure, the min-symbol averaging, and the post-flip events have disappeared. The mechanism is that *every* joining edge lands in  $X_A$  (Lemma 11.2), each marked- $A$  instance receiving exactly two (one from an  $A$ -join, one from its switched partner), while  $X_B$  is never reached: the  $B_1, B_2$  bookkeeping of [3] was the joining-side accounting of  $|X_B|$ . On  $X_B$  itself the only known measure-preserving involution is the cross-switch of [3, Def. 3.9]. By [3, Lem. 3.10] it toggles the status of every pair  $\{\omega^k(j), \omega^{k'}(j')\}$  with  $1 \leq k < \alpha$ ,  $1 \leq k' < \beta$  — a family containing *both* shifted pairs, at  $(k, k') = (1, 1)$  and  $(\alpha - 1, \beta - 1)$  — while (empirically: all 2880 marked- $B$  instances at  $n = 5$ ) fixing the marked status itself. This yields  $\mathbb{P}(B_1 \mid \text{marked } B) = \mathbb{P}(B_2 \mid \text{marked } B) = \frac{1}{2}$  exactly, explaining the extraordinary balance of the post-flip events observed in §10.

**Remark 11.6** (the case  $\alpha = \beta$ , and Remark 10.2). Nothing in the proof distinguishes  $\alpha = \beta$ : marked pairs  $\{j, \sigma^\alpha(j)\}$  in a  $2\alpha$ -cycle are counted once ( $\lambda_m \alpha$  per square), and (20) holds with the  $\alpha = \beta$  conventions. Exhaustive verification at  $n = 6$ :  $\mu = (3, 3)$  gives  $|X_B|/|X_A| = 622080/829440 = 3/4$ , exactly the value  $2C_6((6))/C_6((3, 3)) - 1$  that the joining-side bookkeeping of Remark 10.2 failed to reproduce. The  $\alpha = \beta \geq 3$  anomaly is thus resolved: the identity was never about the joining events.

**Verification.** By complete enumeration (`pf.bridge.py`, `xside.c`): at  $n = 5$ ,  $\mu = (2, 3)$ :  $|J_A| = |J_B| = |X_A| = 1440$ ,  $|X_B| = 2880$ , ratio  $2 = \bar{q}$  exactly. At  $n = 6$ :  $(2, 4) \rightarrow (6)$ :  $1520640/1520640/1520640$ ,  $|X_B| = 1382400$ , ratio  $10/11$ ;  $(2, 2, 2) \rightarrow (2, 4)$ : ratio  $4/7$ ;  $(3, 3) \rightarrow (6)$ : ratio  $3/4$  — all equal to  $2C_6(\lambda)/C_6(\mu) - 1$  exactly. At  $n = 7$ , fixing one representative  $\sigma_0$  per type (conjugation symmetry) and enumerating all  $\approx 6.6 \times 10^6$  completions per type (`xside2.c`), the  $X$ -side odds reproduce  $2C_7(\lambda)/C_7(\mu) - 1$  exactly for all four admissible merges:  $(2, 5) \rightarrow (7)$ :  $22848000/23116800 = 85/86$ ;  $(3, 4) \rightarrow (7)$ :  $23063040/22901760 = 143/142$  (note the  $B$ -excess);  $(2, 2, 3) \rightarrow (2, 5)$ :  $107/108$ ;  $(2, 2, 3) \rightarrow (3, 4)$ :  $35/36$ . The bijections themselves (unflip status, roundtrips, hit multiplicities, orphan structure) were verified instance-by-instance at  $n = 5, 6$ .

## 12 A reduction theorem

**Theorem 12.1.** *Assume  $\text{EQ}(\delta)$  with  $\delta = \delta(n) \rightarrow 0$ . Then*

$$d_{\text{TV}}(P_n, Q_n) = O(\delta \log n + n^{-2/3}).$$

*In particular  $\text{EQ}(o(1/\log n))$  implies Conjecture 1.2, hence also that almost all loops have multiplication group  $S_n$  and trivial character theory.*

*Proof.* Call  $\lambda$  *plain* if not all its parts are equal (so the only non-plain types are  $(\ell^\kappa)$ ,  $\kappa \geq 2$ ).

*Routing.* Every plain  $\lambda$  with  $\kappa = \kappa(\lambda)$  parts is reachable from  $(n)$  by a chain of  $\kappa - 1$  splits, each with  $\alpha \neq \beta$ . Build the chain in reverse, splitting parts of  $\lambda$  off a shrinking residual, in a suitable order; a collision means splitting a part  $p$  off a residual  $R$  with  $p = R - p$ . If the two largest parts of  $\lambda$  are unequal, split smallest-first and let them be the final pair: while at least three parts remain, the split-off minimum  $p$  satisfies  $R - p \geq 2p > p$  (all parts are  $\geq 2$  and at least two other parts remain), so no collision occurs, and the final split is unequal by hypothesis. If instead the two largest parts are equal, say to  $\ell_{\max}$ , then  $\ell_{\max} \neq n/2$  (else  $\lambda = (n/2, n/2)$  is not plain), so we may split one copy of  $\ell_{\max}$  off first without collision, then proceed smallest-first among the remaining parts, reserving for the final split the pair  $(b, \ell_{\max})$  where  $b < \ell_{\max}$  is the smallest part: at every intermediate step the residual minus the split-off minimum still contains the reserved  $\ell_{\max}$  and the reserved  $b$ , hence exceeds the minimum (also when a further copy of  $\ell_{\max}$  is the split-off minimum:  $R - p \geq \ell_{\max} + b > p$ ), and the final split is unequal.

For each step of such a chain, Theorem 10.1 and  $\text{EQ}(\delta)$  give  $|S(p_{i+1})|/|S(p_i)| = (\gamma(p_{i+1})/\gamma(p_i))(1 + \varepsilon_i)$  with  $|\varepsilon_i| \leq \delta$ . Multiplying along the chain,

$$\frac{P_n(\lambda)}{P_n((n))} = \frac{\gamma(\lambda)}{\gamma((n))} (1 + \varepsilon)^{\pm(\kappa-1)}, \quad |\varepsilon| \leq \delta,$$

so for all plain  $\lambda$  with  $\kappa(\lambda) \leq K$ ,  $P_n(\lambda) = Q_n(\lambda) Z^{-1}(1 + O(K\delta))$  where  $Z > 0$  is a normalising constant independent of  $\lambda$  (namely  $D_n C_n((n))/\sum \gamma C$ ).

Let  $K = \lceil A \log n \rceil$  with  $A$  a large absolute constant, and let  $\mathcal{B}$  be the set of types that are either non-plain or have  $\kappa > K$ . Under  $Q_n$ , Markov's inequality with the generating function  $\sum_{\pi \in \mathcal{D}_n} 2^{\kappa(\pi)} = n! [x^n] e^{-2x} (1-x)^{-2} \leq n \cdot n!$  (cf. [3, proof of Lemma 4.4]) gives  $Q_n(\kappa > K) \leq e n 2^{-K} = O(n^{-2})$  for  $A \log 2 \geq 3$ ; also  $Q_n(\text{non-plain}) \leq \sum_{\ell|n, \ell < n} \gamma((\ell^{n/\ell}))/D_n = O(n^{-1})$ . Under  $P_n$ , Proposition 19.11 gives  $P_n(\kappa > K) = O(n^{-2} \log n)$  for  $A \geq 16$  and  $P_n(\text{non-plain}) = O(n^{-1+o(1)})$  without any input from [3] beyond the identity of Theorem 11.4; the original argument, kept for the record, used the unconditional CGW bound [3, Cor. 4.5] (whose proof relies on the 3/2 direction of their Lemma 3.12, see Remark 19.10),  $P_n(\lambda) \leq n^{1/3}(\gamma(\lambda)/n!) 2^{\kappa(\lambda)}$ , combined with  $\sum_{\pi \in \mathcal{D}_n} 4^{\kappa(\pi)} = n! [x^n] e^{-4x} (1-x)^{-4} \leq \binom{n+3}{3} n!$ , gives

$$P_n(\kappa > K) \leq n^{1/3} 2^{-K} \sum_{\lambda} \frac{\gamma(\lambda)}{n!} 4^{\kappa(\lambda)} = O(n^{10/3} 2^{-K}) = O(n^{-2/3})$$

for  $A \log 2 \geq 4$ , and similarly  $P_n(\text{non-plain}) \leq n^{1/3} \sum_{k \geq 2, k|n} 4^k / (k! (n/k)^k) = O(n^{-5/3})$ .

On the complement of  $\mathcal{B}$ , summing  $P_n = Q_n Z^{-1}(1 + O(K\delta))$  over  $\lambda \notin \mathcal{B}$  and using the tail bounds shows  $Z^{-1} = 1 + O(K\delta) + O(n^{-2/3})$ , whence  $\sum_{\lambda \notin \mathcal{B}} |P_n - Q_n| = O(K\delta) + O(n^{-2/3})$ . Combining with the tails,  $d_{\text{TV}}(P_n, Q_n) = O(\delta \log n + n^{-2/3})$  (with Proposition 19.11 in place of [3, Cor. 4.5] the error term becomes  $O(\delta \log n + n^{-1+o(1)})$ ). The loops statement follows from (1), Remark 1.3 (the passage to reduced squares is exact), and §1.3 as in [3, Lemma 6.2].  $\square$

**Remark 12.2.** The proof only uses EQ for splits along the specific chains, a strictly weaker hypothesis; and the numerically observed truth is  $\text{EQ}(Cn^{-3})$  on those chains. The hypothesis is a statement about one bit of structure — *rows 1 and 2 lie in the same column cycle of a shifted post-flip column pair* — for a uniformly random element of  $S(\lambda)$ ; it seems genuinely more tractable than the conjecture itself, being an equidistribution assertion for a single event rather than a global comparison of measures. A natural route is a second-moment/switching argument within  $S(\lambda)$  itself; another is the transpose duality: the  $B$ -status of column pairs of  $L$  is the same-cycle statistic of row pairs of  $L^\top$ , whose cycle-type law is again  $P_n$  — a self-referential structure that may admit a bootstrap.

### 13 Monte Carlo verification at larger $n$

We implemented the Jacobson–Matthews chain [8] (with the  $\pm 1$ -tensor moves; uniform stationary distribution on proper squares) and measured the quantities of §10 at orders beyond the exact tables. Two methodological points deserve record.

1. *Sampling bias of “first proper state after  $T$  moves”.* Emitting the first proper state reached after a fixed number of moves is *not* uniform: the entry distribution into the proper set is biased by the improper-excursion flux, which varies across squares (squares rich in intercalates admit more proper-to-proper moves). At  $n = 4$  this distorts the type- $(2, 2)$  probability from  $1/2$  to  $0.3887$ ; at  $n = 8$  it produced a cycle-type  $\chi^2 \approx 700$  on 6 d.o.f. against the exact distribution. Correct procedure: sample on a fixed grid of times and *discard* improper checkpoints. With this correction the  $n = 4$  marginal is uniform to  $5 \times 10^{-4}$  over  $2 \times 10^6$  samples.
2. *Validation against exact data.* With the corrected sampler: at  $n = 8$  ( $4 \times 10^6$  samples) the seven cycle-type frequencies match the exact distribution with  $\chi^2 = 2.7$  on 6 d.o.f.; at  $n = 11$  ( $2 \times 10^6$  samples) all fourteen types give  $\chi^2 = 10.4$  on 13 d.o.f.

| $n$ | samples           | $\mathbb{P}(B_1)$ | $\mathbb{P}(B_2)$ | $\bar{q}$ (measured) | $\bar{q}$ (exact, Thm 10.1) |
|-----|-------------------|-------------------|-------------------|----------------------|-----------------------------|
| 8   | $4.0 \times 10^6$ | 0.4971(5)         | 0.4961(5)         | 0.9932(8)            | 0.993345                    |
| 11  | $2.0 \times 10^6$ | 0.4987(9)         | 0.4992(9)         | 0.9979(15)           | 0.998080                    |
| 12  | $2.0 \times 10^6$ | 0.5009(10)        | 0.4986(10)        | 0.9995(15)           | [0.99850 predicted]         |
| 16  | $1.5 \times 10^6$ | 0.5001(13)        | 0.4983(13)        | 0.9984(21)           | [0.99937 predicted]         |

(Joins  $\mu = (2, n - 2) \rightarrow (n)$  throughout; “predicted” extrapolates  $\bar{q} = 1 - 2c_2n^{-3}$ .) At  $n = 8$  the measurement is an  $8\sigma$  detection of  $\bar{q} < 1$ , in exact agreement with the identity; at  $n = 11$  it again matches the exact value; from  $n = 12$  on, the conjectured deviation  $1 - \bar{q} \asymp 2.6n^{-3}$  sits below Monte Carlo resolution, and the measurements are consistent both with it and with Hypothesis 10.4 at the precision  $|\bar{q} - 1| \lesssim 2 \times 10^{-3}$ .

For a *generic* admissible pair the measured  $B$ -probabilities are  $0.4993(3)$ ,  $0.4997(4)$ ,  $0.5002(4)$ ,  $0.5007(4)$  at  $n = 8, 11, 12, 16$ ; exact enumeration gives  $3/5$  at  $n = 5$  and  $37/78 \approx 0.474$  at  $n = 6$ . Two marked points of a uniform derangement share a cycle with probability  $s_n = \frac{1}{2} + (1 + o(1))/n$  ( $\approx 0.55$  at  $n = 20$ ), so a naive analogy would predict an excess  $\approx +1/n$ ; but a uniform admissible column pair excludes the  $\approx n$  pairs  $\{j, \omega(j)\}$ , which are *always* same-cycle, and this exclusion cancels the  $1/n$  term exactly, leaving  $\mathbb{P}(B \mid \text{admissible}) = \frac{1}{2} + O(n^{-2})$  — as observed. The post-flip pairs of §10 are then closer still to balance, with deviation  $\Theta(n^{-3})$ : a three-level hierarchy  $\frac{1}{2} + \Theta(n^{-1}) \rightarrow \frac{1}{2} + O(n^{-2}) \rightarrow \frac{1}{2} - \Theta(n^{-3})$  of same-cycle statistics.

## 14 Can the modern machinery reach the conjecture?

The strongest current toolkit for random Latin squares is that of [12, 9]: comparison with the triangle-removal process (TRP), and re-randomisation via intercalate switchings.

**Transfer costs.** The TRP comparison theorems transfer events at multiplicative cost  $e^{O(n \log^2 n)}$ . This is decisive for large-deviation-scale statements (probabilities  $e^{-\Theta(n^{4/3} \log n)}$  or smaller) but is far too lossy for Conjecture 3.2, whose content is at scale  $1 \pm n^{-3}$ ; even Conjecture 1.2 (constant-scale total variation) is out of its direct reach.

**Re-randomisation.** More promising is the [9] re-randomisation philosophy. An intercalate switch involving row 2 and one other row composes  $\sigma_{1,2}$  with a transposition, i.e. acts on the cycle type by an elementary split or merge — these are exactly degenerate (length-4 column cycle) cases of the CGW flip. The row-parity theorem of [9] shows that enough disjoint, well-separated (“stable”) intercalates exist, and mix well enough, to drive a  $\pm 1$  functional (the parity) to near-perfect equidistribution. The cycle type of  $\sigma_{1,2}$  is a richer functional, but the induced dynamics on types is a split/merge chain whose stationary law on cycle types, were the switch statistics exactly balanced, would be  $Q_n$ -like.

**A concrete programme via the marked-pair identity.** Theorem 11.4 makes the re-randomisation target precise and, we believe, attackable with the machinery of [9]. The target bit is now: for uniform  $L \in S(\lambda)$  and a marked pair  $P = \{j, \sigma^\alpha(j)\}$ , whether rows 1, 2 lie in one column cycle of  $P$ . Three structural facts line up favourably.

(i) *The conditioning is free.* Any intercalate avoiding rows 1, 2 can be switched without touching  $\sigma$ : subset-switches of a stable collection  $\mathcal{I}_0(L)$  of disjoint intercalates in rows  $3..n$  (in the sense of [9, §7]: the collection is invariant under such switches) act *within*  $S(\lambda)$ , preserve the marking, and re-randomise exactly ( $\mathbb{P}(L \rightarrow L') = 2^{-|\mathcal{I}_0|}$  both ways): conditionally on the orbit, the configuration is uniform on  $\{0, 1\}^{\mathcal{I}_0}$ . No comparison between  $S(\lambda)$  and the uniform measure is needed at this step.

(ii) *Switch action on the bit.* An intercalate on rows  $\{r, r'\} \subseteq \{3..n\}$  meeting exactly one of the columns  $j, j'$  acts on the pair permutation  $\pi_P$  by precomposition with  $\tau_{rr'}$ : it splits or joins the  $\pi_P$ -cycles through  $r, r'$ , and flips the target bit precisely when  $r, r'$  “interleave” rows 1, 2 (one lies on the segment between them). Intercalates meeting *both* or *neither* of the columns act by conjugation and fix the bit. So the bit is non-linear in the switch configuration but locally computable, and each useful intercalate flips it with probability of constant order in a random-like configuration.

(iii) *The quantifiers fit the budget.* The intercalate-expander lemma of [9, Lem. 8.3, Def. 8.1] (with  $\ell = \log^{11} n$ ) implies that all but at most  $\ell$  columns carry at least  $6\ell$  disjoint stable intercalates (their construction of the row set  $R$ , transposed), with rows and partner columns avoidable in prescribed  $\beta n$ -sets — in particular avoiding rows 1, 2 and column  $j'$ . The exceptional fraction of marks is  $O(\log^{11} n/n) = o(1/\log n)$ : exactly within the  $\text{EQ}(o(1/\log n))$  budget that the weak conjecture requires. Moreover their approximation costs are affordable here:  $\mathbb{P}(S(\lambda)) \geq e^{-O(n \log n)}$  for every type  $\lambda$  (worst case all-2-cycles), while the expander lemma fails with probability  $e^{-\omega(n \log^2 n)}$ , so the expander property holds for  $L$  conditioned on  $S(\lambda)$ , for every  $\lambda$ , by a union-of-costs argument.

What remains — the genuinely open core — is the *non-linear orbit-mixing lemma*: for a  $2^{\mathcal{I}_0}$ -uniform configuration and  $k \geq 6 \log^{11} n$  disjoint stable intercalates through column  $j$  avoiding rows 1, 2, show that the conditional law of the bit given the orbit is  $\frac{1}{2} + o(1/\log n)$  for all but an  $o(1/\log n)$  fraction of orbits. Unlike the parity bits of [9] the bit is not  $\mathbb{F}_2$ -linear in the configuration, so their kernel computation must be replaced by a mixing argument for the cycle-structure bit under random subsets of transpositions — the orbit dynamics on  $\pi_P$  is a coagulation–fragmentation process driven by (a random subset of) disjoint transpositions, for which the classical toolbox is that of random transposition walks [13] — but the per-switch flip probability is of constant order and the budget  $o(1/\log n)$  is far weaker than their polynomial error terms. We record this as the sharpest currently-visible route to Conjecture 1.2.

[Session-6 update: the non-linear orbit-mixing lemma just described has been reformulated exactly. §19.5 proves that the conditional bit given an orbit is an explicit  $\mathbb{F}_2$ -rank event in a fixed interlacement matrix (Theorem 15.6), derives the supply and conditional expander lemmas, shows the orbit-by-orbit form of the mixing lemma stated above is too strong, and replaces it by the annealed Problem 15.8. Session-13 update: in the stationary-frame model Problem 15.8 is proved with the sharp exponent (§18); what is open is the transfer to the real orbit measure, §17.12.9.]

## 15 The orbit bit is a rank condition: an exact interlacement formula

The open core of the transfer programme of §14 is the “non-linear orbit-mixing lemma”: conditionally on a switching orbit, the marked bit — rows 1, 2 in one column cycle of the marked pair — is a cycle-structure functional of a product of a fixed permutation with a uniformly random subset of disjoint transpositions, and one must show its conditional law is close to  $\text{Ber}(1/2)$ . In this section we prove that this functional is, exactly, a rank condition over  $\mathbb{F}_2$  in a matrix determined by the orbit — so the “non-linear” bit is a (non-linear but completely explicit) linear-algebraic event after all, directly comparable to the kernel computation that powers the parity theorem of [9] (their Lemma 8.5). The formula turns the orbit-mixing problem into a question about ranks of random principal submatrices of a fixed interlacement matrix; we then (i) derive an exact identity for the conditional bias, (ii) exhibit the extremal configurations, (iii) show computationally that in generic position the bias decays like  $k^{-1/2}$  in the number  $k$  of available switches — far inside the  $o(1/\log n)$  budget when  $k \geq 6 \log^{11} n$  — and (iv) reformulate the open core accordingly, in an *annealed* form which the present data indicate is both sufficient and true, while the orbit-by-orbit form stated at the end of §14 is provably too strong. We also record the two supporting lemmas of the programme, the supply lemma (§15.6) and the conditional expander (§15.7), whose proofs are short.

Throughout,  $u = \text{row } 1$  and  $v = \text{row } 2$ ;  $\pi_P$  is the pair permutation of a column pair  $P$  on the row set (§11), whose cycles are the column cycles of  $P$ ; and the marked bit of an instance is  $[u \sim v \text{ in } \pi_P]$ , i.e. the indicator of  $B$ -status.

### 15.1 Chords, interlacement, and the Cohn–Lempel theorem

Let  $\gamma$  be a cyclic permutation of a finite set  $V$  (“the circle”). A *chord* is an unordered pair  $\{a, b\} \subseteq V$ ; a set of chords is *disjoint* if the pairs are pairwise disjoint. Chords  $\{a, b\}$  and  $\{c, d\}$  *interlace* (on  $\gamma$ ) if  $c, d$  lie in different arcs of the circle determined by  $a, b$ ; a chord  $\{a, b\}$  *separates*

points  $u, v \notin \{a, b\}$  if  $u, v$  lie in different arcs. For a set  $T$  of disjoint chords, identified with the product  $\tau_T = \prod_{\{a,b\} \in T} (ab)$  of the corresponding commuting transpositions, the *interlacement matrix*  $A_T \in \mathbb{F}_2^{T \times T}$  has  $(A_T)_{ij} = 1$  iff chords  $i \neq j$  interlace, and zero diagonal.

**Theorem 15.1** (Cohn–Lempel [4]). *Let  $\gamma$  be a single cycle on  $V$  and  $T$  a set of disjoint chords. Then the number of cycles of  $\gamma \tau_T$  equals  $\text{null}_{\mathbb{F}_2}(A_T) + 1$ .*

(We re-verified Theorem 15.1 instance-by-instance over random circles and all activation subsets; `pf_cl.py`.)

The matrix  $A_T$  is alternating (symmetric, zero diagonal), so  $z^\top A_T z = 0$  for every  $z$ ; this is what drives the next lemma.

**Lemma 15.2** (bordered rank). *Let  $A \in \mathbb{F}_2^{m \times m}$  be alternating and  $x \in \mathbb{F}_2^m$ , and let  $B = \begin{pmatrix} A & x \\ x^\top & 0 \end{pmatrix}$ . If  $x \in \text{col}(A)$  then  $\text{null}(B) = \text{null}(A) + 1$ ; otherwise  $\text{null}(B) = \text{null}(A) - 1$ .*

*Proof.* If  $x = Ay$ , add the  $y$ -combination of the first  $m$  columns to the last column and the  $y$ -combination of the first  $m$  rows to the last row: the border becomes  $(0, \dots, 0, y^\top Ay) = (0, \dots, 0, 0)$  since  $A$  is alternating; hence  $\text{rank}(B) = \text{rank}(A)$  and  $\text{null}(B) = (m+1) - \text{rank } A = \text{null}(A) + 1$ . If  $x \notin \text{col}(A)$ , then  $\text{rank}[A \ x] = \text{rank}(A) + 1$ , and the last row  $(x^\top, 0)$  is independent of the rows of  $[A \ x]$ : a relation  $x^\top = z^\top A$  would give  $x = Az$  by symmetry. So  $\text{rank}(B) = \text{rank}(A) + 2$  and  $\text{null}(B) = \text{null}(A) - 1$ .  $\square$

**Theorem 15.3** (the bit as a rank condition; single cycle). *Let  $\gamma$  be a single cycle on  $V$ , let  $u, v \in V$ , and let  $T$  be a set of disjoint chords avoiding  $u, v$ . Let  $x \in \mathbb{F}_2^T$  be the separation vector,  $x_i = [\text{chord } i \text{ separates } u, v]$ . Then*

$$u \sim v \text{ in } \gamma \tau_T \iff x \in \text{col}(A_T).$$

*Proof.* Let  $e$  denote the chord  $\{u, v\}$ ; since  $T$  avoids  $u, v$ , the set  $T \cup \{e\}$  is disjoint, its interlacement matrix is the bordered matrix of Lemma 15.2 (chord  $i$  interlaces  $e$  iff it separates  $u, v$ ), and multiplication of any permutation  $\rho$  by  $(uv)$  splits the cycle through  $u, v$  if  $u \sim v$  in  $\rho$  and joins two cycles otherwise. Thus, writing  $\rho = \gamma \tau_T$ , we have  $u \sim v$  in  $\rho$  iff  $c(\rho(uv)) = c(\rho) + 1$ , iff (Theorem 15.1 twice)  $\text{null}(A_{T \cup e}) = \text{null}(A_T) + 1$ , iff (Lemma 15.2)  $x \in \text{col}(A_T)$ .  $\square$

Both directions of the split/join dichotomy for  $(uv)$  are the switch-toggle mechanism of Lemma 11.3 in disguise: adjoining the virtual chord  $e$  is the CGW switch at the marked pair.

## 15.2 Arbitrary base permutations and two-sided products

Two reductions extend Theorem 15.3 to the situation actually produced by an orbit: the base permutation  $\pi_0 = \pi_P$  has many cycles, and the switch actions multiply on both sides.

**Lemma 15.4** (passenger construction). *Let  $\pi_0$  have cycles  $C_1, \dots, C_p$  (written as sequences), let  $s_1, t_1, \dots, s_{p-1}, t_{p-1}$  be  $2(p-1)$  fresh points, let*

$$\gamma = (C_1 \ s_1 \ C_2 \ t_1 \ s_2 \ C_3 \ t_2 \ \cdots \ s_{p-1} \ C_p \ t_{p-1})$$

*be the single cycle obtained by the indicated concatenation, and let  $B = \{\{s_i, t_i\} : 1 \leq i \leq p-1\}$  (the bridges; pairwise disjoint). Then  $\gamma \tau_B$  has cycles  $C_1 \cup \{s_1, \dots, s_{p-1}\}$  (with the  $s_i$  inserted consecutively after the last element of  $C_1$ ) and  $C_i \cup \{t_{i-1}\}$  for  $2 \leq i \leq p$ ; in particular, deleting the fresh points (first-return map) recovers  $\pi_0$  exactly, and for any set  $T$  of chords on  $V$  avoiding the fresh points,  $u \sim v$  in  $\pi_0 \tau_T$  iff  $u \sim v$  in  $\gamma \tau_{B \cup T}$ .*

*Proof.* Direct computation of successors in  $\gamma\tau_B$ : for  $x \in C_i$  not last,  $\gamma\tau_B(x) = \gamma(x)$  is the successor in  $C_i$ ; for the last element  $x$  of  $C_i$  ( $i \geq 2$ ),  $\gamma(x) = t_{i-1}$  and  $\gamma\tau_B(t_{i-1}) = \gamma(s_{i-1})$  is the first element of  $C_i$ , closing the cycle  $C_i \cup \{t_{i-1}\}$ ; for the last element of  $C_1$ ,  $\gamma(x) = s_1$ ,  $\gamma\tau_B(s_i) = \gamma(t_i) = s_{i+1}$  for  $i < p-1$ , and  $\gamma\tau_B(s_{p-1}) = \gamma(t_{p-1}) =$  first element of  $C_1$  (the wrap), closing  $C_1 \cup \{s_1, \dots, s_{p-1}\}$ . Since first return commutes with right multiplication by transpositions avoiding the deleted points, the last claim follows.  $\square$

**Lemma 15.5** (point splitting for two-sided products). *Let  $\pi_0$  be a permutation of  $V$ , and let  $\rho = \tau_R$ ,  $\mu = \tau_M$  be products of transpositions where the chords within  $R$  are pairwise disjoint and likewise within  $M$ , all avoiding  $u, v$ , but chords of  $R$  may meet chords of  $M$ . Form the doubled circle: replace each point  $x$  by two consecutive copies  $(x, 0), (x, 1)$  in each cycle of  $\pi_0$ ; lift each chord  $\{a, b\} \in M$  to  $\{(a, 0), (b, 0)\}$  and each  $\{a, b\} \in R$  to  $\{(a, 1), (b, 1)\}$ . Then all lifted chords are pairwise disjoint, and the permutation  $\tilde{P} = \tilde{\pi}_0 \tau_{\tilde{R} \cup \tilde{M}}$  of the doubled point set (where  $\tilde{\pi}_0$  is the doubled cyclic structure) satisfies: the first-return map of  $\tilde{P}$  on the copies  $\{(x, 0)\}$  is  $(x, 0) \mapsto (\pi_0 \rho \mu(x), 0)$ . In particular  $u \sim v$  in  $\pi_0 \rho \mu$  iff  $(u, 0) \sim (v, 0)$  in  $\tilde{P}$ .*

*Proof.* Compute  $\tilde{P}$  on the two kinds of points. On  $(x, 0)$ : the only lifted chord that can contain  $(x, 0)$  is the lift of the  $M$ -chord through  $x$  (if any), so  $\tau((x, 0)) = (\mu(x), 0)$  in every case, and  $\tilde{\pi}_0((\mu(x), 0)) = (\mu(x), 1)$ : hence  $\tilde{P}((x, 0)) = (\mu(x), 1)$ . On  $(x, 1)$ : the only chord containing  $(x, 1)$  is the lift of the  $R$ -chord through  $x$  (if any), so  $\tau((x, 1)) = (\rho(x), 1)$  and  $\tilde{\pi}_0((\rho(x), 1)) = (\pi_0 \rho(x), 0)$ : hence  $\tilde{P}((x, 1)) = (\pi_0 \rho(x), 0)$ . Composing,  $\tilde{P}^2 \upharpoonright_{\text{copies } 0}: (x, 0) \mapsto (\pi_0 \rho \mu(x), 0)$ , and every second point of each  $\tilde{P}$ -cycle is a copy-0 point, so the first-return map is as claimed and cycle membership transfers.  $\square$

### 15.3 The master formula for the orbit bit

Fix an orbit: a base square  $L_0 \in S(\lambda)$  (any point of the orbit), a marked pair  $P = \{j, j'\}$ , the stable collection  $\mathcal{I}_0$  of disjoint stable intercalates in rows  $3..n$  (in the sense of [9, §7], with respect to a fixed template), and the uniform configuration  $\omega \in \{0, 1\}^{\mathcal{I}_0}$ . Partition the *useful* coordinates by column incidence with  $P$ :  $\mathcal{I}_j$  (meets column  $j$  only),  $\mathcal{I}_{j'}$  (column  $j'$  only),  $\mathcal{I}_{jj'}$  (both). As computed in §14: a switch in  $\mathcal{I}_j$  with rows  $\{r, r'\}$  precomposes  $\pi_P$  with  $(r r')$ ; one in  $\mathcal{I}_{j'}$  postcomposes (left-multiplies); one in  $\mathcal{I}_{jj'}$  conjugates; all other coordinates fix  $\pi_P$ . Two stable intercalates meeting a common column cannot share a row (they would share a cell, contradicting isolation), so the chords within each of the families  $R := \mathcal{I}_j \cup \mathcal{I}_{jj'}$  (right side) and  $M := \mathcal{I}_{j'} \cup \mathcal{I}_{jj'}$  (left side) are pairwise disjoint; between the two sides rows may be shared. Writing  $\pi(\omega) = M_\omega \pi_0 R_\omega$  and conjugating by  $M_\omega^{-1}$  (which fixes  $u, v$ : all supports avoid rows  $1, 2$ ),

$$\text{bit}(\omega) = [u \sim v \text{ in } \pi_0 R_\omega M_\omega].$$

Now apply Lemma 15.5 (doubling;  $M$ -chords to copies 0,  $R$ -chords to copies 1), then Lemma 15.4 (bridges  $B$ ) to the doubled multi-cycle structure, then Theorem 15.3 on the resulting single circle:

**Theorem 15.6** (master formula). *For each orbit there is an explicitly computable alternating matrix  $\bar{A}$  over  $\mathbb{F}_2$ , indexed by the lifted chords of the useful coordinates together with the bridge set  $B$  ( $|B| = \#\text{cycles}(\pi_0) - 1$ ), and a vector  $\bar{x}$  (separation of  $(u, 0), (v, 0)$  on the augmented circle), both determined by the cyclic-order positions of the intercalate rows in the cycles of  $\pi_0 = \pi_P(L_0)$ , such that for every configuration  $\omega$ ,*

$$\text{bit}(\omega) = [\bar{x}_{\varphi(\omega) \cup B} \in \text{col}(\bar{A}_{\varphi(\omega) \cup B})],$$

where  $\varphi(\omega)$  is the set of lifted chords of the active useful coordinates (a coordinate in  $\mathcal{I}_{jj'}$  contributes its two lifts together). Consequently the conditional law of the marked bit given the orbit is the law of a rank event of a uniformly random principal submatrix (containing the frozen block  $B$ ) of the fixed matrix  $\bar{A}$ .

*Status: PROVED (Theorem 15.1 cited; all other ingredients proved above), and VERIFIED instance-by-instance: `pf_cl.py` checks (CL), (BIT), the passenger construction, and randomized two-sided configurations with shared rows and conjugating coordinates (1262 cases, all activation subsets, exhaustively) against direct computation.*

The non-linearity of the bit is thus precisely the non-linearity of  $S \mapsto \text{null}(\bar{A}_S)$ ; the  $\mathbb{F}_2$ -linear parity functionals of [9] correspond to the special case where the relevant matrix acts by a fixed linear map on the configuration. The coagulation–fragmentation dynamics ([13]) has been integrated out: no walk analysis is needed at fixed orbit.

## 15.4 The exact bias identity and extremal configurations

Let  $k$  be the number of useful coordinates and, for  $S$  a uniform random subset, let  $\text{bias} = \mathbb{P}_S[\text{bit}] - \frac{1}{2}$  (conditional on the orbit).

**Proposition 15.7** (bias identity). *With  $e$  the virtual chord  $\{(u, 0), (v, 0)\}$  adjoined to the ground set,*

$$\text{bias} = \frac{1}{2} \mathbb{E}_S [\text{null } \bar{A}_{S \cup B \cup e} - \text{null } \bar{A}_{S \cup B}]$$

— *the expected nullity increment of the virtual chord. Moreover, for any single coordinate  $i$ , the toggle status of  $i$  (whether flipping  $\omega_i$  flips the bit) is a function of  $\omega_{-i}$  alone, and  $|\text{bias}| \leq \frac{1}{2} \mathbb{P}[i \text{ does not toggle}]$ .*

*Proof.* The first claim is Lemma 15.2 applied at each  $S$  (the increment is  $+1$  exactly on the event  $\text{bit} = 1$ ). For the second: multiplying by the transposition of coordinate  $i$  splits or joins according to the current configuration of the others, and the toggle relation is symmetric between  $\omega_i = 0, 1$ ; conditionally on  $\omega_{-i}$  with  $i$  toggling, the bit is exactly Bernoulli( $1/2$ ).  $\square$

Three exactly solvable families calibrate what balance requires (all verified in `pf_bias.py`):

1. **Ladders.**  $r$  separating chords, pairwise interlacing ( $\bar{A} = J + I$  on the ladder,  $x = \mathbf{1}$ ), no other chords: the kernel computation gives  $\text{bit}(S) = [|S| \text{ even}]$  — the bit collapses to an  $\mathbb{F}_2$ -linear parity and  $\text{bias} = 0$  exactly, for every  $r \geq 1$ .
2. **Nested families.**  $r$  separating chords, pairwise non-interlacing ( $\bar{A} = 0$ ,  $x = \mathbf{1}$ ):  $\text{bit}(S) = [S = \emptyset]$ ,  $\text{bias} = 2^{-r} - \frac{1}{2}$ . This is the worst case, and is the exact analogue of the failure mode that the expander hypothesis exists to exclude.
3. **Cut chords.** A single separating chord  $c$  interlacing no other chord (an “uncrossed cut”) forces  $\text{bit} = 0$  whenever  $c$  is active:  $e_c$  is then a kernel vector with  $x_c = 1$ , killing the rank event. Hence, exactly,  $\text{bias} = \frac{1}{2} \text{bias}' - \frac{1}{4} \in [-\frac{1}{2}, 0]$ , where  $\text{bias}'$  is the conditional bias given  $c$  inactive. In particular good structure elsewhere cannot restore balance: with a ladder in the remainder ( $\text{bias}' = 0$ ),  $\text{bias} = -\frac{1}{4}$  exactly (verified). Balance is therefore *not* implied by the presence of good structure; it requires the *absence* of bad structure — every separating chord must itself be crossed, recursively.

Item 3 has a consequence for the shape of the mixing lemma: no argument that only *finds* favourable switches (a ladder) can close the programme; the statement must exploit the measure on orbits.

### 15.5 Generic position: $k^{-1/2}$ decay, and the annealed reformulation

The handoff formulation of the mixing lemma (“conditional bit-law =  $\frac{1}{2} + o(1/\log n)$  for all but  $o(1/\log n)$  of orbits”) is *too strong*: if rows 1, 2 are at distance  $O(1)$  along their common  $\pi_0$ -cycle and no intercalate row falls in the short arc (an event of probability  $\Theta(1/k)$  under any spread-out measure, and of constant conditional probability on some  $S(\lambda)$ -positive events), then no available chord separates  $u, v$ ,  $x = 0$ , and  $\text{bit} \equiv 1$ : bias =  $+\frac{1}{2}$  on an orbit set of measure  $\asymp 1/k$ , not  $o(1/\log n)$  small in general. The correct target is the *annealed* (orbit-averaged) statement, which is also all that Theorem 11.4 requires:

**Problem 15.8** (annealed orbit-mixing — the revised open core). For  $L$  uniform on  $S(\lambda)$  restricted to the intercalate-expander event,  $j$  a supplied column (Lemma 15.9), show

$$\mathbb{E}_L |\text{bias}(\text{orbit}(L), P)| = o(1/\log n) \quad (\text{averaged over marks as in §11}).$$

Computationally, generic position gives far more than is needed. For the ensemble “ $u, v$  and  $k$  disjoint chords in uniformly random position on a circle” (all activation subsets exact for  $k \leq 20$ , Monte Carlo beyond; `bias_scale.c`):

| $k$                       | 8      | 16     | 32     | 64     | 128    |
|---------------------------|--------|--------|--------|--------|--------|
| $\mathbb{E} \text{bias} $ | 0.124  | 0.099  | 0.075  | 0.044  | 0.032  |
| median                    | 0.102  | 0.082  | 0.053  | 0.028  | 0.018  |
| mean (signed)             | +0.015 | +0.000 | −0.000 | +0.000 | +0.007 |

i.e.  $\mathbb{E}|\text{bias}| \approx 0.35 k^{-1/2}$ , with the signed mean consistent with  $O(1/k)$ . The dependence on the  $u$ – $v$  arc length is flat except when the arc contains  $O(1)$  chord endpoints (measured: at  $k = 64$ , mean bias +0.12 with  $\approx 1$  endpoint-slot in the arc, indistinguishable from generic already at  $\approx 3$ ); the short-arc contribution to the annealed average is therefore  $O(1/k)$ , self-regularizing — no separate “join-then-toggle chain argument” is required in the annealed formulation. With  $k \geq 6 \log^{11} n$  (the supply lemma), a bound of quality  $\mathbb{E}|\text{bias}| \leq k^{-c}$  for *any* constant  $c > 1/11$  closes Problem 15.8; the generic-position value  $c = 1/2$  has a margin of a factor 5 in the exponent.

*Status: the reduction of the weak conjecture to Problem 15.8 together with Lemmas 15.9–15.11 is PROVED modulo the two bookkeeping items recorded in Remark 15.10 [session 13: item (i) is closed by Proposition 17.39]; the  $k^{-1/2}$  scaling is COMPUTED (not proved) and refers to the iid-position ensemble, not yet to the true orbit ensemble.*

What would a proof of Problem 15.8 need? By Proposition 15.7 it suffices to show that under the true orbit measure the chord configuration is *generic*: e.g., (i) with probability  $1 - o(1/\log n)$  every separating chord is interlaced by some other available chord (excluding cut chords and nested families), and (ii) a quantitative rank-balance for the resulting random interlacement matrices. Both are statements about the joint law of the positions, in the cycles of  $\pi_P$ , of the stable-intercalate rows of a uniform  $L \in S(\lambda)$  — a concentration/anti-concentration problem, not a worst-case one. The exchangeability of rows 3.. $n$  (which acts on everything by conjugation, though not template-equivariantly) and the Harer–Zagier-type computation of expected nullity

for random chord systems are the natural tools. [Session-7 update: conditions of type (i) are provably insufficient at any fixed crossing multiplicity (Proposition 16.5 and the exhaustive scan of §16.2); the productive formalisation of (ii) is the second-moment factorization of §16.1, which reduces the iid model to Problem 16.4.]

## 15.6 The supply lemma

**Lemma 15.9** (supply). *Let  $H$  be a partial Latin square that is an  $(\ell, \beta)$ -intercalate-expander (Definition 8.1 of [9]) on  $n$  symbols,  $n$  large — in the application,  $H = T \cap L$ ; we avoid the letter  $P$ , which denotes the marked pair — and let  $u, v$  be two rows,  $j'$  a column, and  $t \geq 1$  with  $4t\ell + 2t + \ell + 2 \leq \beta n/2$  and  $\beta \leq \frac{1}{2}$  (for general  $\beta$  replace the hypothesis by  $2t\ell + 2 \leq (1 - \beta)n$ ,  $t\ell + \ell + 1 \leq (1 - \beta)n$ ; the application has  $\beta = 1/10$ ). Then for all but fewer than  $t\ell$  columns  $c \neq j'$  there exist  $t$  stable intercalates of  $H$  through  $c$ , pairwise disjoint in rows, partner columns and symbols, all avoiding rows  $\{u, v\}$  and column  $j'$ . Moreover all stable intercalates through a common column are automatically pairwise row-disjoint (two of them sharing a row would share a cell in that column, contradicting isolation). With  $\ell = \log^{11} n$  and  $t = 6\ell$ : all but  $6 \log^{22} n$  columns carry at least  $6 \log^{11} n$  such intercalates.*

*Proof.* Process rounds  $\tau = 1, \dots, t$ ; every column starts alive with an empty family, and in round  $\tau$  each alive column attempts to add one intercalate to its family subject to the constraints; a column that fails is declared dead and never retried. Suppose  $\ell$  columns die in some round  $\tau$ . Each dead column has a current family of  $\tau - 1 \leq t$  intercalates. Apply the expander property with:  $C^* =$  the  $\ell$  dead columns;  $R = R^* =$  all rows except  $u, v$  and the  $\leq 2t\ell$  rows used by the dead columns' families (size  $\geq n - 2 - 2t\ell \geq \beta n$ );  $C =$  all columns except  $C^*$ ,  $j'$ , and the  $\leq t\ell$  used partner columns (size  $\geq n - \ell - 1 - t\ell \geq \beta n$ );  $S = S^* =$  all symbols except the  $\leq 2t\ell$  used ones. The guaranteed stable intercalate has one column  $c_0 \in C^*$  and avoids every constraint of  $c_0$ 's attempted extension — contradicting the death of  $c_0$ . So fewer than  $\ell$  columns die per round, hence fewer than  $t\ell$  die overall, and every surviving column finishes with a full family of size  $t$ .  $\square$

Note that Definition 8.1 is monotone in the five  $\beta n$ -sets (a larger set contains a  $\beta n$ -subset), which the proof uses freely.

**Remark 15.10** (bookkeeping for exceptional columns; template averaging). Two quantifier items remain to make the assembly fully rigorous, both of which we record with proposed fixes rather than proofs. (i) *Exceptional marks.* The  $t\ell$  exceptional columns of Lemma 15.9 are square-dependent; for merges whose  $m$ -cycles occupy few columns ( $\lambda_m m = n^{o(1)}$ , say) the per-square fraction of bad marks is not automatically  $o(1/\log n)$ . [Session-6 text proposed column-relabelling (template) averaging as a fix. *Withdrawn:* Proposition 16.6 shows that averaging over relabelled templates  $g(T)$  leaves the  $X$ -averaged bad-mark fraction exactly unchanged, by joint equivariance. The genuine requirement is the supply-mark decoupling stated as Problem 16.7.] (ii) *Rows-1, 2 conditioning.* The expander Lemma 8.3 of [9] is stated for uniform  $L$ ; conditioning on  $S(\lambda)$  is handled by Lemma 15.11 below, but the supplied intercalates must additionally avoid rows 1, 2, which Lemma 15.9 already builds in through the row sets  $R, R^*$ .

## 15.7 The conditional expander

**Lemma 15.11** (expander conditioned on the type). *Let  $T$  be the template of [9, Lem. 8.3] and  $\ell = \log^{11} n$ . For every type  $\lambda$ ,*

$$\mathbb{P}[T \cap L \text{ is an } (\ell, \beta)\text{-intercalate-expander} \mid L \in S(\lambda)] \geq 1 - e^{-\omega(n \log^2 n)}.$$

*Proof.*  $\mathbb{P}[\neg\text{expander} \mid S(\lambda)] \leq \mathbb{P}[\neg\text{expander}]/\mathbb{P}[S(\lambda)]$ , and  $\mathbb{P}[\neg\text{expander}] \leq e^{-\omega(n \log^2 n)}$  by [9, Lem. 8.3], so it suffices that  $\mathbb{P}[S(\lambda)] \geq e^{-O(n \log n)}$  for every  $\lambda$ . Writing  $\mathbb{P}[S(\lambda)] = D_\lambda N(\lambda) / \sum_\nu D_\nu N(\nu)$ : every  $D_\nu / D_\lambda \leq n! / D_\lambda \leq e^{O(n \log n)}$  (the worst type, all 2-cycles, has  $D_\lambda = n! / (2^{n/2} (n/2)!) \geq n! e^{-O(n \log n)}$ ), the number of types is  $e^{O(\sqrt{n})}$ , and  $N(\nu) / N(\lambda) \leq 3^n$  for all  $\nu, \lambda$  by composing the two-sided merge bounds ( $\frac{1}{2} \leq C_n(\text{merge}) / C_n(\text{split}) \leq \frac{3}{2}$ : the upper bound is [3, Thm. 3.13], the lower is that theorem or Theorem 11.4) along a path of at most  $n$  merges through the one-part type. Combining,  $\mathbb{P}[S(\lambda)] \geq e^{-O(n \log n)}$ .  $\square$

In particular no transfer through the comparison machinery ([9, Lem. 4.7]) is needed for the conditioning step: direct division of probabilities suffices, for every type simultaneously.

## 15.8 Summary of the revised programme

1. Conditioning is free; orbits re-randomise exactly (§14, unchanged).
2. Supply: for all but  $o(1/\log n)$  of marks, both marked columns carry  $\geq 6 \log^{11} n$  disjoint stable switches avoiding rows 1, 2 (Lemma 15.9 + Lemma 15.11). [PROVED for marks in classes of size  $\lambda_m m \geq \log^{24} n$ , where the deterministic bound  $|E| \leq 6 \log^{22} n$  gives bad fraction  $O(1/\log^2 n)$ ; for smaller classes OPEN as Problem 16.7 — template averaging does not close it, Proposition 16.6.]
3. The conditional bit given the orbit is an explicit  $\mathbb{F}_2$  rank event in a fixed interlacement matrix (Theorem 15.6). [PROVED.]
4. Open core: the annealed bias bound (Problem 15.8). Generic position gives  $\mathbb{E}|\text{bias}| \approx 0.35k^{-1/2}$ , a factor-5 exponent margin over the budget; extremal counterexamples (nested families, cut chords) show the bound must come from the measure, not from worst-case structure. [OPEN; scaling COMPUTED.]
5. EQ( $o(1/\log n)$ ) then follows by Theorem 11.4, and the weak conjecture by Theorem 12.1.

## 16 The annealed bound: factorization of the second moment, refutation of local genericity, and real-ensemble data

This section records the session-7 progress on Problem 15.8. The positive result is a *two-point factorization* of the annealed second moment (Theorem 16.1), which reduces the iid model of the annealed bias bound to a single one-parameter mixing statement — polynomial  $L^2$  decay under a “fresh-chord averaging” Markov operator (Problem 16.4) — verified numerically with a large margin. The negative results sharpen the target: every local genericity condition of the kind contemplated at the end of §15.5 is refuted by an explicit family (Proposition 16.5 and an exhaustive scan), and the template-averaging fix proposed in Remark 15.10(i) is shown to be *exactly neutral* (Proposition 16.6), which re-opens the exceptional-mark bookkeeping as a

genuine (if presumably easier) probabilistic sub-problem, stated as Problem 16.7. Finally, the master formula and the bias statistics are grounded on *real* random squares at  $n = 30\text{--}100$  via Jacobson–Matthews sampling.

### 16.1 The two-point factorization of the annealed second moment

Work in the *iid ensemble*:  $u, v$  fixed on the circle, and chords  $Z_1, \dots, Z_k$  with iid endpoint pairs, uniform on the circle (a.s. disjoint); this is the continuum version of the generic-position ensemble of §15.5, and the argument below uses nothing about the position law except that the  $Z_i$  are iid given  $(u, v)$ . For a finite set  $W$  of disjoint chords let

$$\varepsilon(W) = c(\gamma_W \tau_W(uv)) - c(\gamma_W \tau_W) \in \{\pm 1\},$$

the cycle increment of the virtual chord:  $\varepsilon(W) = +1$  iff  $u \sim v$  in  $\gamma_W \tau_W$ , i.e.  $\varepsilon = 2 \text{ bit} - 1$ . For  $S \subseteq [k]$  write  $\varepsilon_S = \varepsilon(\{Z_i : i \in S\})$ . Then bias  $= \frac{1}{2} \mathbb{E}_S[\varepsilon_S]$  conditionally on the positions, so with  $S, T$  independent uniform subsets of  $[k]$ ,

$$4 \mathbb{E}[\text{bias}^2] = \mathbb{E}_{S,T} \mathbb{E}_{\text{pos}}[\varepsilon_S \varepsilon_T]. \quad (21)$$

**Theorem 16.1** (two-point factorization). *Fix  $S, T \subseteq [k]$  and let  $j = |S \cap T|$ ,  $d_1 = |S \setminus T|$ ,  $d_2 = |T \setminus S|$ . Let  $C = (Z_i)_{i \in S \cap T}$  and define the fresh-averaged sign*

$$M_d(C) := \mathbb{E}[\varepsilon(C \cup F_d) \mid C], \quad F_d = d \text{ fresh iid chords}.$$

Then

$$\mathbb{E}_{\text{pos}}[\varepsilon_S \varepsilon_T] = \mathbb{E}_C[M_{d_1}(C) M_{d_2}(C)].$$

Consequently, with  $\psi(j, d) := \mathbb{E}_{C_j}[M_d(C_j)^2]$  ( $C_j = j$  iid chords),

$$4 \mathbb{E}[\text{bias}^2] \leq \max_{j \leq k, d \geq k/8} \psi(j, d) + 2e^{-k/32}.$$

*Proof.* Given  $C$ , the position blocks  $(Z_i)_{i \in S \setminus T}$  and  $(Z_i)_{i \in T \setminus S}$  are independent, and  $\varepsilon_S$  is a function of  $C$  and the first block only,  $\varepsilon_T$  of  $C$  and the second only: so  $\mathbb{E}[\varepsilon_S \varepsilon_T \mid C] = M_{d_1}(C) M_{d_2}(C)$ ; take expectations. For the second claim: by Cauchy–Schwarz,  $\mathbb{E}_C[M_{d_1} M_{d_2}] \leq \psi(j, d_1)^{1/2} \psi(j, d_2)^{1/2} \leq \max_{d \in \{d_1, d_2\}} \psi(j, d)$ . In (21),  $(j, d_1, d_2)$  is multinomial with cell probabilities  $1/4$ , so  $d_1, d_2 \geq k/8$  outside probability  $2e^{-k/32}$  (Chernoff), where  $|\varepsilon_S \varepsilon_T| \leq 1$  absorbs the exceptional mass.  $\square$

**Corollary 16.2.** *If  $\psi(j, d) \leq K d^{-2c}$  for all  $j \leq k$ ,  $d \geq k/8$ , then  $\mathbb{E}|\text{bias}| \leq \mathbb{E}[\text{bias}^2]^{1/2} \leq K' k^{-c}$ . With the supply  $k \geq 6 \log^{11} n$  of Lemma 15.9, any  $c > 1/11$  closes the iid model of Problem 15.8.*

The quantity  $M_d$  has a Markov-semigroup description: if  $P$  denotes the operator that adds one fresh active chord and averages, i.e.  $(Pf)(C) = \mathbb{E}_z[f(C \cup z)]$ , then by exchangeability of the fresh chords

$$M_d = P^d \varepsilon, \quad \psi(j, d) = \|P^d \varepsilon\|_{L^2(\mu_j)}^2,$$

$\mu_j$  the law of  $j$  iid chords. Adding a fresh chord is one step of a coagulation–fragmentation dynamics on the cycle structure of  $\gamma \tau$  (cf. [13]); the one-point function  $\mathbb{E}[\varepsilon(F_m)]$  is a Harer–Zagier-type quantity [7]. The open core of the iid model is thus:

The array  $\psi$  is monotone in a useful way:

**Lemma 16.3** (freezing dominates averaging).  $\psi(j, d+1) \leq \psi(j+1, d)$  for all  $j, d \geq 0$ ; hence  $\psi(j, d) \leq \psi(j+d-d', d')$  for all  $0 \leq d' \leq d$ , and  $\bar{\psi}(d) := \sup_j \psi(j, d)$  is non-increasing in  $d$ .

*Proof.* Conditional Jensen:  $(Pf)(C)^2 = \mathbb{E}_z[f(C \cup z)]^2 \leq \mathbb{E}_z[f(C \cup z)^2] = P(f^2)(C)$ . Hence  $\psi(j, d+1) = \mathbb{E}_{\mu_j}[(P(P^d \varepsilon))^2] \leq \mathbb{E}_{\mu_j}[P((P^d \varepsilon)^2)] = \mathbb{E}_{\mu_{j+1}}[(P^d \varepsilon)^2] = \psi(j+1, d)$ , since integrating one fresh chord into  $C \sim \mu_j$  produces exactly  $\mu_{j+1}$ . Iterate.  $\square$

**Problem 16.4** (polynomial  $L^2$  mixing of fresh-chord averaging). Show that  $\bar{\psi}(d) \leq Kd^{-2c}$  for some  $c > 1/11$  (numerically true with  $c \approx 0.52$ : the measured  $\psi(256, d)$  fits  $d^{-1.05}$ , and the  $j$ -dependence saturates; see below).

*Status:* Theorem 16.1 and Corollary 16.2 are PROVED (three-line conditioning argument; exact in the iid ensemble). Problem 16.4 is OPEN; the numerics below (COMPUTED, `s7.twopoint.c`) give  $\psi \approx 0.8/d$  with at most slow growth in  $j$ :

| $\psi(j, d)$ | $d = 4$ | $d = 16$ | $d = 64$ | $d = 256$ |
|--------------|---------|----------|----------|-----------|
| $j = 16$     | 0.124   | 0.019    | 0.0017   | 0.0004    |
| $j = 64$     | 0.174   | 0.041    | 0.0084   | 0.0010    |
| $j = 256$    | 0.197   | 0.050    | 0.0114   | 0.0027    |
| $j = 1024$   | —       | 0.064    | 0.0128   | —         |

Directly measured (cycle route, independent of the rank route):  $4\mathbb{E}[\text{bias}^2] \cdot k = 0.95, 1.08, 1.22, 1.28, 1.27, 1.44$  at  $k = 8, 16, 32, 64, 128, 256$ , i.e.  $\mathbb{E}[\text{bias}^2] \approx 0.33/k$ , consistent with  $\mathbb{E}[\text{bias}] \approx 0.35k^{-1/2}$ ; the signed one-point function  $\mathbb{E}[\varepsilon(F_m)]$  is  $+0.024, +0.005, +0.001$  at  $m = 4, 8, 16$  and  $\leq 10^{-3}$  (statistically 0) for  $m \geq 16$ .

Two further remarks. (i) The factorization argument uses exactly one probabilistic fact: the positions attached to disjoint index sets are independent. Transferring it to the true orbit measure — where the chord positions are the stable-intercalate row positions in the cycles of  $\pi_P$ , exchangeable-in-rows but not independent — therefore requires either a conditional-resampling argument (given the shared intercalates, re-randomise the rest of the square) or a quantitative correlation bound; this is the transfer problem 7a(i), not an additional obstacle discovered here. (ii) The short-arc orbits are visible in the factorization as the atom of  $C$ -configurations with no fresh-chord separation available; in the antipodal continuum ensemble this atom is absent, and in the true ensemble it contributes the  $O(1/k)$  term already accounted for in §15.5.

## 16.2 No local genericity condition suffices

The discussion at the end of §15.5 contemplated a genericity condition of the form “with high probability every separating chord is interlaced by some other available chord”. This is *qualitatively insufficient*, in the strongest possible sense: crossing multiplicity of any fixed order does not help.

**Proposition 16.5** (pendant stars). *For every  $m \geq 1$  and  $r \geq 1$  there is a configuration of  $k = r(m+1)$  disjoint chords in which every separating chord is interlaced by exactly  $m$  other chords, and*

$$\text{bias} = (1 - 2^{-(m+1)})^r - \frac{1}{2} \xrightarrow{r \rightarrow \infty} -\frac{1}{2}.$$

*Proof.* Take  $r$  pairwise non-interlacing (nested) separating chords  $c_1, \dots, c_r$ , and for each  $i$  a cluster of  $m$  pairwise nested “pendant” chords straddling one endpoint of  $c_i$ , short enough to interlace  $c_i$  and nothing else, and to separate nothing. The interlacement graph is a disjoint union of  $r$  stars  $K_{1,m}$  with separation vector  $e_{\text{center}}$  on each;  $\bar{A}_S$  is block diagonal, so the bit is the product of the block bits. In a block,  $x_S \in \text{col}(A_S)$  fails iff the center is active and all  $m$  pendants are inactive (probability  $2^{-(m+1)}$ ): the block bit has  $\mathbb{P} = 1 - 2^{-(m+1)}$ , and the blocks are independent.  $\square$

(Verified exactly for  $m \leq 3$ ,  $r \leq 4$ : `s7_bias_lab.py pendant`.) To keep  $|\text{bias}| \leq k^{-c}$  in this family one needs  $m \gtrsim \log r$ : no condition “every separating chord is crossed by  $\geq m$  others” with  $m$  fixed (or growing slower than logarithmically) can imply the annealed bound. An exhaustive scan of *all* chord configurations with  $k \leq 6$  chords (`s7_bias_lab.py exhaust`;  $10,395$  matchings  $\times$  all  $13$   $v$ -positions at  $k = 6$ ) confirms the moral. Within the classes cut by the natural local predicates — number  $r$  of separating chords, minimum crossing number of a separating chord (by all chords, or by separating chords only), connectivity or  $\mathbb{F}_2$ -rank of the separating-chord interlacement subgraph —  $|\text{bias}|$  vanishes identically only on ladders and on the class “ $r = 1$  with the separating chord crossing *nothing*”, where the cut-chord identity of §15.5 gives  $\text{bias} = \frac{1}{2} \cdot (+\frac{1}{2}) - \frac{1}{4} = 0$  exactly (the remainder has empty separation vector, hence conditional bias  $+\frac{1}{2}$ ). At  $k = 6$ , every other class with  $r \geq 1$  contains configurations with  $|\text{bias}| \geq 1/16$ ; the only classes whose maxima stay below  $15/64$  are the all-separating ( $r = k$ ) high-crossing classes, and the class maxima grow with  $k$  (e.g. the class “ $r = 2$ , minimum crossing 1” has maxima  $0.188, 0.219, 0.234$  at  $k = 4, 5, 6$ ). The bound must come from the measure; the moment route of §16.1 is the appropriate formalisation.

Relatedly, the single-coordinate toggle bound of Proposition 15.7 cannot by itself produce decay: in the generic ensemble the measured  $\mathbb{E}[\min_i \mathbb{P}(i \text{ does not toggle})]$  is  $\approx 0.35$ – $0.40$  and *non-decreasing* over  $k = 4, \dots, 12$  (`s7_bias_lab.py toggle`), while  $\mathbb{E}|\text{bias}|$  already decays over the same range.

### 16.3 Template averaging is exactly neutral

Remark 15.10(i) proposed to handle the square-dependence of the exceptional columns of Lemma 15.9 by relabelling the template by a uniform column permutation  $g$  and averaging. This does not work:

**Proposition 16.6** (vacuity of template averaging). *Fix  $n, \lambda, \alpha$  and a template  $T$ . For a mark  $(L, P)$  and a template  $T'$ , let  $\text{bad}(T'; L, P) \in \{0, 1\}$  indicate that a column of  $P$  fails the supply property of Lemma 15.9 for the partial square  $T' \cap L$  (with the exceptional set defined intrinsically: the set of columns  $c$  that do not carry  $t$  disjoint stable intercalates avoiding rows 1, 2 and the partner column). Then for every column permutation  $h$ ,*

$$\mathbb{E}_{(L,P) \sim \text{Unif}(X)} [\text{bad}(h(T); L, P)] = \mathbb{E}_{(L,P) \sim \text{Unif}(X)} [\text{bad}(T; L, P)].$$

*In particular averaging over uniform  $g$  leaves the  $X$ -averaged bad-mark fraction unchanged, and the same holds verbatim when  $T$  is itself random with a column-exchangeable law (e.g. the iid Bernoulli templates behind [9, Lem. 8.3]).*

*Proof.* Column permutations act on squares by  $h(L)(r, h(c)) = L(r, c)$ . This action fixes  $\sigma$  as a permutation of symbols (the pair  $(L(1, c), L(2, c))$  moves with the column), hence preserves the type and maps  $S(\lambda)$  bijectively onto itself; marked pairs move covariantly,  $P \mapsto h(P)$ ,

so  $(L, P) \mapsto (h(L), h(P))$  is a bijection of  $X$ . Moreover  $h(T) \cap h(L) = h(T \cap L)$ , and the supply property is intrinsic (rows are untouched), so  $\text{bad}(h(T); h(L), h(P)) = \text{bad}(T; L, P)$ . Substituting  $(L, P) = (h(L_0), h(P_0))$  in the left-hand average gives the claim.  $\square$

The remark’s proposed fix is therefore withdrawn. What the assembly actually needs is a *decoupling* between the supply structure and the mark distribution:

**Problem 16.7** (supply–mark decoupling). Show that  $\mathbb{P}_{(L,P) \sim \text{Unif}(X(n,\lambda;\alpha,\beta))}[\text{bad}(T; L, P)] = o(1/\log n)$ , uniformly over the admissible  $(\lambda, \alpha, \beta)$  — including types whose  $m$ -cycle class occupies as few as  $m = O(1)$  columns, where the deterministic bound  $|E| \leq 6 \log^{22} n$  of Lemma 15.9 is vacuous.

Heuristically Problem 16.7 should hold with much room: the supply structure of  $T \cap L$  is (almost) a function of rows  $3..n$ , while the class membership of a column is a function of rows  $1, 2$ . A concrete framing: condition on the bottom  $(n-2) \times n$  rectangle  $L_\downarrow$ . This fixes the “leftover” 2-regular bipartite graph on columns whose cycles are completed to rows  $1, 2$  by independent binary orientation choices (tilted by the  $S(\lambda)$  conditioning), so the  $m$ -class columns are a function of  $L_\downarrow$  plus these orientations, while the exceptional set is a function of  $L_\downarrow$  up to the weak dependence of stability (isolation and non-criticality in [9, §7]) on intercalates of  $T \cap L$  meeting rows  $1, 2$ . Quantifying either the orientation-randomness or a per-column failure probability is open; note that a per-column version of Lemma 15.9 cannot be extracted from the expander property alone, which bottoms out at exceptional sets of size  $\ell$ , nor from conditional division as in Lemma 15.11, since the per-tuple failure rates inside [9, Lem. 8.3] are far too weak to survive division by  $\mathbb{P}[S(\lambda)] \geq e^{-O(n \log n)}$ .

*Status: Proposition 16.6 PROVED; Problem 16.7 OPEN (new, replaces Remark 15.10(i)’s proposed fix). Session-8 empirical support (s8\_real\_bias.py, s8\_dec{30,50}.csv): on JM-uniform squares the supply  $k$  of a marked instance is statistically independent of the  $\sigma$ -cycle length  $m$  of its marked pair —  $\text{corr}(k, m) = +0.006 \pm 0.018$  at  $n = 30$  (3200 instances),  $-0.034 \pm 0.022$  at  $n = 50$  (2000), with  $\mathbb{E}[k \mid m]$  flat across the full range of  $m$  and no supply-poor instances at all ( $\mathbb{P}[k \leq 4] = 0$  in-sample) — consistent with the heuristic that supply is a rows- $3..n$  property and class membership a rows- $1, 2$  property. The programme summary of §19.5 now carries two open items: Problem 15.8 (main) and Problem 16.7 (bookkeeping).*

## 16.4 Real-ensemble grounding at $n = 30$ – $100$

Finally, the master formula and the bias statistics were taken to *real* random squares (s7\_real\_bias.py; uniform sampling by Jacobson–Matthews [8], marked pairs  $P = \{j, \sigma^\alpha(j)\}$  sampled uniformly among admissible instances, greedy maximal cell-disjoint intercalate families through the marked columns in rows  $3..n$ , orbit data built exactly as in pf\_master.py).

| $n$ | inst. | $\bar{k}$   | $\mathbb{E} \text{bias} $ | $\mathbb{E} \text{bias}  (r>0)$ | $\mathbb{P}[r=0]$ | signed mean | $\mathbb{E}[\text{bias}^2] \cdot \bar{k}$ |
|-----|-------|-------------|---------------------------|---------------------------------|-------------------|-------------|-------------------------------------------|
| 30  | 4200  | 12.9        | 0.168                     | 0.145                           | 0.065             | +0.038      | 0.67                                      |
| 50  | 3300  | 22.6        | 0.123                     | 0.108                           | 0.039             | +0.023      | 0.75                                      |
| 100 | 1800  | 26 (capped) | 0.114                     | 0.101                           | 0.029             | +0.013      | 0.73                                      |

(Families capped at 26 coordinates;  $k > 14$  instances by MC over 1500–2000 activation subsets.) Findings:

1. **The master formula holds on real squares at  $n \leq 100$ :** 3250 random activation subsets across  $n = 30, 50, 100$  checked against the actually-switched square — 0 mismatches.
2. **The annealed decay survives on the real ensemble:**  $\mathbb{E}[\text{bias}^2] \cdot \bar{k} \approx 0.7$  across  $n$ , i.e. the  $1/k$  second-moment law of the iid ensemble holds with a constant  $\approx 2.2\times$  larger; equivalently  $\mathbb{E}|\text{bias}| \approx 1.5\text{--}1.6\times$  the iid value at the same  $k$ . At fixed  $k = 26$  the value is the same at  $n = 50$  and  $n = 100$ : the elevation is a property of the orbit position law at given supply, not an  $n$ -effect.
3. **The short-arc atom behaves exactly as predicted in §15.5:** the orbits with no separating chord ( $r = 0$ ,  $\text{bias} = +\frac{1}{2}$  exactly) have measure  $\approx 0.85/\bar{k}$  (6.5%, 3.9%, 2.9%), and the *signed* mean bias is accounted for by this atom alone ( $+0.038$  vs.  $\text{atom} \times \frac{1}{2} = +0.032$ ;  $+0.023$  vs.  $+0.020$ ;  $+0.013$  vs.  $+0.015$ ): the annealed average is self-regularizing, no separate treatment needed.
4. Cut-chord configurations occur at frequency 2.1%, 1.6%, 1.1% (decreasing in supply), and no new pathology class appeared: the per-instance bias histogram is consistent with the iid shape plus the  $r = 0$  atom.

**SESSION-13 CORRECTION.** *The Jacobson–Matthews sampler emits unnormalised squares (row 1 is not the identity), and `s7_real_bias.py/s8_real_bias.py` take the symbols of a  $\sigma$ -cycle and use them directly as column indices. The “marked pairs” of this subsection are therefore column pairs  $(s, \sigma^\alpha(s))$  with  $s$  read as a column, i.e. essentially uniformly random column pairs, not marks: in particular rows 1, 2 are adjacent in the frame ( $\pi_P(1) = 2$  or  $\pi_P(2) = 1$ ) in a fraction  $\approx 2/n$  of these instances, which never happens for a true mark (§17.12.10). Everything verified here (the master formula on the switched square, the  $1/k$  law, the  $r = 0$  atom) is insensitive to this, and the corrected pipeline (`s13_real.py`, symbol  $\rightarrow$  column via row 1) gives the same picture, but the numbers below are for random column pairs. The bug was found because the trade chain of §17.12.11 froze the bit on exactly the adjacent instances.*

*Status: COMPUTED (tables above; per-instance data in `s7_inst{30,50,100}.csv`); the ground-truth re-verification of the master formula at  $n = 30, 50, 100$  is VERIFIED (0/3250). Session-8 extension to  $n = 200$  (`s8_inst200.csv`, 144+ instances at  $k = 26$  capped, JM thinning  $2n^3$ ):  $\mathbb{E}|\text{bias}| = 0.118$ ,  $\mathbb{E}[\text{bias}^2] \cdot k = 0.87 \pm 0.15$ , signed mean  $-0.006 \pm 0.014$  — the fixed- $k$  values agree with  $n = 50, 100$  (0.107–0.116): the elevation is confirmed to be a  $k$ -level property out to  $n = 200$ , and the  $1/k$  law shows no  $n$ -drift.*

## 17 The mixing problem as a marked split–merge chain: symmetry, factorization, and the exact rate from below

This section records the session-8 progress on Problem 16.4. The fresh-chord averaging operator  $P$  of §16.1 is identified *exactly* with one step of the uniform split–merge (coagulation–fragmentation) dynamics on measured partitions of the circle, carrying two marked points (Lemma 17.1); the chain admits an exact sign-reversing involution (Theorem 17.2) which factors the fresh-averaged sign as  $M_d = \varepsilon_0 \cdot \Phi_d(\text{geometry})$  (Corollary 17.3) and removes the label from the problem; the stationary law of the relevant “gap” coordinate is computed exactly (linear density; Proposition 17.5); and these combine into a *proved* lower bound  $\bar{\psi}(d) \geq c/d$  (Proposition 17.6): the numerically measured rate  $\psi \approx 1.0/d$  is sharp, and  $c = 1/2$  is the true exponent

of Problem 16.4 if the matching upper bound holds. The upper bound is reduced to two sharply scoped sub-problems (Problems 17.7 and 17.9), each with the mechanism confirmed numerically, plus a spectral route via reversibility (§17.5). *Session 9 (§17.8) shows the first of the two is unnecessary over the horizon  $j \leq 8d$  relevant to Theorem 16.1: any  $\alpha > 2/9$  in Problem 17.9 closes the iid model. Session 10 (§17.9) shows the environment must enter that problem through recharge, proves that the block count on the horizon is  $O(\log d)$  (the Harer–Zagier law of the number of curves), so the environment is healthy where it is needed, and reduces Problem 17.9 to two conditions (R1)–(R2) on the log-macroscopic post-flip chain via an exact Markov-renewal identity.*

Throughout, the ensemble is that of §16.1:  $u, v$  fixed on the circle of unit circumference ( $u$  at 0,  $v$  at  $1/2$  in the code), chords = iid uniform endpoint pairs.

### 17.1 Identification with the uniform split–merge chain

For a finite chord set  $W$ , the cycles of  $\gamma\tau_W$  trace out a partition of the circle (minus the endpoint set) into finite unions of arcs (“curves”); each curve carries its Lebesgue measure, and its internal cyclic order (the order in which  $\gamma\tau_W$  traverses its arcs) transports Lebesgue measure to a uniform measure along the curve. The *marked state* is

$$X = \left( \{\text{curve masses}\}, \text{curves of } u, v, \begin{cases} (x, y) & u \sim v : \text{masses of the two } u \rightarrow v \text{ curve arcs,} \\ (p, q) & u \not\sim v : \text{masses of the curves through } u, v, \end{cases} \right)$$

and  $\varepsilon(W) = +1$  iff  $u \sim v$ .

**Lemma 17.1** (fresh chord = one split–merge move). *Adding one fresh chord (iid uniform endpoints  $c_1, c_2$ ) maps the marked state by the following rules, which depend on the current state only. Write  $B(c)$  for the curve containing  $c$ ; positions of  $c_i$  along their curves are independent uniform.*

1.  $B(c_1) = B(c_2) = B$  (probability  $|B|^2$ ):  $B$  splits into the two arcs of  $B$  between the cuts. Sub-cases: (a)  $B$  unmarked, or marked by exactly one of  $u, v$  (mass  $p$ ): the mark keeps the piece of mass  $p - |t_1 - t_2|$ ,  $t_i \sim U(0, p)$  iid; the complementary piece of mass  $|t_1 - t_2|$  joins the environment. (b)  $B$  the common block, sides  $(x, y)$ : with the cycle parametrised  $u \rightarrow v$  ( $x$ -side) then  $v \rightarrow u$  ( $y$ -side), cuts uniform on  $(0, x+y)$ : both cuts on one side shrinks that side as in (a); one cut per side (probability  $2xy/(x+y)^2$  given the split) separates  $u, v$ : the new blocks have masses  $q' = t_2 - t_1$  ( $v$ 's) and  $p' = x + y - q'$  ( $u$ 's), i.e.  $(p', q') \stackrel{d}{=} (\alpha + y - \beta, x - \alpha + \beta)$ ,  $\alpha \sim U(0, x)$ ,  $\beta \sim U(0, y)$ .
2.  $B(c_1) \neq B(c_2)$ : the two curves merge (cut-and-join at  $c_1, c_2$ ). Sub-cases: (a) at most one is marked, or one holds  $u$  and the other neither: masses add, marks carried along. (b)  $u$ 's and  $v$ 's blocks (masses  $p, q$ ) merge:  $u, v$  become joined with sides  $(x, y) = (\alpha + \beta, p + q - \alpha - \beta)$ ,  $\alpha \sim U(0, p)$ ,  $\beta \sim U(0, q)$ . (c) the common block (sides  $x, y$ ) merges with an environment block of mass  $z$ : the insertion point is on the  $x$ -side with probability  $x/(x+y)$ , and that side gains  $z$ .

In particular  $P$  (add one fresh chord and average) is the transition operator of this marked uniform split–merge chain, the continuum limit of the cycle structure of the random-transposition walk (cf. [13]), and  $M_d = P^d \varepsilon$  is its  $d$ -step averaged together-indicator sign.

*Proof.* Adding the chord  $z = (c_1, c_2)$  replaces  $\gamma\tau_W$  (computed on the endpoint set enlarged by  $c_1, c_2$ ; refining  $\gamma$  at non-endpoint positions does not change the curve partition) by  $\gamma\tau_W(c_1c_2)$ . Right multiplication by a transposition whose entries lie on one cycle splits it into the two arcs between them, and merges two cycles otherwise — with the traversal orders spliced at the cuts; this is the standard surgery computation. The probabilistic statements:  $c_1, c_2$  are iid Lebesgue, the curves partition the circle, and Lebesgue restricted to a curve is uniform along it; all sub-case laws then read off from the traversal order, e.g. in 1(b) the cut positions along the cycle are iid  $U(0, x+y)$  and  $u, v$  separate iff the cuts straddle both sides.  $\square$

*Status: PROVED. VERIFIED end-to-end (s8\_splitmerge.c): the continuum chain reproduces the discrete-surgery values of  $\psi(j, d)$  ( $fs=1$  unbiased runs;  $\psi(16, 4)$ : 0.1341(22) vs 0.1305(22);  $\psi(64, 16)$ : 0.0391(26) vs 0.0395(26)), the one-point function ( $f(4)$ : +0.0232(4) vs +0.0237(5)), and the Poisson–Dirichlet sanity values at stationarity ( $\mathbb{E} \sum p_i^2 = 0.4979$  vs PD(1)’s  $1/2$ ;  $\mathbb{E}[\text{common-block mass} \mid u \sim v] = 0.6685$  vs  $2/3$ , see Proposition 17.5). NOTE: repeat runs show the session-7  $\psi$  table of §16.1 carries sampling scatter  $\approx \pm 0.005$  (its inner-average standard errors were optimistic); its fitted exponent is unaffected.*

## 17.2 The sign-reversing involution and the quotient chain

**Theorem 17.2** (label-flip symmetry). *Let  $\iota$  be the involution of marked states that swaps the label  $\{u \sim v\} \leftrightarrow \{u \not\sim v\}$  while keeping the environment and the ordered geometry pair fixed:*

$$\iota : (\text{together}, (x, y), \text{env}) \longleftrightarrow (\text{apart}, (p, q) = (x, y), \text{env}).$$

*Then  $\iota$  commutes with the transition kernel of the marked split–merge chain, and  $\varepsilon \circ \iota = -\varepsilon$ . Consequently  $M_d \circ \iota = -M_d$  for every  $d$ .*

*Proof.* Match the transition laws of Lemma 17.1 case by case under the correspondence  $x \leftrightarrow p$  (the  $u$ -coordinate),  $y \leftrightarrow q$  ( $v$ -coordinate). The total marked mass is  $x + y = p + q$  in both states, so the probabilities of the events “both cuts in marked material”, “one cut in marked material, one in a given environment block”, “both in environment” agree, as does the finer classification by  $u$ - vs  $v$ -material ( $x^2/(x+y)^2$  etc., against  $p^2/(p+q)^2$  etc.). Given the event: both cuts in  $u$ -material shrinks  $x$  (resp.  $p$ ) by  $|t_1 - t_2|$  with the same law; a cut pair straddling  $u$ - and  $v$ -material is the *separation* in the together state and the *join* in the apart state — i.e. it flips the label in both — and the outgoing geometry laws agree:  $(\alpha + y - \beta, x - \alpha + \beta) \stackrel{d}{=} (\alpha + \beta', x + y - \alpha - \beta')$  jointly, by  $\beta \mapsto y - \beta'$ ; a single cut in  $u$ -material merges an environment block  $z$  into the  $u$ -coordinate in both states; and environment-only moves are identical. This exhausts all cases, so  $K(\iota s, \iota \cdot) = \iota_* K(s, \cdot)$ ;  $\varepsilon$  flips by definition of  $\iota$ , and  $M_d = P^d \varepsilon$  inherits oddness.  $\square$

*Status: PROVED (case check as above). VERIFIED (s8\_splitmerge iota): from mirrored starts (tog,  $(x, y), z$ ) and (apart,  $(x, y), z$ ), measured  $P_A(u \sim v) + P_B(u \sim v) = 1$  within Monte Carlo error at all tested  $t \leq 32$  and geometries, with identical  $u$ -coordinate marginals.*

**Corollary 17.3** (quotient factorization; the label carries all the sign). *Let  $\Gamma_t = (a_t, b_t; \text{env}_t)$  be the quotient of the marked chain by  $\iota$  (the geometry with the label forgotten;  $a \leftrightarrow x$  or  $p$ ,  $b \leftrightarrow y$  or  $q$ ). Then:*

1.  $\Gamma_t$  is Markov, with the transition laws of Lemma 17.1 read on  $(a, b; \text{env})$ ; the flip moves (cut pair straddling  $a$ - and  $b$ -material) occur with probability  $2a_t b_t$  per step and update  $(a, b) \rightarrow (\alpha + \beta, a + b - \alpha - \beta)$ .

2. The label evolves deterministically along the quotient path:  $\varepsilon_d = \varepsilon_0 \cdot (-1)^{N_d}$  with  $N_d$  the number of flip moves, a functional of the quotient path alone.

3. Hence, with  $\Phi_d(\Gamma) := \mathbb{E}_\Gamma[(-1)^{N_d}]$ ,

$$M_d(C) = \varepsilon(C) \cdot \Phi_d(\Gamma(C)), \quad \psi(j, d) = \mathbb{E}_{\mu_j}[\Phi_d(\Gamma_0)^2].$$

4. Every stationary law of the marked chain is  $\iota$ -invariant wherever uniqueness holds on its support; in particular the label is a fair coin given the geometry at stationarity.

*Proof.* (1),(2): the flip moves are exactly the label-changing moves in both sectors (Theorem 17.2), and their effect on the geometry has the stated common law; all other moves preserve the label. (Flip moves are a.s. identifiable on the quotient path: they are the only moves that change both marked coordinates.) (3): condition on the quotient path. (4):  $\iota$  pushes any stationary law to a stationary law with the same quotient marginal.  $\square$

*Status: PROVED.* Problem 16.4 is thereby freed of the sign structure: it is now a statement about the parity functional  $\Phi_d = \mathbb{E}[(-1)^{N_d}]$  of an autonomous positive-recurrent chain. Consistency check: at stationarity the label-fairness (4) predicts  $P(u \sim v) = 1/2$ ; measured 0.5065(29) at burn-in 8192.

**Remark 17.4** (renewal structure at flip times). Let  $\tau$  be the first flip move. Strong Markov gives the exact renewal identity

$$\Phi_d(\Gamma) = \mathbb{P}_\Gamma[\tau > d] - \mathbb{E}_\Gamma[\Phi_{d-\tau}(\Gamma_\tau); \tau \leq d].$$

The flip clock runs at rate  $2a_t b_t$ ; after a flip the  $u$ -coordinate is *recharged* ( $a' = \alpha + \beta$  is macroscopic unless both  $a, b$  were small). The correlation-survival mechanism is therefore gap-trapping:  $\mathbb{P}[\tau > d]$  is dominated by trajectories on which  $g_t = \min(a_t, b_t)$  stays  $O(1/d)$ , since an untouched small side freezes ( $\mathbb{P}[\text{the } g\text{-material is hit in } d \text{ steps}] \leq 4gd$ ), while a touched one is typically recharged to macroscopic size by a size-biased merge.

### 17.3 The stationary gap law, and the exact rate from below

**Proposition 17.5** (stationary gap law). Let  $\pi$  be the stationary law of the marked chain (the  $n \rightarrow \infty$  limit of the cycle structure of a uniform permutation of  $n$  circle points with two marked points; top of the size spectrum  $\text{PD}(1)$ ). Under  $\pi$ :  $\mathbb{P}[u \sim v] = \frac{1}{2}$ ; given  $u \sim v$  the common-block mass  $s$  has density  $2s$  on  $(0, 1)$  (so  $\mathbb{E}[s \mid u \sim v] = 2/3$ ) and  $x \mid s \sim U(0, s)$ ; given  $u \not\sim v$  the pair  $(p, q)$  is uniform on the triangle  $\{p, q > 0, p + q < 1\}$  (so the joint density of  $(\text{apart}, p, q)$  is 1 there, mass  $\frac{1}{2}$ ). Consequently, with  $g = \min(x, y)$  resp.  $\min(p, q)$ ,

$$\mathbb{P}_\pi[g \leq \delta] = 4\delta + O(\delta^2) \quad (\delta \rightarrow 0),$$

and the density of  $g$  at  $0^+$  is 4.

*Proof.* At  $n$  points: for uniform  $\sigma$ ,  $\mathbb{P}[u, v \text{ in a common } m\text{-cycle}] = (m-1)/(n(n-1))$  (choose and order the cycle), summing to  $1/2$ ; given the cycle, the  $u \rightarrow v$  distance is uniform on  $\{1, \dots, m-1\}$ ; if apart,  $\mathbb{P}[|C_u| = m_1, v \notin C_u] = \frac{1}{n} \cdot \frac{n-m_1}{n-1}$  and given this  $|C_v|$  is uniform on  $\{1, \dots, n-m_1\}$ , so the joint law of  $(p, q) = (m_1, m_2)/n$  converges to  $(1-p) dp \cdot \frac{dq}{1-p} = dp dq$  on the triangle. Pass to the limit ( $m/n \rightarrow s$ ). The gap: given together,  $\mathbb{P}[\min(x, y) \leq \delta] = \int_0^1 2s \min(1, 2\delta/s) ds = 4\delta - 4\delta^2$ ; given apart,  $\mathbb{P}[\min(p, q) \leq \delta] = 4\delta + O(\delta^2)$  likewise; average the two sectors.  $\square$

*Status: PROVED at finite  $n$  with a routine limit exchange (the marked chain is the lumping of the transposition walk, see §17.5; over any fixed horizon the kernels converge). VERIFIED: measured dyadic masses at stationarity ( $\mathbb{P}[g \in (2^{-6}, 2^{-5})]$ ): together 0.0297 vs predicted 0.0312, apart 0.0288 vs 0.0312; lower bins consistent with density 4 throughout), and the label is a fair coin in every gap bin.*

**Proposition 17.6** (the mixing rate is at most  $1/d$ : lower bound). *There are  $c_0 > 0$ ,  $d_0$  such that the stationary-start mixing functional obeys*

$$\psi_\pi(d) := \mathbb{E}_\pi[M_d^2] \geq \frac{c_0}{d}, \quad d \geq d_0.$$

*Hence  $\bar{\psi}(d) \geq c_0/d$  along any initial family  $\{\mu_j\}$  whose gap law dominates a multiple of the stationary one (numerically true for the iid-chord  $\mu_j$  already at  $j = 16$ ), and no exponent  $c > 1/2$  is attainable in Problem 16.4.*

*Proof.* Fix  $K$  and let  $G = \{g_0 \leq 1/(Kd)\}$ . On  $G$ , the  $g$ -material (the short side if together, the small block if apart) is hit by no cut during  $d$  steps except with probability  $\leq 4g_0d \leq 4/K$ : each step's two cuts avoid it with probability  $\geq 1 - 4g_0$  while it remains unhit, and while unhit its mass is frozen. If it is never hit there is no flip move (a flip requires a cut in both  $a$ - and  $b$ -material), so  $\varepsilon_d = \varepsilon_0$  and, for the conditional expectation,  $|M_d| = |\mathbb{E}[\varepsilon_d \mid X_0]| \geq 1 - 8/K$  on  $G$ . Therefore  $\psi_\pi(d) \geq \mathbb{E}[M_d^2; G] \geq (1 - 8/K)^2 \pi(G)$ , and  $\pi(G) \geq c_1/(Kd)$  by Proposition 17.5. Take  $K = 16$ .  $\square$

*Status: PROVED (given Proposition 17.5). This settles the sharpness question left open in §16.1: the measured  $\psi \approx 0.8/d$  is the true order, the budget margin of Corollary 16.2 is exactly  $c = 1/2$  vs  $c > 1/11$  (factor 5.5 in the exponent), and no route can do better than  $1/d$  — in particular any proof of Problem 16.4 must tolerate the gap-trapped trajectories, they are really there.*

## 17.4 What remains for the upper bound

The decomposition  $\psi(j, d) = \mathbb{E}[\Phi_d^2; g_0 < \varepsilon] + \mathbb{E}[\Phi_d^2; g_0 \geq \varepsilon]$  with  $\varepsilon \asymp 1/d$ , together with  $|\Phi_d| \leq 1$ , reduces the upper bound to two independent statements.

**Problem 17.7** (uniform gap law). Show  $\mathbb{P}_{\mu_j}[g_0 \leq \varepsilon] \leq C\varepsilon^\kappa$  (some  $\kappa \in (0, 1]$ , uniformly in  $j$ ) for the iid-chord initial laws  $\mu_j$  — equivalently for the chain run  $j$  steps from the one-block start.

*Partial progress:* (i) at stationarity this is Proposition 17.5 with  $\kappa = 1$ ; (ii) the per-step creation flux into  $\{g \leq \varepsilon\}$  has the universal bound  $\mathbb{P}[\text{step creates } g' \in d\varepsilon] \leq C\varepsilon d\varepsilon$  for every current state (each channel — shrink of a marked coordinate, flip recharge landing small — has outgoing density  $\leq 2\varepsilon$  at  $\varepsilon$ : e.g. a split of the  $u$ -block of mass  $p$  leaves  $u$  with  $p - |t_1 - t_2|$ , of density  $2\varepsilon'/p^2 \cdot p^2$  at  $\varepsilon'$ ); (iii) escape from  $\{g \leq \varepsilon\}$  happens at rate  $\geq 2g(1 - g) \cdot \mathbb{P}[\text{merge partner macroscopic}]$ , so a renewal bound gives  $\kappa = 1$  given a uniform-in-time “no-dust” estimate (the environment retains a macroscopic block); it is the no-dust step that is open, and only it. Numerically the dyadic gap masses under  $\mu_j$  match the stationary density already at  $j = 16$ .

**Lemma 17.8** (recharge estimates). *For the quotient chain: (i) flip moves preserve the marked total  $s = a + b$ ; (ii) a flip from  $(a, b)$  lands at gap  $g' = \min(\alpha + \beta, s - \alpha - \beta)$  with  $\mathbb{P}[g' \leq \eta] \leq \eta^2/(ab)$  for  $\eta \leq \min(a, b)$ ; (iii) from any state, each move channel sends a marked coordinate into  $(0, \eta]$*

from above with probability  $\leq \eta^2$  per step (e.g. the shrink channel: both cuts in the  $a$ -material has probability  $a^2$ , and  $|t_1 - t_2| \in [a - \eta, a)$  has conditional probability  $\leq \eta^2/a^2$ ); so the total creation flux into  $\{g \leq \eta\}$  is  $\leq 4\eta^2$  per step.

*Proof.* (i) is immediate from the flip law. (ii):  $\alpha + \beta \leq \eta$  has probability  $\eta^2/(2ab)$  ( $\alpha \sim U(0, a)$ ,  $\beta \sim U(0, b)$  independent), likewise the other end. (iii): direct integration of the stated densities, channel by channel (shrink of  $a$ , shrink of  $b$ , and the two flip ends).  $\square$

*Status: PROVED. Together with Remark 17.4 this pins the intended endgame: from a good start the first-flip time  $\tau$  has tail  $\asymp 1/t$  (trapping is the only way to stall, and Lemma 17.8(iii) prices its creation), each flip recharges the gap with quadratic safety (ii), so the flips form an approximate renewal process with infinite-mean, regularly-varying inter-arrival tail  $\sim c/t$ ; for such renewals the alternating sum  $\mathbb{E}[(-1)^{N_d}]$  collapses to  $\Theta(\text{tail}(d)) = \Theta(1/d)$  (the signed renewal measure  $\sum_k (-1)^k G^{*k}$  has total mass  $\frac{1}{2}$ ). Two genuine gaps remain: the inter-flip times are state-dependent rather than iid, and the escape-rate lower bound  $\geq cg$  behind the  $1/t$  tail needs the environment to offer a macroscopic merge partner — a “no-dust” estimate; at stationarity dust configurations have probability  $\approx$  the Dickman mass  $\rho(8) \sim 10^{-8}$ , small but not  $o(1)$  in  $d$ , so the assembly must let dust episodes delay rather than veto escape (dust coarsens under merges). Note also the second trap coordinate: the marked total  $s$  itself can be small (then flips are rare no matter the gap); but  $\mathbb{P}_\pi[s \leq \delta] = O(\delta^2)$  in both sectors (the together mass  $s$  has density  $2s$ , and  $(p, q)$  is uniform on the triangle), so the  $s$ -trap contributes only  $O(1/d^2)$  and rides along harmlessly in the renewal bookkeeping.*

**Problem 17.9** (parity decay off small gaps). Show  $\mathbb{E}_{\mu_j}[\Phi_d(\Gamma_0)^2; g_0 \geq \varepsilon] \leq C(\varepsilon d)^{-2\alpha}$  for some  $\alpha > 0$  (uniformly in  $j$ ), i.e. trajectories started with both marked coordinates  $\geq \varepsilon$  lose the parity memory at a polynomial rate in  $\varepsilon d$ .

Any pair  $(\kappa, \alpha)$  solves Problem 16.4: from  $\psi \leq C(\varepsilon^\kappa + (\varepsilon d)^{-2\alpha})$ , optimising at  $\varepsilon = d^{-2\alpha/(\kappa+2\alpha)}$  gives  $\psi \leq Cd^{-2c'}$  with  $c' = \alpha\kappa/(\kappa + 2\alpha)$ ; the budget needs only  $c' > 1/11$ , e.g.  $(\kappa, \alpha) = (1, 1/2)$  gives  $c' = 1/4$ , and any  $\kappa = \alpha > 3/10$  still closes ( $c' = \kappa/3 > 1/11$  there). *The mechanism behind Problem 17.9 is confirmed numerically: binning  $\psi$ -contributions by  $g_0$  (`s8_splitmerge psibar`) shows  $\mathbb{E}[\Phi_d^2 | g_0]$  is a scaling function of  $g_0 d$  alone —  $\approx 1$  for  $g_0 d \lesssim 1/4$ ,  $\lesssim 0.05$  for  $g_0 d \gtrsim 4$  — so the conditional decay is fast; what is missing is a proof device for the parity cancellation (the renewal identity of Remark 17.4 plus recharge is the intended route: after the first flip the state is macroscopic w.h.p., flips then recur at rate  $\Theta(1)$ , and the alternating sum cancels to the heavy-tail term  $\mathbb{P}[\tau > d]$ ).*

## 17.5 The spectral route

The discrete precursor of the marked chain — right multiplication of  $\sigma_t \in S_n$  by uniform transpositions, marked cycle structure observed — is a strongly lumpable function of a chain that is reversible with respect to the uniform measure; the lumped chain is therefore reversible with respect to its stationary law, and this persists in the  $n \rightarrow \infty$  limit chainwise (the continuum statement deserves its own write-up; nothing below is used elsewhere). Granting it:  $\psi_\pi(d) = \int \lambda^{2d} d\nu_\varepsilon(\lambda)$  for the spectral measure  $\nu_\varepsilon$  of  $\varepsilon$  under the self-adjoint  $P$ , and Problem 16.4 (stationary start) becomes:  $\nu_\varepsilon$  has at most polynomial mass near  $|\lambda| = 1$ , with Proposition 17.6 asserting at least linear mass near  $\lambda = 1$ . Two structural facts make this attractive. (i) By

Theorem 17.2 the sign lives in the  $\iota$ -odd sector, where the Dirichlet form has a *massive* term on flip moves: for  $f = \varepsilon \cdot F(\Gamma)$ ,

$$\mathcal{E}(f, f) = \frac{1}{2} \mathbb{E}_\pi \left[ (F - F')^2; \text{non-flip} \right] + \frac{1}{2} \mathbb{E}_\pi \left[ (F + F')^2; \text{flip} \right],$$

so a low-energy odd function must be small wherever the flip rate  $2ab$  is not, i.e. must localise on  $\{g \lesssim \delta\}$ , whose mass is  $\asymp \delta$  (Proposition 17.5): this is precisely the geometric picture of a linear spectral density, and trial functions  $F = \mathbf{1}\{g \leq \delta\}$  realise it (Rayleigh quotient  $\asymp \delta$  at mass  $\asymp \delta$ ), reproving Proposition 17.6 spectrally. (ii) The converse direction — *every* odd  $F$  with  $\mathcal{E} \leq \delta \|F\|^2$  has  $|\langle F, \varepsilon \rangle|^2 \leq C\delta \|F\|^2$  — is a weighted Hardy-type inequality in the gap coordinate and is the cleanest formulation of the missing upper bound. A caveat for whoever takes this up: the chain is 2-periodic ( $(-1)^{\#\text{blocks}}$  flips deterministically, an exact  $\lambda = -1$  eigenfunction, and it is  $\iota$ -odd since  $\iota$  changes the block count by one), so the top-of-spectrum analysis must be run on  $P^2$  or per parity sector;  $\psi$  decays, so  $\langle \varepsilon, (-1)^{\#\text{blocks}} \rangle_\pi = 0$ , which is consistent with phase symmetry of  $\pi$ , but intermediate inequalities must not silently assume aperiodicity.

## 17.6 Transfer to the orbit measure: the mixture route

The factorization proof of Theorem 16.1 uses exactly one probabilistic input — chords attached to disjoint index sets are conditionally independent given the shared ones — and the session-7 caution stands: this fails per-instance. The split-merge picture suggests the correct weakening. The orbit chord positions  $(Z_1, \dots, Z_k)$  are exchangeable given the frame (the cycles of  $\pi_P$ , the points  $u, v$ ); a finitely exchangeable vector drawn from a population of  $N$  position slots is a mixture of iid( $\theta$ ) vectors up to total variation  $O(k^2/N)$  (Diaconis–Freedman), with  $\theta$  the (random) empirical position law of the instance, and the cell-disjointness of the intercalate family is a pairwise hard-core correction of the same  $O(k^2/N)$  order. Two observations make the mixture usable:

1. Conditionally on  $\theta$ , Theorem 16.1 applies verbatim (its proof needs only iid given  $(u, v, \theta)$ ), giving  $4 \mathbb{E}[\text{bias}^2] \leq \mathbb{E}_\theta [\max_{j, d \geq k/8} \psi^\theta(j, d)] + 2e^{-k/32} + O(k^2/N)$ .
2. The entire analysis of §§17.1–17.3 is *measure-covariant*: for chords with iid  $\theta$ -distributed endpoints, the induced chain is the split-merge chain in which every rate is read off the  $\theta$ -masses of curves and sides (cuts land in a curve with probability its  $\theta$ -mass, uniformly in its  $\theta$ -parametrisation). In particular Lemma 17.1, Theorem 17.2, Corollary 17.3 and Lemma 17.8 hold verbatim with “mass” =  $\theta$ -mass, provided  $\theta$  is atomless; atoms of mass  $m$  act as permanent traps of that scale and cap the decay at  $\psi^\theta \gtrsim \max_i \theta(\{i\})$ , which for orbit frames ( $N \asymp n$  near-uniform slots) is  $O(1/n)$  — far below the  $k^{-2c}$  budget,  $k = \Theta(\log^{11} n)$ .

So the transfer problem 7a(i) is reduced to (a) a quantitative exchangeable-to-mixture step for the greedy maximal family under the orbit measure (the hard-core correction; KSS-type concentration should control the conflict density), and (b) the  $\theta$ -uniform version of the mixing bound, i.e. Problem 17.9 with constants depending only on  $\max_i \theta(\{i\})$  and not otherwise on  $\theta$  (Proposition 17.10 is already  $\theta$ -covariant: Lemma 17.8(iii) reads verbatim in  $\theta$ -mass for atomless  $\theta$ ). Empirically the conclusion already holds on real squares: binning the session-7 per-instance data (`s7_inst{30,50,100}.csv`) by  $k$  gives  $\mathbb{E}[\text{bias}^2 \mid k] \cdot k = 0.6\text{--}0.75$ , flat across  $k = 8\text{--}26$  and  $n = 30\text{--}100$  (and `s8_inst200.csv` extends this, see §17.7): the  $1/k$  law transfers with a constant  $\approx 2.2\times$  the iid one, no drift.

*Status: reduction formulated; both steps OPEN. This replaces the vaguer options  $(\alpha)/(\beta)$  of the session-7 list.*

## 17.7 Numerics at large $d$

All session-8 runs, continuum chain unless stated (`s8_psibar.log`, `s8_4eb2.big.log`, `s8_onept.log`); COMPUTED.

(a) *The stationary rate is  $1/d$  on the nose.* Stationary-start (burn-in 16384) values of  $\psi_\pi(d) \cdot d$ : 0.96 ( $d=64$ ), 1.06 ( $d=128$ ), 1.11 ( $d=256$ ), 0.93 ( $d=512$ ), 0.87 ( $d=1024$ ; errors grow from  $\pm 0.05$  to  $\pm 0.25$  along the list) — constant  $\approx 1.0$ , fitted exponent  $1.03 \pm 0.09$  over  $d = 64$ – $1024$  (and 1.05 on the session-7  $j$ -table route), no logarithmic growth, matching Proposition 17.6 from below and pinning the conjectured upper bound at  $c = 1/2$  exactly.

(b) *The scaling collapse.* Binning the  $\psi$ -contribution by the initial gap:  $\mathbb{E}[\Phi_d^2 \mid g_0]$  is a function of  $u = g_0 d$  alone across  $d = 64, 128, 512$  — the dyadic profile  $\varphi(u) \approx 0.06, 0.23, 0.49, 0.69, 0.82, 0.9, \rightarrow 1$  at  $u \approx 0.7, 0.35, 0.18, 0.09, 0.04, 0.02, \dots$ , and  $\varphi \leq 0.02$  for  $u \geq 1.4$  (consistent with fast decay); the transition sits at  $g_0 \approx 1/d$  as the trap picture requires, the together and apart rows agree bin by bin (the  $\iota$ -symmetry made visible), and summing bins reproduces  $\psi \approx 1.0/d$  with the  $\{g_0 \lesssim 1/d\}$  bins carrying essentially all of it. Small- $u$  behaviour  $1 - \varphi(u) \approx u^{1/2-0.6}$  (not the naive linear touch-count) — unexplained, low stakes.

(c) *The iid-ensemble constant flattens.* Extending the session-7 drift  $4\mathbb{E}[\text{bias}^2] \cdot k = 0.95 \rightarrow 1.44$  ( $k = 8.256$ ):  $k = 512$ :  $1.448 \pm 0.229$ ;  $k = 1024$ :  $1.171 \pm 0.324$  (discrete code, 20M/40M trials) — constant  $\approx 1.3$ – $1.45$ , no  $\log k$ ; the budget question of session 7 (8d(b)) is settled: no log factor.

(d) *The one-point exponent is exactly 2.* From the clean start (single curve,  $x = y = \frac{1}{2}$ ):  $f(m) = +0.006280(50), +0.002951(41), +0.001713(35), +0.000814(29), +0.000458(25)$  at  $m = 8, 12, 16, 24, 32$ ;  $f(m) \cdot m^2 = 0.402, 0.425, 0.438, 0.469, 0.469$  and the local exponent reaches 2.00 on  $m = 24 \rightarrow 32$ :  $f(m) \rightarrow 0.47 m^{-2}$ . This confirms the trap-creation mechanism: a clean start must *create* its trap (Lemma 17.8(iii) prices a depth-1/ $m$  trap at  $\asymp 1/m^2$ ), whereas the stationary start carries a ready-made linear atom — hence  $1/m^2$  against the stationary  $1/d$ ; it also resolves session 7's 8d(c) (the value was below MC resolution there). *Session-9 correction*: the same data at odd  $m$  shows  $f(m)$  *alternates* in sign, so the leading  $1/m^2$  term is the alternating one and not the frozen-label trap term; see Lemma 17.13(a) and Theorem 17.14.

## 17.8 Session 9: the gap law is not needed; parity; an exact one-point formula

### 17.8.1 Short horizon: Problem 17.7 is unnecessary

The reduction of §17.4 asked for a gap law uniform in  $j$ . But in Theorem 16.1 the initial law  $\mu_j$  enters only for  $j \leq k$  while  $d \geq k/8$ , i.e.  $j \leq 8d$ : the chain has run for at most  $8d$  steps before the  $d$  fresh steps begin, and over such a short horizon the state-independent creation flux of Lemma 17.8(iii) alone controls the gap.

**Proposition 17.10** (short-horizon gap bound). *Let  $g_0^{\text{ini}} = \min(|u - v|, 1 - |u - v|)$  be the initial separation ( $= \frac{1}{2}$  in the antipodal ensemble) and let  $g_j$  be the gap of the marked state after  $j$  iid chords. Then for every  $\varepsilon > 0$  and every  $j \geq 0$ ,*

$$\mathbb{P}_{\mu_j}[g_j \leq \varepsilon] \leq \mathbf{1}\{g_0^{\text{ini}} \leq \varepsilon\} + 4\varepsilon^2 j.$$

*Proof.* If  $g_j \leq \varepsilon < g_0^{\text{ini}}$  there is a first step  $t \leq j$  at which the gap enters  $(0, \varepsilon]$  from above; by Lemma 17.8(iii) each step does so with probability at most  $4\varepsilon^2$  whatever the current state (the four channels — shrink of either coordinate, either end of a flip — each contribute at most  $\varepsilon^2$ ; merges only increase coordinates and environment-only moves leave them unchanged). Sum over  $t \leq j$ .  $\square$

*Status: PROVED. VERIFIED (s9\_survival flux):* at  $j = 8$  the ratio  $\mathbb{P}[g_j \leq \varepsilon]/(4\varepsilon^2 j)$  is  $1.00 \pm 0.05$  for  $\varepsilon \leq 2^{-8}$  — the bound is sharp at small  $\varepsilon$  (each channel's entry probability is exactly  $\varepsilon^2$  while all coordinates exceed  $\varepsilon$ , and nothing has had time to escape); at  $j = 64$  and  $j = 512$  the ratio is  $\leq 0.93$  resp.  $\leq 0.92$  in every bin and decreases to the stationary linear law  $4\varepsilon$  at moderate  $\varepsilon$ .

**Corollary 17.11** ( $\kappa$ -free budget). *Suppose Problem 17.9 holds with exponent  $\alpha$ :  $\mathbb{E}_{\mu_j}[\Phi_d^2; g_j \geq \varepsilon] \leq C(\varepsilon d)^{-2\alpha}$  for all  $j \leq 8d$ . Then, in the antipodal ensemble,*

$$\psi(j, d) \leq 32\varepsilon^2 d + C(\varepsilon d)^{-2\alpha} \leq C' d^{-\alpha/(1+\alpha)} \quad (j \leq 8d),$$

*i.e. Problem 16.4 holds with  $c' = \alpha/(2(1+\alpha))$ , and the budget  $c' > 1/11$  of Corollary 16.2 is met as soon as  $\alpha > 2/9$ . In particular the expected value  $\alpha = 1/2$  gives  $c' = 1/6$  (against  $c' = 1/4$  with a linear gap law), and Problem 17.7 is not needed for the iid model.*

*Proof.* Combine Proposition 17.10 ( $j \leq 8d$ ) with  $|\Phi_d| \leq 1$  in the decomposition of §17.4 and optimise  $\varepsilon = d^{-(1+2\alpha)/(2+2\alpha)}$ ;  $c' > 1/11$  iff  $11\alpha > 2 + 2\alpha$ .  $\square$

*Status: PROVED.* The whole “no-dust”/escape discussion of §17.4 is thereby demoted to a route for the sharp exponent  $c = 1/2$  (or for the stationary-start statement); the iid model needs only Problem 17.9 with any  $\alpha > 2/9$ . In the true ensemble the indicator term is the short-arc atom already accounted for as the  $O(1/k)$  term of §15.5.

**Lemma 17.12** ( $s$ -trap flux). *From any state, the marked total  $s = a + b$  enters  $(0, \sigma]$  from above with probability at most  $2\sigma^2$  per step; hence  $\mathbb{P}_{\mu_j}[s_j \leq \sigma] \leq 2\sigma^2 j$  in the antipodal ensemble.*

*Proof.* Flips and merges do not decrease  $s$ ; a shrink of  $a$  lands at  $s' = a' + b \leq \sigma$  only if  $b \leq \sigma$  and  $a' \leq \sigma - b$ , which has probability  $\leq (\sigma - b)^2 \leq \sigma^2$  (the shrink channel of Lemma 17.8(iii)); likewise for  $b$ .  $\square$

*Status: PROVED.* Consequently in Corollary 17.11 one may also assume  $s_0 \geq \sigma$  at cost  $16\sigma^2 d$ , so the target estimate of Problem 17.9 need only be proved for starts with  $a_0, b_0 \geq \varepsilon$  and  $s_0 \geq \sigma \asymp \varepsilon$ .

## 17.8.2 Parity: the label is a function of time and the environment block count

**Lemma 17.13** (block-count parity). *Let  $B_t$  be the total number of curves and  $E_t$  the number of environment curves (those through neither  $u$  nor  $v$ ) at time  $t$ . Every step is a split or a merge, so  $B_t \equiv B_0 + t \pmod{2}$ ; since  $B_t = E_t + 1 + \mathbf{1}\{u \not\sim v\}$ ,*

$$\varepsilon_t = -(-1)^{B_0+t} (-1)^{E_t}, \quad (-1)^{N_t} = (-1)^t (-1)^{E_t-E_0},$$

*with  $N_t$  the number of flip moves. Hence  $\Phi_d(\Gamma) = (-1)^d \mathbb{E}_\Gamma[(-1)^{E_d-E_0}]$ .*

*Proof.* Splits add a curve, merges remove one; a flip (separate or join) changes  $B$  by one and leaves  $E$  unchanged, every other move leaves the label and changes  $E$  by one.  $\square$

*Status: PROVED (trivial, but structurally decisive). Three consequences.*

(a) *The sign of the one-point function alternates.* From the clean start,  $f(m) = \mathbb{E}[\varepsilon(F_m)]$  obeys  $f(m) \cdot (-1)^m > 0$ : measured  $f(7) = -0.00753(22)$ ,  $f(8) = +0.00599(22)$ ,  $f(9) = -0.00526(22)$ ,  $f(11) = -0.00353$ ,  $f(12) = +0.00295$ ,  $f(13) = -0.00275$ ,  $f(15) = -0.00192$ ,

$f(16) = +0.00213$ ,  $f(17) = -0.00149$  ( $2 \times 10^7$  trials each). Session 8 sampled only even  $m$  and read  $f(m) \approx 0.47/m^2$  as the frozen-label trap term; that reading is *incomplete*: the leading term is the alternating one,  $f(m) \approx (-1)^m c_1/m^2$  with  $m^2|f(m)|$  increasing through 0.40, 0.43, 0.44, 0.47, 0.47 ( $m = 8..32$ ) and plausibly  $c_1 = \frac{1}{2}$  (Theorem 17.14 below gives exactly  $\frac{1}{2}$  for the  $(u, v)$ -averaged function), while the non-alternating (trap) part,  $\frac{1}{2}(f(m-1) + f(m+1)) + f(m)$  at even  $m$ , is  $\lesssim 2 \times 10^{-4}$  at  $m = 8$ . Nothing downstream is affected ( $\psi = \mathbb{E}\Phi^2$  is sign-blind), but the smooth object is  $(-1)^d \Phi_d$ , not  $\Phi_d$ , and Remark 17.4's heuristic " $\Phi_d \approx \mathbb{P}[\tau > d]$ " must be read with this sign.

(b) *An obstruction for coupling proofs.* Any two words of the same length whose final environments have the same block-count parity have the same flip parity; in particular two trajectories from the same start that agree in marked geometry *and* environment at time  $d$  have the same label. A coupling proof of Problem 17.9 must therefore produce, at time  $d$ , environments of *different* block-count parity with the same law — e.g. one carrying an extra tiny block. The natural “swap a flip for a shrink-re-merge pair and let the environment take a tiny extra split” coupling has success probability  $O(1/d)$  per step and is repeated  $d$  times, so it yields only  $|\Phi_d| \leq 1 - c$ : no decay. Same-time  $\iota$ -mirroring is impossible ( $\iota$  changes  $B$  by one); a one-step shifted mirror gives  $\Phi_d \approx -\Phi_{d+1}$ , which is the alternation (a), not decay. The parity of  $N_d$  is thus a genuinely “continuous-state” functional and the proof of Problem 17.9 has to go through the renewal or spectral route.

(c) *First-flip tail from good starts is  $1/t^2$ , not  $1/t$ .* From a stationary environment conditioned on  $g_0 \in [1/16, 1/8)$  (`s9_survival phic`),  $t^2 \mathbb{P}[\tau > t] \rightarrow 11.1$  ( $t = 128, 256$ ; 16.1 at  $t = 64$ , 29 at  $t = 32$ ); from the clean start with  $g_0 = 1/4$ ,  $t \mathbb{P}[\tau > t]$  halves per doubling of  $t$  from  $t = 32$  on. The renewal heuristic explains this: a trap at depth  $\eta \leq \varepsilon$  is created at rate  $8\eta d\eta$  (Lemma 17.8(iii)) and survives  $m$  further steps without flipping with probability  $\approx e^{-2\eta sm}$ , so the kernel  $K(m) = \int_0^\varepsilon 8\eta e^{-2\eta sm} d\eta \approx 2/(s^2 m^2)$  for  $m \gg 1/(\varepsilon s)$ , with total mass  $\approx 4\varepsilon/s < 1$ : the first-flip time is a *subcritical* renewal sum with a  $1/t^2$  kernel, hence  $\mathbb{P}[\tau > t] \asymp t^{-2}$  and, if the alternating-renewal cancellation holds,  $|\Phi_d| \lesssim (\varepsilon d)^{-2}$  off small gaps — i.e. the true  $\alpha$  in Problem 17.9 is plausibly 2, far above the  $2/9$  needed. Consistently,  $\mathbb{E}[\Phi_d^2 | g_0 = u/d]$  is at the noise floor ( $\lesssim 10^{-3}$ , often negative estimates) for every bin  $u \geq 3$  in the session-8 `psibar` logs at  $d = 128\text{--}1024$ . COMPUTED, not proved.

### 17.8.3 An exact one-point formula

Averaging the marked points over the circle turns the one-point function into a class function of the discrete precursor, and hook characters compute it exactly.

**Theorem 17.14** (exact  $(u, v)$ -averaged one-point function). *Let  $\sigma_m = \gamma \tau_1 \cdots \tau_m$  with  $\gamma$  a uniform  $n$ -cycle and  $\tau_i$  iid uniform transpositions, and let  $c_i$  be its cycle lengths. Then*

$$\mathbb{E}\left[\sum_i c_i(\sigma_m)^2\right] = \frac{n(n+1)}{2} + \sum_{k=1}^{n-1} (n-k) \left(1 - \frac{2k}{n-1}\right)^m,$$

so that for uniform distinct  $u, v$ ,  $\mathbb{P}[u \sim v \text{ in } \sigma_m] = (\mathbb{E} \sum c_i^2 - n)/(n(n-1))$ . Consequently, for the continuum uniform split-merge chain  $(p_i(m))$  started from a single block,

$$\mathbb{E}\left[\sum_i p_i(m)^2\right] = \frac{1}{2} + \frac{1}{4} \left[ \frac{1 + (-1)^m}{m+1} + \frac{1 - (-1)^m}{m+2} \right] = \begin{cases} \frac{1}{2} + \frac{1}{2(m+1)}, & m \text{ even,} \\ \frac{1}{2} + \frac{1}{2(m+2)}, & m \text{ odd,} \end{cases}$$

and the  $(u, v)$ -averaged one-point function  $\bar{f}(m) = \mathbb{E}_{u,v}[\varepsilon(F_m)] = 2\mathbb{E} \sum p_i^2 - 1$  equals  $1/(m+1)$  for even and  $1/(m+2)$  for odd  $m$ :  $\bar{f}(m) = \frac{2m+3}{2(m+1)(m+2)} + (-1)^m \frac{1}{2(m+1)(m+2)}$ .

*Proof.* Right multiplication by a uniform transposition acts on the  $\lambda$ -isotypic component of  $L^2(S_n)$  by  $r_\lambda = \chi_\lambda(\tau)/d_\lambda$ , and the law of  $\gamma$  is the class measure of  $n$ -cycles, so for any  $F$ ,  $\mathbb{E}F(\sigma_m) = \sum_\lambda \rho_\lambda r_\lambda^m \langle \chi_\lambda, F \rangle$  with  $\rho_\lambda = \chi_\lambda(n\text{-cycle})$  and  $\langle \chi, F \rangle = \frac{1}{n!} \sum_\sigma \chi(\sigma)F(\sigma)$  (Fourier inversion for class measures). Only hooks  $\lambda_k = (n-k, 1^k)$  have  $\rho_\lambda \neq 0$ , with  $\rho_{\lambda_k} = (-1)^k$ , and their content sum gives  $r_{\lambda_k} = [(n-k)(n-k-1) - k(k+1)]/(n(n-1)) = 1 - 2k/(n-1)$ . For  $F = \sum_i c_i^2$  use the hook generating function  $\sum_k \chi_{\lambda_k}(\mu)t^k = (1+t)^{-1} \prod_i (1 - (-t)^{\mu_i})$  and the exponential formula with one marked part:

$$\sum_n x^n \sum_k t^k \langle \chi_{\lambda_k}, F \rangle = \frac{1}{1+t} \left[ \sum_{j \geq 1} j (1 - (-t)^j) x^j \right] \exp \left( \sum_{j \geq 1} \frac{(1 - (-t)^j) x^j}{j} \right) = \frac{1}{1+t} \left[ \frac{x}{(1-x)^2} + \frac{tx}{(1+tx)^2} \right] \frac{1+tx}{1-x}.$$

Extracting  $[x^n]$ : the first term gives  $[(\binom{n+1}{2} + t\binom{n}{2})/(1+t)]$ , the second  $t(1 - (-t)^n)/(1+t)^2$ ; the coefficient of  $t^k$  ( $1 \leq k \leq n-1$ ) is  $(-1)^k n + (-1)^{k-1} k = (-1)^k (n-k)$ , and  $\binom{n+1}{2}$  at  $k=0$ . Multiply by  $\rho_{\lambda_k} = (-1)^k$  and  $r^m$ . The continuum limit is the Riemann sum  $\int_0^1 (1-x)(1-2x)^m dx$ , evaluated by  $y = 1-2x$ ; the identification of  $\sigma_m/n$  with the split-merge chain from one block is Lemma 17.1 (the  $n$  points carry arcs of length  $1/n$ ).  $\square$

*Status: PROVED. VERIFIED: the finite- $n$  formula (s9\_hooks.py, exact rationals; the inner products  $(-1)^k(n-k)$  were also checked against the full character sums for  $n \leq 12$ ) gives at  $n = 40$ : 0.6007, 0.5567, 0.5314 at  $m = 4, 8, 16$  against the continuum simulator's 0.6005, 0.5561, 0.5297 (s8\_splitmerge traj), and the continuum values  $m = 1 : 2/3$ ,  $m = 2 : 2/3$  agree with direct hand integration. Remarks: (i) this is the  $(u, v)$ -averaged version of the one-point function of §17.7(d); its non-alternating part  $\approx 1/m$  is the frozen-label trap term fed by the linear initial gap density (2 on  $(0, \frac{1}{2})$ ) of a random separation, absent for antipodal  $u, v$ , and its alternating part  $(-1)^m/(2(m+1)(m+2))$  has exactly the coefficient  $\frac{1}{2}$  towards which the antipodal  $m^2|f(m)|$  appears to converge; (ii) in spectral terms the averaged start has spectral density  $(1+y)/4$  on  $y = 1-2x \in [-1, 1]$  — linear at both ends  $y = \pm 1$ , the two ends being the trap ( $y = +1$ ) and the alternating ( $y = -1$ , the  $(-1)^B$  eigenfunction) edges; (iii) the same technique cannot reach  $\psi$  (a two-time quantity is not a class function of the increment), but it does prove the averaged one-point decay  $\bar{f}(m) \leq 1/(m+1)$  outright, and gives the exact law of  $\mathbb{E} \sum p_i^2$  along the whole approach to PD(1).*

#### 17.8.4 The spectral route on the quotient: twisted kernel

Lemma 17.13 and Corollary 17.3 make the spectral formulation of §17.5 concrete on the quotient chain. Write  $P = P_{\text{nf}} + P_{\text{f}}$  for the non-flip and flip parts of the quotient kernel and let  $P_\pi := P_{\text{nf}} - P_{\text{f}}$  (the kernel twisted by  $(-1)^F$ ). Then  $\Phi_d = P_\pi^d \mathbf{1}$ , and since flips are reversed by flips,  $P_\pi$  is self-adjoint on  $L^2(\tilde{\pi})$  ( $\tilde{\pi}$  the quotient stationary law), so

$$\psi_\pi(d) = \langle \mathbf{1}, P_\pi^{2d} \mathbf{1} \rangle = \int_{[-1,1]} \lambda^{2d} \nu(d\lambda), \quad \nu = \text{spectral measure of } \mathbf{1} \text{ under } P_\pi, \quad \nu([-1,1]) = 1,$$

and Problem 16.4 at stationarity reads  $\nu(\{|\lambda| \geq 1 - \delta\}) \leq C\delta^\beta$  ( $\psi_\pi(d) \leq Cd^{-\beta}$ ). Two facts, one numerical and one a proof sketch:

(a) *No mass at the  $-1$  end at stationarity (COMPUTED).*  $(-1)^E$  is an exact  $\lambda = -1$  eigenfunction of  $P_\pi$  (Lemma 17.13), so the 2-periodicity caveat of §17.5 is a priori relevant. But  $\langle P^d \varepsilon, P^{d+1} \varepsilon \rangle_\pi = \int \lambda^{2d+1} d\nu$  versus  $\int \lambda^{2d} d\nu$  (s9\_survival oddeven):  $0.0559/0.0575 = 0.971$  at  $d = 16$  and  $0.0146/0.0150 = 0.974$  at  $d = 64$  (both  $\pm 0.03$ ), so the relative mass of  $\nu$  near  $-1$  is  $\lesssim 3\%$  and compatible with zero; in the continuum  $E = \infty$  under  $\tilde{\pi}$  and the eigenfunction  $(-1)^E$  is not in  $L^2(\tilde{\pi})$ . The sign alternation of §17.8.2(a) is a finite-block (clean-start) phenomenon, invisible at stationarity.

(b) *A Cauchy–Schwarz/smoothing route to  $\beta = 1/2$  (SKETCH).* The Dirichlet form of  $P_\pi$  is  $\mathcal{E}(F) = \frac{1}{2}\mathbb{E}[(F - F')^2; \text{nf}] + \frac{1}{2}\mathbb{E}[(F + F')^2; \text{f}]$ . Let  $k(x) = 2ab$  be the flip rate and  $Q(x, \cdot) = P_{\text{f}}(x, \cdot)/k(x)$  the normalised flip kernel, which acts on each segment  $\{(a', s - a'; \text{env})\}$  separately by the trapezoidal density  $a' \mapsto |\{\alpha \in (0, a) : a' - \alpha \in (0, s - a)\}|/(a(s - a))$ . By Cauchy–Schwarz in the flip term,  $\mathcal{E}(F) \geq \frac{1}{2} \int \tilde{\pi}(dx) k(x) (F + QF)^2(x)$ , so  $\mathcal{E}(F) \leq \delta \|F\|^2$  forces  $F \approx -QF$  in  $L^2(\tilde{\pi}k)$ , hence  $F \approx Q^2 F$ :  $F$  is nearly constant, then nearly equal to minus itself, on every segment away from its ends — i.e.  $\int_{\{ab \geq \eta\}} F^2 d\tilde{\pi} \leq C\delta \|F\|^2/\eta$  once the one-dimensional smoothing inequality for  $Q^2$  on a segment is quantified (the trapezoid density is  $\geq c/s$  away from the ends,  $\asymp a'/s^2$  near them). With  $\tilde{\pi}(ab < \eta) \leq C\eta$  (Proposition 17.5 plus the quadratic  $s$ -law),  $\langle F, \mathbf{1} \rangle^2 \leq C\|F\|^2(\delta/\eta + \eta)$ , and  $\eta = \sqrt{\delta}$  gives  $\nu([1 - \delta, 1]) \leq C\sqrt{\delta}$ , i.e.  $\psi_\pi(d) \leq Cd^{-1/2}$ :  $c = 1/4 > 1/11$  at stationarity. Gaps: (i) the segment smoothing inequality (a weighted Poincaré for  $Q^2$  with weight  $a'(s - a')$ ); (ii) the  $-1$  end, to be handled either by (a) made rigorous or by running the argument for  $\langle F, (I + P_\pi)F \rangle$ ; (iii) the passage from  $\pi$  to  $\mu_j$ ,  $j \leq 8d$  — the short-horizon Proposition 17.10 handles the gap coordinate but not the rest of the state, and Lemma 16.3 only trades  $d$  for  $j$ . (iii) is the real obstacle for the iid model; a dust-insensitive comparison of  $\mu_j$  with  $\pi$  for  $j \geq d/2$  (a block of mass  $w$  influences  $d$  steps with probability  $\leq 2wd$ ) is the natural tool.

## 17.9 Session 10: the block count is Harer–Zagier, the environment is healthy on the short horizon, the exact size-biased law, and the Markov-renewal form of the survival problem

Session 10 was to fix the exact statement behind Problem 17.9 (TODO 10a, step 0) and choose a route. The first finding is negative and structural: *the environment cannot be avoided* — without merges the parity of the flip count from a start with  $\min(a_0, b_0) = \varepsilon$  decays on the time scale  $\varepsilon^{-2}$ , not  $\varepsilon^{-1}$ , and the budget of Corollary 17.11 forces  $\varepsilon \ll d^{-1/2}$ , so a proof must use the recharge of small marked coordinates by environment blocks (§17.9.1). The second finding removes the obstacle this creates. The number of curves  $B_t$  after  $t$  chords has an *exact* law — it is the Harer–Zagier vertex-count distribution of a random gluing of a  $2t$ -gon — with  $\mathbb{E}[3^{B_t}] = 3 + 4t + 2t^2$ ; a Chernoff bound then gives  $B_t \leq C \log d$  simultaneously for all  $t \leq 9d$  except with probability  $d^{-\gamma}$  (§17.9.2). On that event every state on the horizon has  $\sum p_i^2 \geq 1/(C \log d)$  and an environment block of mass  $\geq (1 - s)/(C \log d)$ : the “no-dust” question is closed, up to logarithms, in exactly the form the short horizon needs. The same hook computation gives the exact law of the mass of the curve through a uniform point at every time (§17.9.3; this contains Theorem 17.14 and closes TODO 10b(b)). Finally the survival problem is put in Markov-renewal form (§17.9.5): an exact operator identity for the generating function  $\sum_d \Phi_d z^d$  separates the two remaining contents — moments of the first-flip time (traps) and invertibility of  $I + K_z$  (cancellation) — and a conditional decay proposition states precisely what is left.

### 17.9.1 The environment cannot be avoided

Consider the marked pair alone, with merges suppressed (an adversarial “dust” environment): per step,  $a \rightarrow a\sqrt{U}$  with probability  $a^2$ ,  $b \rightarrow b\sqrt{U}$  with probability  $b^2$ , a flip  $(a, b) \rightarrow (\alpha + \beta, s - \alpha - \beta)$  with probability  $2ab$ , and nothing otherwise. Since merges only increase marked coordinates and are the only moves that do, this is the chain that any argument using the environment solely through upper bounds would have to control. Its scaling is elementary: all rates are quadratic in the coordinates, so from  $(a_0, b_0) = (\varepsilon, \varepsilon)$  nothing happens for  $\asymp \varepsilon^{-2}$  steps, and from  $(c_0, \varepsilon)$  the large coordinate decays like  $t^{-1/2}$  before a flip (rate  $2\varepsilon b$ ) can occur, the first flip arriving only at  $t \asymp \varepsilon^{-2}$ . Measured (`s10_nomerge`,  $\Phi_t = \mathbb{E}[(-1)^{N_t}]$ ): from  $(\frac{1}{4}, \frac{1}{4})$ ,  $\Phi_t = 0.130, 0.031, 0.0070, 0.0024$  at  $t = 8, 16, 32, 64$  (then noise floor  $10^{-3}$ ;  $t^2\Phi_t \approx 8$ – $10$  on  $16 \leq t \leq 64$ ) — a healthy  $1/t^2$ -like decay without any environment; but from  $(\frac{1}{4}, 0.02)$ ,  $\Phi_t = 0.53, 0.25, 0.088, 0.040$  at  $t = 64, 256, 1024, 4096$ , and from  $(0.02, 0.02)$ ,  $\Phi_t = 0.90, 0.68, 0.30, 0.13$  at  $t = 64, 256, 1024, 8192$ : decay only on the scale  $\varepsilon^2 t$ , and slow even there. In the budget of Corollary 17.11 the trap-entry flux forces  $\varepsilon \ll d^{-1/2}$ , hence  $\varepsilon^2 d \ll 1$ : a bound depending on the environment only through upper bounds is worthless. *Status: COMPUTED (the scaling statement is trivial). Consequence: any proof of Problem 17.9 must lower-bound the masses of environment blocks over the horizon — the recharge mechanism of Remark 17.4. The next subsection supplies exactly this.*

### 17.9.2 The exact law of the number of curves

Let  $B_t$  be the number of curves after  $t$  iid chords (the total block count of the marked chain from the one-block start; it does not depend on the marks).

**Theorem 17.15** (the block count is Harer–Zagier). *For every  $x$ ,*

$$\sum_{t \geq 0} \mathbb{E}[x^{B_t}] z^t = \frac{1}{2z} \left[ \left( \frac{1+z}{1-z} \right)^x - 1 \right], \quad \text{i.e.} \quad \mathbb{P}[B_t = m] = [z^{t+1}] \frac{(2 \operatorname{artanh} z)^m}{2m!},$$

so  $B_t \equiv t+1 \pmod{2}$ ,  $\mathbb{P}[B_t = t+1-2g] = \varepsilon_g(t)/(2t-1)!!$  with  $\varepsilon_g(t)$  the Harer–Zagier numbers [7], and for integer  $x \geq 1$  the expectation  $\mathbb{E}[x^{B_t}]$  is a polynomial in  $t$  of degree  $x-1$ :

$$\mathbb{E}[2^{B_t}] = 2+2t, \quad \mathbb{E}[3^{B_t}] = 3+4t+2t^2, \quad \mathbb{E}[4^{B_t}] = 4+\frac{20}{3}t+4t^2+\frac{4}{3}t^3 = \frac{4}{3}(t+1)(t^2+2t+3), \quad \mathbb{P}[B_t = 1] = \frac{1+}{2(t+1)}$$

*Proof.* The  $2t$  endpoints are iid uniform on the circle, a.s. distinct; by exchangeability, conditionally on their positions the pairing into chords is uniform among the  $(2t-1)!!$  pairings. Refining  $\gamma$  at the endpoints does not change the curve partition (Lemma 17.1), so  $B_t$  is the number of cycles of  $\gamma_{2t}\tau$ , with  $\gamma_{2t}$  the rotation of the  $2t$  endpoints and  $\tau$  the pairing involution — the number of vertices of the surface obtained by gluing the sides of a  $2t$ -gon along a uniform pairing. Harer–Zagier’s formula [7] is  $1 + 2 \sum_{t \geq 0} [\sum_g \varepsilon_g(t) N^{t+1-2g}] z^{t+1} / (2t-1)!! = ((1+z)/(1-z))^N$ ; dividing by  $(2t-1)!!$  turns the bracket into  $\mathbb{E}[N^{B_t}]$ , which is the displayed identity for  $x = N$ , and both sides are polynomials in  $N$  for each  $t$ . Expanding  $((1+z)/(1-z))^x = \exp(2x \operatorname{artanh} z)$  in powers of  $x$  gives the law; the polynomials are the coefficients of  $(1-z)^{-x} \sum_k \binom{x}{2k+1} z^{2k}$ .  $\square$

*Status: PROVED (the statement is classical once the chord diagram is identified; the identification is the content). A second proof runs through the finite- $n$  hook machinery of Theorem 17.14:  $x^{c(\sigma)} = \sum_{\lambda} s_{\lambda}(1^x) \chi_{\lambda}(\sigma)$ , so  $\mathbb{E}[x^{c(\sigma_t)}] = \sum_k (-1)^k r_k^t s_{(n-k, 1^k)}(1^x)$  with  $s_{(n-k, 1^k)}(1^x) =$*

$x(x+1)\cdots(x+n-k-1) \cdot (x-1)\cdots(x-k)/(n(n-k-1)!k!)$  by hook-content; for integer  $x$  only  $k \leq x-1$  survive, e.g.  $\mathbb{E}[2^{c(\sigma_t)}] = (n+1) - (n-1)r_1^t$ ,  $\mathbb{E}[3^{c(\sigma_t)}] = \frac{1}{2}(n+1)(n+2) - (n^2 - 1)r_1^t + \frac{1}{2}(n-1)(n-2)r_2^t$ , whose  $n \rightarrow \infty$  limits are  $2+2t$  and  $3+4t+2t^2$  (`s10_blocks.py`, exact rationals, Lagrange-extrapolated in  $1/(n-1)$ , gives the polynomials for  $x \leq 5$ ). VERIFIED: the full law against a direct simulation of the split-merge chain (`s10_blocksim.py`,  $2 \times 10^5$  trials,  $t \leq 6$ ): every probability agrees to  $\pm 0.001$ , e.g.  $\mathbb{P}[B_6 = 1, 3, 5, 7] = 0.1429, 0.6222, 0.2222, 0.0127$  predicted vs.  $0.1431, 0.6230, 0.2212, 0.0127$  measured.

**Corollary 17.16** (block count over the horizon). *For all  $T \geq 1$  and  $K \geq 1$ ,*

$$\mathbb{P}\left[\max_{0 \leq t \leq T} B_t \geq K\right] \leq 3^{1-K}(T+1)^3.$$

*In particular, for  $\gamma > 0$  and  $K_\gamma(T) := \lceil (3+\gamma) \log_3(T+1) \rceil + 1$ , the event  $H_T = \{B_t < K_\gamma(T) \forall t \leq T\}$  has probability  $\geq 1 - (T+1)^{-\gamma}$ , and on  $H_T$ , for every  $t \leq T$ : the total number of blocks is  $< K_\gamma(T)$ ,  $\sum_i p_i(t)^2 \geq 1/K_\gamma(T)$ , the environment has fewer than  $K_\gamma(T)$  blocks and hence a block of mass  $\geq (1-s_t)/K_\gamma(T)$ .*

*Proof.* Markov with  $x = 3$ :  $\mathbb{P}[B_t \geq K] \leq 3^{-K}(3+4t+2t^2) \leq 3^{1-K}(t+1)^2$ ; sum over  $t \leq T$ . With  $K = K_\gamma(T)$ ,  $3^{1-K} \leq (T+1)^{-3-\gamma}$ . The consequences are Cauchy-Schwarz ( $\sum p_i^2 \geq 1/\#\text{blocks}$ ) and pigeonhole.  $\square$

*Status: PROVED. In Problem 17.9 the start  $\Gamma_0 \sim \mu_j$  ( $j \leq 8d$ ) followed by  $d$  steps is the chain from one block over at most  $9d$  steps, so  $H_{9d}$  has probability  $\geq 1 - (9d+1)^{-\gamma}$  for any fixed  $\gamma$  at the price  $K = O_\gamma(\log d)$ . This is the “no-dust” estimate that §17.4 and the session-8/9 handoffs listed as the open ingredient of every route: over the short horizon it holds for free, up to a logarithm, because the chain simply has not had time to create many blocks —  $\mathbb{E}[B_t] = [z^t] (1+z) \operatorname{artanh}(z)/(z(1-z)) = \log t + O(1)$  and the pgf makes the tail geometric. (At stationarity the count is infinite and the analogous statement is false as stated; the short horizon of Theorem 16.1 is again what makes the iid model tractable.) Since the required exponent in Corollary 17.11 is only  $\alpha > 2/9$ , losing logarithms is harmless.*

### 17.9.3 The exact law of the block of a uniform point

**Theorem 17.17** (size-biased block mass at all times). *Let  $U$  be uniform on the circle, independent of the chords, and let  $\zeta_m$  be the mass of the curve through  $U$  after  $m$  chords (chain from one block). Then*

$$\mathbb{P}[\zeta_m \in dx] = (1 - (1-2x)^m) dx \quad \text{on } (0,1), \quad \mathbb{P}[\zeta_m = 1] = \frac{1 + (-1)^m}{2(m+1)},$$

*and consequently, for every  $r \geq 1$ ,*

$$\mathbb{E}\left[\sum_i p_i(m)^r\right] = \mathbb{E}[\zeta_m^{r-1}] = \frac{1}{r} + \int_0^1 (1-x^{r-1})(1-2x)^m dx.$$

*By rotation invariance the same law governs the mass of the curve through any fixed point, in particular through  $u$  (and through  $v$ ). At finite  $n$ :  $\mathbb{P}[|C_{\sigma_m}(1)| = j] = (1 - r_j^m)/n$  for  $j < n$  and  $\mathbb{P}[|C_{\sigma_m}(1)| = n] = \frac{1}{n} \sum_{k=0}^{n-1} r_k^m$ , with  $r_k = 1 - 2k/(n-1)$ ; equivalently  $(-1)^k \langle \chi_{(n-k,1^k)}, \sum_i c_i^r \rangle = n^{r-1} - k^{r-1}$  for  $1 \leq k \leq n-1$  and  $\sum_{j \leq n} j^{r-1}$  for  $k = 0$ .*

*Proof.* For  $F$  on  $\{1, \dots, n\}$  put  $G_F(\sigma) = \sum_u F(|C_\sigma(u)|) = \sum_i c_i F(c_i)$ . The exponential formula with one marked cycle, as in the proof of Theorem 17.14 (weight  $jF(j)$  for a marked cycle of length  $j$  replaces  $j \cdot j$  there), gives

$$\sum_n x^n \sum_k t^k \langle \chi_{(n-k, 1^k)}, G_F \rangle = \frac{1}{1+t} \left[ \sum_{j \geq 1} F(j) (1 - (-t)^j) x^j \right] \frac{1+tx}{1-x}.$$

Extracting  $[x^n]$ :  $\sum_{j < n} F(j)(1 - (-t)^j)(1+t) + F(n)(1 - (-t)^n)$ ; dividing by  $1+t$  and extracting  $[t^k]$  gives, for  $1 \leq k \leq n-1$ ,  $-(-1)^k F(k) + (-1)^k F(n)$ , and  $\sum_{j \leq n} F(j)$  for  $k=0$ . Hence, by the Fourier inversion of Theorem 17.14,  $\mathbb{E}[G_F(\sigma_m)] = \sum_{j \leq n} F(j) + \sum_{k=1}^{n-1} r_k^m (F(n) - F(k)) = \sum_{j < n} F(j)(1 - r_j^m) + F(n) \sum_{k=0}^{n-1} r_k^m$ . Since  $G_F(\sigma)/n = \mathbb{E}_U[F(|C_\sigma(U)|)]$  for a uniform point  $U$ , this is the finite- $n$  law;  $F(j) = j^{r-1}$  gives the hook inner products, and  $j = nx$ ,  $r_j^m \rightarrow (1-2x)^m$  gives the continuum statements (Lemma 17.1 for the identification).  $\square$

*Status: PROVED. VERIFIED: the hook identity  $(-1)^k \langle \chi_k, \sum c^r \rangle = n^{r-1} - k^{r-1}$  by brute-force character sums for  $r \leq 6$ ,  $n \leq 9$  (`s10.hooks34.py`); the continuum law and the moments  $r = 2, 3, 4$  against simulation (`s10.sizebiased.py`,  $2 \times 10^5$  trials): at  $m = 8$  the ten decile masses agree to  $\pm 0.001$ , the atom  $0.1115$  vs  $1/9$ , and  $\mathbb{E} \sum p^{2,3,4} = 0.5561, 0.3947, 0.3140$  vs  $0.5556, 0.3939, 0.3131$ . Remarks. (i)  $r = 2$  is Theorem 17.14; the atom is  $\mathbb{P}[B_m = 1]$  of Theorem 17.15, as it must be. (ii) The drift identity  $\mathbb{E}[\Delta L] = 2L^2 - \frac{7}{3} \sum p_i^4$  ( $L = \sum p_i^2$ ) is now closed in expectation along the whole approach to PD(1):  $\mathbb{E} \sum p_i^4(m) = \frac{1}{4} + \int_0^1 (1-x^3)(1-2x)^m dx$ . (iii) Small marked blocks:  $\mathbb{P}[\text{mass of } u\text{'s curve} \leq \eta] = \int_0^\eta (1 - (1-2x)^m) dx \leq \min(\eta, m\eta^2)$  — the  $m\eta^2$  regime is the creation flux of Lemma 17.8(iii) made exact for the block through a fixed point, and the crossover at  $\eta \asymp 1/m$  is the trap scale. (iv) The density  $1 - (1-2x)^m$  is  $\geq 1 - e^{-2mx}$  on  $(0, \frac{1}{2}]$ : the size-biased pick is macroscopic with probability bounded below uniformly in  $m \geq 1$  — the quantitative handle for recharge that TODO 10a(A)(iii) asked for, now superseded by Corollary 17.16.*

#### 17.9.4 Recharge under the block bound

Fix  $K \geq 3$  and let  $H = H_T$  be the event of Corollary 17.16 with  $K_\gamma(T)$  replaced by  $K$ .

**Lemma 17.18** (recharge of the marked total). *Let  $K \geq 3$ ,  $s_0 \geq \varepsilon$  with  $\varepsilon \leq 1/(2K)$ . Then for every  $T \geq 1$ ,*

$$\mathbb{P} \left[ \max_{t \leq T} s_t < \frac{1}{2K} \right] \leq e^{-\varepsilon T/K} + 2\varepsilon^2 T + \mathbb{P}[H^c].$$

*The same holds for each marked coordinate separately: if  $a_0 \geq \varepsilon$  then  $\mathbb{P}[\max_{t \leq T} a_t < 1/(2K)] \leq e^{-\varepsilon T/(3K)} + 2\varepsilon^2 T + \mathbb{P}[H^c]$ , and likewise for  $b$ .*

*Proof.* On  $H \cap \{s_t < 1/(2K)\}$  the environment has mass  $> 1 - 1/(2K) \geq \frac{1}{2}$  in fewer than  $K$  blocks, so it has a block of mass  $z \geq 1/(2K)$ ; the chord with one cut in the marked material and one in that block has probability  $2s_t z \geq \varepsilon/K$  as long as  $s_t \geq \varepsilon$ , and it raises  $s$  by  $z \geq 1/(2K)$ . Flips and merges never decrease  $s$ , and  $s$  enters  $(0, \varepsilon]$  from above with probability  $\leq 2\varepsilon^2$  per step (Lemma 17.12); so the event  $\{\max_{t \leq T} s_t < 1/(2K)\}$  is contained in  $\{s_t < \varepsilon \text{ for some } t \leq T\} \cup H^c \cup \{T \text{ consecutive failures of an event of probability } \geq \varepsilon/K\}$ . For a single coordinate  $a$ : if  $s_t \leq \frac{1}{2}$  the same merge argument applies to  $a$  (probability  $\geq 2a \cdot 1/(2K) \geq \varepsilon/K$ ); if  $s_t > \frac{1}{2}$  and  $a_t < 1/(2K)$  then  $b_t \geq \frac{1}{3}$ , a flip has probability  $2a_t b_t \geq 2\varepsilon/3$  and lands at  $a' = \alpha + \beta \geq \beta \sim U(0, b_t)$ , so  $a' \geq 1/(2K)$  with conditional probability  $\geq 1 - 3/(2K) \geq \frac{1}{2}$ : success probability

$\geq \varepsilon/3 \geq \varepsilon/(3K)$  per step in both regimes; the entry flux into  $(0, \varepsilon]$  is  $\leq 2\varepsilon^2$  per step (shrink of  $a$ , flip landing small; Lemma 17.8(iii)).  $\square$

*Status: PROVED.* With  $T = (3K/\varepsilon) \log(1/\delta)$  the failure probability is  $\delta + 6\varepsilon K \log(1/\delta) + \mathbb{P}[H^c]$ , small precisely when  $\varepsilon \ll 1/K$  — the regime in which recharge is needed at all (for  $\varepsilon \geq 1/(2K)$  the start is already log-macroscopic). At the optimising  $\varepsilon = d^{-(1+2\alpha)/(2+2\alpha)} \in [d^{-3/4}, d^{-1/2})$  of Corollary 17.11 ( $\alpha < 1$ ) and  $K = O(\log d)$ ,  $T = O(\varepsilon^{-1} \log^2 d) \ll d$  and the failure probability is  $O(d^{-\gamma}) + O(\varepsilon \log^2 d)$ , which is  $o((\varepsilon d)^{-2\alpha}) = o(d^{-\alpha/(1+\alpha)})$  since  $\varepsilon^{1+2\alpha} d^{2\alpha} = d^{-1/(2+2\alpha)}$ .

**Lemma 17.19** (log-drift of the marked total). *On  $H$ , for a state with  $s \leq \frac{1}{2}$ ,*

$$\mathbb{E}[\log s_{t+1} - \log s_t \mid \Gamma_t] \geq \frac{s}{K} \log\left(1 + \frac{1}{2Ks}\right) - \frac{s^2}{2},$$

*which is  $\geq s(\log 2 - \frac{1}{4})/K > 0$  for  $s \leq 1/(2K)$ ; the negative jumps of  $\log s$  are those of  $\frac{1}{2} \log U$  (exponential tails), the positive ones are unbounded above only through merges.*

*Proof.* Merges: with probability  $\geq 2sz$  the total gains  $z$ , and the largest environment block has  $z \geq (1-s)/K \geq 1/(2K)$ ;  $\log(1+z/s)$  is increasing in  $z$ . Shrinks:  $a \rightarrow a\sqrt{U}$  with probability  $a^2$  changes  $\log s$  by  $\log(1 - (a/s)(1 - \sqrt{U})) \geq \log \sqrt{U}$ , of mean  $-\frac{1}{2}$ ; likewise  $b$ ;  $a^2 + b^2 \leq s^2$ . Flips preserve  $s$ ; environment-only moves do not touch it.  $\square$

*Status: PROVED (a calculation).* It says that, once the block count is bounded, the marked total is a one-dimensional process with positive log-drift below the level  $\asymp 1/K$  and light-tailed down-jumps; standard drift arguments (Lyapunov function  $s^{-\theta}$ , small  $\theta$ ) then bound the occupation time of  $\{s < c/K\}$  over any window by a small fraction with polynomially small failure probability. This is the “marked pair health” input for the tail of the first-flip time in §17.9.5; it is NOT written out.

### 17.9.5 The Markov-renewal form of Problem 17.9

Let  $\tau$  be the first flip time of the quotient chain and, for  $|z| \leq 1$  and bounded measurable  $F$ ,

$$(K_z F)(x) := \mathbb{E}_x[z^\tau F(\Gamma_\tau)], \quad \|K_z\|_{L^\infty \rightarrow L^\infty} \leq \sup_x \mathbb{E}_x|z|^\tau \leq |z|,$$

since  $\tau \geq 1$ .  $K_1$  is the post-flip kernel (Markov whenever  $\tau < \infty$  a.s.).

**Proposition 17.20** (generating function of the parity). *For  $|z| < 1$ , as bounded functions of the initial state,*

$$\sum_{d \geq 0} \Phi_d z^d = \frac{1}{1-z} (I + K_z)^{-1} (I - K_z) \mathbf{1}.$$

*Proof.* With flip times  $\tau_1 < \tau_2 < \dots$ ,  $\Phi_d = \sum_{k \geq 0} (-1)^k \mathbb{P}_x[\tau_k \leq d < \tau_{k+1}]$  and  $\sum_d z^d \mathbf{1}\{\tau_k \leq d < \tau_{k+1}\} = (z^{\tau_k} - z^{\tau_{k+1}})/(1-z)$ ; by the strong Markov property at the flip times  $\mathbb{E}_x[z^{\tau_k}] = (K_z^k \mathbf{1})(x)$ . Sum the Neumann series, which converges since  $\|K_z\| \leq |z| < 1$ .  $\square$

*Status: PROVED.* It is exact and separates the problem: the factor  $(I - K_z) \mathbf{1} = \mathbb{E}.[1 - z^\tau]$  carries the first-flip tail (traps), the resolvent  $(I + K_z)^{-1}$  carries the cancellation (the parity of the number of flips per unit time). Note  $(I - K_1) \mathbf{1} = 0$  wherever  $\tau < \infty$  a.s. (on  $\{ab > 0\}$ , since  $\sum_t 2a_t b_t = \infty$  a.s. there — taken for granted below), so the right side is a difference quotient at  $z = 1$ : the decay of  $\Phi_d$  is the regularity of  $G(z) := (I + K_z)^{-1} (I - K_z) \mathbf{1}$  on the closed disc.

**Proposition 17.21** (conditional decay). *Let  $\mathcal{B}$  be a Banach space of functions on the state space containing  $\mathbf{1}$ , on which the evaluations  $F \mapsto F(x)$ ,  $x \in S'$ , are uniformly bounded, and let  $\gamma \in (0, 1)$ . Suppose*

- (R1)  $z \mapsto K_z \in \mathcal{L}(\mathcal{B})$  is  $C^{1,\gamma}$  on  $\{|z| \leq 1\}$  (for a weighted sup-norm space  $\mathcal{B} = \{F : \sup |F|/V < \infty\}$  it suffices that  $\sup_x \mathbb{E}_x[\tau^{1+\gamma} V(\Gamma_\tau)]/V(x) < \infty$ , by  $|\tau z_1^{\tau-1} - \tau z_2^{\tau-1}| \leq 2\tau^{1+\gamma}|z_1 - z_2|^\gamma$ );
- (R2)  $I + K_z$  is invertible on  $\mathcal{B}$  for every  $|z| \leq 1$ , with  $\sup_{|z| \leq 1} \|(I + K_z)^{-1}\| < \infty$ .

Then  $\sup_{x \in S'} |\Phi_d(x)| \leq C d^{-\gamma}$ .

*Proof.* Under (R1)–(R2),  $G$  is  $C^{1,\gamma}$  on the closed disc with  $G(1) = 0$ , so  $\hat{\Phi}(z) = (G(z) - G(1))/(1 - z)$  extends to a  $C^{0,\gamma}$  function on the closed disc; it is analytic inside, so its boundary Fourier coefficients are the  $\Phi_d$ , and the Fourier coefficients of a Hölder- $\gamma$  function are  $O(d^{-\gamma})$ . Evaluate at  $x \in S'$ .  $\square$

**SESSION-11 CORRECTION.** Condition (R2) is FALSE at  $z = \pm 1$  for every function space containing the bounded functions:  $K_1 \varepsilon = -\varepsilon$ , because exactly one flip has occurred at time  $\tau$  (Proposition 17.24). Proposition 17.21 is therefore vacuous as stated. The same holds in the label-free bookkeeping, with  $\kappa_{-1}[(-1)^E] = -(-1)^E$ . It is repaired in §17.10: (R2) is replaced by the two conditions (C1) and (C2) of §17.10.4, both statements about the single Markov kernel  $\kappa_1$ , and everything on the closed disc away from  $z = \pm 1$  is proved unconditionally by Theorem 17.27. The discussion of (R1) below is unaffected, and so is the reduction paragraph at the end of this subsection. Read the rest of this paragraph with that substitution.

Status: PROVED as a reduction; (R1) and (R2) are OPEN. What each needs, and what is in hand:

(R1) — traps. A uniform bound  $\sup_{x \in S'} \mathbb{P}_x[\tau > t] \leq C_K t^{-2}$  (measured:  $t^2 \mathbb{P}[\tau > t] \rightarrow 11$  from stationary environments, §17.8.2(c)) gives every moment  $\tau^{1+\gamma}$ ,  $\gamma < 1$ , hence  $\gamma$  arbitrarily close to 1 in Proposition 17.21, far above the 4/9 [session-13: 1/11, Corollary 17.36] that Corollary 17.11 needs. Ingredients: the state-independent trap-creation flux (Lemma 17.8(iii)); the recharge of Lemma 17.18 and the drift of Lemma 17.19, both available on  $H$  by Corollary 17.16; and the fact that a trap of depth  $g$  is left by a flip alone, at rate  $2g(s - g)$ , needing no environment. The proof of the  $1/t^2$  tail is a renewal computation with these inputs (the subcritical kernel  $\approx 2/(s^2 m^2)$  of §17.8.2(c)), which we have not written.

(R2) — cancellation. This is the parity content and cannot be had from trap accounting. It is needed on the whole closed disc: the trivial bound  $\|K_z\| \leq |z|$  gives nothing on  $|z| = 1$ . At  $z = 1$  it is the statement that  $-1$  is not in the spectrum of the post-flip chain  $K_1$  on  $\mathcal{B}$ , with a uniform inverse; near  $z = 1$  it follows by perturbation under (R1). At other points of the circle it is the non-lattice condition of Markov renewal theory — no  $F \neq 0$  with  $\mathbb{E}_x[z^\tau F(\Gamma_\tau)] = -F(x)$  — and at  $z = -1$ , where  $K_{-1}F = \mathbb{E}_x[(-1)^\tau F(\Gamma_\tau)]$ , it says the parity of the inter-flip times is not asymptotically deterministic; this is where the  $(-1)^d$  alternation of §17.8.2(a) lives and it needs its own statement. The Doeblin structure is on the marked pair: two consecutive flips from any  $(a, b)$  with total  $s$  produce  $a'$  whose law dominates  $\frac{1}{4} \cdot \text{Unif}[s/4, 3s/4]$  (the trapezoid of Lemma 17.8(ii) is symmetric and unimodal, so the first flip lands in  $[s/4, 3s/4]$  with probability  $\geq \frac{1}{2}$ , and from there the second has density  $\geq 1/s$  on that interval). This is a minorisation on the fibre  $\{a' + b' = s, \text{env}\}$  only; the base  $(s, \text{env})$  is not regenerated, so  $\mathcal{B}$  must be a weighted space in which the post-flip chain is  $V$ -geometrically ergodic in the base coordinates as well (Meyn–Tweedie), with the weight matched to the moment condition in (R1). On  $H$  the base is finite-dimensional ( $< K$  blocks), which is what makes this plausible. Because the label parity is

determined by the environment block-count parity (Lemma 17.13), no argument can avoid the environment in (R2) either; but the environment enters only through the ergodicity of the base chain between flips, not through any fine property of PD(1).

*Reduction to Problem 17.9.* Take  $S' = S'_K := \{a, b \geq 1/(16K)\}$  and enforce  $H_{9d}$  (so  $C = C(K)$  in Proposition 17.21 may depend on the block bound, which holds up to time  $d$ ). Granting Proposition 17.21 on  $S'_K$  with  $C(K)$  polynomial in  $K$ , and granting the occupation-time consequence of Lemma 17.19 (that after  $s$  first reaches  $1/(2K)$  it stays  $\geq 1/(4K)$  for at least half of the next  $T_2$  steps, except with polynomially small probability; NOT written out), the chain from  $\Gamma_0 \sim \mu_j$  with  $a_0, b_0 \geq \varepsilon$  enters  $S'_K$  at a stopping time  $T'$ : first  $s$  reaches  $1/(2K)$  within  $T_1 = O(\varepsilon^{-1}K \log d)$  steps (Lemma 17.18); then, on the times with  $s_t \geq 1/(4K)$  and  $\min(a_t, b_t) \geq \varepsilon$  (the latter failing with probability  $\leq 4\varepsilon^2(T_1 + T_2)$  by the flux bound), a flip occurs with probability  $\geq 2\varepsilon(s_t - \varepsilon) \geq \varepsilon/(4K)$  per step and lands with  $\min(a', b') \geq s_t/4 \geq 1/(16K)$  with conditional probability  $\geq \frac{1}{2}$  (the trapezoid is symmetric and unimodal), so  $T' \leq T_1 + T_2$  with  $T_2 = O(\varepsilon^{-1}K \log d)$  except with probability  $O(d^{-\gamma}) + O(\varepsilon K \log d)$ . Note that a single application of Lemma 17.18 to each coordinate does not suffice: it makes  $a$  and  $b$  large at different times, and a coordinate of size  $1/(2K)$  shrinks on the scale  $K^2 \ll \varepsilon^{-1}K$ . By the strong Markov property  $|\Phi_d(\Gamma_0)| \leq \mathbb{E}|\Phi_{d-T'}(\Gamma_{T'})| + \mathbb{P}[T' > T_1 + T_2] \leq C(K)(d/2)^{-\gamma} + O(d^{-\gamma}) + O(\varepsilon K \log d)$ , and with  $K = O(\log d)$  and  $\varepsilon$  as in Corollary 17.11 this is  $\leq C(\varepsilon d)^{-2\alpha}$  for  $\alpha < \gamma/2$  (for  $\varepsilon \geq 1/(16K)$  no recharge is needed). [**SESSION-13 CORRECTION:** this sentence bounds  $|\Phi_d|$  where Problem 17.9 asks for  $\Phi_d^2$ , and discards the  $\varepsilon$  in  $(\varepsilon d)^{-2\alpha}$ ; both losses are unnecessary, and the supremum over  $S'_K$  taken two lines above is not needed either. The correct accounting is Proposition 17.34 and Corollary 17.36: the budget is  $\gamma > 1/11$ , not  $\gamma > 4/9$ .] Thus Problem 17.9, hence the iid model, is reduced to (R1) and (R2) on the log-macroscopic set  $S'_K$  plus the occupation-time step, with the environment controlled throughout by Corollary 17.16.

## 17.10 Session 11: (R2) is false; the idle loop proves its repair on the whole disc except two points; and the residual condition is one scalar statement about the parity of the environment block count

Session 11 was directed at condition (R2) of Proposition 17.21 — uniform invertibility of  $I + K_z$  on the closed unit disc — under the instruction that a fourth reformulation of the cancellation core would not count as progress. There are four findings, in logical order.

- (1) **The label is a passenger.** The quotient chain  $\bar{\Gamma} = (p_1, p_2; \text{env})$  — the two marked masses and the environment, with the label  $\varepsilon$  forgotten — is *autonomous*: its transition law does not depend on  $\varepsilon$ , and the involution  $J$  that swaps the two sectors keeping  $\bar{\Gamma}$  fixed commutes with the chain (Lemma 17.22). Hence  $K_z = \kappa_z \oplus (-\kappa_z)$  for the quotient first-flip kernel  $\kappa_z$ , and  $\text{spec } K_z = \pm \text{spec } \kappa_z$ . One line of this gives  $\mathbb{P}_\pi[u \sim v] = \frac{1}{2}$ .
- (2) **(R2) is false.** With the label kept,  $K_1\varepsilon = -\varepsilon$  — exactly one flip has occurred at time  $\tau$  — so  $I + K_1$  has a bounded null vector. On the quotient, where that artefact is absent,  $I + \kappa_{-1}$  still has the bounded null vector  $(-1)^E$  ( $E$  = number of environment curves), by Lemma 17.13. Either way  $\sup_{|z| \leq 1} \|(I + K_z)^{-1}\| = \infty$  and Proposition 17.21 is vacuous as stated (Proposition 17.24).
- (3) **An unconditional contraction off  $z = \pm 1$ .** The chain can waste exactly two steps by splitting a curve and merging the two pieces it just created, returning to the same state without flipping. This “idle loop” has probability  $\lambda(\Gamma) = \frac{1}{3} \sum_j r_j^4$  — an exact formula,  $r_j$

the arc-refined mass list — and it forces  $\|\kappa_z\|_\infty \leq (1 - \lambda)/|1 - \lambda z^2| \leq 1/(1 + \lambda(1 - \Re z^2))$  for all  $|z| \leq 1$  (Theorem 17.27). On the block-bounded state space this is uniform, with  $\lambda \geq \frac{1}{3}K^{-3}$ . The non-lattice condition of Markov renewal theory on the *whole* circle — Pitfall 45, TODO 11b(v), and the third name of the cancellation core — is therefore *proved*, unconditionally and with explicit constants.

- (4) **What is left is two scalar conditions on one Markov kernel.** After (3),  $\widehat{\Phi}(z) = (1 - z)^{-1}(I + \kappa_z)^{-1}(I - \kappa_z)\mathbf{1}$  can fail to be regular only at  $z = 1$  and  $z = -1$ . At  $z = -1$  the  $\bar{M}$ -conjugation of Lemma 17.23 turns the problem into the ordinary renewal pole at  $+1$  applied to the test function  $(-1)^E$ , and the condition is

$$(C2) \quad \|\kappa_1^k(-1)^E\|_\infty \leq C\rho^k \quad \text{for some } \rho < 1,$$

i.e. *the parity of the environment block count decorrelates geometrically along the post-flip chain*. At  $z = 1$  the condition is  $-1 \notin \text{spec } \kappa_1$  with a uniform inverse (C1), and we prove that a single Doeblin minorisation shared by the  $n$ -th and  $(n+1)$ -st post-flip laws implies it, with  $\|(I + \kappa_1)^{-1}\| \leq (2n + 1)/(2\delta)$  (Proposition 17.28). Neither condition contains a  $z$ , a phase, or an alternating sign that is not the sign of a state function. (C2) is exactly what `s11_kflip` measures, and it holds numerically with  $\rho \approx 0.3$  from both a macroscopic and a deeply trapped start (§17.10.5).

### 17.10.1 The label is a passenger

Recall the state: a partition of the circle into curves with masses, two marks  $u, v$ , and the pair  $(p_1, p_2)$  — the two arcs  $(x, y)$  of the marked curve when  $u \sim v$ , the two marked curve masses  $(a, b)$  when  $u \not\sim v$  — with  $s = p_1 + p_2$ , and  $\varepsilon = \pm 1$  the label. Let  $\bar{\Gamma} = (p_1, p_2; \text{env})$  be the state with the label forgotten and  $\pi_* : \Gamma \mapsto \bar{\Gamma}$  the projection.

**Lemma 17.22** (label autonomy). *The transition kernel of  $\Gamma$  projects to a kernel on  $\bar{\Gamma}$ , and in both sectors the moves and their probabilities are*

*flip:*  $2p_1p_2, \quad p_i \rightarrow p_i\sqrt{U} : p_i^2, \quad p_i \rightarrow p_i + q_j : 2p_iq_j, \quad \text{environment moves: unchanged,}$

*a flip replacing  $(p_1, p_2)$  by  $(V, s - V)$  with  $V \sim U(0, p_1) \star U(0, p_2)$  and toggling  $\varepsilon$ . Consequently:*

- (i) *the involution  $J$  of the state space that swaps the two sectors keeping  $\bar{\Gamma}$  fixed commutes with the transition kernel and fixes every flip time;*
- (ii) *in the decomposition  $F = g \circ \pi_* + \varepsilon \cdot (f \circ \pi_*)$  of bounded functions,  $K_z = \kappa_z \oplus (-\kappa_z)$ , where  $(\kappa_z g)(\bar{x}) = \mathbb{E}_{\bar{x}}[z^\tau g(\bar{\Gamma}_\tau)]$ ; hence  $\text{spec } K_z = \text{spec } \kappa_z \cup (-\text{spec } \kappa_z)$  and  $\widehat{\Phi}(z) = (1 - z)^{-1}(I + \kappa_z)^{-1}(I - \kappa_z)\mathbf{1}$ ;*
- (iii) *if  $\bar{\pi}$  is a stationary law of  $\bar{\Gamma}$  then the stationary label is uniform:  $\mathbb{P}[u \sim v] = \frac{1}{2}$ .*

*Proof.* A step picks two uniform points of the circle; if they lie in the same curve it is a split, otherwise a merge. In the label-+ sector the marked curve has mass  $s$  and is picked twice with probability  $s^2$ ; given that, the two cuts lie on different arcs with probability  $2xy/s^2$  (a flip, separating  $u$  from  $v$ ) and on the same arc  $i$  with probability  $p_i^2/s^2$ , in which case the arc loses a factor  $\sqrt{U}$  in law ( $1 - |U_1 - U_2| \sim \sqrt{U}$ ) and the removed piece becomes an unmarked curve. A merge of the marked curve with an environment curve  $q_j$  has probability  $2sq_j$ , and the mass

$q_j$  is attached to arc  $i$  with probability  $p_i/s$ , i.e. rate  $2p_iq_j$ . In the label— sector the same four rates hold with  $(a, b)$  in place of  $(x, y)$ : picking  $u$ 's and  $v$ 's curves is a merge — a flip, joining them — with probability  $2ab$ ; picking  $u$ 's curve twice splits it,  $a \rightarrow a\sqrt{U}$ ; picking  $u$ 's curve and  $q_j$  merges them,  $a \rightarrow a + q_j$ . The post-flip law is the same trapezoid  $U(0, p_1) \star U(0, p_2)$  in both directions (Lemma 17.8(ii)). This is the displayed list, and it does not mention  $\varepsilon$ . (i) is immediate; (ii) follows because  $\varepsilon(\Gamma_\tau) = -\varepsilon(x)$  while the law of  $\bar{\Gamma}_\tau$  depends only on  $\bar{x}$ ; (iii) because  $J$  preserves the stationary law — it commutes with the kernel and fixes  $\bar{\Gamma}$  — while exchanging the two sectors.  $\square$

*Status: PROVED. Implicit in the phrase “quotient chain” used since session 8 but never stated; it is worth stating because it is what makes the next two subsections possible. Two consequences. (a) The first-flip time  $\tau$  has the same law from  $x$  and from  $Jx$ : the alternating renewal governing  $\Phi_d$  is symmetric, which is why  $\Phi_d \rightarrow 0$  rather than to the generic limit  $(\mathbb{E}_+\tau - \mathbb{E}_-\tau)/(\mathbb{E}_+\tau + \mathbb{E}_-\tau)$  of an asymmetric alternating renewal. This is the structural reason behind  $\mathbb{P}_\pi[\text{tog}] = \frac{1}{2}$ , and it is exactly the residue that has to vanish at  $z = 1$  below. (b) All the parity content therefore lives in  $\kappa_z$ , a kernel on a space where  $\varepsilon$  does not exist.*

### 17.10.2 (R2) is false, in both bookkeepings

Write  $E(\Gamma)$  for the number of environment curves and let  $\bar{M}$  be multiplication by  $(-1)^E$ , an isometric involution of every weighted sup-norm space;  $N$  denotes multiplication by  $\varepsilon$ .

**Lemma 17.23** (sign conjugations). *For every  $z \in \mathbb{C}$ ,*

$$(i) \quad NK_zN = -K_z, \quad (ii) \quad \bar{M}\kappa_z\bar{M} = -\kappa_{-z}.$$

*Hence  $\text{spec } K_z = -\text{spec } K_z$  and  $\text{spec } \kappa_{-z} = -\text{spec } \kappa_z$ , and the resolvents  $(I \pm \kappa_{\pm z})^{-1}$  have equal norms in pairs.*

*Proof.* (i) Exactly one flip has occurred at time  $\tau$ , so  $\varepsilon(\Gamma_\tau) = -\varepsilon(x)$  a.s. (ii) By Lemma 17.13,  $(-1)^{N_t} = (-1)^t(-1)^{E_t - E_0}$ ; at  $t = \tau$  we have  $N_\tau = 1$ , so  $(-1)^\tau = -(-1)^{E_\tau - E_0}$ . Therefore  $(\bar{M}\kappa_z\bar{M}g)(\bar{x}) = (-1)^{E(\bar{x})}\mathbb{E}_{\bar{x}}[z^\tau(-1)^{E(\bar{\Gamma}_\tau)}g(\bar{\Gamma}_\tau)] = -\mathbb{E}_{\bar{x}}[(-z)^\tau g(\bar{\Gamma}_\tau)]$ .  $\square$

*Remark (the label-carrying form of (ii)). Since  $B$  changes by exactly  $\pm 1$  at every step,  $(-1)^\tau = (-1)^{B(\Gamma_\tau) - B(x)}$  directly, so with  $M = \text{multiplication by } (-1)^B$  one has the cleaner-looking  $MK_zM = K_{-z}$  on the label-carrying space, and  $K_{-1}[(-1)^B] = +(-1)^B$ . That eigenvalue is  $+1$  and is harmless; the obstruction to (R2) is the eigenvalue  $-1$  of  $\kappa_{-1}$  at  $(-1)^E$  in Proposition 17.24. The two are consistent:  $(-1)^B = -\varepsilon(-1)^E$ , and  $NK_zN = -K_z$  supplies the sign. Keeping these apart matters: in the label-carrying picture  $\mathbb{E}_x[(-1)^{\tau_k}]$  has no a-priori reason to decay, while in the label-free picture its decay is exactly (C2).*

**Proposition 17.24** (failure of (R2)). *Assume  $\tau < \infty$  a.s. (as is taken for granted after Proposition 17.20), so that  $\kappa_1$  is Markov. Then*

$$K_1\varepsilon = -\varepsilon \quad \text{and} \quad \kappa_{-1}[(-1)^E] = -(-1)^E.$$

*Hence  $I + K_z$  is not injective at  $z = 1$  in the label-carrying bookkeeping and  $I + \kappa_z$  is not injective at  $z = -1$  in the label-free one, on every function space containing the bounded functions. In particular condition (R2) of Proposition 17.21 never holds, and Proposition 17.21 is vacuous as stated.*

*Proof.* The first identity is Lemma 17.23(i) applied to  $\mathbf{1}$  together with  $K_1\mathbf{1} = \mathbf{1}$ . For the second, by Lemma 17.23(ii) with  $z = 1$ ,  $\kappa_{-1}\bar{M}\mathbf{1} = -\bar{M}\kappa_1\mathbf{1} = -\bar{M}\mathbf{1}$ .  $\square$

*Status: PROVED.* The failure is structural, not technical: the null vectors  $\varepsilon$  and  $(-1)^E$  are bounded by 1 and lie in every space in sight, so no choice of weight, state space or test class removes them. Session 10 did not see it because Proposition 17.20 applies  $(I + K_z)^{-1}$  only to  $(I - K_z)\mathbf{1}$ , which degenerates at the same points; the whole question is whether the vanishing of the numerator beats the blow-up of the resolvent, and that is not what (R2) says. The two failures are the same failure in the two bookkeepings, since  $\text{spec } K_z = \pm \text{spec } \kappa_z$  by Lemma 17.22(ii).

**Proposition 17.25** (the two singular points, exactly). *For  $|z| < 1$ ,  $\hat{\Phi}(z) = (1 - z)^{-1}(I + \kappa_z)^{-1}(I - \kappa_z)\mathbf{1}$ , and for every  $w$ ,*

$$(I + \kappa_{-w})^{-1}(I - \kappa_{-w})\mathbf{1} = \bar{M}(I - \kappa_w)^{-1}(I + \kappa_w)\bar{M}\mathbf{1}.$$

*Thus near  $z = 1$  the singular factor is  $(I + \kappa_z)^{-1}$  acting on the numerator  $(I - \kappa_z)\mathbf{1} = \mathbb{E}[1 - z^\tau] = O(|1 - z|)$ , while near  $z = -1$  it is the ordinary renewal resolvent  $(I - \kappa_w)^{-1}$ ,  $w \rightarrow 1$ , acting on  $(I + \kappa_w)(-1)^E \rightarrow 2(-1)^E$ .*

*Proof.* The first statement is Lemma 17.22(ii) substituted in Proposition 17.20; the second is Lemma 17.23(ii) together with  $\bar{M}^2 = I$ .  $\square$

*Status: PROVED (algebra).* This is the sharp form of the correction. At  $z = -1$  the singularity is the classical simple pole of a Markov renewal resolvent at  $w = 1$ , whose residue is the stationary projection  $\langle \bar{\pi}, \cdot \rangle \mathbf{1}$ ; the residue of  $\hat{\Phi}$  at  $z = -1$  is therefore proportional to  $\langle \bar{\pi}, (-1)^E \rangle$ , and  $\Phi_d$  carries a non-decaying alternating part  $c(-1)^d$  unless that pairing vanishes. That is precisely the alternation found in Lemma 17.13(a) — where it is measured to decay like  $(-1)^d c_1/d^2$ , i.e. the residue does vanish. At  $z = 1$  the numerator vanishes and the residue to be beaten is the residue of a symmetric alternating renewal, zero by Lemma 17.22(a). Both residues therefore vanish for structural reasons; what is unproved is the quantitative, uniform version, and that is (C1) and (C2) of §17.10.4.

### 17.10.3 The idle loop: an unconditional contraction off $z = \pm 1$

Write the *arc-refined mass list*  $r(\Gamma) = (r_j)_j$  for the masses of the unmarked curves together with  $p_1$  and  $p_2$ ; so  $\sum_j r_j = 1$  and  $r$  has at most  $B(\Gamma) + 1$  entries,  $B$  the total number of curves.

**Lemma 17.26** (the idle loop). *Let  $L$  be the event that step 1 splits some curve and step 2 merges the two pieces that step 1 created, the returned piece being re-attached to the arc it came from if the split curve was the marked one in the label-+ sector. On  $L$  no flip occurs and  $\bar{\Gamma}_2 = \bar{\Gamma}_0$ ; up to a null set  $L = \{\bar{\Gamma}_2 = \bar{\Gamma}_0\}$ ; and*

$$\lambda(\Gamma) := \mathbb{P}_\Gamma[L] = \frac{1}{3} \sum_j r_j(\Gamma)^4 \geq \frac{1}{3(B(\Gamma) + 1)^3}.$$

*Proof.* Fix an entry  $r$  of the arc-refined list; the events for distinct entries are disjoint.

$r = q$ , an unmarked curve. Step 1 picks it twice with probability  $q^2$  and cuts it at two independent uniform points, giving pieces  $qV$  and  $q(1 - V)$  with  $V = |U_1 - U_2|$ , density  $2(1 - v)$ . Step 2 picks those two curves with probability  $2qV \cdot q(1 - V)$  and merges them back into a curve of mass  $q$ . Total  $q^2 \cdot 2q^2 \mathbb{E}[V(1 - V)] = q^4/3$ , since  $\mathbb{E}[V(1 - V)] = \int_0^1 v(1 - v) 2(1 - v) dv = \frac{1}{6}$ .

$r = a = p_1$  in the label-— sector. Identical, the mark staying in the piece of mass  $a(1 - V)$ : total  $a^4/3$ .

$r = x$ , one arc in the label+ sector. Both cuts must fall in that arc, probability  $x^2$  — and this is a split, not a flip, precisely because a flip needs the cuts on *different* arcs. The piece has mass  $w = xV$ ; the marked curve keeps  $s - w$  and the arc becomes  $x - w$ . Step 2 merges the piece back with probability  $2w(s - w)$ , and the returned mass is re-attached to the arc it came from with conditional probability  $(x - w)/(s - w)$ . The factor  $s - w$  cancels:  $x^2 \mathbb{E}[2xV \cdot x(1 - V)] = 2x^4 \mathbb{E}[V(1 - V)] = x^4/3$ .

Summing gives  $\lambda = \frac{1}{3} \sum_j r_j^4$ . Conversely a merge at step 1 followed by a split at step 2 reproduces the two merged masses with probability 0, and a split followed by the merge of two other curves reproduces the mass list only on a null set. Finally, with  $p \leq B + 1$  entries, Cauchy–Schwarz twice gives  $\sum r_j^4 \geq (\sum r_j^2)^2/p \geq p^{-3}$ .  $\square$

*Status: PROVED. VERIFIED (s11.ztau,  $2 \times 10^6$  two-step trials per start, exact state comparison at tolerance  $10^{-12}$ ): one curve, arcs  $(\frac{1}{2}, \frac{1}{2})$ :  $\lambda$  measured 0.041530 vs.  $\frac{1}{3} \cdot 2 \cdot 2^{-4} = 0.041667$ ;  $s_0 = 0.6$ , arcs  $(0.1, 0.5)$ , environment 0.4: 0.029459 vs. 0.029400;  $s_0 = 0.5$ , arcs  $(0.02, 0.48)$ , environment 0.5: 0.038556 vs. 0.038528. The formula is an identity, not an estimate, and it is the arc-refined list that appears: splitting the marked curve across its two arcs is a flip, not an idle loop. The quantity  $\sum_j r_j^4$  is the one whose expectation Theorem 17.17 computes exactly,  $\mathbb{E} \sum p^4(m) = \frac{1}{4} + \int_0^1 (1 - x^3)(1 - 2x)^m dx \rightarrow \frac{1}{4}$ , so  $\mathbb{E} \lambda \rightarrow \frac{1}{12}$ ; the block bound is needed only to turn that average into a uniform bound.*

**Theorem 17.27** (idle-loop contraction). *Let  $\lambda_* = \inf_{\bar{x}} \lambda(\bar{x})$  over the state space on which  $\kappa_z$  acts. For all  $|z| \leq 1$ ,*

$$\|\kappa_z\|_{\infty \rightarrow \infty} \leq \sup_{\bar{x}} \frac{1 - \lambda(\bar{x})}{|1 - \lambda(\bar{x})z^2|} \leq \frac{1}{1 + \lambda_*(1 - \Re z^2)}.$$

*Consequently if  $1 - \Re z^2 \geq t_0$  then  $I \pm \kappa_z$  and  $I - \kappa_z^2$  are invertible with  $\|(I \pm \kappa_z)^{-1}\| \leq 1 + (\lambda_* t_0)^{-1}$ . On the state space  $\Omega_K = \{B \leq K\}$  — the chain killed at the first split that would make  $B = K + 1$ , whose  $\Phi_d$  differs from the true one by at most  $\mathbb{P}[H^c] \leq (9d + 1)^{-\gamma}$  (Corollary 17.16) — one has uniformly*

$$\|\kappa_z\|_{\infty} \leq 1 - \min\left(\frac{t_0}{6K^3}, \frac{1}{2K}\right), \quad \text{so} \quad \|(I \pm \kappa_z)^{-1}\| \leq \max\left(\frac{6K^3}{t_0}, 2K\right).$$

*Proof.*  $L$  is measurable with respect to the first two steps; on  $L$  no flip occurs and  $\bar{\Gamma}_2 = \bar{x}$ . By the Markov property at time 2,  $\mathbb{E}_{\bar{x}}[z^\tau g(\bar{\Gamma}_\tau) \mathbf{1}_L] = z^2 \lambda(\bar{x})(\kappa_z g)(\bar{x})$ , whence

$$(1 - \lambda z^2)(\kappa_z g)(\bar{x}) = \mathbb{E}_{\bar{x}}[z^\tau g(\bar{\Gamma}_\tau) \mathbf{1}_{L^c}], \quad |(\kappa_z g)(\bar{x})| \leq \frac{(1 - \lambda)\|g\|_{\infty}}{|1 - \lambda z^2|}.$$

For  $|z| \leq 1$ ,  $|1 - \lambda z^2| \geq \Re(1 - \lambda z^2) = (1 - \lambda) + \lambda t$  with  $t = 1 - \Re z^2$ , and  $u \mapsto (1 - u)/(1 - u + ut)$  has derivative  $-t/(\cdot)^2 < 0$ , so the supremum over  $\bar{x}$  is attained at  $\lambda_*$  and equals  $1/(1 + \lambda_* t/(1 - \lambda_*)) \leq 1/(1 + \lambda_* t)$ . The resolvent bounds are Neumann series.

For the killed chain, split the states. If  $B(\bar{x}) \leq K - 1$  the idle loop raises  $B$  only to  $B + 1 \leq K$ , so it stays inside  $\Omega_K$  and is not killed; Lemma 17.26 gives  $\lambda \geq \frac{1}{3}K^{-3}$ , and  $1/(1 + u) \leq 1 - u/2$  for  $u \leq 1$  gives the first branch. If  $B(\bar{x}) = K$  then *every* split kills, and a split occurs at the next step with probability  $\sum_i (\text{curve mass})^2 \geq 1/K$ , so  $|\kappa_z g(\bar{x})| \leq \|g\|_{\infty} \mathbb{P}_{\bar{x}}[\tau < \text{kill}] \leq (1 - 1/K)\|g\|_{\infty}$ . Take the worse of the two branches.  $\square$

*Status: PROVED. VERIFIED (s11\_ztau):* from each of the three deterministic starts above,  $|\mathbb{E}_{\bar{x}}[z^\tau]| = |\kappa_z \mathbf{1}(\bar{x})|$  was measured at  $\theta/\pi = 0.1, \dots, 0.9$  ( $2 \times 10^5$  trials each) and compared with  $(1 - \lambda)/|1 - \lambda z^2|$ : the bound holds at every point, with margin (at  $\theta = \pi/2$  from the  $(\frac{1}{2}, \frac{1}{2})$  start, measured 0.446 against the bound 0.920). The bound is crude — it uses one two-step return and nothing else — but it is uniform in the state, which is the entire difficulty; and it is the right mechanism, since  $1 - \lambda z^2$  is exactly the Euler factor of the geometric component that the idle loop puts into the law of  $\tau$  at lag 2, which is what an aperiodicity or non-lattice hypothesis is normally assumed to supply.

Three items on the open list are settled by Theorem 17.27 and Lemma 17.23. (i) Pitfall 45 and TODO 11b(v) — the non-lattice condition “on the whole circle, not only near  $z = 1$ ” — are PROVED, with rate  $1 - \Re z^2$  and constant  $3K^3$ . (ii) TODO 11b(iv), “ $-1 \notin \text{spec } K_1$ ”, is DISPROVED and was the wrong question (Proposition 17.24). (iii) The point  $z = -1$ , treated as a separate problem in every handoff since session 8, is by Proposition 17.25 the ordinary renewal point  $w = 1$  with the test function  $(-1)^E$ : not a separate problem, and not a place where anything new has to be invented.

#### 17.10.4 What is left: (C1) and (C2)

By Theorem 17.27,  $\hat{\Phi}$  is regular on  $\{|z| \leq 1\} \setminus \{\pm 1\}$  with bounds polynomial in  $K$ . Two conditions remain, both about the single Markov kernel  $\kappa_1$  — the post-flip quotient chain — on  $\Omega_K$ :

- (C1)  $-1 \notin \text{spec } \kappa_1$ , with  $\|(I + \kappa_1)^{-1}\| \leq C(K)$ ;
- (C2)  $\|\kappa_1^k(-1)^E\|_\infty \leq C(K)\rho^k$  for some  $\rho = \rho(K) < 1$ .

With (R1) and these two, the proof of Proposition 17.21 goes through: (C1) gives regularity at  $z = 1$ , where the numerator has its zero; (C2) gives, through Proposition 17.25, both the vanishing of the residue at  $z = -1$  — since  $\langle \bar{\pi}, (-1)^E \rangle = \lim_k \kappa_1^k(-1)^E = 0$  — and a Hölder modulus there. *Neither contains a  $z$ , a phase, or an alternating sign that is not the sign of a state function: the parity content of the problem has been reduced to the single statement that the environment block count has no asymptotic parity bias along the post-flip chain.*

*Neither (C1) nor (C2) is false for the reason (R2) was. The only deterministically-toggled state function available on the quotient is  $(-1)^E$  (flips leave  $E$  fixed; every non-flip step changes it by  $\pm 1$ ), and by Lemma 17.23(ii)  $\kappa_1[(-1)^E] = -(-1)^E \mathbb{E}[(-1)^\tau]$ , so  $(-1)^E$  is an eigenfunction of  $\kappa_1$  if and only if  $\mathbb{E}_x[(-1)^\tau]$  is constant in  $x$  — which it measurably is not ( $-0.1306 \pm 0.003$  from the macroscopic start against  $-0.0009 \pm 0.005$  from the trapped start, §17.10.5).*

**Proposition 17.28** (a minorisation suffices for (C1)). *Suppose there are  $n \geq 1$ ,  $\delta > 0$  and a probability measure  $\nu$  with*

$$\mathbb{P}_{\bar{x}}[\bar{\Gamma}_{\tau_n} \in \cdot] \geq \delta \nu(\cdot) \quad \text{and} \quad \mathbb{P}_{\bar{x}}[\bar{\Gamma}_{\tau_{n+1}} \in \cdot] \geq \delta \nu(\cdot) \quad \text{for every } \bar{x}.$$

*Then  $I + \kappa_1$  is invertible on  $L^\infty$  with  $\|(I + \kappa_1)^{-1}\| \leq (2n + 1)/(2\delta)$ .*

*Proof.* Let  $(I + \kappa_1)F = G$  and  $m = \|F\|_\infty$ . Iterating  $F = G - \kappa_1 F$  gives  $F = H_j + (-1)^j \kappa_1^j F$  with  $H_j = \sum_{i < j} (-1)^i \kappa_1^i G$ ,  $\|H_j\| \leq j\|G\|$ . Write  $\kappa_1^n F = \delta c + (1 - \delta)\rho_x^1(F)$  and  $\kappa_1^{n+1} F = \delta c + (1 - \delta)\rho_x^2(F)$  with the same  $c = \nu(F)$  and probability measures  $\rho_x^i$ . Multiplying the  $j = n$  identity by  $(-1)^n$  and the  $j = n + 1$  identity by  $(-1)^{n+1}$  and subtracting the results eliminates the  $\delta c$  terms’ difference and leaves

$$2F(\bar{x}) = \pm(H_n + H_{n+1})(\bar{x}) + (1 - \delta)(\rho_x^2 - \rho_x^1)(F),$$

so  $2m \leq (2n+1)\|G\| + 2(1-\delta)m$ , i.e.  $2\delta m \leq (2n+1)\|G\|$ .  $\square$

The proof uses no property of  $\kappa_1$  beyond  $\|\kappa_1\| \leq 1$ : it is valid verbatim for the sub-Markov  $\kappa_1$  of the chain killed on  $\Omega_K$ , the residual measures  $\rho_{\bar{x}}^i$  then having mass  $\leq 1-\delta$  rather than exactly  $1-\delta$ . This matters because  $\Omega_K$  is where Theorem 17.27 lives.

**SESSION-12 CORRECTION.** The hypothesis of this proposition is never satisfied: for every  $n$  there is no common minorant of the  $n$ -th post-flip laws over  $\Omega_K$ , because a curve of mass  $\varepsilon$  survives the first  $n$  flips untouched with probability  $\rightarrow 1$  (Proposition 17.30). The proposition stays correct and vacuous; TODO 12b is closed.

Status: PROVED as a reduction; the minorisation is OPEN. It is the honest remaining difficulty and it is the classical one: the post-flip chain regenerates on the fibre — two consecutive flips produce a  $p_1''$  whose law dominates  $\frac{1}{4}\text{Unif}[s/4, 3s/4]$  (§17.9.5) — but not in the base  $(s, \text{env})$ , which is carried along. On  $\Omega_K$  the base has at most  $K$  curves, and the numerics of §17.10.5 show it equilibrating in about four flips from a deliberately bad start, so a minorisation with  $n = O(1)$  and a  $\delta$  polynomial in  $1/K$  is the natural target; the  $K$ -dependence of  $\delta$  may be as bad as one likes, polynomially.

Routes to (C2). (a) Uniform ergodicity plus a parity symmetry: the same minorisation gives  $\|\kappa_1^k(F - \bar{\pi}F)\| \leq C\rho^k$  for all bounded  $F$ , reducing (C2) to  $\langle \bar{\pi}, (-1)^E \rangle = 0$  — the statement that the post-flip stationary law on  $\Omega_K$  has no environment block-count parity bias. On  $\Omega_K$  that is a statement about a finite-dimensional chain and is the cleanest target for session 12. (b) Directly: by Lemma 17.23(ii),  $\kappa_1^k(-1)^E = (-1)^E(-1)^k\kappa_{-1}^k\mathbf{1}$  and  $\kappa_{-1}^k\mathbf{1}(\bar{x}) = \mathbb{E}_{\bar{x}}[(-1)^{\tau_k}]$ , so (C2) says exactly that the parity of the  $k$ -th flip time decorrelates geometrically — which is what `s11_kflip` measures. A proof along (b) needs a state-uniform lower bound on the probability that two consecutive inter-flip times have opposite parity; the idle loop supplies a uniform event at lag 2, but it changes  $\tau$  by an even amount, so a genuinely odd perturbation is still needed. That is the one place where an odd/even asymmetry must be produced rather than conjugated away, and it is the sharpest statement of the residue of the whole problem.

A Lyapunov remark. The proof of Theorem 17.27 degrades only through  $\lambda^{-1} = 3(\sum_j r_j^4)^{-1}$ , whose expectation along the chain is known exactly (Theorem 17.17). A drift inequality  $\kappa_1 V \leq \varrho V + b$  with  $V = (\sum_j r_j^4)^{-\theta}$ ,  $\theta > 0$  small, would replace the killing at  $B = K$  by a weighted space and supply the “base” half of the minorisation of Proposition 17.28 at the same time. It is a finite computation of the same kind as Lemma 17.19 and is the recommended first task of session 12.

### 17.10.5 Numerics along the post-flip chain

TODO 11d asked for  $\mathbb{E}_{\bar{x}}[(-1)^{\tau_k}]$  along the post-flip chain, predicting geometric decay under (R2). The prediction needed correcting before the run could be interpreted — under (R2) as stated there is no such prediction, (R2) being false — but the quantity is, by route (b) above, *exactly* the residual condition (C2). Measured (`s11_kflip`: stationary environment burnt in over 4096 steps, accepted on the initial gap,  $k \leq 16$ , no censoring):

Macroscopic start, gap in  $[\frac{1}{8}, \frac{3}{8}]$ ,  $1.2 \times 10^5$  samples (s.e. 0.0029):  $\mathbb{E}[(-1)^{\tau_k}] = -0.1306, +0.0263, -0.0100, +0.0009, \dots$  and at the noise floor for every  $k \geq 4$ : geometric decay with ratio  $\approx 0.2-0.4$  and no persistent  $(-1)^k$  component. Trap start, gap in  $[10^{-3}, 10^{-2}]$ ,  $4 \times 10^4$  samples (s.e. 0.0050):  $-0.0009, +0.0018, +0.0024, \dots$ , at the noise floor throughout — one long inter-flip time randomises the parity immediately. **This is (C2), measured, with  $\rho \approx 0.3$ .**

**SESSION-12 CORRECTION.** The “ $\rho \approx 0.3$ ” is the transient. With  $2 \times 10^7$  samples from fixed low-block-count starts,  $a_k = (-1)^k(2.5 \pm 0.1)k^{-3}$  for  $7 \leq k \leq 12$ : the decay is polynomial, not geometric, and the stationary-start average above hides it because it averages the sign  $(-1)^{B(\bar{x})}$  over the start (§17.11.4). (C2) is replaced by the polynomial (C2') of §17.11.5, which suffices for a Hölder exponent  $(p-1)/(p+1) > 4/9$  at  $z = -1$  when  $p > 13/5$ . [SESSION-13 CORRECTION: the budget is  $\gamma > 1/11$ , so  $p > 6/5$  suffices; Corollary 17.36.]

Inter-flip parity  $\mathbb{E}[(-1)^{\tau_k - \tau_{k-1}}]$  settles at  $-0.170 \pm 0.003$  for  $k \geq 3$  from both starts: bounded away from  $\pm 1$ , as (C2) needs, and strikingly stable.

The post-flip chain equilibrates in about four flips. From the trap start,  $\mathbb{E}[\text{gap}]$  at  $\tau_k$  runs 0.1995, 0.2375, 0.2468, 0.2484, 0.2491, ... and  $\mathbb{E}[s]$  runs 0.715, 0.774, 0.790, 0.797, 0.798, matching the macroscopic start's 0.2498 and 0.799 by  $k = 4$ ; the correlations  $\mathbb{E}[\text{sgn}(g_0 - \frac{1}{4}) \text{sgn}(g_k - \frac{1}{4})]$  decay 0.317, 0.090, 0.022, 0.012, ..., geometrically with ratio  $\approx 0.3$ . This is direct evidence for the minorisation of Proposition 17.28 with  $n$  of order 4.

First-flip tails along the chain:  $t^2 \mathbb{P}[\tau_k - \tau_{k-1} > t]$  lies in 15–20 on  $16 \leq t \leq 128$  for  $k = 2, 4, 8$  from both starts (and  $\approx 10^4$  for  $k = 1$  from the trap start, consistent with the  $g^{-2}$  prefactor), so (R1) holds uniformly along the post-flip chain and not merely at the start: TODO 11d(iii), answered in the affirmative.

Finally, **s11.kflip** asserts  $B(\Gamma_t) - B(\Gamma_0) \equiv t \pmod{2}$  at every flip of every one of the  $1.6 \times 10^5$  trajectories — the hypothesis of Lemma 17.23(ii), via Lemma 17.13 — and it holds.

## 17.11 Session 12: the minorisation route is closed by inert dust; (C2) is polynomial, not geometric

Session 12 was directed at (C1)+(C2) of §17.10.4, through TODO 12a (a Lyapunov drift for  $\lambda$ ), 12b (the minorisation of Proposition 17.28) and 12c (an odd perturbation), with the instruction to return one of (A1)–(A3) or a clean negative (B). The outcome is (B), in a form sharper than the one the brief anticipated. Four findings, in logical order.

- (1) **The environment carries inert dust, and this closes every Doeblin-type route.** A curve of mass  $\varepsilon$  is touched by a chord with probability  $\leq 2\varepsilon$  per step, so it survives *unchanged* through the first  $n$  flips with probability  $\rightarrow 1$  as  $\varepsilon \rightarrow 0$  (Lemma 17.29). Consequently the post-flip laws from the states  $x_\varepsilon$  (a fixed macroscopic marked pair, plus one dust curve of mass  $\varepsilon$ ) are supported, up to  $o(1)$ , on the pairwise disjoint sets “some curve has mass exactly  $\varepsilon$ ”. No common minorant exists: the hypothesis of Proposition 17.28 fails on  $\Omega_K$  for *every*  $n$  (Proposition 17.30), and so does uniform ergodicity of  $\kappa_1$  in  $L^\infty(\Omega_K)$  and  $V$ -uniform ergodicity for *every* weight  $V$  (Corollary 17.31). Route (a) of §17.10.4 and TODO 12b are therefore closed, not deferred. The proof uses only  $\tau_n < \infty$  a.s.
- (2) **TODO 12a fails on the full state space.** From a state whose non-dust material has total mass  $m$  and whose remaining mass sits in  $N \gg m^{-3}$  equal pieces, one has  $\sum_j r_j^4 \leq 17m^4$  deterministically for a long time, so  $\kappa_1^k V \geq \frac{3}{4}(17m^4)^{-\theta}$  while  $V \leq m^{-4\theta}$ : no inequality  $\kappa_1 V \leq \varrho V + b$  with  $\varrho < 1$  can hold (Proposition 17.32). As instructed, the fallback is the killed chain on  $\Omega_K$ , where  $V$  is bounded anyway.
- (3) **(C2) is true but polynomial.** With  $2 \times 10^7$  samples from the one-curve start,  $a_k(x) = \mathbb{E}_x[(-1)^{\tau_k}] = (-1)^k(2.5 \pm 0.1)k^{-3}$  for  $7 \leq k \leq 13$ , and with  $10^8$  samples the local exponent over  $8 \leq k \leq 20$  is  $2.7 \pm 0.3$ ; a tail of the same form,  $k^3|a_k| \rightarrow 2.0\text{--}2.4$ , from the three-curve

start  $(0.3, 0.3; \{0.4\})$ , and the same tail *emerging* after a transient of  $\approx 8$  flips from a nine-curve start. Session 11's " $\rho \approx 0.3$ " was the transient above a noise floor of 0.003. Since  $a_k = \pm \mathbb{E}_x[(-1)^{B(\bar{\Gamma}_{\tau_k})}]$  is the parity bias of the block count at the  $k$ -th flip, the mechanism is the slow (log) growth of  $B$ : the block-count parity is a slow mode of  $\kappa_1$  at eigenvalue  $+1$ , not a mode near  $-1$  (§17.11.4). The polynomial decay is insensitive to inserting dust into the start (identical to two standard errors) and is *weaker* from starts with 9 or 17 curves.

- (4) **What the reduction actually needs is polynomial.** At  $z = -1$  Proposition 17.21 needs only  $G \in C^{0,\gamma}$ ,  $\gamma > 4/9$  [session-13:  $\gamma > 1/11$ , Corollary 17.36], and a polynomial decay of the twisted parity  $\mathbb{E}_x[w^{\tau_k}(-1)^{E_{\tau_k}}]$  with exponent  $p$ , uniformly for  $w$  near 1, gives  $\gamma = (p-1)/(p+1)$  by a crude bound (Proposition 17.33), so the budget needs  $p > 13/5$ ; if the parity memory factorises from the time weighting, as it does numerically, the exponent improves to  $\min(1, p-1)$  and the budget needs only  $p > 13/9$ . The measured  $p = 2.7 \pm 0.3$  meets the second comfortably and the first marginally. (C2) is accordingly replaced by (C2'') in §17.11.5. Any proof must be *dust-blind*, i.e. work in a topology in which  $x$  and  $x_\varepsilon$  are close; we record why the natural transport metric is not contracted by the dynamics and why an adversarial environment cannot be allowed (§17.11.5).

### 17.11.1 Inert dust

Throughout, the chain is realised on the circle: the state is a partition of the circle into curves (each a cyclically ordered union of arcs, the cyclic order being that of the chord diagram), a step picks two independent uniform points, and the move is determined by the curves the points lie in and their positions along those curves.

**Lemma 17.29** (inert dust coupling). *Let  $y$  be a state,  $D$  a sub-arc of the circle of mass  $\varepsilon$  lying inside one curve of  $y$  and containing neither mark, and let  $y_\varepsilon$  be the state obtained from  $y$  by declaring  $D$  a separate (unmarked) curve. Drive the chains from  $y$  and from  $y_\varepsilon$  by the same uniform points. Then, up to the first step at which a point falls in  $D$ , the state of the  $y_\varepsilon$ -chain is the state of the  $y$ -chain with  $D$  declared a separate curve; in particular the two chains have the same flip times, and the  $y_\varepsilon$ -chain contains a curve of mass exactly  $\varepsilon$ . The first point in  $D$  falls at a step of probability at most  $2\varepsilon$ , so for every  $T$ ,*

$$\mathbb{P}[\text{the coupling holds through time } T] \geq 1 - 2\varepsilon T.$$

*Proof.*  $D$  is a contiguous segment of the cyclic order of the curve  $C \supset D$  containing it. A split of  $C$  at two points outside  $D$  cuts the cycle into two arcs, one of which contains  $D$ ; in the  $y_\varepsilon$ -chain the same cuts split  $C \setminus D$  into the same two arcs with  $D$  excised. A merge of  $C$  with another curve at points outside  $D$  splices the two cycles at those points; in the  $y_\varepsilon$ -chain the same splice is applied to  $C \setminus D$ . Moves not involving  $C$  are identical. A flip is a split or merge decided by which marked material the two points lie in; the marked material of the two chains differs only by  $D$ , which contains no point. Induction on the steps.  $\square$

**Proposition 17.30** (no state-uniform minorisation). *Let  $K \geq 4$  and assume  $\tau_n < \infty$  a.s. from the state  $x_0 = (\frac{1}{4}, \frac{1}{4}; \{\frac{1}{2}\})$  (apart sector). Then there are no  $n \geq 1$ ,  $\delta > 0$  and probability measure  $\nu$  on  $\Omega_K$  with*

$$\mathbb{P}_{\bar{x}}^{\Omega_K} [\bar{\Gamma}_{\tau_n} \in \cdot] \geq \delta \nu(\cdot) \quad \text{for every } \bar{x} \in \Omega_K \cap S'_K,$$

*for the chain killed on  $\Omega_K$  or for the unkilled chain. A fortiori the hypothesis of Proposition 17.28 is never satisfied.*

*Proof.* Suppose such  $n, \delta, \nu$  exist. For  $i \geq 3$  let  $\varepsilon_i = 2^{-i}$ , let  $x_i$  be  $x_0$  with a sub-arc  $D_i$  of mass  $\varepsilon_i$  of its environment curve declared a separate curve (so  $x_i = (\frac{1}{4}, \frac{1}{4}; \{\frac{1}{2} - \varepsilon_i, \varepsilon_i\})$ ,  $B(x_i) = 4$ ,  $x_i \in \Omega_K \cap S'_K$ ), and let  $S_i = \{\bar{x} \in \Omega_K : \text{some curve of } \bar{x} \text{ has mass exactly } \varepsilon_i\}$ . Apply the minorisation at  $x_i$  to the set  $\Omega_K \setminus S_i$ . By Lemma 17.29 with  $T = T_i$ , on the event that the coupling holds through  $T_i$  and  $\tau_n \leq T_i$  for the unkilld  $x_0$ -chain, the  $x_i$ -chain performs its  $n$ -th flip at the same time and, if it is still alive, its state then contains  $D_i$  intact; if it has been killed, its state is the cemetery, which is not in  $\Omega_K \setminus S_i$ . Hence

$$\delta \nu(\Omega_K \setminus S_i) \leq 2\varepsilon_i T_i + \mathbb{P}_{x_0}[\tau_n > T_i] \quad \text{for every } T_i,$$

and  $T_i = \varepsilon_i^{-1/2}$  gives  $\nu(S_i) \rightarrow 1$  as  $i \rightarrow \infty$ . But a configuration in  $\Omega_K$  has at most  $K$  curves, hence at most  $K$  distinct masses, hence lies in at most  $K$  of the sets  $S_i$ ; so  $\sum_i \nu(S_i) = \int \#\{i : \bar{x} \in S_i\} \nu(d\bar{x}) \leq K < \infty$ , contradicting  $\nu(S_i) \rightarrow 1$ .  $\square$

*Status: PROVED. Note what the proof does not use: no moment of  $\tau$ , no property of the environment beyond the trivial one that a curve of mass  $\varepsilon$  is rarely hit, and no bound on  $K$  beyond  $K \geq 4$ . It also shows that the difficulty is not the one anticipated in TODO 12b (“ $K^{-2K} \rightarrow K^{-O(1)}$ ”): the obstruction is not the cost of driving the environment to a reference configuration but the fact that no finite sequence of steps can remove a curve that is never hit. The same applies with the  $(n, n+1)$  pair of Proposition 17.28, to any weighted space, and to any “small set” containing a state with a dust curve — which every set of positive  $\bar{\pi}$ -measure does, since dust is created at every scale at rate  $\asymp 2\eta d\eta$  per step.*

**Corollary 17.31** (no uniform or  $V$ -uniform ergodicity). *Let  $K \geq 4$  and let  $\kappa_1$  be the post-flip kernel of the unkilld chain.*

- (i) *There are no probability measure  $\bar{\pi}$ ,  $C < \infty$  and  $\rho < 1$  with  $\sup_{\bar{x} \in \Omega_K} |\kappa_1^k F(\bar{x}) - \bar{\pi} F| \leq C\rho^k \|F\|_\infty$  for all bounded  $F$  and all  $k$ .*
- (ii) *Let  $\mathcal{Y} = \{(a', \frac{1}{2} - a'; \{\frac{1}{2}\}) : a' \in [\frac{1}{8}, \frac{3}{8}]\}$  and assume  $(M_p)$ :  $\sup_{y \in \mathcal{Y}} \mathbb{E}_y[\tau_k] \leq Ck^p$  for some  $C, p < \infty$ . Then there is no  $V \geq 1$  for which  $\kappa_1$  is  $V$ -uniformly ergodic: no  $\bar{\pi}$ ,  $C'$ ,  $\rho < 1$  with  $|\kappa_1^k F(\bar{x}) - \bar{\pi} F| \leq C'\rho^k V(\bar{x}) \|F\|_V$  for all  $F$  with  $\|F\|_V = \sup |F|/V < \infty$  and all  $\bar{x}$  with  $B(\bar{x}) \leq 4$ .*

*Proof.* (i) With  $F_i = \mathbf{1}_{S_i}$  (notation of the previous proof) and  $x_i$  as there, Lemma 17.29 gives  $\kappa_1^k F_i(x_i) \geq 1 - 2\varepsilon_i T - \mathbb{P}_{x_0}[\tau_k > T]$ , while  $\kappa_1^k F_i(x_0) = 0$  because from  $x_0$  every curve mass at every time is either a sum of some of  $\frac{1}{4}, \frac{1}{4}, \frac{1}{2}$  or has an atomless law, so no curve ever has mass exactly  $\varepsilon_i \leq \frac{1}{8}$  a.s. Let  $i \rightarrow \infty$ : the oscillation of  $\kappa_1^k F_i$  over  $\Omega_K$  tends to 1 for each fixed  $k$ , contradicting  $2C\rho^k < 1$ .

(ii) Suppose such  $V, \bar{\pi}, C', \rho$  exist. First,  $\Pi = \mathbf{1} \otimes \bar{\pi}$  is bounded on the  $V$ -space, so  $\bar{\pi}V < \infty$  and  $\kappa_1 V \leq (C'\rho + \bar{\pi}V)V$ ; in particular  $\kappa_1 V(x_0) < \infty$ . Second, for  $y \in \mathcal{Y}$  and  $y_\varepsilon$  obtained by carving a sub-arc of mass  $\varepsilon$  from its environment curve, the argument of (i) at  $y$  gives  $\kappa_1^k F_i(y_\varepsilon) \geq 1 - 2\varepsilon T - \mathbb{P}_y[\tau_k > T]$  and  $\bar{\pi}F_i = \lim_k \kappa_1^k F_i(x_0) = 0$ , so with  $T = T_k := 4Ck^p$  and  $\varepsilon \leq \varepsilon_k := 1/(8T_k)$  (Markov and  $(M_p)$ ),

$$V(y_\varepsilon) \geq \frac{1 - \frac{1}{4} - \frac{1}{4}}{C'\rho^k} = \frac{1}{2C'\rho^k} \quad (y \in \mathcal{Y}, \varepsilon \leq \varepsilon_k).$$

Third, from  $x_0$  the first step carves a piece of mass in  $d\varepsilon$  from the environment curve with probability  $\geq \frac{1}{2} d\varepsilon$  (pick that curve twice,  $\frac{1}{4}$ ;  $|t_1 - t_2|/\frac{1}{2}$  has density  $\geq 2$  near 0) and the second

step is a flip landing in  $\mathcal{Y}$  with probability  $\geq \frac{1}{8} \cdot \frac{3}{4}$ ; the landing state is then some  $y_\varepsilon$ ,  $y \in \mathcal{Y}$ . So

$$\kappa_1 V(x_0) \geq \frac{3}{64} \sum_{k \geq 1} \frac{\varepsilon_k - \varepsilon_{k+1}}{2C' \rho^k} \geq c \sum_k \frac{k^{-p-1}}{\rho^k} = \infty,$$

contradicting the first step.  $\square$

*Status: PROVED ((ii) under  $(M_p)$ , which (R1) along the chain implies with  $p = 1$  and which is measured:  $\mathbb{E}[\tau_k] \approx 6k$ ). For the chain killed on  $\Omega_K$  the same argument runs up to  $k \leq c3^{K/(3p)}$ , so any  $V$ -uniform ergodicity of the killed chain has  $(1 - \rho)^{-1} \log(C'V(x_0)\bar{\pi}V) \geq c3^{K/(3p)}$ : exponential in  $K$ , i.e. fatal in the budget of the session-11 handoff. This settles, negatively, the sentence of §17.9.5 that asked for a weighted space “in which the post-flip chain is  $V$ -geometrically ergodic in the base coordinates”: there is none. It does not touch (C1) or (C2) themselves, whose null vectors would have to sit near  $-1$ , whereas the dust modes sit near  $+1$ .*

### 17.11.2 The drift for $\lambda$ fails on the full state space (TODO 12a)

**Proposition 17.32** (no Lyapunov drift for  $(\sum r_j^4)^{-\theta}$ ). *Let  $V = (\sum_j r_j^4)^{-\theta}$  with  $0 < \theta \leq \frac{1}{2}$ , on the full (unkilled) state space, and assume  $\tau_k < \infty$  a.s. from every state. There are no  $\varrho < 1$  and  $b < \infty$  with  $\kappa_1 V \leq \varrho V + b$ .*

*Proof.* Suppose the inequality holds. Iterating,  $\kappa_1^k V \leq \varrho^k V + b/(1 - \varrho)$  for all  $k$ . Fix  $k$  with  $\varrho^k \leq \frac{3}{8} 17^{-\theta}$  and  $m \in (0, \frac{1}{4})$ .

*The isolated process.* Let  $\bar{x}$  be the apart-sector state with  $a = m$ ,  $b = m/10$  and one environment curve of mass  $1 - 1.1m$ , and let  $P^{\text{iso}}$  be the law of the chain from  $\bar{x}$  modified so that a chord with one point in the original environment curve (or in any curve descended from it by splits and merges among such curves) and one point elsewhere does nothing. Under  $P^{\text{iso}}$  the “non-dust” material of total mass  $1.1m$  evolves as a split–merge chain in its own right, slowed by the factor  $(1.1m)^2$ , with the marked pair inside it; its flip times are a.s. finite, so there is  $T = T(m, k)$  with  $P^{\text{iso}}[\tau_k > T] \leq \frac{1}{8}$ .

*The states  $x_{m,N}$ .* Let  $x_{m,N}$  be  $\bar{x}$  with the environment curve cut into  $N$  equal pieces,  $N \geq 16(T+1)^3 \max(m^{-1}, m^{-4/3})$ . Drive the chain from  $x_{m,N}$  and the isolated process from  $\bar{x}$  by the same points (the dust pieces of  $x_{m,N}$  are sub-arcs of the environment curve of  $\bar{x}$ ; dust–dust chords split or merge dust material in both and never touch the non-dust material). The two processes make the same non-dust moves except at chords joining dust to non-dust, of which there are at most  $T$  up to time  $T$ ; each such chord appends to one non-dust curve of the  $x_{m,N}$ -chain a dust curve, i.e. a union of at most  $T+1$  original pieces or fragments of them, of mass  $\leq (T+1)/N$ . As in Lemma 17.29 (with the appended material in the role of  $D$ , now inside a curve of one chain and absent from the other), the non-dust configurations of the two processes then agree up to the appended material, and their flip times agree, until a point falls in appended material, which has probability  $\leq 2T(T+1)/N$  per step. Hence  $\mathbb{P}_{x_{m,N}}[\tau_k > T] \leq \frac{1}{8} + 2T^2(T+1)/N \leq \frac{1}{4}$ .

*The deterministic bound.* Along the  $x_{m,N}$ -chain up to time  $T$ , the non-dust mass is at most  $1.1m + T(T+1)/N \leq 2m$  and every dust curve has mass  $\leq (T+1)/N \leq m^{4/3}$ , so

$$\sum_j r_j(\bar{\Gamma}_t)^4 \leq (2m)^4 + \left( \max_{\text{dust}} r_j \right)^3 \sum_{\text{dust}} r_j \leq 16m^4 + m^4 = 17m^4 \quad (t \leq T),$$

whence  $\kappa_1^k V(x_{m,N}) \geq \mathbb{P}_{x_{m,N}}[\tau_k \leq T] (17m^4)^{-\theta} \geq \frac{3}{4} 17^{-\theta} m^{-4\theta}$ , while  $V(x_{m,N}) \leq m^{-4\theta}$  because the  $u$ -curve alone contributes  $m^4$ . The drift bound gives  $\frac{3}{4} 17^{-\theta} m^{-4\theta} \leq \frac{3}{8} 17^{-\theta} m^{-4\theta} + b/(1-\varrho)$ , i.e.  $m^{-4\theta} \leq \frac{8}{3} 17^\theta b/(1-\varrho)$  for every  $m \in (0, \frac{1}{4})$ : false.  $\square$

*Status: PROVED. This is the fallback the brief provided for: the obstruction is the family “all non-dust mass  $\asymp m$ , the rest in ultra-fine mass-carrying dust”, which is of vanishing probability under the chain from one curve (Corollary 17.16) but is a legitimate state, and on it  $\lambda$  cannot recover within one inter-flip period because there is no macroscopic partner to merge with. On  $\Omega_K$ ,  $V \leq (2K)^{4\theta}$  is bounded and the drift is vacuous. The killing at  $B = K$  in Theorem 17.27 therefore stays, and it is harmless: it costs  $(9d+1)^{-\gamma}$  over the horizon. The second purpose of TODO 12a — supplying the base half of the minorisation — is void by Proposition 17.30. A quick numerical check of the mechanism (*s12\_driftfail*): from  $(0.3, 0.03; 3 \times 10^4 \text{ equal pieces})$ ,  $\mathbb{E}[(\sum r^4(\bar{\Gamma}_\tau)/\sum r^4(x))^{-\theta}] = 1.19$  ( $\theta = 0.1$ ),  $1.71$  ( $\theta = 0.3$ ), against  $0.91, 0.77$  with a single environment curve of the same mass.*

### 17.11.3 TODO 12c: the stated criterion does not give (C2)

The session-11 handoff proposed, as route 12c, a state-uniform lower bound  $c(K)$  on the probability that two consecutive inter-flip times have opposite parity, “which gives (C2) by a standard two-block argument”. It does not: if the inter-flip times were the deterministic sequence  $1, 2, 1, 2, \dots$  then consecutive parities are always opposite ( $c = 1$ ) while  $(-1)^{\tau_k}$  is periodic with period 4 and does not decay. What the two-block argument needs is the *operator* statement  $\|\kappa_{-1}^2\|_\infty < 1$ , and that is false exactly:  $\kappa_{-1}(-1)^E = -(-1)^E$  (Proposition 17.24), so  $\|\kappa_{-1}^m\|_\infty = 1$  for every  $m$ . Any odd perturbation that is exact returns a state with a different block-count parity, which is visible; the only odd perturbations that are invisible are dust insertions, whose weight at scale  $\eta$  is  $\asymp 2\eta$  per step and whose lifetime is  $\asymp 1/(2\eta)$ , and one checks in one line that they give a contraction  $1 - O(\eta)$  per flip and nothing geometric. Route 12c is closed as stated; the honest form of (C2) is in §17.11.4 and §17.11.5.

### 17.11.4 The decay of the parity along the post-flip chain is polynomial

By Lemma 17.13,  $(-1)^{\tau_k} = (-1)^{B(\bar{\Gamma}_{\tau_k}) - B(\bar{x})}$  exactly, so

$$a_k(\bar{x}) := \mathbb{E}_{\bar{x}}[(-1)^{\tau_k}] = (-1)^{B(\bar{x})} \sum_{b \geq 1} (-1)^b \mathbb{P}_{\bar{x}}[B(\bar{\Gamma}_{\tau_k}) = b], \quad (22)$$

the parity bias of the block count at the  $k$ -th flip, and  $\kappa_1^k(-1)^E = (-1)^E(-1)^k a_k$ , so (C2) is the statement that (22) decays geometrically, uniformly in  $\bar{x}$ .

*Measurement (s12\_akdecay, s12\_akB; fixed deterministic starts,  $2 \times 10^7$  independent trajectories each, standard error  $2.2 \times 10^{-4}$ , no censoring).* From the one-curve start (arcs  $(\frac{1}{2}, \frac{1}{2})$ ):

$$a_k = -0.4003, +0.1530, -0.0674, +0.0335, -0.0186, +0.0111, -0.0073, +0.0048, -0.0034, +0.0022, -0.0019,$$

so that  $k^3|a_k| = 2.51, 2.48, 2.49, 2.20, 2.54, 2.54$  for  $k = 7, \dots, 12$ :

$$a_k(\text{one curve}) = (-1)^k (2.5 \pm 0.1) k^{-3} \quad (7 \leq k \leq 12),$$

with the successive ratios  $|a_{k+1}/a_k|$  rising through  $0.38, 0.44, 0.50, 0.56, 0.60, 0.66, 0.66, 0.70$  — no geometric rate. A  $10^8$ -sample run to  $k = 20$  (standard error  $10^{-4}$ ) gives  $|a_k| = 0.00495, 0.00349, 0.00271, 0.00195, 0$

for  $k = 8, \dots, 20$ , all with sign  $(-1)^k$ ;  $k^3|a_k|$  stays in 2.4–2.7 up to  $k = 13$  and fluctuates in 2.8–3.9 (two to three standard errors) for  $k = 14$ –20; the least-squares exponent over  $8 \leq k \leq 20$  is  $2.7 \pm 0.3$ . From the three-curve apart start  $(0.3, 0.3; \{0.4\})$ :  $a_1 = -0.004$ , then  $-0.0606, +0.0385, -0.0219, +0.0132, -0.0086, +0.0060, -0.0041, +0.0031, -0.0020, +0.0018, -0.0014$  ( $k = 2, \dots, 12$ ), with  $k^3|a_k| \rightarrow 2.0$ –2.4: the same tail, sign  $(-1)^{k+B_0+1}$ . Inserting a dust curve of mass  $10^{-4}$  into either start (mass taken from an environment curve or from an arc) reproduces every  $a_k$  to within two standard errors, as Lemma 17.29 predicts. From a 9-curve start  $(0.3, 0.3; 7 \times 0.057)$ ,  $10^8$  samples:  $a_k$  passes through zero at  $k = 3$  and then *grows* in absolute value to 0.00145 at  $k = 7$ –8 before decaying, with the sign  $(-1)^{k+1}$  at every  $k = 4, \dots, 20$  (seventeen consecutive alternations);  $k^3|a_k|$  rises from 0.5 at  $k = 7$  to 2–3 at  $k = 15$ –20. So the tail is universal —  $Ck^{-3}$  with  $C \approx 2.5$ –3 — and is reached after a start-dependent transient, of about eight flips when the environment starts with nine curves. From a 17-curve start ( $5 \times 10^6$  samples)  $|a_k| \leq 7 \times 10^{-4}$  for  $3 \leq k \leq 16$ , the transient not yet over. From the burnt-in “stationary” start of session 11 (gap window  $[\frac{1}{8}, \frac{3}{8})$ ,  $5 \times 10^6$  samples),  $a_k(\mu)$  is at the noise floor  $4.5 \times 10^{-4}$  for  $k \geq 5$  — but that average carries the factor  $(-1)^{B(\bar{x})}$  of (22) inside the mixture, which cancels the tail; the sign-blind  $\mathbb{E}_\mu[a_k(\bar{x})^2]$  is at  $(4 \pm 5) \times 10^{-4}$  for  $k = 5..8$ , consistent with the fixed-start values. The decomposition of (22) by  $b$  from the one-curve start (s12\_akB) shows the mechanism:  $\mathbb{P}[B_{\tau_k} = 1]$  decays like  $c/k$  with  $c \approx 0.5$  (the chain merges back to one curve with probability  $\asymp 1/t$ , and there  $\tau_k$  is even), the law of  $B_{\tau_k}$  spreads only like  $\log k$ , and the alternating sum of a lattice law that spreads logarithmically decays polynomially.

*Status: MEASURED. Conclusions.* (i) (C2) in the geometric form of §17.10.4 is false on the full state space, and on  $\Omega_K$  it holds only with  $1 - \rho(K) \leq e^{-cK}$  (the rate the killing provides), which is outside the budget; the session-11 value  $\rho \approx 0.3$  described the transient  $k \leq 4$ . (ii) The polynomial decay is a +1 phenomenon, not a −1 phenomenon: the generating function  $\sum_k a_k w^k$  is singular at  $w = -1$ , i.e. at the eigenvalue  $-1$  of  $\kappa_{-1}$ , which is the eigenvalue +1 of  $\kappa_1$  (the trivial one), approached sub-geometrically by the slow variable  $B$ . Nothing in the measurement bears on (C1). (iii) The decay is sign-alternating in  $k$  from every start tested, with the sign  $(-1)^{k+B_0+1}$ ; this is the statement that  $B_{\tau_k} \equiv k+1$  is slightly favoured, i.e. that the block count at the  $k$ -th flip remembers that it started odd — a memory that log-growth of  $B$  erases only polynomially.

### 17.11.5 What is left, restated

**Proposition 17.33** (polynomial parity decay suffices at  $z = -1$ ). *Fix  $K$  and  $S'_K$ . Suppose*

(C2'') *there are  $C, p$  and  $\delta_0 > 0$  with  $\sup_{\bar{x} \in S'_K} |\mathbb{E}_{\bar{x}}[w^{\tau_k} (-1)^{E_{\tau_k}}]| \leq Ck^{-p}$  for all  $k \geq 1$  and all  $|w| \leq 1$  with  $|1 - w| \leq \delta_0$ ;*

(M1)  $\sup_{\bar{x} \in S'_K} \mathbb{E}_{\bar{x}}[\tau_k] \leq C_1 k$  *for all  $k$ .*

*Then  $G(z) = (I + \kappa_z)^{-1}(I - \kappa_z)\mathbf{1}$ , evaluated on  $S'_K$ , is Hölder continuous of exponent  $(p-1)/(p+1)$  on  $\{|z| \leq 1, |1+z| \leq \delta_0\}$ , with constant  $\leq C_2(C+C_1)$ . In particular  $p > 13/5$  gives an exponent  $> 4/9$  [session-13: the budget is 1/11, met for  $p > 6/5$ ; Corollary 17.36].*

*Proof.* By Proposition 17.25, for  $z = -w$ ,  $G(-w) = \bar{M}(I - \kappa_w)^{-1}(I + \kappa_w)\bar{M}\mathbf{1} = \mathbf{1} + 2\bar{M} \sum_{k \geq 1} g_k(w)$  with  $g_k(w)(\bar{x}) = \mathbb{E}_{\bar{x}}[w^{\tau_k} (-1)^{E_{\tau_k}}]$  (Neumann series,  $\|\kappa_w\| \leq |w| < 1$ ; the boundary values are limits). For  $|w|, |w'| \leq 1$ ,  $|w^n - w'^n| \leq n|w - w'|$ , so  $|g_k(w) - g_k(w')| \leq \min(2Ck^{-p}, C_1 k|w - w'|)$  by (C2'') and (M1). Summing with the cut at  $k_* = |w - w'|^{-1/(p+1)}$ :  $\sum_{k \leq k_*} C_1 k|w - w'| + \sum_{k > k_*} 2Ck^{-p} \leq C_2(C+C_1)|w - w'|^{(p-1)/(p+1)}$ .  $\square$

*Status: PROVED as a reduction; the hypotheses are OPEN. Two comments. (a) The exponent  $(p-1)/(p+1)$  is crude: it bounds  $|g_k(w) - g_k(w')|$  by the trivial Lipschitz bound and never uses that  $g_k(1) = (-1)^E(-1)^k a_k$  has one sign for all  $k$ . If the parity memory factorises from the time weighting,*

$$(C2'') \quad \mathbb{E}_{\bar{x}}[w^{\tau_k}(-1)^{\tau_k}] = \mathbb{E}_{\bar{x}}[w^{\tau_k}] a_k(\bar{x}) (1 + O(|1-w|k)),$$

*then  $\sum_k g_k(w) \approx (-1)^E \sum_k Ck^{-p} \mathbb{E}[w^{\tau_k}]$ , and with  $\mathbb{E}[w^{\tau_k}] \approx w^{6k}$  this is  $C \text{Li}_p(w^6)$  up to smoother terms, whose modulus of continuity at  $w = 1$  is  $|1-w|^{\min(1,p-1)}$  (up to a logarithm at integer  $p$ ): the budget then needs only  $p > 13/9$ .  $(C2'')$  was measured (`s12_twist`, one-curve start,  $3 \times 10^7$  samples,  $w = 0.99, 0.95, 0.90$  real): the exact factorisation fails — the twisted parity  $\mathbb{E}[w^{\tau_k}(-1)^{\tau_k}]$  exceeds  $\mathbb{E}[w^{\tau_k}] a_k$  by a factor 1.2 at  $|1-w|k = 0.08$  and 2–4 at  $|1-w|k \approx 0.4$ –0.8 (short paths carry more parity memory) — but the uniform bound  $(C2'')$  holds with the same constant,  $|\mathbb{E}[w^{\tau_k}(-1)^{\tau_k}]| \leq |a_k|$  at every  $k \leq 16$  and every  $w$  tested. More usefully, the sum itself was measured: with  $\Sigma(w) := \sum_k (-1)^k E_x[w^{\tau_k}(-1)^{\tau_k}]$  (all terms of one sign),  $\Sigma(1) - \Sigma(w) = 0.026, 0.085, 0.143$  at  $|1-w| = 0.01, 0.05, 0.10$  (tail beyond  $k = 16$  estimated from  $2.5k^{-3}$ ), i.e. a modulus of continuity  $\asymp |1-w|^{0.75}$  over that range — above the crude  $(p-1)/(p+1) \approx 0.5$ , below the  $\min(1, p-1)$  of the factorised heuristic, and far above the  $4/9$  [session-13: 1/11] the budget needs. The same on the unit circle near  $w = 1$  is the first numerical task of session 13. With the measured  $p = 2.7 \pm 0.3$  the crude route is marginal ( $p > 13/5$  required); the measured modulus is not. (b)  $(C2')$  at  $w = 1$  is exactly the polynomial  $(C2')$ :  $\sup_{S'_K} |a_k| \leq Ck^{-p}$ . Away from  $w = 1$  on the circle Theorem 17.27 gives a geometric rate;  $(C2')$  asks that the polynomial rate survive a small complex twist near  $w = 1$ , which is a damping of long paths and should not spoil a parity cancellation that is produced by the block count. This is measurable and is the first numerical task for session 13.*

The open conditions are now:

**(C1)**  $\|(I + \kappa_1)^{-1}\|_{L^\infty(\Omega_K)} \leq C(K)$  — equivalently  $\|(I - \kappa_{-1})^{-1}\| \leq C(K)$ , the summability of the signed renewal series  $\sum_k \kappa_{-1}^k G$  for every bounded  $G$ ;  $(C2')$  is the case  $G = \mathbf{1}$ . Unchanged, not contradicted, and no longer reachable by any Doeblin/ergodicity argument (Proposition 17.30, Corollary 17.31).

**(C2'')** as in Proposition 17.33, with  $p > 13/5$  and  $C = C(K)$  polynomial; measured  $p \approx 3$  at  $w = 1$ .

*What a proof must look like. (i) It must be dust-blind: by Lemma 17.29,  $a_k(y_\varepsilon) = a_k(y) + O(\varepsilon \mathbb{E}_y \tau_k)$  while  $(-1)^E$  changes sign, so any argument that treats  $(-1)^E$  as a function to be contracted in a sup-type norm on configurations is doomed, and any argument that works must use a topology in which  $y_\varepsilon \rightarrow y$ . (ii) The natural candidate — the transport distance between mass configurations, under the coupling of Lemma 17.29 — is not contracted: when the dust is finally hit, the two coupled configurations differ by a macroscopic merge (“ $r + r_0$ ” against “ $r + \varepsilon, r_0 - \varepsilon$ ”), so a discrepancy of size  $\varepsilon$  is amplified to size  $O(1)$  rather than resolved. A contracting metric would have to compare coarse laws rather than configurations. (iii) The environment’s own randomness is needed: an adversary allowed to choose the masses gained by the marked pair at the merge steps (whose timing it cannot control) can bias the parity of each inter-flip time toward a target by a fixed amount, since a gain changes the flip hazard  $2p_1p_2$  by a bounded amount at a time of its choosing among the merge steps; so no “uniformly over environments” statement can be true, and the proof must use that the true gains are size-biased picks from a partition whose count parity is invisible to them. (iv) Given (22), the cleanest target is a statement about the*

block count alone: the parity of  $B$  at the  $k$ -th flip equidistributes polynomially. Since the total block count from one curve has the exact law of Theorem 17.15 at fixed times, and the flip times are a renewal-like sampling of the time axis, this may be provable from the exact formulas by an averaging argument, at least for the annealed environment — which is the form the downstream reduction (Problem 17.9, whose start is  $\mu_j$ ) actually consumes.

## 17.12 Session 13: audit of the reduction chain

Session 13 opened with the audit recommended in the session-11 health note: nothing in the chain Theorem 11.4  $\rightarrow$  Problem 15.8  $\rightarrow$  Theorem 16.1  $\rightarrow$  Problem 16.4  $\rightarrow$  Problem 17.9  $\rightarrow$  (R1)+(C1)+(C2'') had been re-read since it was laid, and two carried conditions had just been found false in consecutive sessions. The audit produced three corrections to the *form* of the bottom of the chain, none of which changes what is proved above it:

- (1) the quantity the programme consumes is an  $L^2$  average of the quenched parity  $\Phi_m$  over the start, not a supremum over states (Proposition 17.34); the state-uniform target (S) of §17.11.5 was introduced by the strong-Markov step of the session-10 reduction, not demanded by anything above it, and the inert-dust obstruction of §17.11.1 does not touch the  $L^2$  form (Remark 17.35);
- (2) the exponent budget carried since session 10 ( $\gamma > 4/9$  at  $z = -1$ , hence  $p > 13/5$  in Proposition 17.33) is too demanding by a factor of about five: the correct budget is  $\gamma > 1/11$ , hence  $p > 6/5$  (Corollary 17.36). The measured  $p \approx 2.7$  and modulus 0.75 are far inside it, not marginal;
- (3)  $\tau_k < \infty$  a.s. from every state with  $ab > 0$ , taken for granted since session 10, has a four-line proof (Lemma 17.37).

The occupation-time step and (M<sub>1</sub>) are stated exactly and left open (§17.12.4, §17.12.5).

### 17.12.1 The reduction consumes an $L^2$ average, not a supremum

Recall (Corollary 17.3) that  $\psi(j, d) = \mathbb{E}_{\mu_j}[\Phi_d(\Gamma_0)^2]$  with  $\Phi_m(\Gamma) = \mathbb{E}_\Gamma[(-1)^{N_m}]$  the parity of the number of flips in  $m$  steps of the quotient chain, and that Theorem 16.1 needs  $\psi(j, d) \leq Kd^{-2c}$  for  $j \leq 8d$  only. Fix  $K \geq 3$  and  $S'_K = \{a, b \geq 1/(16K)\}$  as in §17.9.5.

**Proposition 17.34** ( $L^2$  form of the reduction to  $S'_K$ ). *Let  $d \geq 1$ ,  $j \leq 8d$ ,  $0 < \varepsilon \leq 1/(2K)$ , and let  $T^* = T^*(\varepsilon, K, d) < d/2$  be any deterministic time. Let  $T' = \inf\{t \geq 0 : \Gamma_t \in S'_K\}$  and let  $\nu = \nu_{j, \varepsilon, K, T^*}$  be the joint law of  $(\Gamma_{T'}, T')$  under  $\mathbb{P}_{\mu_j}(\cdot; g_0 \geq \varepsilon, T' \leq T^*)$ , a sub-probability measure on  $S'_K \times \{0, \dots, T^*\}$ ; write  $\nu_t = \nu(\cdot \times \{t\})$  and, when  $\nu_t \neq 0$ , let  $\bar{\nu}_t = \nu_t/\nu_t(S'_K)$  be the time- $t$  entrance law, a probability measure on  $S'_K$ . Then*

$$\mathbb{E}_{\mu_j}[\Phi_d(\Gamma_0)^2; g_0 \geq \varepsilon] \leq 2 \int \Phi_{d-t}(x)^2 \nu(dx, dt) + 2 \mathbb{P}_{\mu_j}[T' > T^*; g_0 \geq \varepsilon] \leq 2 \sup_{t \leq T^*} \mathbb{E}_{\bar{\nu}_t}[\Phi_{d-t}^2] + 2 \mathbb{P}_{\mu_j}[T' > T^*];$$

*Proof.*  $T'$  is a stopping time and  $T' \wedge T^* \leq T^* < d$ . By the strong Markov property at  $T' \wedge T^*$  and the additivity of the flip count,  $\mathbb{E}[(-1)^{N_d - N_{T'}} \mid \mathcal{F}_{T'}] = \Phi_{d-T'}(\Gamma_{T'})$  on  $\{T' \leq T^*\}$ , so

$$\Phi_d(\Gamma_0) = \mathbb{E}_{\Gamma_0}[(-1)^{N_{T'}} \Phi_{d-T'}(\Gamma_{T'}); T' \leq T^*] + \mathbb{E}_{\Gamma_0}[(-1)^{N_d}; T' > T^*] =: A(\Gamma_0) + B(\Gamma_0).$$

Then  $|B| \leq \mathbb{P}_{\Gamma_0}[T' > T^*] \leq 1$ , so  $B^2 \leq |B|$ , and by Cauchy–Schwarz  $A^2 \leq \mathbb{E}_{\Gamma_0}[\Phi_{d-T'}(\Gamma_{T'})^2; T' \leq T^*] \cdot \mathbb{P}_{\Gamma_0}[T' \leq T^*] \leq \mathbb{E}_{\Gamma_0}[\Phi_{d-T'}(\Gamma_{T'})^2; T' \leq T^*]$ . Use  $(A + B)^2 \leq 2A^2 + 2B^2$  and integrate over  $\Gamma_0 \sim \mu_j$  on  $\{g_0 \geq \varepsilon\}$ ; the first integral is, by definition of  $\nu$ ,  $\int \Phi_{d-t}(x)^2 \nu(dx, dt) = \sum_{t \leq T^*} \nu_t(S'_K) \mathbb{E}_{\bar{\nu}_t}[\Phi_{d-t}^2]$ , and  $\sum_t \nu_t(S'_K) \leq 1$ . (The supremum over  $t$  must stay inside the sum in this way: a bound in terms of the time-marginal  $\sum_t \nu_t$  alone would cost a factor  $T^*$ .)  $\square$

*Status: PROVED. Compare the session-10 reduction paragraph at the end of §17.9.5, which at the same point bounds  $|\Phi_d(\Gamma_0)| \leq \mathbb{E}|\Phi_{d-T'}(\Gamma_{T'})| + \mathbb{P}[T' > T^*]$  and then takes the supremum of  $|\Phi_m|$  over  $S'_K \cap H$ . The supremum is where the state-uniform target (S) entered the programme; nothing above this point asks for it. What the programme needs on  $S'_K$  is therefore:*

(S<sub>2</sub>)  $\mathbb{E}_{\bar{\nu}_t}[\Phi_m^2] \leq C(K) m^{-2\gamma}$  for every  $t \leq T^*$  and  $m = d - t \in [d/2, d]$ , for the time- $t$  entrance laws  $\bar{\nu}_t = \bar{\nu}_{j,\varepsilon,K,T^*;t}$  of  $S'_K$ ,  $j \leq 8d$ , with  $K = K_{\gamma'}(9d) = O(\log d)$  and  $C(K)$  polynomial in  $K$ ,

together with the entrance estimate  $\mathbb{P}_{\mu_j}[T' > T^*; g_0 \geq \varepsilon] \leq O(d^{-\gamma'}) + O(\varepsilon K \log d)$  for  $T^* = O(\varepsilon^{-1} K \log d)$ , which is the content of the session-10 reduction paragraph (Lemma 17.18 plus the occupation-time step of §17.12.4). The block-bound event  $H_{9d}$  of Corollary 17.16 is also handled in  $L^2$  without any state-by-state argument: with  $H = \{B_r < K, r \leq m\}$  for the chain from  $x$ , writing  $\Phi_m = \Phi_m^H + \Phi_m^{H^c}$  with  $|\Phi_m^{H^c}(x)| \leq \mathbb{P}_x[H^c]$ , one has  $\mathbb{E}_{\bar{\nu}_t}[\Phi_m^2] \leq 2\mathbb{E}_{\bar{\nu}_t}[(\Phi_m^H)^2] + 2\mathbb{E}_{\bar{\nu}_t}[\mathbb{P}_x[H^c]]$ , and  $\sum_t \nu_t(S'_K) \mathbb{E}_{\bar{\nu}_t}[\mathbb{P}_x[H^c]] \leq \mathbb{P}_{\mu_j}[H_{9d}^c] \leq (9d + 1)^{-\gamma'}$  because  $\nu$  is a sub-law of the path measure from  $\mu_j$ , whose clock reaches at most  $j + d \leq 9d$ .

**Remark 17.35** (what the dust obstruction does and does not touch). (a) Proposition 17.30 and Corollary 17.31 are statements about the supremum norm on  $\Omega_K$ : they show that no Doeblin-type argument can give a bound on  $\sup_{x \in S'_K} |\Phi_m(x)|$  (or on  $\|\kappa_1^k F - \bar{\pi} F\|_\infty$  for any weight). They say nothing against (S<sub>2</sub>), which is an average over laws  $\bar{\nu}_t$  that are themselves marginals of the chain run from one block. (b) Under the coupling of Lemma 17.29,  $\Phi_m$  is Lipschitz in the dust:  $|\Phi_m(y_\varepsilon) - \Phi_m(y)| \leq 4\varepsilon m$ , since the flip counts agree on the coupling event. So  $\Phi_m$  is a continuous function in the coarse topology at scale  $1/m$  — the topology in which  $y_\varepsilon \rightarrow y$  — and the dust of Proposition 17.30, which lives at scales  $\varepsilon_i \ll 1/T_i$ , is invisible to the functions that (S<sub>2</sub>) is about. This is the precise sense in which §17.11.5(i) asked for a “dust-blind” statement, and (S<sub>2</sub>) is one. (c) (S<sub>2</sub>) is *not* an annealed statement in the naive sense: it bounds the second moment of the quenched parity, so a proof must still show that  $\Phi_m(x)$  is small for  $\bar{\nu}_t$ -typical  $x$ , with the environment of  $x$  frozen; only the *uniformity* over  $x$  has been removed. The quantity  $\mathbb{E}_{\bar{\nu}_t}[\Phi_m]$  (parity averaged over the environment of the start) is not what is needed and is not sufficient. (d) The natural function space for (S<sub>2</sub>) is  $L^2(\bar{\pi})$ , in which the chain is self-adjoint (§17.5); the spectral route of session 8, shelved in favour of the Markov-renewal route, is the dust-blind route. What it additionally needs is a comparison of the entrance laws  $\bar{\nu}_t$  with  $\bar{\pi}$  on the functions  $\Phi_m^2$ ; this is the first sub-problem of §17.12.6.

## 17.12.2 The exponent budget, recomputed

**Corollary 17.36** (the budget is  $\gamma > 1/11$ , not  $\gamma > 4/9$ ). *Suppose (S<sub>2</sub>) holds with exponent  $\gamma \in (0, \frac{1}{2})$  and the entrance estimate holds as stated after Proposition 17.34. Then  $\psi(j, d) \leq C'(\log d)^{O(1)} d^{-2\gamma}$  for all  $j \leq 8d$ , i.e. Problem 16.4 holds with  $c = \gamma$ , and Corollary 16.2 closes the iid model as soon as  $\gamma > 1/11$ . The same holds if (S<sub>2</sub>) is replaced by the state-uniform (S) with exponent  $\gamma$ . In particular, in Proposition 17.33 the crude route needs only  $p > 6/5$ , and (R1) needs only  $C^{1,\gamma}$  regularity with  $\gamma > 1/11$ , i.e. moments  $\mathbb{E}_x[\tau^{1+\gamma}]$  with  $\gamma > 1/11$ .*

*Proof.* Set  $\varepsilon := d^{-1/2-\gamma}$  and  $T^* := C\varepsilon^{-1}K \log d$  with  $K = K_{\gamma'}(9d)$ ,  $\gamma' \geq 2\gamma$ ; then  $T^* = O(d^{1/2+\gamma} \log^2 d) < d/2$  for  $d$  large because  $\gamma < \frac{1}{2}$ , and  $\varepsilon \leq 1/(2K)$ . By Proposition 17.10 ( $j \leq 8d$ ),  $\mathbb{P}_{\mu_j}[g_0 < \varepsilon] \leq 4\varepsilon^2 j \leq 32\varepsilon^2 d = 32d^{-2\gamma}$ . By Proposition 17.34 and (S<sub>2</sub>) (with  $m = d - t \geq d/2$ ),  $\mathbb{E}_{\mu_j}[\Phi_d^2; g_0 \geq \varepsilon] \leq 2C(K)(d/2)^{-2\gamma} + 2(O(d^{-\gamma'}) + O(\varepsilon K \log d))$ , and  $\varepsilon K \log d = O(d^{-1/2-\gamma} \log^2 d) = o(d^{-2\gamma})$  since  $\gamma < \frac{1}{2}$ . Add, and use  $|\Phi_d| \leq 1$  on  $\{g_0 < \varepsilon\}$ . The budget: Corollary 16.2 needs  $c > 1/11$ . For Proposition 17.33:  $(p-1)/(p+1) > 1/11 \iff p > 6/5$ .  $\square$

*Status: PROVED (arithmetic). Where the factor was lost. The session-10 reduction paragraph ends by bounding  $|\Phi_d(\Gamma_0)|$ , not  $\Phi_d(\Gamma_0)^2$ , by  $C(\varepsilon d)^{-2\alpha}$  — the form of Problem 17.9, which is a bound on the second moment — and then replaces  $(\varepsilon d)^{-2\alpha}$  by  $d^{-2\alpha}$ . The first step costs a factor 2 in the exponent, the second a further factor  $2(1+\alpha) = 22/9$ ; together they turn the true requirement  $\gamma > 1/11$  into  $\gamma > 2\alpha > 4/9$ . Problem 17.9 and Corollary 17.11 are correct as stated and, with the correct accounting, lossless: from  $\Phi_d^2 \leq C(K)d^{-2\gamma}$  after the entrance one gets Problem 17.9 with  $(\varepsilon d)^{-2\alpha} = d^{-\alpha/(1+\alpha)}$  at the optimising  $\varepsilon$  for  $\alpha = 2\gamma/(1-2\gamma)$ , and then  $c' = \alpha/(2(1+\alpha)) = \gamma$  exactly. The loss was entirely in the two steps just named. Every downstream statement that quotes “ $\gamma > 4/9$ ” (§17.9.5 (R1) discussion, §17.10.4, Proposition 17.33 and §17.11.5, the handoff documents of sessions 10–12) should be read with  $1/11$  in its place. Consequences: the “marginal” verdict on the crude route of Proposition 17.33 ( $p > 13/5$  against a measured  $2.7 \pm 0.3$ ) is withdrawn — the crude route needs  $p > 1.2$ ; (R1) needs barely more than a first moment of  $\tau$  along the chain (the measured tail is  $t^{-2}$ ); and Proposition 17.6’s sharp rate  $c = 1/2$  exceeds the budget by a factor 5.5, as §16.1 already noted for  $\bar{\psi}$ . The programme has a wide margin everywhere except in the one place where it has no proof.*

The budget chain, for reference (each line is proved above it):

| statement                                                        | exponent                         | needs           |
|------------------------------------------------------------------|----------------------------------|-----------------|
| $\mathbb{E} \text{bias}  = o(1/\log n)$ , $k \geq 6 \log^{11} n$ | $k^{-c}$                         | $c > 1/11$      |
| $\psi(j, d) \leq Kd^{-2c}$ , $j \leq 8d$ (Thm. 16.1)             | $c$                              | $c > 1/11$      |
| (S)/(S <sub>2</sub> ) on $S'_K$ (Cor. 17.36)                     | $\gamma = c$                     | $\gamma > 1/11$ |
| $G \in C^{0,\gamma}$ at $z = -1$ (Prop. 17.21)                   | $\gamma$                         | $\gamma > 1/11$ |
| (C2'') via Prop. 17.33, crude                                    | $(p-1)/(p+1)$                    | $p > 6/5$       |
| (C2'') via Prop. 17.33, factorised                               | $\min(1, p-1)$                   | $p > 12/11$     |
| measured (§17.11.4)                                              | $p = 2.7 \pm 0.3$ , modulus 0.75 |                 |

### 17.12.3 Flips recur: $\tau_k < \infty$ a.s.

**Lemma 17.37** (flips recur). *From every state of the quotient chain with  $a_0 b_0 > 0$ ,  $\sum_t 2a_t b_t = \infty$  a.s.; consequently flips occur infinitely often a.s., and  $\tau_k < \infty$  a.s. for every  $k$ .*

*Proof.* Write  $m_t = \min(a_t, b_t)$ ,  $M_t = \max(a_t, b_t)$  and  $p_t = 2a_t b_t = 2m_t M_t$ , the conditional probability of a flip at step  $t+1$ . By Lemma 17.1 (read on the quotient, Corollary 17.3), at a non-flip step a marked coordinate either stays fixed, increases (a merge), or is multiplied by  $\sqrt{U}$ ,  $U \sim U(0,1)$  (a shrink: both cuts in that coordinate’s material, probability equal to the square of the coordinate). Hence  $m_t$  is non-decreasing between the steps at which it strictly decreases, and at a non-flip step  $m$  can decrease in exactly two ways: a shrink of the smaller coordinate, probability  $m_t^2$ ; or a shrink of the larger coordinate below  $m_t$ , probability  $M_t^2 \cdot \mathbb{P}[M_t \sqrt{U} < m_t] = M_t^2 (m_t/M_t)^2 = m_t^2$ . So the event  $E_{t+1} = \{\text{step } t+1 \text{ is not a flip and } m_{t+1} < m_t\}$  has  $\mathbb{P}[E_{t+1} \mid \mathcal{F}_t] \leq 2m_t^2 \leq 2m_t M_t = p_t$ .

Let  $\Omega_0 = \{\sum_t p_t < \infty\}$ . By Lévy's conditional Borel–Cantelli lemma, on  $\Omega_0$  almost surely only finitely many flips occur and only finitely many  $E_t$  occur. All marked coordinates stay strictly positive a.s. ( $U > 0$ , flips land in the open interval), so after the last flip and the last  $E_t$  the process  $m_t$  is non-decreasing and bounded below by some  $m_* > 0$ ; then  $p_t \geq 2m_*^2$  for all large  $t$  and  $\sum_t p_t = \infty$ , contradicting  $\Omega_0$ . Hence  $\mathbb{P}[\Omega_0] = 0$ , i.e.  $\sum_t p_t = \infty$  a.s., and Lévy's lemma in the other direction gives flips infinitely often a.s.  $\square$

*Status: PROVED. This is the hypothesis of Propositions 17.24, 17.30, 17.32 and of the remark after Proposition 17.20; it needs nothing about the environment — in particular it holds with an all-dust environment — because the only mechanism that decreases the smaller marked coordinate has probability at most the flip probability. It gives no rate; rates are (R1).*

#### 17.12.4 The occupation-time step, stated

The session-10 reduction (end of §17.9.5) and the entrance estimate after Proposition 17.34 use, without proof, the following statement.

**Lemma 17.38** (occupation time of the marked total; OPEN). *Let  $K \geq 3$  and let  $H = H_T$  be the block-bound event with bound  $K$ . There are  $c_K, C_K > 0$  such that for every start with  $s_0 \geq 1/(2K)$ , every  $\varepsilon \leq 1/(8K)$  and every  $T \geq 1$ ,*

$$\mathbb{P}\left[\#\{t \leq T : s_t < \tfrac{1}{4K}\} > \tfrac{T}{2}\right] \leq C_K e^{-c_K \varepsilon T/K} + 2\varepsilon^2 T + \mathbb{P}[H^c].$$

*Status: OPEN; believed standard. What it is used for: in the entrance estimate, after  $s$  first reaches  $1/(2K)$  (within  $T_1 = O(\varepsilon^{-1} K \log d)$  steps by Lemma 17.18), one needs, on at least half of the next  $T_2 = O(\varepsilon^{-1} K \log d)$  steps,  $s_t \geq 1/(4K)$  and  $\min(a_t, b_t) \geq \varepsilon$  (the second from the flux bound, cost  $4\varepsilon^2 T_2$ ); on those steps a flip has probability  $\geq 2\varepsilon(s_t - \varepsilon) \geq \varepsilon/(4K)$  and lands in  $S'_K$  with conditional probability  $\geq \frac{1}{2}$  (the trapezoid of Lemma 17.8(ii) is symmetric and unimodal), so  $T' \leq T_1 + T_2$  except with probability  $O(d^{-\gamma'}) + O(\varepsilon K \log d)$ . What fails if it is false: the entrance estimate, hence the reduction of  $(S_2)$  to the log-macroscopic set  $S'_K$ ;  $(S_2)$  would then have to be proved for the laws  $\mu_j$  restricted to  $\{g_0 \geq \varepsilon\}$  with  $\varepsilon = d^{-1/2-\gamma}$ , i.e. with the trap accounting inside the  $L^2$  statement. Nothing above Problem 17.9 depends on it. Intended proof: Lemma 17.19 gives  $\mathbb{E}[\Delta \log s \mid \Gamma_t] \geq c s_t/K$  on  $H \cap \{s_t \leq 1/(2K)\}$ , with down-jumps those of  $\frac{1}{2} \log U$ ; so for small  $\theta > 0$ ,  $V = s^{-\theta}$  is a supermartingale on  $\{s \leq 1/(2K)\} \cap H$  with drift  $\leq -\theta c' s^{1-\theta}/K$ , and the occupation time of  $\{\varepsilon \leq s < 1/(4K)\}$  over  $T$  steps is at most  $\varepsilon^{-(1-\theta)}$  times the accumulated drift; the time below  $\varepsilon$  is priced by Lemma 17.12; excursions above  $1/(2K)$  are handled by stopping  $V$  at  $1/(2K)$ . The missing piece is only the bookkeeping of the re-entries into  $\{s < 1/(2K)\}$  from above, each of which starts  $V$  at  $\leq (2K)^\theta \cdot 2^\theta$  (a single shrink from  $\geq 1/(2K)$  lands at  $\geq s\sqrt{U}$ , and  $\mathbb{P}[\sqrt{U} \leq \frac{1}{2}] = \frac{1}{4}$ ) and of which there are at most  $O(T/K^2)$  in expectation (shrinks have probability  $\leq s^2$ ). We have not written it out.*

#### 17.12.5 What $(M_1)$ needs

Proposition 17.33 assumes  $(M_1)$ :  $\sup_{\bar{x} \in S'_K} \mathbb{E}_{\bar{x}}[\tau_k] \leq C_1 k$ . By the strong Markov property at the flip times,  $\mathbb{E}_{\bar{x}}[\tau_k] = \sum_{i < k} \mathbb{E}_{\bar{x}}[\mathbb{E}_{\bar{\Gamma}_{\tau_i}}[\tau]]$ , so  $(M_1)$  follows from a bound on the mean first-flip time that is uniform on average over post-flip states. The post-flip state has marked total  $s$  (preserved by the flip) and landing gap  $g'$  with  $\mathbb{P}[g' \leq \eta] \leq \eta^2/(ab)$  (Lemma 17.8(ii)), so

$\mathbb{E}[1/g'] \leq \int_0^s \eta^{-2} \mathbb{P}[g' \leq \eta] d\eta \cdot (\text{const}) < \infty$  is automatic from a macroscopic pre-flip state; what is needed is therefore

$$(R1') \quad \mathbb{E}_{\bar{x}}[\tau] \leq C(K) (1 + 1/g(\bar{x})) \quad \text{on } H, \text{ uniformly over the environment,}$$

together with control of the pre-flip state at each flip (so that  $1/(ab)$  in the landing bound is  $O(K^2)$  on average), which is the recurrence of the marked pair to  $S'_K$  — the same recharge machinery as the entrance estimate.  $(R1')$  is the first-moment part of TODO 11a (which asks for the tail  $\mathbb{P}_x[\tau > t] \leq C_K t^{-2}$ , measured in §17.8.2(c)); by Corollary 17.36,  $(R1)$  itself now needs only a  $(1 + \gamma)$ -th moment with  $\gamma > 1/11$ , so a tail bound  $t^{-1-\delta}$  for any  $\delta > 1/11$  would suffice for both.  $(M_p)$  in Corollary 17.31(ii) is used only in a negative result and needs no further attention.

### 17.12.6 Where this leaves the programme

Items (I)–(III) of the overview are unchanged in substance, but (I) is now: prove  $(S_2)$  with any  $\gamma > 1/11$ , for the time- $t$  entrance laws  $\bar{\nu}_t$  of  $S'_K$ , with  $C(K)$  polynomial. Two routes remain, neither of which is Doeblin-type:

- (i) *The spectral route* (§17.5): the quotient chain is reversible for  $\bar{\pi}$ ; for the odd function  $f = \varepsilon F$  the Dirichlet form has the massive flip term, and polynomial decay of  $\psi_{\bar{\pi}}(d) = \|P^d \varepsilon\|_{L^2(\pi)}^2$  with exponent  $\beta$  is equivalent to the spectral measure of  $\varepsilon$  having mass  $O(\delta^\beta)$  within  $\delta$  of  $\pm 1$ . A sufficient functional inequality is

$$(H_\beta) \quad |\mathbb{E}_{\bar{\pi}}[F]|^2 \leq C \mathcal{E}_{\text{odd}}(F)^\beta \|F\|_{L^2(\bar{\pi})}^{2(1-\beta)} \quad \text{for all } F \in L^2(\bar{\pi}),$$

with  $\mathcal{E}_{\text{odd}}(F) = \frac{1}{2} \mathbb{E}_{\bar{\pi}}[(F - F')^2; \text{non-flip}] + \frac{1}{2} \mathbb{E}_{\bar{\pi}}[(F + F')^2; \text{flip}]$ : applied to  $F = \Pi_\delta \varepsilon$  (spectral projection onto  $[1 - \delta, 1]$ ) it gives  $\nu_\varepsilon([1 - \delta, 1]) \leq C\delta^\beta$ . By Proposition 17.6,  $\beta \leq 1$ ; the budget needs only  $\beta > 2/11$ . (The limiting case  $\beta = 1$  is a Hardy inequality in the gap coordinate and is false by a logarithm: test functions  $F = g^{-1} \log^{-2}(1/g) \mathbf{1}\{g \leq \delta\}$  have  $|\mathbb{E}F|^2 / \mathcal{E}_{\text{odd}}(F) \asymp \log(1/\delta)$  — consistent with  $\psi_{\bar{\pi}} \asymp 1/d$ , since linear spectral density gives  $\int d\nu_\varepsilon/(1 - \lambda) = \infty$ .) The periodicity caveat of §17.5 applies: the same inequality is needed for  $(-1)^B F$ , i.e. near  $\lambda = -1$ . What the spectral route does *not* give by itself is  $(S_2)$  for the entrance laws  $\bar{\nu}_t$ : these are singular with respect to  $\bar{\pi}$  (finitely many blocks against infinitely many), and the comparison must be made at a coarse scale — the blocks of mass  $\gtrsim 1/(m \log m)$ , which are the only ones  $\Phi_m$  sees (Remark 17.35(b)) — where  $\bar{\nu}_t$  has had time of order  $m \log m \gg$  the relaxation time  $1/(2\eta)$  of each scale  $\eta$  in that range. This start-comparison is the first concrete sub-problem.

- (ii) *The exact-law route* of §17.11.5(iv): polynomial decay of the block-count parity at flip times from the exact fixed-time laws, from the one-curve start, extended to the  $\mu_j$  starts. Its target is now  $p > 6/5$  in Proposition 17.33, i.e.  $a_k = O(k^{-6/5})$ , far weaker than the measured  $k^{-3}$ . Since  $B(\bar{\Gamma}_{\tau_k})$  spreads only like  $\log k$ , the decay cannot come from the width of its law; it comes from the smoothness of the law of  $\tau_k$  on the time axis (the parity of  $B$  at a *fixed* time is deterministic, Lemma 17.13), and quantifying that smoothness is exactly what (ii) must do.

Both routes consume the same two inputs that the audit has left open — the occupation-time lemma and  $(R1')$  — and both are now aimed at a target with a fivefold margin.

Numerical task 3 of the session-12 handoff (pin p) was run (`runs/s13/akB_0.p.log`,  $4 \times 10^8$  samples, one-curve start):  $a_k$  alternates in sign for all  $8 \leq k \leq 32$ ,  $k^3|a_k|$  drifts from 2.5 to 3–4.7, and the fit of  $\log|a_k|$  over  $10 \leq k \leq 32$  gives  $p = 2.4$ , twice the corrected budget 6/5. Task 2 (9- and 17-curve starts to  $k = 40$ ,  $10^8$  samples):  $|a_k|$  below  $2 \times 10^{-4}$  for  $k \geq 30$  with the same alternation. Both are superseded by §18, which removes (C2') from the programme.

### 17.12.7 Scouting (III): Problem 16.7 is not needed

Problem 16.7 asks that the fraction of marks whose column lacks intercalate supply be  $o(1/\log n)$  uniformly over all admissible  $(\lambda, \alpha, \beta)$ , including merges whose  $m$ -cycle class occupies  $O(1)$  columns. But Theorem 12.1 uses EQ only along its routing chains, and on those chains the merged part is always large.

**Proposition 17.39** (no decoupling is needed on the routing chains). *Let  $K = \lceil A \log n \rceil$  be as in the proof of Theorem 12.1. Every split  $\lambda \rightarrow \mu$  used by its routing chains (for plain types with  $\kappa(\lambda) \leq K$  parts) merges a part  $m \geq n/K$ . Consequently, on the intercalate-expander event, the number of marks  $(L', P)$  of  $X(n, \lambda; \alpha, \beta)$  whose column  $j$  fails the supply property of Lemma 15.9 is at most  $14 \log^{22} n$  per square, out of  $\lambda_m m \geq n/K$  marks per square: the exceptional-mark fraction is at most  $14A \log^{23} n/n = n^{-1+o(1)}$ , for every chain and every square.*

*Proof.* The routing of Theorem 12.1 builds the chain in reverse by splitting parts off a residual  $R$  smallest-first; the part merged at each step of the forward chain is the residual  $R$  at that step. In the case “two largest parts unequal”,  $R$  always contains the largest part  $p_{\max} \geq n/\kappa \geq n/K$ ; in the case “two largest parts equal to  $\ell_{\max}$ ”, the first split has  $R = n$  and every later residual contains the reserved copy of  $\ell_{\max} \geq n/\kappa$ . So  $m = R \geq n/K$ , and the  $m$ -class has  $\lambda_m m \geq m \geq n/K$  columns.

For the count: apply Lemma 15.9 *without* the constraint “avoid column  $j'$ ” (drop  $j'$  from its hypotheses; the proof is unchanged), with  $\ell = \log^{11} n$ ,  $t = 6\ell + 1$ : all but fewer than  $t\ell < 7 \log^{22} n$  columns  $c$  carry  $t$  stable intercalates avoiding rows 1, 2, pairwise disjoint in rows, partner columns and symbols. For a mark  $P = \{j, j'\}$  with  $j$  outside this exceptional set  $E$ , at most one of the  $t$  intercalates through  $j$  has partner column  $j'$  (the partner columns are distinct), and discarding it leaves  $6\ell$  useful ones avoiding  $j'$ : the mark is good. So a bad mark has a column in  $E$ . For  $\alpha \neq \beta$  the  $m$  marks  $\{j, \sigma^\alpha(j)\}$  of an  $m$ -cycle are distinct (coincidence would need  $\sigma^{2\alpha}(j) = j$ , i.e.  $m \mid 2\alpha$ , impossible for  $0 < \alpha < m$ ,  $2\alpha \neq m$ ) and each column of the cycle lies in exactly two of them; hence at most  $2|E| < 14 \log^{22} n$  marks are bad. (For  $\alpha = \beta$ , which the routing avoids, each column lies in one mark and the bound is  $|E|$ .) With  $t = 6\ell + 1$  in place of  $6\ell$  the hypothesis  $4t\ell + 2t + \ell + 2 \leq \beta n/2$  of Lemma 15.9 still holds for large  $n$ .  $\square$

*Status: PROVED. Consequence: the exceptional-mark bookkeeping of Remark 15.10(i) is closed for the purposes of Theorem 12.1, with room  $n^{1-o(1)}$  to spare, and Problem 16.7 can be struck from the programme. The types with small  $m$ -classes — the ones on which the deterministic bound is vacuous and on which Problem 16.7 is a genuine question — are exactly those the routing never uses, or reaches only through a large merge. (The heuristic in §16.3, that class membership is a rows-1, 2 property, is in any case not right: the cycle type of  $\sigma$  is determined by the bottom rectangle  $L_\downarrow$  alone, since a leftover cycle of length  $2\ell$  of the missing-pair graph contributes an  $\ell$ -cycle of  $\sigma$  whichever way it is oriented; the orientations are fair coins that change nothing. So the question was never about rows 1, 2 against rows 3..n; but it does not need an answer.)*

### 17.12.8 Scouting (II): the transfer from the iid model to the orbit measure

Item (II) was carried since session 7 as “transfer problem 7a(i)” and reformulated in §17.6 as (a) an exchangeable-to-mixture step plus (b) a  $\theta$ -covariant mixing bound. Reading Theorem 15.6 against Theorem 16.1 with fresh eyes gives a different classification. What the true orbit supplies is: a *frame*  $\pi_0 = \pi_P(L_0)$ , a permutation of the rows whose cycles are the curves; the marked points  $u, v = \text{rows } 1, 2$ ; and a *pool* of  $k$  chords, the row pairs of the useful stable intercalates, in positions determined by  $L_0$ . The iid model replaces the frame by a single circle with  $u, v$  antipodal and the pool by iid uniform chords. Three separate statements are therefore being modelled, of very different status.

**(T3) The start is stationary, not one-block.** The frame is the cycle structure of the pair permutation of two columns of a random Latin square with two marked rows: in the continuum it is the stationary law  $\pi$  of the marked chain (Proposition 17.5), not the one-curve state. Two things follow. First, the “free” short-horizon inputs of §17.8–17.9 — Proposition 17.10 (gap law  $4\varepsilon^2 j$ ) and Corollary 17.16 (block count  $O(\log d)$  from the one-block start) — are artefacts of the model. Under the true start the gap law is the linear stationary one,  $\mathbb{P}[g_0 \leq \varepsilon] \asymp 4\varepsilon$ , so the decomposition of §17.4 gives  $\psi \lesssim \varepsilon + C(K)(d - T^*)^{-2\gamma}$  with  $T^* = \varepsilon^{-1}$  polylog, i.e.  $\psi \lesssim d^{-\min(1, 2\gamma)}$  polylog: the budget  $c > 1/11$  is *unaffected* (and the trap term reproduces the true rate  $1/d$  of Proposition 17.6). The block bound is replaced by the stationary one: blocks of mass  $\geq \eta$  number  $O(\log(1/\eta))$  with exponential tails under PD(1), and blocks below  $\eta = 1/(k \log^2 n)$  carry total mass  $\asymp \eta$  and are inert over the horizon (Lemma 17.29: hit with probability  $\leq 2k\eta = o(1/\log n)$ ). Second, for the spectral route of §17.12.6 this is good news: the start-comparison sub-problem disappears, because the quantity the true ensemble needs is  $\psi_\pi(d)$  itself, coarse-grained. (S<sub>2</sub>) and the entrance estimate of §17.12.1 are posed for the entrance laws of the one-block-started  $\mu_j$  and must be re-posed for the stationary start; the arithmetic above is what that re-posing costs. Note also that “the frame is stationary” is itself a statement about  $\text{Unif}(S(\lambda))$  and the special pair  $(j, \sigma^\alpha(j))$  — it is (T1) below, and indeed  $\mathbb{P}[u \sim v] = \frac{1}{2}$  for the frame is the weak conjecture’s own bit, since the base point of an orbit is a uniform orbit point. [Referee note, end of session 13: for the spectral route of §18 this is *not* a matter of “health” only — Corollary 18.3 uses the exact uniform law of the frame. What (T) below needs from the frame law is that it be close to uniform *on the test function*  $\alpha \mapsto \varepsilon(\alpha)(P^M \varepsilon)(\alpha)$ , a function of the cycle type and of the cycles of  $u, v$ ; the quenched one-point function of §§17.9–17.11 is thereby relocated into (II), not removed. Proposition 17.40 proves the trapping half of this.]

*Numerical check (s13\_frame.c, 10<sup>6</sup> instances per point, N = 4096 slots, k iid chords, 4E[bias<sup>2</sup>] = E[ε<sub>S</sub>ε<sub>T</sub>]):*

| $\mathbb{E}[\text{bias}^2] \cdot k$                     | $k = 8$ | 16   | 32   | 64   | 128  | 256  |
|---------------------------------------------------------|---------|------|------|------|------|------|
| one block, $v$ antipodal (the model of §16.1)           | 0.24    | 0.28 | 0.29 | 0.32 | 0.25 | 0.34 |
| one block, $v$ uniform                                  | 0.47    | 0.55 | 0.61 | 0.62 | 0.65 | 0.62 |
| random-permutation frame, $v$ uniform                   | 0.73    | 0.85 | 0.93 | 0.97 | 0.96 | 1.15 |
| real squares, §16.4: $k \approx 10, 13, 17, 22, 25, 26$ | 0.64    | 0.67 | 0.60 | 0.74 | 0.72 | 0.73 |

(standard errors 0.002 at  $k = 8$  rising to 0.06 at  $k = 256$  in the model rows; the real-square row lists its own  $k$ -bins, with standard errors 0.03–0.06, from  $n = 30, 30, 30, 50, 50, 100$ ; the random-frame values at  $N = 50, 100, 200$  agree with  $N = 4096$  to 0.01, so there is no finite- $N$  correction).

The factor 2–3 between the iid model and real squares, unexplained since session 7, is the start: the stationary frame accounts for a factor  $\approx 3$  at  $k \leq 32$ , of which a factor 2 is the gap law ( $v$  uniform vs antipodal) and 1.5 the multi-curve environment. Real squares sit 15–20% below the random-frame model at matched  $k$  (0.76, 0.81, 0.86, 0.90 at  $k = 10, 13, 17, 24$ ), plausibly from shared rows between the  $R$ - and  $M$ -sides at  $k^2/n \gg 1$ ; within the precision of §16.4 the real pool behaves as a generic pool on a stationary frame. The  $1/k$  law and its constant are thus consistent with (T2) below being true.

**(T1) Health of the frame under the orbit measure.** The chain analysis needs the frame to have  $O(\log k)$  blocks above scale  $1/(k \log^2 n)$  and  $u, v$  untrapped except on a set of marks of measure  $O(\text{polylog}/k)$  — properties that a uniform random permutation has. The frame is the pair permutation of the columns  $j, \sigma^\alpha(j)$  of  $L \sim \text{Unif}(S(\lambda))$ : by transposition symmetry its cycle type has, for a *uniform* square and a *uniform* column pair, exactly the law  $P_n$  — the object of the conjecture. What is needed is far weaker (a polylogarithmic block count at coarse scale, a trapped fraction  $O(1/k)$ ), but it is needed under the conditioning on  $S(\lambda)$  and for the special pair  $(j, \sigma^\alpha(j))$ , and it cannot be imported from the uniform measure by the union-of-costs device of Lemma 15.11: the device transfers only properties whose failure probability is  $e^{-\omega(n \log^2 n)}$ , while “more than  $2n/k$  marks are trapped” fails with probability of order  $e^{-n/k}$  at best. Status: the trapping half is PROVED by a switching argument (Proposition 17.40, written after this paragraph); the closeness of the frame law to uniform on the test function  $\varepsilon \cdot P^M \varepsilon$  is OPEN.

**(T2) Genericity of the pool.** This is the real content of (II). Step (a) of §17.6 is routine: take the supplied family through  $j$  with  $t \asymp n/\log^{11} n$  members (Lemma 15.9 allows  $t \leq \beta n/(8\ell)$ ) and use a uniformly random  $k$ -subset of it as the pool; by Diaconis–Freedman the  $k$  chords are, conditionally on  $L$ , iid from the empirical pair law  $\theta_L$  of the family up to total variation  $k^2/(2t) = n^{-1+o(1)}$ , and the hard-core correction is absent because the family is already pairwise disjoint. Step (b) is *not* a mixing statement: the chain analysis is covariant under reparametrisation of the circle by an atomless measure  $m$  only when the two endpoints of a chord are *independent* draws from  $m$ ;  $\theta_L$  is an empirical law on *pairs* of frame positions, and if (say) the intercalate rows  $r, r'$  were always adjacent in the cycles of  $\pi_0$  the dynamics would be degenerate and the bit frozen. What is needed is a coarse product structure: at scale  $\eta = 1/(k \log^2 n)$ , the fraction of the family with  $(\text{pos } r, \text{pos } r') \in A \times B$  is  $|A||B|$  up to  $\delta \ll \eta^2/k$  for all arcs  $A, B$  (a coupling of the  $\theta_L$ -chain with the uniform chain over  $k$  steps then costs  $O(k\delta/\eta^2)$ ). Quantitatively this is mild — a family of  $n^{1-o(1)}$  random-like points in the square has discrepancy  $n^{-1/2+o(1)} \ll \eta^2/k = \log^{-O(1)} n$  — but it is a *distributional* statement about where the stable intercalates of column  $j$  sit relative to the pair permutation of columns  $j, j'$ , under  $\text{Unif}(S(\lambda))$ , and like (T1) it is out of reach of union-of-costs. The only route visible is the one [9] use for the expander itself: demand the property for all but  $\ell = \log^{11} n$  columns, so that failure requires  $\ell$  simultaneous failures and costs  $e^{-\ell \cdot n\delta^2/\text{polylog}} = e^{-\omega(n \log^2 n)}$  if the per-column failures decouple in the triangle-removal model — a KPS-scale sub-project (their Lemma 8.3 machinery re-run for a two-column pair permutation), not a problem of the continuum chain.

**Classification.** (III): *closed* (Proposition 17.39). (II)(a): routine. (T3): a model hypothesis, part of (II); given it, the arithmetic is routine and it improves the position of (I). (II)(T1)–(T2): *hard*, of a different nature from (I): they need distributional control of the intercalate structure relative to a pair permutation under the type-conditioned measure, which the existing toolkit (existence statements transferred by union of costs) cannot give; the plausible route

is a KPS-style “all but  $\ell$  columns” robustness argument in the triangle-removal model. The numerics say the statements are true with a comfortable margin. First concrete sub-problem: state the minimal coarse product property of (T2) precisely and determine whether its “all but  $\ell$  columns” version has failure probability  $e^{-\omega(n \log^2 n)}$  in the triangle-removal process; a positive answer would also deliver (T1) by the same device. Second: redo the bookkeeping of §17.8–§17.9 from the stationary start (T3), which is the honest form of (I).

### 17.12.9 The transfer (II): exact conditional structure, and the real-square test

With §18 in hand, the whole programme rests on (T1)–(T2). This subsection makes them precise and tests them.

**The conditional structure.** Fix the mark  $P = \{j, j'\}$  and write  $L_{\text{rest}}$  for the  $n \times (n - 2)$  Latin rectangle obtained by deleting columns  $j, j'$  from  $L$ ; for each row  $r$  let  $\text{miss}(r)$  be the pair of symbols absent from row  $r$  of  $L_{\text{rest}}$ . Then:

- (i) *The frame is a function of  $L_{\text{rest}}$ .*  $\{L(r, j), L(r, j')\} = \text{miss}(r)$ , and  $\pi_P(r)$  is the row  $r'$  with  $L(r', j') = L(r, j)$ , i.e. the other row whose missing pair contains  $L(r, j)$ . So the cycles of  $\pi_P$  are exactly the cycles of the 2-regular “leftover” graph  $G$  on rows ( $r \sim r'$  iff  $\text{miss}(r) \cap \text{miss}(r') \neq \emptyset$ ), traversed in a direction fixed by the orientation of each leftover cycle; the orientations are the  $2^{\#\text{cycles}}$  completions of  $L_{\text{rest}}$ , fair coins except on the (at most two) cycles through rows 1, 2, which the conditioning on  $S(\lambda)$  and the mark constrain. The bit  $[u \sim v]$  does not depend on the orientations.
- (ii) *The pool is a coin-flip thinning of a candidate set determined by  $L_{\text{rest}}$ .* An intercalate  $(r, r', j, c)$  with  $c \notin \{j, j'\}$  requires  $L(r', c) = L(r, j) \in \text{miss}(r)$  and  $L(r, c) = L(r', j) \in \text{miss}(r')$ . Call  $(r, r', c)$  a *candidate* if  $L(r', c) \in \text{miss}(r)$  and  $L(r, c) \in \text{miss}(r')$  (a property of  $L_{\text{rest}}$ ). A candidate is an intercalate through  $j$  iff both rows place the relevant symbol in column  $j$ , through  $j'$  iff both place it in  $j'$ , and otherwise it is nothing; so it is realised on the  $j$ -side with probability  $\frac{1}{4}$  and on the  $j'$ -side with probability  $\frac{1}{4}$  when  $r, r'$  lie on different leftover cycles, and with probabilities  $(\frac{1}{2}, \frac{1}{2})$  or  $(0, 0)$  when they lie on the same cycle. The conjugating coordinates ( $c = j'$ ) are exactly the 2-cycles of  $\pi_P$ . Each row has about four candidate partners (a missing symbol of  $r$  sits in row  $r'$  in a unique column  $c$ , and  $L(r, c)$  must then be missing from  $r'$ , probability  $\approx 2/n$  for each of the  $n - 2$  rows  $r'$  and each of the two missing symbols), so there are  $\approx 2n$  candidates and  $\approx n/2$  pool chords on each side, as observed.
- (iii) *Local exclusion.* A non-conjugating pool chord  $(r, r')$  never has  $\pi_P(r) = r'$  or  $\pi_P(r') = r$ :  $L(r', c) = L(r, j)$  and  $L(r', j') = L(r, j)$  would put one symbol twice in row  $r'$ . Rows adjacent along a cycle of  $\pi_P$  therefore never form a chord, and in particular no chord has both ends in a cycle of length  $\leq 3$  (a 2-cycle is conjugating, a 3-cycle has all its pairs adjacent).

Thus (T1)–(T2) are statements about  $L_{\text{rest}}$  alone: the cycle structure of its leftover graph  $G$ , and the position of the candidate graph  $H$  ( $r \sim r'$  iff  $(r, r', c)$  is a candidate for some  $c$ ) relative to the cycles of  $G$ .

**The test.** `s13_real.py` takes uniform squares from the Jacobson–Matthews sampler, samples marks  $(j, \sigma^\alpha(j))$  with  $\alpha \geq 2$  in  $\sigma$ -cycles of length  $\geq 4$ , builds the frame  $\pi_P$  and a randomised greedy cell-disjoint pool of all intercalates through  $j$  or  $j'$  in rows  $3..n$  (so  $k \approx n/2$ , no cap), computes the bit as  $[u \sim v \text{ in } M_\omega \pi_P R_\omega]$  (verified against the switched square on 60 random subsets: 0 mismatches), and measures, with independent uniform subsets  $S, T$  and uniform  $\omega$ :  $\mathbb{E}[\varepsilon_S \varepsilon_T]$ ; the level statistics  $\mathbb{E}[\varepsilon(\omega) \varepsilon(\omega \Delta D)]$  for  $|D| = 1, 2, 3$  non-conjugating coordinates (the model predicts  $\langle \varepsilon, P^{|D|} \varepsilon \rangle$  up to  $O(k^2/n)$ ); the frame  $(\# \text{cycles}, \mathbb{P}[u \sim v], |C_u|)$ ; and the pool's position relative to the frame (number of chords with both ends in one  $\pi_P$ -cycle against the null  $\sum_i |C_{r_i} \setminus \{r_i, u, v\}|/(n-3)$ , and the relative position  $d(r, r')/|C_r|$  of the second end in four bins). For every real instance a *matched model instance* is generated: a uniform derangement frame and uniform disjoint chords on rows  $3..n$  with the same side counts  $(|\mathcal{I}_j|, |\mathcal{I}_{j'}|, |\mathcal{I}_{jj'}|)$ , and the same statistics are computed.

**Results (corrected pipeline, true marks).** ( $n = 50$ : 600 squares, 2400 instances, 16  $(S, T)$ -pairs and 16 level samples per instance; standard errors in parentheses.)

| $n = 50$  | $k'$ | $\mathbb{E}[\varepsilon_S \varepsilon_T]$ | lev 1     | lev 2     | lev 3     | $\mathbb{P}[u \sim v]$ | $\mathbb{P}[ C_u  \leq 5]$  |
|-----------|------|-------------------------------------------|-----------|-----------|-----------|------------------------|-----------------------------|
| real      | 22.7 | 0.0928 (62)                               | 0.608 (5) | 0.445 (5) | 0.333 (6) | 0.497                  | 0.071                       |
| model     | 22.7 | 0.1338 (73)                               | 0.644 (5) | 0.488 (6) | 0.382 (6) | 0.518                  | 0.085                       |
| $n = 100$ | $k'$ | $\mathbb{E}[\varepsilon_S \varepsilon_T]$ | lev 1     | lev 2     | lev 3     | $\mathbb{P}[u \sim v]$ | $\mathbb{P}[ C_u  \leq 10]$ |
| real      | 47.8 | 0.0431 (82)                               | 0.643 (6) | 0.477 (7) | 0.370 (8) | 0.496                  | 0.083                       |
| model     | 47.8 | 0.0512 (86)                               | 0.656 (6) | 0.497 (7) | 0.398 (8) | 0.517                  | 0.080                       |

( $n = 100$ : 300 squares, 1200 instances; the real-minus-model gap at each level is about half of its  $n = 50$  value, as a  $1/n$  effect should be.) Pool position: real 27847 same-cycle chords against a null of 28586 ( $-2.6\%$ ), model 29182 against 29176; position bins real  $[5947, 7734, 7892, 6274]$ , model  $[6869, 7489, 7715, 7109]$ . The real squares decorrelate *faster* than the model, at every level, by 10–30% at  $n = 50$ ; the mechanism is the exclusion (iii): the model's pool contains the  $\approx |C|$  adjacent pairs of each cycle, which never change the bit, while the real pool does not, which raises the bit-changing fraction of a chord by a factor  $\approx (|C| - 1)/(|C| - 3)$  — a  $1/n$ -type effect, in the favourable direction. (With the symbol/column bug the two effects of order  $1/n$  — this one and the frozen adjacent instances — happened to cancel, which is why the first run showed real  $\approx$  model.) The frame statistics agree with the model and the trapping fraction  $\mathbb{P}[|C_u| \leq 5]$  is 0.071 against the model's 0.085 (and against the bound  $36 \cdot 5/50$  of Proposition 17.40).

**What is now needed, restated.** By Corollary 18.3 the programme needs, for  $L \sim \text{Unif}(S(\lambda))$  on the expander event and the marks of Theorem 11.4:

$$(T) \quad \mathbb{E}[\varepsilon(\alpha) \varepsilon(\alpha \rho_D)] = o(1/\log^2 n), \quad \alpha = \pi_P(L), \ D \text{ a uniform subset of the pool},$$

and the model says the left side is  $\approx 2/k'$ . Two remarks change the shape of the problem. (a) *Large pools.* Nothing forces  $k' = \Theta(\log^{11} n)$ : Lemma 15.9 supplies  $t \leq \beta n/(8\ell)$  intercalates per column, and the real pool has  $\approx n/2$ . With  $k' = n/\text{polylog}$  the trapped marks are those with row 1 or 2 in a  $\pi_P$ -cycle of length  $\leq \text{polylog } n$ , so (T1) becomes a statement about *short* column cycles of a random Latin square —  $\mathbb{E}[C_\ell(\pi_P)] \leq C/\ell$  for  $\ell \leq \text{polylog } n$  — rather than about cycles of length  $n/\text{polylog}$ ; for bounded  $\ell$  this is the Poisson statistics of short cycles of two

columns, which is classical for uniform squares. (b) *What union-of-costs cannot do.* Neither (T1) nor (T2) is an  $e^{-\omega(n \log^2 n)}$ -robust property, so neither transfers from the uniform measure to  $\text{Unif}(S(\lambda))$  by Lemma 15.11; the natural tool is a switching argument on  $L_{\text{rest}}$  that preserves  $S(\lambda)$  (switches avoiding rows 1, 2 preserve  $\sigma$ , the leftover graph  $G$ , and hence the frame, while moving the candidate graph  $H$ ), i.e. a proof that, conditionally on  $G$ , the candidate graph is in generic position. This is the open problem of the project; it is a problem about  $n \times (n - 2)$  Latin rectangles, and the numerics say its conclusion holds with margin.

**First concrete sub-problems.** (1) *(T1) at polylog scale by join switchings.* Kwan–Sudakov [11, Thm. 3 and Fact 10] bound the expected number of 2-cycles of two rows of a uniform Latin square by  $1 + o(1)$  with a switching (the CGW flip joining a 2-cycle to another cycle, with the “turn” to make the pair flippable), counting the ways in and out. The same switching applied to a column pair  $(j, j')$  joins the cycle through row 1 to a cycle avoiding rows 1, 2; it changes  $L(1, j), L(1, j')$  and hence  $\sigma$ , but only by a merge or split of  $\sigma$ -cycles whose effect on  $|S(\lambda)|$  is controlled by Theorem 11.4 within a factor  $[\frac{1}{2}, \frac{3}{2}]$ . Target:  $\mathbb{E}_{\text{Unif}(X)}[\#\{\text{rows in } \pi_P\text{-cycles of length } \leq \ell\}] \leq C\ell$  uniformly for  $\ell \leq n^{1/2}$  and all admissible marks, which gives  $\mathbb{P}[\text{trapped}] \leq C\ell/n$  — far more than the  $o(1/\log n)$  needed, with constant-factor losses affordable. (2) *(T2) as a switching statement on  $L_{\text{rest}}$ .* For a window of relative positions  $W \subset (0, 1)$  along a leftover cycle, the number of candidates  $(r, r')$  with  $r'$  in the window of  $r$ 's cycle, against the null  $|W| \cdot (\text{cycle length}) \cdot (\text{candidate degree})/n$ : a switching that moves one end of a candidate (exchange the symbol  $L(r', c)$  with another occurrence by an intercalate switch in  $L_{\text{rest}}$ ) should show the ratio is  $1 + O(1/\text{cycle length})$ . What the final argument needs is then  $\mathbb{E}_\alpha[\mathbb{E}[\varepsilon(\alpha\rho_D) \mid \alpha] - (P^M\varepsilon)(\alpha)] = o(1/\log^2 n)$ , a statement comparing the conditional law of the pool given the frame with iid chords through the single test function  $\varepsilon$ .

### 17.12.10 (T1) is a switching lemma: trapped marks are rare

Fix  $n$ , a plain type  $\lambda$  and a split  $m = \alpha + \beta$  with  $\alpha \neq \beta$ , and let  $(L, P)$  be uniform on  $X = X(n, \lambda; \alpha, \beta)$ ,  $P = \{j, j'\}$ . Write  $\pi = \pi_P$  for the frame (a derangement of the rows),  $C_r$  for the  $\pi$ -cycle of a row  $r$ . Two elementary moves on Latin squares are used; both are involutions.

- (TRADE) For rows  $x \neq y$  let  $\rho_{x,y}$  be the permutation of the columns with  $L(y, \rho_{x,y}(q)) = L(x, q)$ . For a cycle  $Q$  of  $\rho_{x,y}$ , exchanging the entries of rows  $x$  and  $y$  in the columns of  $Q$  gives a Latin square (the two rows carry the same symbols on  $Q$ ). If  $j' \in Q \not\ni j$  the new frame is  $(xy) \circ \pi$ ; if  $j \in Q \not\ni j'$  it is  $\pi \circ (xy)$ ; if both or neither lie in  $Q$  the cycle partition of  $\pi$  is unchanged. Call  $(x, y)$  *flippable* if  $j, j'$  lie in different cycles of  $\rho_{x,y}$ , and let  $Q_{x,y}$  be the cycle through  $j'$ .
- (TURN) For a cycle  $c$  of  $\pi$ , exchanging the entries of columns  $j$  and  $j'$  in the rows of  $c$  gives a Latin square; the frame is inverted on  $c$  (Lemma 11.2) and its cycle partition is unchanged. If  $x \in c \not\ni y$ , the turn replaces  $\rho_{x,y}$  by  $\rho_{x,y} \circ (j j')$  (if instead  $y \in c \not\ni x$ , by  $(j j') \circ \rho_{x,y}$ ), so it toggles the flippability of  $(x, y)$ .

A trade at rows  $x, y \notin \{1, 2\}$  leaves rows 1, 2, hence  $\sigma$  and the mark, untouched. A turn of a cycle  $c \not\ni 1, 2$  likewise. A turn of  $C_1$  when  $2 \notin C_1$  replaces  $\sigma$  by  $\sigma \circ (j j')$  (reading  $j, j'$  as the row-1 symbols), which splits the  $m$ -cycle of  $\sigma$  through  $j$  into cycles of lengths  $\alpha$  and  $\beta$  and sends  $(L, P)$  into the joining space  $J = J(n, \mu; \alpha, \beta)$  of Definition 11.1; a turn of a cycle containing both rows 1 and 2 leaves  $\sigma$  unchanged.

**Proposition 17.40** (short frame cycles through a marked row are rare). *For  $(L, P)$  uniform on  $X$  and every  $2 \leq \ell \leq n/2$ ,*

$$\mathbb{P}[|C_1| = \ell] \leq \frac{18}{n - \ell - 1}, \quad \text{hence} \quad \mathbb{P}[|C_1| \leq \ell] \leq \frac{36\ell}{n} \quad (\ell \leq n/2 - 1),$$

and the same bounds hold for  $C_2$ .

*Proof.* Let  $E_\ell = \{(L, P) \in X : |C_1| = \ell\}$ . (The case  $C_1 = \{1, 2\}$  cannot occur: it would mean  $\sigma(j) = j'$  and  $\sigma(j') = j$ , but  $j' = \sigma^\alpha(j)$  lies in an  $m$ -cycle with  $m \geq 4$ .)

*Forward moves.* From  $(L, P) \in E_\ell$  choose  $y \in C_1 \setminus \{1, 2\}$  and  $x \notin C_1 \cup \{2\}$ : at least  $(\ell - 2)(n - \ell - 1)$  pairs  $((\ell - 1)(n - \ell - 1)$  if  $2 \notin C_1$ ; for  $\ell = 2$  this is  $n - 3$ ). If  $(x, y)$  is flippable, trade  $Q_{x,y}$ ; otherwise turn  $C_1$  (which makes  $(x, y)$  flippable, as  $y \in C_1 \not\ni x$ ) and then trade  $Q_{x,y}$  in the turned square. In both cases the resulting frame is  $(xy) \circ \pi^\pm$  with  $\pi^\pm$  equal to  $\pi$  or to  $\pi$  inverted on  $C_1$ , so the cycle of row 1 becomes  $C_1 \cup C_x$ , of length  $\ell + |C_x| > \ell$ ; rows 1, 2 are untouched by the trade, and the result is in  $X$  (no turn, or  $2 \in C_1$ ) or in  $J$  (turn with  $2 \notin C_1$ ).

*Backward moves.* A square  $(L', P) \in X \cup J$  is reached from  $(L, P) \in E_\ell$  by the move at  $(x, y)$  only if  $x, y$  lie in the cycle  $C'_1$  of row 1 in  $L'$ , and undoing the trade (the same trade, since  $\rho'_{x,y} = \rho_{x,y}^{-1}$  on  $Q_{x,y}$ ) splits  $C'_1$  into the arcs  $\{\pi'(y), \dots, x\}$  and  $\{\pi'(x), \dots, y\}$ , of which the one containing row 1 must have length exactly  $\ell$ : given  $L'$ , that arc is determined by its position relative to row 1 ( $\ell$  choices), and the arc determines  $\{x, y\}$ . With the binary choice “turn back or not” this gives at most  $2\ell$  preimages per  $(L', P)$ .

*Counting.* The map  $((L, P), \text{move}) \mapsto ((L', P), \text{inverse move})$  is injective, so  $|E_\ell|(\ell - 2)(n - \ell - 1) \leq 2\ell(|X| + |J|) \leq 6\ell|X|$ , using  $|J| = 2|X_A| \leq 2|X|$  (Theorem 11.4). For  $\ell \geq 3$  this is the claim; for  $\ell = 2$  the forward count is  $n - 3$  and the backward count 4, giving  $12/(n - 3)$ . The bound for  $C_2$  is proved by the same argument with the roles of rows 1 and 2 exchanged: a turn of  $C_2 \not\ni 1$  replaces  $\sigma$  by  $(\sigma(j)\sigma(j')) \circ \sigma$ , which also splits the  $m$ -cycle into lengths  $\alpha, \beta$  and lands in  $J$ .  $\square$

*Status: PROVED. This is the trapping half of (T1): with a pool of  $k'$  chords the marks for which the chain cannot move row 1 or row 2 within the horizon are those with  $|C_1|$  or  $|C_2|$  below  $\asymp n/k'$ , of total probability  $O(1/k')$  — exactly as for a uniform frame, up to the constant 36. The proof is a Kwan–Sudakov style switching [11] in which the only non-trivial step, the turn through row 1, is priced by the marked-pair identity itself: the turn is the CGW switch at the mark, and  $|J| \leq 2|X|$  is Theorem 11.4. It uses nothing about intercalates. A VERIFICATION on real squares is in §17.12.9 ( $\mathbb{P}[|C_1| \leq 10]$  at  $n = 100$ : 0.089 real against 0.083 model).*

### 17.12.11 A third route: the trade chain

The moves of §17.12.10 suggest replacing the intercalate orbit of [9] altogether. Consider the Markov chain  $T$  on  $X$  which, from  $(L, P)$ , with probability  $\frac{1}{2}$  picks a uniform pair of rows  $x \neq y$  in  $\{3, \dots, n\}$  and trades  $Q_{x,y}$  if  $(x, y)$  is flippable (else does nothing), and with probability  $\frac{1}{2}$  picks a uniform row  $x \in \{3, \dots, n\}$  and turns  $C_x$  if  $C_x \not\ni 1, 2$  (else does nothing). Every move is an involution, the probability of a move equals that of its inverse (flippability and  $|C_x|$  are preserved), so  $T$  is symmetric and  $\text{Unif}(X)$  is stationary;  $T$  never touches rows 1, 2, so it preserves  $S(\lambda)$  and the mark exactly, with no expander hypothesis, no stable families and no orbit bookkeeping. Along  $T$  the frame evolves as  $\pi \mapsto (xy) \circ \pi$  at the flippable trades (turns do not change the bit), i.e. as the random-transposition walk *thinned by flippability*, and the

marked bit is  $\varepsilon(\pi)$ . Hence, by stationarity, for every  $M$ ,

$$\mathbb{E}_X[\varepsilon] = \mathbb{E}_X[(T^M \varepsilon)(L, P)], \quad (T^M \varepsilon)(L, P) = \mathbb{E}[\varepsilon(\pi_P(L_M)) \mid L_0 = L],$$

and, since  $\mathbb{E}_X \varepsilon = 2(\mathbb{P}_X[B] - \frac{1}{2})$ , EQ(4 $\delta$ ) follows from  $\mathbb{E}_X |T^M \varepsilon| \leq \delta$  for some  $M$ . If every pair were flippable, the frame would perform the walk of left-multiplication by uniform transpositions of  $\{3, \dots, n\}$ , which preserves  $\pi^{-1}(1)$  and  $\pi^{-1}(2)$  and is the ordinary transposition walk on  $S_{n-2}$  after contracting the arcs  $\pi^{-1}(1) \rightarrow 1$  and  $\pi^{-1}(2) \rightarrow 2$  to single points; on every such fibre  $\{\pi^{-1}(1) = a, \pi^{-1}(2) = b\}$  (not on the parity classes, where the fraction is  $\frac{1}{2} \pm 1/((n-2)(n-3))$ )  $\mathbb{P}[u \sim v] = \frac{1}{2}$  exactly (the fibre is the set of permutations of  $n-2$  points with no fixed point outside  $\{a, b\}$ , not all of  $S_{n-2}$ ; left multiplication by  $(ab)$  is an involution of it toggling  $u \sim v$ ), so  $T^M \varepsilon$  would be bounded by twice the total-variation distance of that walk from the uniform law on the fibre, which is  $o(1)$  after  $Cn \log n$  trades by Diaconis–Shahshahani (the turn steps supply aperiodicity) — EQ( $o(1)$ ) with no knowledge of the frame law, no spectral measure, and no genericity of anything. The entire difficulty is the thinning: which pairs are flippable depends on  $L$ , so the frame is not Markov by itself. What is needed is that, for  $\text{Unif}(X)$ -typical squares, the flippable graph on  $\{3, \dots, n\}$  is dense and expanding and only locally correlated with the frame (an adjacent pair  $\pi(x) = y$  is never flippable, by §17.12.9(iii)); stationarity then makes it typical at every time, and a comparison of the thinned walk with the full one (Aldous’ spectral-gap theorem for the interchange process on a dense expander, or a direct path argument) would give  $T^M \varepsilon \rightarrow 0$ . This is a second, independent formulation of (II), with the advantage that its hypotheses are about pairs of rows of a random Latin square — objects the switching method of Proposition 17.40 can reach — rather than about the joint position of intercalates. Numerical test: run  $T$  on real squares and compare the autocorrelation of the bit, as a function of the number of trades performed, with  $\langle \varepsilon, P^m \varepsilon \rangle$  of Theorem 18.1; and measure the flippable fraction and its correlation with the frame.

**The trade chain, continued: what a proof needs.** Three observations sharpen the route.

(i) *EQ from the autocorrelation.* Since  $T$  is symmetric,  $\mathbb{E}_X[\varepsilon(L_0)\varepsilon(L_{2M})] = \|T^M \varepsilon\|_{L^2(X)}^2 \geq \langle \varepsilon, 1 \rangle^2 = (\mathbb{E}_X \varepsilon)^2$ . So EQ( $\delta$ ) follows from  $\mathbb{E}_X[\varepsilon(L_0)\varepsilon(L_{2M})] \leq \delta^2$  for a single  $M$  — a stationary two-time statement, the exact analogue for the trade chain of Corollary 18.2 for the walk, and the quantity `s13.trade.py` measures.

(ii) *Which frame moves are free.* A turn of  $C_x$  toggles the flippability of  $(x, y)$  only when  $y \notin C_x$ ; when  $x, y$  lie in one cycle the turn conjugates  $\rho_{x,y}$  by  $(j j')$  and flippability is unchanged. Hence combining a turn with a trade frees exactly the *merges*  $(x, y)$  for which  $C_x$  or  $C_y$  avoids rows 1, 2; *every split* is constrained (needs flippability), and so is the merge of  $C_1$  with  $C_2$  in the apart state. In the together state the bit-changing moves are the separating splits of the common cycle; in the apart state they are the merges  $C_1$ – $C_2$ . [An earlier version of this paragraph claimed that turning the common cycle frees the separating splits; it does not.]

(iii) *Flippability, exactly.* Let  $a = L(x, j)$ ,  $b' = L(y, j')$  and  $q^*$  the column where row  $x$  holds  $b'$  ( $q^* \neq j$  when  $\pi(x) \neq y$ ). Following the  $\rho_{x,y}$ -cycle of  $j$ , column  $j'$  is reached exactly at the step after  $q^*$ ; so  $(x, y)$  is flippable iff  $q^*$  does not lie on the  $\rho_{x,y}$ -cycle of  $j$ . For a random-like  $\rho_{x,y}$  this has probability  $\frac{1}{2}$ , which is what is measured (0.500–0.511 in every class: separating, non-separating, merging, other; and exactly 0 for adjacent pairs). A turn of  $C_1$  (the move  $X \rightarrow J$  of Proposition 17.40) toggles the flippability of *every* merging pair simultaneously, so  $\mathbb{P}_X[\text{no flippable merge} \mid \text{apart}] \leq 2\mathbb{P}_J[\text{all merges flippable}]$ ; this does not by itself show either is small.

What a proof must supply is therefore: (F1) under  $\text{Unif}(X)$ , a positive fraction of the bit-changing pairs — the merging pairs  $C_1 \times C_2$  in the apart state and the separating pairs of the common cycle in the together state — and of the other splitting pairs, are flippable, except on a set of squares of measure  $o(1/\log^2 n)$ ; and (F2) a mixing argument for the frame walk in which all moves are available at rate  $\asymp 1/n^2$  except the constrained ones, which are available at rate  $\geq c/n^2$  on the event (F1). Both are statements about a pair of rows and a pair of columns of a random Latin square, with no intercalates, orbits or expanders; (F1) is the kind of statement the switching method reaches. Measured on real squares with true marks ( $n = 30$ , 900 instances, `runs/s13/trade30.log`): flippable fraction 0 for adjacent pairs and 0.5005, 0.4999 for same-cycle and different-cycle pairs; the autocorrelation of the bit after  $m$  trades is 0.559, 0.413, 0.307, 0.250, 0.170, 0.121, 0.069, 0.048, 0.025, 0.007, 0.004, 0.002 at  $m = 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 64$ , against 0.617, 0.454, 0.345, 0.279, 0.192, 0.143, 0.086, 0.058, 0.026, 0.011, 0.003, 0.001 for the walk with iid transpositions avoiding  $u, v$  from a uniform derangement frame: the chain decorrelates slightly faster (the adjacency exclusion again) and reaches the noise floor by  $m \approx 32$ , with no plateau. [With the symbol/column bug the same experiment showed a plateau at  $\approx 0.09$ : the frozen instances were exactly those with  $\pi(1) = 2$  or  $\pi(2) = 1$ , impossible for a true mark, on which the chain — which preserves  $\pi^{-1}(1)$  and  $\pi^{-1}(2)$  — cannot move the bit.]

**(F1): what was tried, and the enlarged chain.** A direct switching proof of (F1) was attempted and not found: the natural moves at a non-flippable pair  $(x, y)$  (trading the  $\rho_{x,y}$ -cycle through  $j$ , which now contains  $j'$  too) conjugate the frame by  $(xy)$  and change the flippability of every pair involving  $x$  or  $y$  in an uncontrolled way, so a graded double count on the number of flippable bit-changing pairs does not close. Three observations frame what is true. (a) *Conjugating trades are legal moves.* Adding to  $T$  the trades at non-flippable pairs (both  $j, j'$  in the traded cycle) keeps the chain symmetric; such a trade exchanges the positions of  $x$  and  $y$  in the frame without changing the bit, and changes rows  $x, y$ . The enlarged chain  $T''$  therefore relabels the rows sitting in each frame position (by the conjugations) while merging and splitting only at flippable pairs; a bit-changing position (a separating pair of positions of the common cycle, or a pair of positions in  $C_1 \times C_2$ ) is visited by many different row pairs over time, so for the bit to stay frozen *every* row pair that ever occupies such a position must be non-flippable. In the extreme, a square in which every pair of rows  $x, y \notin \{1, 2\}$  is non-flippable (“locked”:  $j, j'$  lie in a common cycle of  $\rho_{x,y}$  for all  $x, y$ ) has a  $T''$ -component on which the bit is constant; locked squares should be rare beyond any polynomial, but no bound is proved. (b) *Components.* Since  $T''$  is symmetric,  $\lim_M \mathbb{E}_X[\varepsilon(L_0)\varepsilon(L_{2M})] = \mathbb{E}_X[\mathbb{E}[\varepsilon \mid \text{component}]^2]$ , so EQ reduces to the conditional fairness of the bit on the components of  $T''$  — a purely combinatorial statement about the orbits of a group of involutions on  $X$ , each of which fixes rows 1, 2 and the mark. (c) *Numerics* ( $n = 30$ , 240 instances, true marks): the flippable fraction of all pairs of rows  $3..n$  is 0.467 on average (the adjacency exclusion accounts for the deficit from  $\frac{1}{2}$ ) with minimum 0.415 and maximum 0.550 over instances — tightly concentrated; exactly one instance had bit-changing pairs with none flippable (an apart state with a tiny  $C_1 \times C_2$ ). So (F1) is true with margin in the data; what is missing is a structural reason, and the cleanest target is (b).

### 17.12.12 The components of the trade chain: exact enumeration at $n = 6, 7$

The component question (b) above — restated application-free in `paper/fairness.question.tex` — was settled numerically by exact enumeration (`s13.comp7.c`; logs in `runs/s13/comp7/`). For a fixed first-row pair  $R$  and columns  $j, j'$  satisfying  $\pi(1) \neq 2$ ,  $\pi(2) \neq 1$ , all completions  $L \in S(R)$

were generated ( $|S(R)| \approx 6.6 \cdot 10^6$  at  $n = 7$ ), and union-find was run over the moves: trades at all  $x < y$  in  $\{3, \dots, n\}$  along the  $\rho_{x,y}$ -cycle through  $j'$  (a left multiplication  $(xy) \circ \pi$  when flippable, a conjugation otherwise), turns of the  $\pi$ -cycles avoiding rows 1, 2 (move set A), and additionally trades along the  $\rho_{x,y}$ -cycle through  $j$  when flippable (right multiplication; move set B).

**The rectangle picture.** Delete columns  $j, j'$ . Each row of the  $n \times (n-2)$  rectangle misses two symbols and each symbol is missing from two rows, so the leftover row-symbol graph is a union of even cycles; its cycles are the frame cycles, the completions of the rectangle are the  $2^{\#\text{cycles}}$  orientations, a turn is a change of orientation, and the bit is a function of the rectangle. A trade acts on the rectangle by swapping rows  $x, y$  along a path of their row-pair structure;  $(x, y)$  is flippable iff the two  $(x, y)$ -paths pair their ends “parallel” ( $a_x \rightarrow a_y$ , with  $a_x = L(x, j')$ ) rather than “crossed”. Consequently, for  $x, y$  in different frame cycles, turning either cycle toggles flippability: the thinning is immaterial for every pair meeting a free cycle, and is a genuine constraint only for pairs inside  $C_1 \cup C_2$ , i.e. for the bit-changing pairs, where it is an intrinsic property of the rectangle. Applied to the pair (1, 2) itself: the mark lies inside a  $\sigma$ -cycle iff (1, 2) is crossed iff the (1, 2)-trade is a conjugation; for cross-cycle marks the (1, 2)-trade flips the bit, and composed with the isotopy (reflect the  $\sigma$ -cycle through  $j$  about  $j$ , adjust symbols) that restores  $R$  it is an exact bit-flipping bijection of  $S(R)$  commuting with all moves.

**$n = 6$  is degenerate.** A trade preserves the row parity of  $L$  (number of odd row permutations mod 2: both traded rows change sign by  $(-1)^{|Q|-1}$ ), while a turn of a cycle  $c$  changes it by  $|c|$ . At  $n = 6$  the exclusion  $\pi(1) \neq 2, \pi(2) \neq 1$  forces every free cycle to be a 2-cycle, so row parity is invariant and  $S(R)$  splits into two classes (sizes e.g.  $3456 + 576 + 576, 2304 + 1920, 2112 + 1728 + 192$ ); the within-cycle values 0.004–0.14 of the left-hand side of the Question at  $n = 6$  are artefacts of this.

**$n = 7$ : one giant component plus frozen dust.** Six instances ( $\sigma$  of types (7) with  $\alpha = 2, 3; (5, 2)$  within and across;  $(4, 3)$  within and across;  $(3, 2, 2)$  across), each of  $6.54\text{--}6.64 \cdot 10^6$  squares:

| type      | mark         | $\mathbb{P}[\beta=1 \mid R]$ | giant (A) | $p(K_{\text{giant}})$ | dust (A)           | LHS (A) | LHS (B) |
|-----------|--------------|------------------------------|-----------|-----------------------|--------------------|---------|---------|
| (7)       | $\alpha = 2$ | 0.4971                       | 0.9912    | 0.4968                | 0.0080 (423 comps) | 0.0076  | 0.0074  |
| (7)       | $\alpha = 3$ | 0.5018                       | 0.9917    | 0.5019                | 0.0075 (402)       | 0.0069  | 0.0068  |
| (5, 2)    | $\alpha = 2$ | 0.4977                       | 0.9960    | 0.4980                | 0.0035 (197)       | 0.0034  | 0.0034  |
| (4, 3)    | $\alpha = 2$ | 0.4930                       | 0.9927    | 0.4932                | 0.0067 (373)       | 0.0069  | 0.0068  |
| (5, 2)    | cross        | 0.5000                       | 0.9956    | 0.5000                | 0.0040 (221)       | 0.0038  | 0.0038  |
| (4, 3)    | cross        | 0.5000                       | 0.9922    | 0.5000                | 0.0072 (393)       | 0.0070  | 0.0069  |
| (3, 2, 2) | cross        | 0.5000                       | 0.9928    | 0.5000                | 0.0068 (369)       | 0.0067  | 0.0067  |

In every instance there is one component containing 99.1–99.7% of  $S(R)$ , on which all three Latin-square parities and  $\text{sgn } \pi$  take all values and  $p(K)$  agrees with  $\mathbb{P}[\beta = 1 \mid R]$  to four decimals; the small deviations of  $\mathbb{P}[\beta = 1 \mid R]$  from  $\frac{1}{2}$  in the within-cycle rows are the finite- $n$  marked-pair ratios of Theorem 11.4, and the cross-cycle rows are exactly  $\frac{1}{2}$  by the involution above. The remaining 0.3–0.8% (“dust”) sits in hundreds of components of sizes 120–960 and contributes essentially the whole left-hand side. Dumping representatives shows what the dust is: squares in which  $\rho_{x,y}$  is a single 7-cycle for all, or all but one or two, pairs of rows (isotopes of  $\mathbb{Z}_7$  and other row-Hamiltonian squares), so that every trade swaps two whole rows (a conjugation), no pair of rows 3..7 is flippable, and the component is the orbit of the  $5!$  permutations of rows

3..7 — the “locked” squares of (a). Move set B merges some dust into the giant component but changes nothing qualitatively.

**Consequences.** (i) The component structure is not where the difficulty lies: for large  $n$  one expects a single component carrying all but a super-polynomially small part of  $S(R)$  (locked squares are an exponentially small fraction of Latin squares), and then  $p(K) = \mathbb{P}[\beta = 1 \mid R] + o(1)$ . The Question with  $R$  fixed is thus the *quenched* statement  $\mathbb{P}[\beta = 1 \mid R] = \frac{1}{2} + o(1)$  uniformly over admissible  $R$ , and the averaged form of Remark 3(i) of the fairness note is exactly EQ; the chain is a tool for proving it (through the bit autocorrelation), not a reformulation that weakens it. (ii) What must be controlled is the dust-like failure of (F1) in a quantitative form: the stationary probability that no bit-changing pair is flippable must be  $o(1/\log^2 n)$ ; at  $n = 7$  it is  $< 0.008$  and consists of near-group squares. (iii) By the rectangle picture this is a statement about the parallel/crossed pairing of row-pair paths inside  $C_1 \cup C_2$  in a random Latin rectangle, which is the natural target for a Kwan–Sudakov switching as in Proposition 17.40.

## 18 The stationary marked chain, solved exactly: the spectral measure of the same-cycle sign

Session 13 took up the spectral route of §17.5 in the form that the scouting of §17.12.8 says is the true one: the frame is a (near-)uniform permutation of the  $n$  rows, the pool is a set of transpositions, and the quantity the second-moment bound of Theorem 16.1 needs is  $\mathbb{E}[\varepsilon_S \varepsilon_T]$  with the start *stationary*. In that setting the chain is the random-transposition walk on  $S_N$  observed through the function “ $u$  and  $v$  lie in a common cycle”, and the walk is diagonalised by the characters of  $S_N$ . It turns out that the spectral measure of this function can be written down in closed form. Everything in this section is at finite  $N$ ; no continuum limit is taken, and nothing from §§17–17.12 is used.

### 18.1 Setting

Let  $N \geq 2$ , let  $u \neq v$  be two fixed points of  $[N]$ , and on  $L^2(S_N)$  (uniform measure, inner product  $\langle f, g \rangle = \frac{1}{N!} \sum_{\sigma} f(\sigma)g(\sigma)$ ) let

$$\varepsilon(\sigma) = 2 \cdot \mathbf{1}\{u \sim_{\sigma} v\} - 1 \in \{\pm 1\}, \quad (Pf)(\sigma) = \binom{N}{2}^{-1} \sum_{\tau \text{ transposition}} f(\sigma\tau).$$

$P$  is the transition operator of the random-transposition walk (right multiplication by a uniform transposition), self-adjoint on  $L^2(S_N)$  with  $\|P\| \leq 1$ . For a partition  $\lambda \vdash N$  let  $\chi_{\lambda}$ ,  $d_{\lambda}$  be its character and dimension, let  $\Pi_{\lambda} = \frac{d_{\lambda}}{N!} \sum_{\rho} \chi_{\lambda}(\rho) R(\rho)$ ,  $(R(\rho)f)(\sigma) = f(\sigma\rho)$ , be the projection onto the  $\lambda$ -isotypic component of the right-regular representation, and let

$$r_{\lambda} = \frac{\chi_{\lambda}(\tau)}{d_{\lambda}} = \frac{2}{N(N-1)} \sum_{(i,j) \in \lambda} (j-i)$$

be the eigenvalue of  $P$  on that component (the normalised character at a transposition; the sum is over the boxes of  $\lambda$ ,  $j-i$  the content of the box). Then  $P = \sum_{\lambda} r_{\lambda} \Pi_{\lambda}$  and for every  $f$ ,

$$\|P^d f\|^2 = \sum_{\lambda} r_{\lambda}^{2d} \|\Pi_{\lambda} f\|^2, \quad \langle f, P^m f \rangle = \sum_{\lambda} r_{\lambda}^m \|\Pi_{\lambda} f\|^2.$$

Write  $\nu_f(\lambda) = \|\Pi_\lambda f\|^2$  for the spectral measure.

The connection with the rest of these notes:  $\langle \varepsilon, P^m \varepsilon \rangle = \mathbb{E}[\varepsilon(\sigma) \varepsilon(\sigma \tau_1 \cdots \tau_m)]$  for  $\sigma$  uniform, which is  $\mathbb{E}_\pi[\Phi_m]$ , the annealed parity of the number of flips in  $m$  steps from the stationary start (the discrete marked chain of §17.5 is the lumping of this walk), and  $\|P^d \varepsilon\|^2 = \psi_\pi(d)$  is the stationary-start mixing functional of Proposition 17.6. Self-adjointness gives  $\psi_\pi(d) = \mathbb{E}_\pi[\Phi_{2d}]$ : at stationarity the second moment of the quenched parity is the first moment at twice the time.

## 18.2 The theorem

**Theorem 18.1** (spectral measure of the same-cycle sign).  $\nu_\varepsilon(\lambda) \neq 0$  only for  $\lambda = (a, b, 1^j)$  with  $a \geq b \geq 1$ ,  $a + b + j = N$ , and for these

$$\nu_\varepsilon(\lambda) = 2c_\lambda^2(1 - r_\lambda), \quad c_\lambda = \frac{a - b + 1}{(a + j + 1)(b + j)}, \quad r_\lambda = \frac{a(a - 1) + b(b - 3) - j(j + 3)}{N(N - 1)}.$$

Here  $c_\lambda = |\langle c, \chi_\lambda \rangle|$  is the Fourier coefficient of the number-of-cycles function  $c(\sigma)$ . In particular  $\sum_\lambda \nu_\varepsilon(\lambda) = 1$  and

$$\mathbb{E}_\pi[\Phi_m] = \langle \varepsilon, P^m \varepsilon \rangle = \sum_{a \geq b \geq 1, a+b+j=N} \frac{2(a - b + 1)^2}{(a + j + 1)^2(b + j)^2} (1 - r_\lambda) r_\lambda^m, \quad \psi_\pi(d) = \mathbb{E}_\pi[\Phi_{2d}].$$

*Proof. Step 1: the sign is a difference of the cycle-count function.* Right multiplication by the transposition  $(uv)$  splits the cycle through  $u$  and  $v$  if  $u \sim_\sigma v$  and merges their two cycles otherwise, so with  $c(\sigma)$  the number of cycles,

$$\mathbf{1}\{u \sim_\sigma v\} = \frac{1}{2}(1 + c(\sigma(uv)) - c(\sigma)), \quad \text{i.e.} \quad \varepsilon = (R((uv)) - I)c.$$

*Step 2: Fourier coefficients of  $c$ .*  $c$  is a class function,  $c = \sum_\lambda \hat{c}_\lambda \chi_\lambda$  with  $\hat{c}_\lambda = \langle c, \chi_\lambda \rangle$ . By the classical identity  $\sum_\sigma x^{c(\sigma)} \chi_\lambda(\sigma) = d_\lambda \prod_{\square \in \lambda} (x + \text{cont}(\square))$  (the content polynomial; both sides equal  $N! s_\lambda(1^x)$ ), differentiating at  $x = 1$  gives

$$\hat{c}_\lambda = \frac{d_\lambda}{N!} \frac{d}{dx} \Big|_{x=1} \prod_{\square \in \lambda} (x + \text{cont}(\square)) = \frac{d_\lambda}{N!} \sum_{\square} \prod_{\square' \neq \square} (1 + \text{cont}(\square')).$$

A factor vanishes at  $x = 1$  exactly for the boxes of content  $-1$ , which are the boxes  $(i + 1, i)$ . If two such boxes lie in  $\lambda$  the derivative vanishes. No such box:  $\lambda = (N)$ . Exactly one, the box  $(2, 1)$ :  $\lambda_2 \geq 1$  and  $\lambda_3 \leq 1$ , i.e.  $\lambda = (a, b, 1^j)$ ,  $a \geq b \geq 1$ , and then  $\hat{c}_\lambda = \frac{d_\lambda}{N!} \prod_{\square \neq (2,1)} (1 + \text{cont}) = \frac{d_\lambda}{N!} a! (b - 1)! (-1)^j j!$  (row 1 has contents  $0, \dots, a - 1$ ; row 2 without its first box  $0, \dots, b - 2$ ; the column  $-2, \dots, -(j + 1)$ ). The hook lengths of  $(a, b, 1^j)$  are  $a + j + 1$  at  $(1, 1)$ ;  $a - c + 2$  at  $(1, c)$  for  $2 \leq c \leq b$ ;  $a - c + 1$  at  $(1, c)$  for  $c > b$ ;  $b + j$  at  $(2, 1)$ ;  $b - c + 1$  at  $(2, c)$ ,  $c \geq 2$ ; and  $j, j - 1, \dots, 1$  down the column. Hence  $N!/d_\lambda = \prod \text{hooks} = (a + j + 1)(b + j) a! (b - 1)! j! / (a - b + 1)$  and

$$\hat{c}_\lambda = (-1)^j \frac{a - b + 1}{(a + j + 1)(b + j)} \quad (\lambda = (a, b, 1^j)), \quad \hat{c}_{(N)} = H_N, \quad \hat{c}_\lambda = 0 \text{ otherwise.}$$

*Step 3: the projections.*  $\Pi_\lambda$  is right-convolution by a central element and so commutes with  $R((uv))$ ; hence  $\Pi_\lambda \varepsilon = (R((uv)) - I) \Pi_\lambda c = \hat{c}_\lambda (R((uv)) - I) \chi_\lambda$ , and for  $\lambda = (N)$  this is 0 (so

$\nu_\varepsilon((N)) = 0$ , consistent with  $\langle \varepsilon \rangle = 0$ :  $\mathbb{P}[u \sim v] = \frac{1}{2}$  exactly). For the norm, with  $\|\chi_\lambda\| = 1$  and the generalised orthogonality relation  $\frac{1}{N!} \sum_\sigma \chi_\lambda(\sigma) \chi_\lambda(\sigma\rho) = \chi_\lambda(\rho)/d_\lambda$ ,

$$\|(R((uv)) - I)\chi_\lambda\|^2 = 2 - 2 \frac{\chi_\lambda((uv))}{d_\lambda} = 2(1 - r_\lambda),$$

so  $\nu_\varepsilon(\lambda) = 2\hat{c}_\lambda^2(1 - r_\lambda)$ . *Step 4: the contents.* Row 1 contributes  $a(a-1)/2$ , row 2 contributes  $\sum_{c=1}^b (c-2) = b(b-3)/2$ , the column  $\sum_{i=1}^j (-1-i) = -j(j+3)/2$ ; multiply by  $2/(N(N-1))$ .  $\sum_\lambda \nu_\varepsilon(\lambda) = \|\varepsilon\|^2 = 1$ .  $\square$

*Status: PROVED. VERIFIED three ways (s13\_spectral.py, s13\_spectral\_big.py): (i) by brute force over  $S_N$  for  $N = 5, 6, 7$ , applying  $P$  directly to  $\varepsilon$  and comparing  $\|P^d \varepsilon\|^2$  and  $\langle \varepsilon, P^d \varepsilon \rangle$  for  $d \leq 4$  with the formula — agreement to all printed digits (e.g.  $N = 7$ ,  $d = 3$ : 0.047319 and 0.155229); (ii)  $\sum \nu = 1$  to  $10^{-12}$  at  $N = 400, 1000, 4000$ ; (iii) against the frame simulation of §17.12.8 (s13\_frame mode 2,  $N = 4096$ ,  $10^6$  instances, disjoint chords): with  $M = |\mathcal{S} \Delta T| \sim \text{Bin}(k, \frac{1}{2})$ ,  $\mathbb{E}[\varepsilon_{\mathcal{S} \Delta T}] = \mathbb{E}_M[\langle \varepsilon, P^M \varepsilon \rangle] = 0.3633, 0.2132, 0.1151, 0.0596, 0.0302$  predicted against 0.3632, 0.2123, 0.1159, 0.0605, 0.0301 measured ( $\pm 0.001$ ) at  $k = 8, 16, 32, 64, 128$ .*

### 18.3 Consequences

**Corollary 18.2** (the stationary parity decays like  $1/m$ , uniformly in  $N$ ). *For all  $N \geq 8$  and  $m \geq 1$ ,*

$$0 \leq \psi_\pi(d) = \mathbb{E}_\pi[\Phi_{2d}] \quad \text{and} \quad |\mathbb{E}_\pi[\Phi_m]| \leq \frac{12}{m} + 32 \left( \frac{4 \log(m+1)}{m} + \frac{3}{N} \right)^4 + \frac{1}{(m+1)^4}.$$

Moreover, as  $N \rightarrow \infty$  the spectral measure converges to the law of  $r = 1 - 2(1-\alpha)(1-\beta)$  under the density  $4(\alpha-\beta)^2/((1-\alpha)(1-\beta)) d\alpha d\beta$  on  $\{\alpha \geq \beta \geq 0, \alpha+\beta \leq 1\}$  (total mass 1), whose mass within  $\delta$  of 1 is  $2\delta(1+o(1))$ ; consequently  $\mathbb{E}_\pi[\Phi_m] \rightarrow \iint 4 \frac{(\alpha-\beta)^2}{(1-\alpha)(1-\beta)} (1 - 2(1-\alpha)(1-\beta))^m d\alpha d\beta$  and  $m \mathbb{E}_\pi[\Phi_m] \rightarrow 2$  as  $m \rightarrow \infty$  after  $N \rightarrow \infty$ .

*Proof.* Index the shapes by  $k = b + j = N - a$  (the number of boxes outside the first row) and  $b$ . *Upper bound on the level mass.* Since  $a - b + 1 \leq a + j + 1$ ,  $c_\lambda \leq 1/k$ ; and  $N(N-1)(1-r_\lambda) = k(2N-k-1) + j^2 + 3j - b^2 + 3b \leq 2kN + k^2 + 3k \leq 3kN$  when  $k \leq N-3$  (using  $j \leq k$ ,  $j+b=k$ ), while for  $k \geq N-2$  the trivial  $1-r_\lambda \leq 2 \leq 6k/N$  holds; so  $1-r_\lambda \leq 6k/N$  and  $\nu_\varepsilon(\lambda) \leq 12/(kN)$ . There are at most  $k$  shapes at level  $k$ , so the level mass is  $\leq 12/N$ . *Lower bound on the gap.* Since  $b \leq k$  and  $b \leq a = N-k$ ,  $b^2 \leq k(N-k)$ , so  $N(N-1)(1-r_\lambda) \geq k(2N-k-1) - b^2 \geq k(2N-k-1) - k(N-k) = k(N-1)$ , i.e.  $r_\lambda \leq 1 - \frac{k}{N}$  for every shape. Hence, summing the non-negative- $r$  part,

$$\sum_{\lambda: r_\lambda \geq 0} \nu_\varepsilon(\lambda) r_\lambda^m \leq \frac{12}{N} \sum_{k \geq 1} e^{-km/N} = \frac{12}{N} \cdot \frac{1}{e^{m/N} - 1} \leq \frac{12}{m}.$$

*The negative part.* For  $r_\lambda < 0$  we bound  $\nu|r|^m \leq \nu(1-\delta)^m$  unless  $1+r_\lambda \leq \delta$ . Now  $N(N-1)(1+r_\lambda) = N(N-1) - j(j+3) + a(a-1) + b(b-3) \geq (a+b)(N+j) - N - 3j - 2 \geq (a+b-1)N - 2$  for  $a+b \geq 3$  (the last step needs  $(a+b-3)j \geq 0$ ; the only shape with  $a+b=2$  is the sign character  $(1^N)$ , with  $r = -1$  and  $\nu = 4/(N^2(N-1)^2)$ , which is included in the count below), so  $1+r_\lambda \leq \delta$  forces  $a+b \leq \delta N + 3$ ; such shapes have  $a \leq \delta N + 2$ ,  $b+j \geq N/2$ ,  $a+j+1 \geq N/2$ , hence  $\nu_\varepsilon(\lambda) \leq 4c_\lambda^2 \leq 64(\delta N + 3)^2/N^4$ , and there are at most  $(\delta N + 3)^2/2$  of them;

so  $\nu_\varepsilon([-1, -1+\delta]) \leq 32(\delta+3/N)^4$ . With  $\delta = 4 \log(m+1)/m$  the remainder is  $(1-\delta)^m \leq (m+1)^{-4}$ . *Continuum limit.* With  $a = \alpha N$ ,  $b = \beta N$ ,  $j = (1 - \alpha - \beta)N$ :  $c_\lambda \rightarrow (\alpha - \beta)/((1 - \beta)(1 - \alpha)N)$ ,  $1 - r_\lambda \rightarrow 2(1 - \alpha)(1 - \beta)$ , and the number of shapes in  $d\alpha d\beta$  is  $N^2 d\alpha d\beta$ ; the mass near  $r = 1$  comes from  $s = 1 - \alpha \leq \delta/2$  with  $\beta \leq s$ , where the density is  $4/s$ , giving  $\int_0^{\delta/2} 4 ds = 2\delta$ , and the same computation gives  $\int 4(1-2s)^m ds = 2/(m+1)$  for the leading term. (The total mass 1 and the values  $m\mathbb{E}[\Phi_m] = 1.634, 1.896, 1.973, 1.993$  at  $m = 8, 32, 128, 512$  were checked by numerical integration.)  $\square$

*Status: PROVED. Remarks.* (i) The rate  $1/m$  is sharp: the lower bound of Proposition 17.6 is reproduced by the linear spectral density, with the exact constant: the top-of-spectrum density is 2 (mass  $2\delta$  within  $\delta$  of 1), against the trial-function bound of §17.5. (ii) The periodicity caveat of §17.5 is void for  $\varepsilon$ : the mass within  $\delta$  of  $-1$  is  $O(\delta^4)$  (numerically  $2 \times 10^{-5}$  at  $\delta = 0.2$ ), so the sign character and its neighbours carry essentially none of  $\varepsilon$ . (iii) The whole spectral measure lives on the two-row-plus-column shapes  $(a, b, 1^j)$ : this is the exact statement behind “the environment enters only through the block count” (§17.9.1) — the function  $\mathbf{1}\{u \sim v\}$  is a first difference of the cycle count, and the cycle count is supported on these shapes. (iv) Nothing here is asymptotic: the bound holds at the finite  $N = n$  of the actual problem, which matters because the frame of an orbit is a permutation of  $n$  rows and the pool has  $k = \Theta(\log^{11} n)$  chords, so “continuum” is not even the right regime; the only asymptotic statement, the continuum limit, is a remark.

**Corollary 18.3** (the stationary iid model of Problem 15.8). *Let  $\sigma$  be uniform on  $S_N$ , let  $\tau_1, \dots, \tau_k$  be  $k$  uniformly random pairwise disjoint transpositions avoiding  $u, v$ , let  $S, T \subseteq [k]$  be independent uniform subsets and  $\varepsilon_S = \varepsilon(\sigma \prod_{i \in S} \tau_i)$ . Then*

$$4\mathbb{E}[\text{bias}^2] = \mathbb{E}[\varepsilon_S \varepsilon_T] \leq \frac{C}{k} + \frac{Ck^2}{N}, \quad \mathbb{E}|\text{bias}| \leq \sqrt{\mathbb{E}[\text{bias}^2]} \leq \frac{C'}{\sqrt{k}} + \frac{C'k}{\sqrt{N}},$$

with absolute constants. The same holds for the two-sided pools of Theorem 15.6 (chords on both columns  $j, j'$ , with shared rows and conjugating coordinates), with  $k$  replaced by  $k' = |\mathcal{I}_j| + |\mathcal{I}_{j'}|$ , the number of non-conjugating useful coordinates.

*Proof.* The  $\tau_i$  commute, so with  $\alpha = \sigma\tau_S$  (uniform, independent of the  $\tau$ 's) one has  $\sigma\tau_T = \alpha\tau_{S\Delta T}$  and  $\mathbb{E}[\varepsilon_S \varepsilon_T] = \mathbb{E}[\varepsilon(\alpha)\varepsilon(\alpha\tau_{S\Delta T})]$ , where  $M = |S\Delta T| \sim \text{Bin}(k, \frac{1}{2})$  and  $\tau_{S\Delta T}$  is a product of  $M$  uniformly random disjoint transpositions avoiding  $u, v$ . Couple these with  $M$  iid uniform transpositions: the iid ones are pairwise disjoint and avoid  $u, v$  except with probability  $\leq M^2 \cdot 4/N + M \cdot 4/N$ , and on that event the products agree. Hence  $\mathbb{E}[\varepsilon_S \varepsilon_T] \leq \mathbb{E}_M \langle \varepsilon, P^M \varepsilon \rangle + 8k^2/N$ , and by Corollary 18.2 and  $\mathbb{P}[M < k/4] \leq e^{-k/8}$  the first term is  $\leq \mathbb{P}[M < k/4] + \max_{m \geq k/4} (12/m + O(m^{-4} \log^4 m) + O(N^{-4})) \leq C/k$ . For the two-sided pool,  $\pi_0 R_S M_S$  is uniform and  $\pi_0 R_T M_T = (\pi_0 R_S M_S) \cdot (M_S R_{S\Delta T} M_S) M_{S\Delta T}$ ; off the  $O(k^2/N)$  event that  $M_S$  shares a row with  $R_{S\Delta T}$  the conjugation is trivial, and a conjugating coordinate  $i \in \mathcal{I}_{jj'} \cap (S\Delta T)$  contributes the same transposition to both  $R_{S\Delta T}$  and  $M_{S\Delta T}$ , where the two copies cancel; so the product is  $|(S\Delta T) \setminus \mathcal{I}_{jj'}| \sim \text{Bin}(k', \frac{1}{2})$  disjoint transpositions, uniform up to the same correction. (With a pool consisting only of conjugating coordinates the bit is frozen and  $\mathbb{E}[\varepsilon_S \varepsilon_T] = 1 - O(k^2/N)$ , so  $k'$  cannot be replaced by  $k$ .)  $\square$

*Status: PROVED. This is the iid-model core of Problem 15.8, in the stationary-frame model that §17.12.8(T3) identifies as the right one, with the sharp exponent  $c = \frac{1}{2}$  (budget:  $c > 1/11$ ).*

It is obtained without Theorem 16.1's Chernoff splitting, without the split-merge chain, without Problems 16.4, 17.9, without (R1), (C1), (C2'), (S<sub>2</sub>), the occupation-time lemma or the Markov-renewal apparatus of §§17.9–17.11: none of those is needed for the stationary start. The one-block-start version of Problem 16.4 ( $\psi(j, d)$  from  $\mu_j$ ) is not covered and is now moot. What remains of item (I) is nothing; what remains of the programme is item (II) — the transfer (T1)–(T2) of §17.12.8: that the frame  $\pi_P(L)$  and the pool under  $\text{Unif}(S(\lambda))$  behave, for the purposes of  $\mathbb{E}[\varepsilon(\alpha)\varepsilon(\alpha\rho)]$ , like a uniform permutation and independent uniform chords. (The supply of Lemma 15.9 is of intercalates through  $j$  avoiding  $j'$ , i.e. of non-conjugating coordinates, so  $k' \geq 6 \log^{11} n$  is what the lemma provides.) The exact formula makes that hypothesis sharply testable on real squares: the model predicts  $\mathbb{E}[\varepsilon_S \varepsilon_T] = \mathbb{E}_M\langle \varepsilon, P^M \varepsilon \rangle$  to three digits.

**Remark 18.4** (what the exact formula says about the earlier picture). (a) The “gap-trapping” mechanism of §17.3 is the top of the spectrum: the shapes  $(N - k, b, 1^j)$  with  $k \ll N$  have  $1 - r_\lambda \approx 2k/N$  and level mass  $\approx 4/N$ , i.e. a linear density 2 at  $r = 1$ , and  $k/N$  is the gap scale. (b) The measured  $\psi(256, 256) \cdot 256 \approx 0.7$  and the real-square constant  $\mathbb{E}[\text{bias}^2]k \approx 0.7\text{--}0.9$  are both  $\frac{1}{4}\mathbb{E}_M[\Phi_M]k$  with  $\mathbb{E}[\Phi_m] \approx 1.9/m$ ,  $M \approx k/2$ :  $\approx 0.95$ , with the finite- $m$  corrections of the table in §17.12.8. (c) The quenched quantities studied in §§17.9–17.11 ( $a_k$ ,  $t_k(w)$ , the post-flip chain) are not needed, but the exact formula gives a handle on them too:  $(P^m \varepsilon)(\sigma_0) = \sum_\lambda r_\lambda^m \hat{c}_\lambda[\chi_\lambda(\sigma_0(uv)) - \chi_\lambda(\sigma_0)]$  for any fixed start  $\sigma_0$ , which for the  $N$ -cycle reduces to hook characters and is Theorem 17.14.

## 19 The ladder identity: the bias is carried by the squares with no flippable ladder pair

Session 13, after the component enumeration of §17.12.12. This section proves an exact identity that replaces the whole chain §18–§17.12.12 (pool, master formula, annealed bias, transfer, trade chain) by a single combinatorial statement. Notation as in §17.12.10:  $(L, P) \in X = X(n, \lambda; \alpha, \beta)$ ,  $P = \{j, j'\}$ , frame  $\pi = \pi_P$  ( $L(\pi(r), j') = L(r, j)$ ),  $C_r$  the  $\pi$ -cycle of row  $r$ ,  $\rho_{x,y}$  the column permutation of rows  $x, y$  ( $L(y, \rho_{x,y}(q)) = L(x, q)$ ),  $(x, y)$  *flippable* iff  $j, j'$  lie in different cycles of  $\rho_{x,y}$ , and the  $j'$ -trade at  $(x, y)$  exchanges rows  $x, y$  on the  $\rho_{x,y}$ -cycle through  $j'$ , which replaces  $\pi$  by  $(xy) \circ \pi$  if  $(x, y)$  is flippable and by  $(xy) \circ \pi \circ (xy)$  otherwise.  $B = \{1 \sim 2\}$  (rows 1, 2 in a common frame cycle),  $A$  its complement,  $\mathbf{1}_B$  the bit; for  $B$ -instances  $d_{12}$  and  $d_{21}$  are the numbers of  $\pi$ -steps from 1 to 2 and from 2 to 1.

**Definition 19.1** (ladder). Put  $K_0 = \min(|C_1|, |C_2|)$  on  $A$  and  $K_0 = \min(d_{12}, d_{21})$  on  $B$ , and for  $1 \leq k < K_0$  let

$$x_k = \pi^{-k}(1), \quad y_k = \pi^{-k}(2), \quad T_k = \text{the } j'\text{-trade at } (x_k, y_k).$$

The pairs  $\{x_k, y_k\}$  are the *ladder* of  $(L, P)$ , and  $\text{Flip}(L, P) = \{k < K_0 : (x_k, y_k) \text{ flippable}\}$ .

The rows  $x_1, \dots, x_{K_0-1}$  are distinct rows of  $C_1 \setminus \{1\}$  (on  $A$ ) or of the arc from 2 to 1 (on  $B$ ), the  $y_k$  likewise in  $C_2 \setminus \{2\}$  or the arc from 1 to 2; so the  $2(K_0 - 1)$  ladder rows are distinct and avoid rows 1, 2, and  $K_0 \geq 2$ : on  $A$  because  $\pi$  is a derangement, on  $B$  by the admissibility of the mark ( $\pi(1) \neq 2$ ,  $\pi(2) \neq 1$ ).

**Theorem 19.2** (ladder identity). For every  $n, \lambda$  and split  $\alpha + \beta = m$  with  $\alpha, \beta \geq 2$ ,

$$\mathbb{P}_X[B] - \mathbb{P}_X[A] = \mathbb{P}_X[B, \text{Flip} = \emptyset] - \mathbb{P}_X[A, \text{Flip} = \emptyset], \quad \text{hence} \quad |\mathbb{P}_X[B] - \tfrac{1}{2}| \leq \tfrac{1}{2} \mathbb{P}_X[\text{Flip} = \emptyset].$$

The same identity holds with  $X$  replaced by  $S(R) \times \{P\}$  for any fixed admissible pair of first rows  $R$  and mark  $P$ .

*Proof.* Fix  $(L, P) \in X$  and write  $K = \{1, \dots, K_0 - 1\}$ . The ladder depends on  $(L, P)$ , so the  $T_k$  are not maps of  $X$ ; the statement to prove is that  $(L, P) \sim T_S(L, P)$ ,  $S \subseteq K$  (the trades at the unordered pairs of  $S$ , in any order), is an equivalence relation on  $X$  whose classes have exactly  $2^{|K|}$  elements and on which the bit is  $\mathbf{1}_B(T_S(L, P)) = \mathbf{1}_B(L, P) + |S \cap \text{Flip}| \bmod 2$ . This follows once we know that each  $T_k$  is an involution of  $X$  which preserves  $K_0$ , the set of unordered ladder pairs and the flippability of each of them, that  $T_k$  toggles the bit exactly when  $k \in \text{Flip}$ , and that trades at disjoint row pairs commute: then the same trades, at the same unordered pairs with the same flippabilities, are available from every point of a class (symmetry and transitivity), the  $2^{|K|}$  points  $T_S(L, P)$  are distinct (rows  $x_k, y_k$  are changed by  $T_k$  and by no other  $T_i$ ), and the bit formula holds. If  $\text{Flip} \neq \emptyset$  exactly half of the subsets  $S$  have  $|S \cap \text{Flip}|$  odd, so  $B$  and  $A$  each contain half of the class; summing over classes gives the identity, and the bound follows from  $\mathbb{P}[B] + \mathbb{P}[A] = 1$ .

$T_k$  is a well-defined involution of  $X$ . The  $j'$ -trade at  $(x_k, y_k)$  changes only rows  $x_k, y_k \notin \{1, 2\}$ , so  $\sigma$  and the mark are untouched and  $(L, P)$  stays in  $X$ ; it is an involution because the traded columns are again the  $\rho'_{x_k, y_k}$ -cycle through  $j'$  (§17.12.10).

*Effect on the frame.* If  $k \in \text{Flip}$  the new frame is  $\pi' = (x_k y_k) \circ \pi$ . On  $A$  this merges  $C_1$  and  $C_2$ : the new cycle runs  $1 \rightarrow \pi(1) \rightarrow \dots \rightarrow x_{k+1} \rightarrow y_k \rightarrow \pi(y_k) \rightarrow \dots \rightarrow 2 \rightarrow \dots \rightarrow y_{k+1} \rightarrow x_k \rightarrow \dots \rightarrow 1$ , so  $d'_{12} = |C_1|$  and  $d'_{21} = |C_2|$  (reading  $x_{|C_1|} = 1, y_{|C_2|} = 2$ ). On  $B$ ,  $x_k$  lies on the arc from 2 to 1 and  $y_k$  on the arc from 1 to 2 (as  $k < \min(d_{12}, d_{21})$ ), and the split into  $(x_k, \pi(x_k), \dots, \pi^{-1}(y_k))$  and  $(y_k, \dots, \pi^{-1}(x_k))$  puts 1 ( $k$  steps after  $x_k$ , fewer than the  $d_{12}$  steps from  $x_k$  to  $y_k$ ) in the first cycle and 2 in the second: the new state is in  $A$  with  $|C'_1| = d_{12}$ ,  $|C'_2| = d_{21}$ . In both cases  $K_0$  is preserved and the bit toggles. If  $k \notin \text{Flip}$  the new frame is  $(x_k y_k) \circ \pi \circ (x_k y_k)$ : the cycle partition is unchanged up to exchanging the labels  $x_k, y_k$ , so  $K_0$  and the bit are preserved.

*The ladder is preserved.* Let  $\pi' = (x_k y_k) \circ \pi$  (the conjugation case is similar and easier). Since  $\pi'^{-1}(r) = \pi^{-1}((x_k y_k)(r))$ , the preimage chains of 1 and 2 under  $\pi'$  agree with those under  $\pi$  up to depth  $k$  and are exchanged beyond it:  $\pi'^{-i}(1) = x_i$  and  $\pi'^{-i}(2) = y_i$  for  $i \leq k$ , while  $\pi'^{-(k+1)}(1) = \pi^{-1}(y_k) = y_{k+1}$  and then  $\pi'^{-i}(1) = y_i$ ,  $\pi'^{-i}(2) = x_i$  for  $k < i < K_0$ . So the unordered pairs  $\{x_i, y_i\}$ ,  $i \in K$ , are the ladder of  $T_k(L, P)$ . In the conjugation case the same computation with  $(x_k y_k) \circ \pi \circ (x_k y_k)$  gives  $\pi'^{-i}(1) = (x_k y_k)(x_i)$ , which again permutes the pairs  $\{x_i, y_i\}$  among themselves (only  $i = k$  is affected, by exchanging  $x_k, y_k$ ).

*Flippability is preserved and the  $T_k$  commute.*  $T_k$  changes rows  $x_k, y_k$  only, and the flippability of  $(x_i, y_i)$ ,  $i \neq k$ , depends only on rows  $x_i, y_i$ , which are different rows. Likewise  $T_i$  and  $T_k$  act on disjoint pairs of rows, each along a column set determined by its own two rows, so  $T_i T_k = T_k T_i$ . (After  $T_k$  the pair  $\{x_i, y_i\}$  may have changed its index, but the trade at an unordered pair does not depend on the index or on the order of the two rows.)

The statement for fixed  $R$  holds because every  $T_k$  fixes rows 1 and 2.  $\square$

**Corollary 19.3** (the conjecture reduces to the ladder). *Write  $f(n, \lambda; \alpha, \beta) = \mathbb{P}_X[\text{Flip} = \emptyset]$ . By Theorem 11.4,  $\bar{q}(\mu \rightarrow \lambda) - 1 = (\mathbb{P}_X[B] - \mathbb{P}_X[A])/\mathbb{P}_X[A]$  and  $\mathbb{P}_X[A] \geq (1 - f)/2$ , so  $|\bar{q} - 1| \leq 2f/(1 - f)$ : EQ( $\delta$ ) holds with  $\delta(n) = \max_{\lambda, \alpha \neq \beta} 2f/(1 - f)$ . If this  $\delta(n) \rightarrow 0$ , Theorem 12.1 gives*

$$d_{\text{TV}}(P_n, Q_n) = O\left(\log n \cdot \max_{\lambda, \alpha \neq \beta} f(n, \lambda; \alpha, \beta) + n^{-2/3}\right).$$

In particular the weak CGW conjecture follows from

$$(L) \quad \mathbb{P}_X[\text{no ladder pair is flippable}] = o(1/\log n) \quad \text{uniformly in } \lambda, \alpha, \beta. \quad (23)$$

**Remark 19.4** (what (L) asks, and what the data say). (i) The ladder consists of  $K_0 - 1$  pairs of *disjoint* rows,  $K_0 = \min(|C_1|, |C_2|)$  resp.  $\min(d_{12}, d_{21})$ , and a pair is flippable iff columns  $j, j'$  lie in different cycles of its row permutation — the complement of the conjugate (rows  $\leftrightarrow$  columns) of the bit itself. (L) does not ask these coins to be fair, only that they do not all fail.

(ii) Exact enumeration at  $n = 7$  (`s13_comp7.c`, four instances) confirms the identity to all digits ( $\mathbb{P}[B] - \mathbb{P}[A] = -0.005848$  and  $\mathbb{P}[B, \text{Flip} = \emptyset] - \mathbb{P}[A, \text{Flip} = \emptyset] = -0.005848$  for  $\lambda = (7)$ ,  $\{\alpha, \beta\} = \{2, 5\}$ ;  $+0.003509$  for  $\{3, 4\}$ ;  $-0.004651$  for  $(5, 2)$ ; 0 for a cross-cycle pair), where however  $\mathbb{P}[\text{Flip} = \emptyset] = 0.40\text{--}0.42$ : the bound only bites for large  $n$ . On uniform squares with true marks (`s13_ladder.py`, `runs/s13/f1/ladder`{30,50,100}.log; 9000, 7200, 3000 instances):

$$\mathbb{P}[\text{Flip} = \emptyset \mid K_0] = 2^{-(K_0-1)}$$

within statistical error for  $K_0 \leq 6$  and consistent with it beyond ( $n = 50$ : 0.480, 0.265, 0.142, 0.070, 0.036, 0.030, 0.0 for  $K_0 = 2..8$  against 0.5, 0.25, 0.125, 0.0625, 0.031, 0.016, 0.008), the ladder coins are flippable with frequency 0.4990, 0.5002, 0.4993, and  $\mathbb{P}[\text{Flip} = \emptyset] = 0.125, 0.079, 0.041$  against  $\mathbb{E}[2^{-(K_0-1)}] = 0.124, 0.077, 0.038$ . Since  $\mathbb{P}[K_0 = k] \approx 4/n$  for small  $k$  and  $\sum_{k \geq 2} 2^{-(k-1)} = 1$ , this is  $\approx 4/n$  (0.133, 0.080, 0.040): if the coins are fair and independent, (L) holds with  $O(1/n)$  and the conjecture would follow with  $d_{TV} = O(\log n/n)$ . The heuristic rests on uniformity: for isotopes of  $\mathbb{Z}_n$  every  $\rho_{x,y}$  is an  $n$ -cycle and  $\text{Flip} = \emptyset$  always.

(iii) *Cost of a short ladder.*  $\mathbb{P}[A, K_0 \leq \ell] \leq \mathbb{P}[|C_1| \leq \ell] + \mathbb{P}[|C_2| \leq \ell] \leq 72\ell/n$  for  $\ell \leq n/2 - 1$  by Proposition 17.40. On  $B$  the ladder identity itself gives  $\mathbb{P}[B, K_0 \leq \ell, \text{Flip} \neq \emptyset] = \mathbb{P}[A, K_0 \leq \ell, \text{Flip} \neq \emptyset] \leq 72\ell/n$ , and the switching of Proposition 17.40 gives  $\mathbb{P}[B, d_{12} = \ell, |C_1| \leq n/2] \leq 4/n$  (and the same for  $d_{21}$ ): from such a square pick  $y$  strictly inside the arc from 1 to 2 ( $\ell - 1$  choices) and  $x$  in a free cycle  $C_x$  ( $n - |C_1| \geq n/2$  choices), turn  $C_x$  if  $(x, y)$  is not flippable, and  $j'$ -trade; the arc becomes  $1 \rightarrow \dots \rightarrow \pi^{-1}(y) \rightarrow x \rightarrow (C_x) \rightarrow y \rightarrow \dots \rightarrow 2$ , so  $d_{12}$  grows by  $|C_x|$  and nothing else changes; given the target, the position of  $x$  on the new arc ( $\ell - 1$  choices) and the binary “turn back” pin the move, so there are at most  $2(\ell - 1)$  preimages. What is missing is  $\mathbb{P}[B, K_0 \leq \ell_0, |C_1| > n/2, \text{Flip} = \emptyset] = O(\ell_0/n)$ : a short arc inside a cycle covering most rows, with no flippable ladder pair; there the free rows may be absent and the flippability of a pair inside  $C_1$  cannot be toggled by a turn. *To do.*

(iv) *Local witnesses.* A ladder pair is flippable as soon as the  $\rho_{x_k, y_k}$ -cycle through  $j$  is short and avoids  $j'$  (or the one through  $j'$  is short and avoids  $j$ ). On real squares the length of the  $j$ -cycle of a ladder pair is spread like a uniform variable on  $\{2, \dots, n\}$  ( $\mathbb{P}[\leq \ell] = 0.021, 0.081, 0.18, 0.38$  for  $\ell = 2, 5, 10, 20$  at  $n = 50$  against  $(\ell-1)/(n-1) = 0.020, 0.082, 0.18, 0.39$ ; 0.010, 0.041, 0.092, 0.19 at  $n = 100$ ), and among instances with  $K_0 \geq n/4$  a flippable ladder pair with  $j$ -cycle of length  $\leq 10$  exists in 93% ( $n = 50$ ) and 95% ( $n = 100$ ),  $\leq 20$  in 99.6% and 99.8%. A non-flippable pair has  $j'$  on its  $j$ -cycle, hence no  $j$ -cycle avoiding  $j'$  of any length, so  $\{\text{Flip} = \emptyset, K_0 \geq \ell_0\}$  is contained in the event below; granting the missing part of (iii) with  $\ell_0 = n/\log^3 n$  (total cost  $O(1/\log^3 n)$ ), (L) follows from

$$\mathbb{P}\left[K_0 \geq \frac{n}{\log^3 n} \text{ and no ladder pair has a } j\text{-cycle of length } \leq \log^5 n \text{ avoiding } j'\right] = o(1/\log n), \quad (24)$$

a statement about at least  $n/\log^3 n$  disjoint row pairs each of which carries such a cycle with probability  $\approx \log^5 n/n$  (expected number  $\gtrsim \log^2 n$  witnesses). Short cycles through a given

column in the row permutation of a given row pair are substructures of polylogarithmic size, so (24) is the kind of statement that the switching method of Kwan–Sudakov [11] and its successors is built for: a second-moment bound needs  $\text{Cov}(W_k, W_{k'}) = o(\mathbb{P}[W_k]\mathbb{P}[W_{k'}]/\log n)$  for disjoint pairs  $k \neq k'$  (with  $\mathbb{E} \sum_k W_k \gtrsim \log^2 n$  the diagonal term is already  $O(1/\log^2 n)$ ), in the space of completions of two rows and two columns. No fairness of any global statistic is needed anywhere.

**Remark 19.5** (relation to the earlier programme). Theorem 19.2 is the orbit identity of the trade chain (§17.12.11(b)) specialised to the abelian subgroup generated by the ladder trades, for which the orbit structure is explicit: on each orbit the bit is *exactly* fair unless no ladder pair is flippable. Compared with the pool of intercalates (Theorem 15.6) the ladder has deterministic bit-changing pairs and an exact bias formula, at the price of needing the pairs to be flippable; compared with the trade chain it needs no mixing. The quenched version (fixed  $R$ ) reduces the fairness Question of `paper/fairness_question.tex` to the quenched form of (L):  $|2\mathbb{P}[B \mid R] - 1| \leq \mathbb{P}[\text{Flip} = \emptyset \mid R]$ , and since the ladder classes refine the components of the trade chain, also  $|2p(K) - 1| \leq \mathbb{P}[\text{Flip} = \emptyset \mid K]$  for every component  $K$ . Items (I), (II), (III), (T), (F1), (F2) and the component question are superseded by (L).

## 19.1 Offset ladders, and what (L) needs

The ladder is one of a family. For an integer  $c$  (an *offset*) put

$$K_c = \begin{cases} \min(|C_1| - c, |C_2|) & \text{on } A, \\ \min(d_{21} - c, d_{12}) & \text{on } B, \end{cases} \quad D_c = \{(x_{c+i}, y_i) = (\pi^{-(c+i)}(1), \pi^{-i}(2)) : 1 \leq i < K_c\},$$

so  $D_0$  is the ladder and for  $c > 0$  the  $x$ -row is  $c$  steps deeper than the  $y$ -row ( $x$ -depths  $c + 1, \dots, c + K_c - 1$  stay inside  $C_1 \setminus \{1\}$ , resp. the arc from 2 to 1;  $y$ -depths  $1, \dots, K_c - 1$  inside  $C_2 \setminus \{2\}$ , resp. the arc from 1 to 2). We take  $c \geq 0$ ; the families with the  $y$ -row deeper are obtained by exchanging the roles of rows 1 and 2 (and of  $C_1, C_2, d_{12}, d_{21}$ ) throughout.

**Theorem 19.6** (offset ladders). *For every offset  $c \geq 0$ , on the domain  $\mathcal{D}_c = \{K_c \geq 2\}$  (where  $D_c \neq \emptyset$ ) the  $j'$ -trades at the pairs of  $D_c$  are commuting involutions that preserve  $K_c$ , the unordered pairs of  $D_c$  and their flippabilities, and toggle the bit iff the traded pair is flippable. Hence, with  $\text{Flip}_c$  the set of flippable pairs of  $D_c$ ,*

$$\mathbb{P}[B, \mathcal{D}_c] - \mathbb{P}[A, \mathcal{D}_c] = \mathbb{P}[B, \mathcal{D}_c, \text{Flip}_c = \emptyset] - \mathbb{P}[A, \mathcal{D}_c, \text{Flip}_c = \emptyset].$$

*Proof.* Identical to Theorem 19.2; only the bookkeeping changes. Let  $\pi' = (x_{c+i} y_i) \circ \pi$ . The preimage chain of 1 under  $\pi'$  agrees with that of  $\pi$  up to depth  $c + i$  and then continues along the chain of 2:  $\pi'^{-(c+i+t)}(1) = y_{i+t}$ ; symmetrically  $\pi'^{-(i+t)}(2) = x_{c+i+t}$  ( $t \geq 1$ ). So the pair of  $D_c$  at  $y$ -depth  $i' > i$  is  $\{y_{i'}, x_{c+i'}\}$ , unchanged as a set, and the pairs at depth  $i' < i$  are untouched:  $D_c$  is preserved. (The same computation shows that trading a pair at depths  $(k, k')$  sends a deeper pair at depths  $(K, K')$  to depths  $(K' + k - k', K - k + k')$ , whose offset is  $2(k - k') - (K - K')$ ; it is preserved iff  $K - K' = k - k'$ , so only families of constant offset, i.e. the  $D_c$ , are preserved.) On  $A$  the trade merges  $C_1, C_2$  into a cycle with  $d'_{12} = |C_1| - (c + i) + i = |C_1| - c$  and  $d'_{21} = |C_2| + c$ , so  $K'_c = \min(d'_{21} - c, d'_{12}) = \min(|C_2|, |C_1| - c) = K_c$ ; on  $B$  the pair separates rows 1, 2 ( $c + i < d_{21}$  puts  $x_{c+i}$  on the arc from 2 to 1,  $i < d_{12}$  puts  $y_i$  on the arc from 1 to 2) and gives  $|C'_1| = d_{12} + c$ ,  $|C'_2| = d_{21} - c$ , again  $K'_c = \min(d_{12}, d_{21} - c) = K_c$ . No pair of  $D_c$  is frame-adjacent:  $y_i = \pi(x_{c+i})$  would need  $d_{21} = c - 1$  (or  $\pi(1) = 2$  when  $c = 0$ ) and  $x_{c+i} = \pi(y_i)$  would need  $d_{21} = c + 1$ , both excluded on  $\mathcal{D}_c$ . Commutation, preservation of flippability and the orbit count are as before.  $\square$

Exact enumeration at  $n = 7$  ( $\lambda = (7)$ ,  $\{\alpha, \beta\} = \{2, 5\}$ ) confirms the identities for  $c = 0, 1, 2, 3$  to all digits ( $\mathbb{P}[B, \mathcal{D}_c] - \mathbb{P}[A, \mathcal{D}_c] = -0.005848, +0.001462, +0.003509, -0.001316$ , equal to the right-hand sides).

**Using the offsets to price short arcs.** Fix  $\ell \geq 2$  and consider  $B$ -instances with  $d_{12} = \ell$ . For  $0 \leq c \leq n/2 - \ell$  and  $d_{21} \geq c + \ell$  one has  $K_c = \ell$ , so  $D_c$  has  $\ell - 1$  pairs, and the offset- $c$  trade at a flippable pair of  $D_c$  lands in  $A$  with  $|C_1| = \ell + c \leq n/2$  and  $|C_2| = d_{21} - c \geq \ell$ . Within a class of  $\mathcal{D}_c$  all  $B$ -members have the same  $(d_{12}, d_{21})$  and all  $A$ -members the same  $(|C_1|, |C_2|) = (d_{12} + c, d_{21} - c)$ , and  $\text{Flip}_c$  is a class invariant, so Theorem 19.6 class by class and Proposition 17.40 give

$$\mathbb{P}[B, d_{12} = \ell, d_{21} \geq c + \ell, \text{Flip}_c \neq \emptyset] = \mathbb{P}[A, |C_1| = \ell + c, |C_2| \geq \ell, \text{Flip}_c \neq \emptyset] \leq \frac{18}{n - \ell - c - 1} \leq \frac{36}{n - 2}. \quad (25)$$

Averaging over  $c$ : if  $\theta(L, P)$  denotes the fraction of offsets  $c \in [0, n/2 - \ell]$  whose diagonal  $D_c$  contains a flippable pair, then for every  $\theta_0 > 0$

$$\mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, \theta \geq \theta_0] \leq \frac{36}{(n - 2)\theta_0}, \quad (26)$$

since on  $|C_1| \geq n/2 + \ell$  one has  $d_{21} = |C_1| - \ell \geq n/2 \geq c + \ell$  for every such  $c$ , so  $\mathbb{E}[\theta \mathbf{1}_E]$  is the average over  $c$  of the left side of (25), and Markov's inequality applies. Together with the splice-in bound  $\mathbb{P}[B, d_{12} = \ell, |C_1| < n/2 + \ell] \leq 4/(n - 2\ell + 1) \leq 8/n$  for  $\ell \leq n/4$  (the argument of Remark 19.4(iii) with  $n - |C_1| > n/2 - \ell$  free rows), this prices every short arc except on the event  $\{\theta < \theta_0\}$ .

**The single remaining input.** Say that a pair of rows  $(x, y)$ ,  $x, y \notin \{1, 2\}$ , carries a *witness* if the  $\rho_{x,y}$ -cycle through column  $j$  has length at most  $\ell' = \log^5 n$  and does not contain  $j'$  (then  $(x, y)$  is flippable). A frame-adjacent pair ( $y = \pi(x)$ , so  $\rho_{x,y}(j) = j'$ ) never carries one; the families below contain no such pair. For a pair of rows of a random Latin square the length of the  $j$ -cycle is spread like a uniform variable on  $\{2, \dots, n\}$  (Remark 19.4(iv)), so a witness has probability  $\approx \ell'/n$ .

**Hypothesis 19.7** (witness lemma). *Let  $(L, P)$  be uniform on  $X$ . Let  $\mathcal{G}$  be one of the following families of blocks of pairs of rows, determined by the frame: (i) the ladder, each pair a block of its own; (ii) for a  $B$ -instance with  $d_{12} = \ell \leq n/4$  and  $|C_1| \geq n/2 + \ell$ , the diagonals  $D_c$ ,  $0 \leq c \leq n/2 - \ell$ , each a block of  $\ell - 1$  pairs. Let  $G_{\mathcal{G}}$  be the number of blocks containing a pair that carries a witness, and  $\mu_{\mathcal{G}} = \sum_{B \in \mathcal{G}} \min(\frac{1}{2}, \ell'|\mathcal{B}|/n)$ . Then there is a constant  $\kappa > 0$  such that*

$$\mathbb{P}[G_{\mathcal{G}} < \kappa \mu_{\mathcal{G}} \text{ and } \mu_{\mathcal{G}} \geq \log^2 n] = o(1/\log n).$$

(This is annealed: an exceptional set of frames of probability  $o(1/\log n)$  is allowed. For a single pair of non-adjacent rows the witness probability is  $\approx \ell'/n$ , Remark 19.4(iv), so  $\kappa \mu_{\mathcal{G}}$  is a constant fraction of the expected count if blocks are nearly independent given their shared rows.)

**Proposition 19.8** ((L) follows from the witness lemma). *Hypothesis 19.7 implies (L), hence the weak CGW conjecture.*

*Proof.* Put  $\ell_0 = n/\log^2 n$ . By Theorem 19.2,  $|\mathbb{P}[B] - \mathbb{P}[A]| \leq \mathbb{P}[\text{Flip} = \emptyset] \leq \mathbb{P}[K_0 \geq \ell_0, \text{Flip} = \emptyset] + \mathbb{P}[A, K_0 < \ell_0] + \mathbb{P}[B, K_0 < \ell_0]$ .

*Long ladders.* On  $\{K_0 \geq \ell_0\}$  take family (i):  $\mu \geq (\ell_0 - 1)\ell'/n \geq \log^2 n$ , so  $G \geq \kappa\mu \geq 1$  except on an event of probability  $o(1/\log n)$ , and  $G \geq 1$  means  $\text{Flip} \neq \emptyset$ .

*Short cycles.*  $\mathbb{P}[A, K_0 < \ell_0] \leq 72\ell_0/n = 72/\log^2 n$  by Proposition 17.40.

*Short arcs.*  $\mathbb{P}[B, K_0 < \ell_0] \leq \sum_{\ell < \ell_0} (\mathbb{P}[B, d_{12} = \ell] + \mathbb{P}[B, d_{21} = \ell])$ ; we treat  $d_{12} = \ell$ , and  $d_{21} = \ell$  is the same argument with the roles of rows 1, 2 exchanged (Proposition 17.40 covers  $C_2$ , and the offset families become those with the  $y$ -row deeper). Fix  $\ell$ . The part with  $|C_1| < n/2 + \ell$  costs  $\leq 8/n$ . On  $|C_1| \geq n/2 + \ell$  take family (ii), with  $N = \lfloor n/2 - \ell \rfloor + 1 \geq n/4$  blocks:  $\mu = N \min(\frac{1}{2}, (\ell - 1)\ell'/n) \geq \frac{1}{4} \min(n/2, (\ell - 1)\ell') \geq \log^2 n$ , so except on an event  $F_\ell$  of probability  $o(1/\log n)$  the fraction of offsets whose diagonal contains a flippable pair is  $\theta = G/N \geq \theta_0(\ell) := \kappa \min(\frac{1}{2}, (\ell - 1)\ell'/n)$ . By (26),

$$\sum_{\ell < \ell_0} \mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, \theta \geq \theta_0(\ell)] \leq \sum_{\ell - 1 \leq n/(2\ell')} \frac{36n}{(n-2)\kappa(\ell-1)\ell'} + \sum_{n/(2\ell') < \ell - 1 < \ell_0} \frac{72}{(n-2)\kappa} = O\left(\frac{\log n}{\ell'} + \frac{\ell_0}{n}\right)$$

and  $\sum_{\ell < \ell_0} 8/n \leq 8/\log^2 n$ . The exceptional events: the family (ii) is a function of the frame, and the events  $\{d_{12} = \ell\}$  are disjoint, so  $\sum_\ell \mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, F_\ell] = \mathbb{P}[\bigcup_\ell \{B, d_{12} = \ell, |C_1| \geq n/2 + \ell, F_\ell\}]$  is the probability of a single failure event of the hypothesis, which is  $o(1/\log n)$ . Altogether  $\mathbb{P}[\text{Flip} = \emptyset] = o(1/\log n)$ .  $\square$

**Remark 19.9** (what the witness lemma is, and is not). (i) It is a statement about substructures of polylogarithmic size (a cycle of length  $\leq \log^4 n$  of a row-pair permutation through a given column), in the space of completions of two rows and two columns, for families of pairs selected by the frame. It asks for a lower bound on their expected number and a lower-tail concentration (a second moment with relative variance  $o(1/\log n)$  suffices) — the territory of the switching method [11]: a switching that shortens or lengthens the  $j$ -cycle of  $\rho_{x,y}$  by a bounded amount, applied in the conditional space, compares the counts of squares with  $j$ -cycle lengths  $m$  and  $m + 1$  and gives the uniform spread, and the same switchings applied to two pairs at once control the covariance. (ii) It asks nothing about long cycles, same-cycle events, or the fairness of any global statistic. (iii) Rigid squares ( $\mathbb{Z}_p$  isotopes and their relatives, where no row pair has a short cycle) enter only through the failure probability, which must be  $o(1/\log n)$ ; for them the hypothesis is simply the statement that they are rare. (iv) The constants:  $\ell' = \log^5 n$  and  $\ell_0 = n/\log^2 n$  are one admissible choice; the proof needs  $\ell_0\ell'/n \geq \log^2 n$ ,  $\ell_0/n = o(1/\log n)$  and  $\log n/\ell' = o(1/\log n)$ . (v) The block form is what the short-arc argument uses: the number of offsets with a witness, not the number of witnesses, since the diagonals of different offsets are dependent through the shared rows  $y_i$ . (vi) The hypothesis is stated only for the two families used; a version for arbitrary frame-determined families would be false, since frame-adjacent pairs never carry a witness.

## 19.2 Status of the witness lemma: literature and data

**Data.** `s13.witness.py` (`runs/s13/f1/witness50.log`: 9000 instances at  $n = 50$ ). The single-pair witness frequency for ladder pairs is 0.079, 0.168, 0.314 for  $\ell' = 5, 10, 20$  against  $(\ell' - 1)/(n - 1) = 0.082, 0.184, 0.388$  (the deficit is the probability that  $j'$  lies on the short cycle,  $\approx \ell'/n$ ). For two disjoint ladder pairs  $\mathbb{E}[WW']/p^2 = 1.03, 1.00, 0.99$  ( $1.7 \cdot 10^5$  samples). The number  $G$  of ladder pairs with a witness and the number  $S$  of rows  $x$  preceding row 1 whose pair with  $y_1 = \pi^{-1}(2)$  carries a witness (the “star” family, all pairs sharing one row) both have variance equal to the binomial value within the resolution of the data (ratios 0.95–1.1 in bins of  $K_0$ , resp.

of the star size; the inflation in the smallest bins is the spread of  $K_0$  within a bin). So at  $n = 50$  the witness indicators behave as independent events of probability  $\approx \ell'/n$ , both across disjoint pairs and across pairs sharing a row, to within a few percent — the hypothesis holds with margin in the data.

**Literature (session-13 scout; see the references).** Nothing in print gives the witness lemma, but the tools are there.

- Cavenagh–Greenhill–Wanless [3] work with switchings inside two rows (flip, backflip, cross-switch), uniformly over an arbitrary prescription of a set of full columns, and prove that the cycle-type law of a row pair is within a factor  $3/2$  of the derangement law per merge/split (their Theorem 3.13), within  $n^{3/2}$  overall for types with  $\leq \frac{6}{5} \log n$  cycles (Corollary 4.6),  $\mathbb{P}[\text{single } n\text{-cycle}] \in [2n^{-2}, 2n^{-2/3}]$  (Lemma 4.4), and whp more than  $(\log n)^{1-c}$  cycles (Theorem 4.11). Their Theorem 3.13 is exactly  $\mathbb{P}_X[A] \geq 1/3$  (§19.3), and summed over types it gives the unconditional lower-bound clause of the witness lemma with the right dependence on  $\ell'$  (Proposition 19.12); what it does not give is the conditional version (two prescribed columns on which the two rows carry distinct symbols) or the concentration.
- Allsop–Morris [10] give universal bounds  $(\delta/n)^{|P|} \leq \mathbb{P}[L \supseteq P] \leq (\Delta/n)^{|P|}$  for partial squares  $P$  on  $\leq \alpha n$  rows and  $\leq \beta n$  columns,  $2\alpha + \beta < 1$ . Summed over the patterns of a  $j$ -cycle of length  $\ell$  this gives  $\mathbb{P}[\text{length } \ell] \in [\delta^{2\ell}, \Delta^{2\ell}]/n$ : the right order  $1/n$  for each  $\ell$ , with an exponential-in- $\ell$  slack; whether  $\delta = \Delta = 1 + o(1)$  for small  $P$  is stated as open there, and that is exactly the witness lemma’s lower-bound clause. Their hypotheses do not allow full prescribed columns together with prescribed rows, so the conditioning on rows 1, 2 and columns  $j, j'$  would have to be handled separately (the switchings never touch those cells, so this should be bookkeeping).
- Lower tails: Kwan–Sudakov and Kwan–Sah–Sawhney bound lower tails of the *total* intercalate count with  $\exp(-n^c)$  errors, by comparison with Latin rectangles and the McKay–Wanless completion ratio  $e^{O(n \log^2 n)}$ ; that technique is useless for a count of polylogarithmic mean. For the witness lemma one needs either a second moment with pairwise covariances of relative size  $o(1/\log n)$  (switchings on two pairs at once, exact to  $1 + O(\text{polylog}/n)$ ) or a Poisson-type lower tail from matched creation/destruction degrees, as in the Kwan–Sudakov ratio method.
- Nothing is known about any “two given columns in different cycles” event with constant probability, in any conjugate form; the witness lemma deliberately avoids such events.

**Verdict.** (L), hence the weak CGW conjecture, is reduced to a local statement (Hypothesis 19.7) whose two clauses —  $\Theta(\ell'/n)$  lower bounds for short  $j$ -cycles of prescribed row pairs in the space of completions of two rows and two columns, and lower-tail concentration of their count across a frame-determined family — are true in the data with margin and are of the kind the two-row switching method of [3, 11] is designed for. The unconditional form of the first clause follows from [3] (Proposition 19.12 below); the conditional form and the concentration are what remains.

### 19.3 The unconditional witness bound follows from CGW’s switching theorem

The lower-bound clause of the witness lemma, for a fixed pair of rows of a uniform square without any conditioning, is a consequence of a theorem that has been in print since 2008. Cavenagh, Greenhill and Wanless [3] (Lemma 3.12, Theorem 3.13) prove, for adjacent types  $\lambda$  (one part  $m = \alpha + \beta$ ) and  $\mu$  (that part split into  $\alpha \neq \beta$ ), that

$$\frac{1}{2} \leq \frac{|S(\lambda)| \lambda_m m}{|S(\mu)| \mu_\alpha \mu_\beta \alpha \beta} \leq \frac{3}{2},$$

uniformly over  $\lambda, \alpha, \beta$  (and over any prescription of a set of full columns on which rows 1, 2 carry the same symbols). In the notation of §11 the middle term is  $|X|/|J|$  for  $X = X(n, \lambda; \alpha, \beta)$  and  $J = J(n, \mu; \alpha, \beta)$ , so by Theorem 11.4 ( $|J| = 2|X_A|$ ) the CGW theorem is exactly

$$\frac{1}{3} \leq \mathbb{P}_X[A] \leq 1 \quad \text{for every } n, \lambda, \alpha, \beta \quad (\text{lower bound: CGW}(3/2), \text{ see Remark 19.10}) \quad (27)$$

(for  $\alpha = \beta$  CGW’s second form of the lemma and the  $\alpha = \beta$  conventions of §11 give the same middle term  $|X|/|J|$ ): rows 1, 2 lie in different frame cycles of a marked pair with probability at least  $1/3$ . The bound is attained: at  $n = 5$ ,  $(5) \rightarrow (2, 3)$ ,  $|X_A| = 1440$ ,  $|X_B| = 2880$ ,  $\mathbb{P}_X[A] = 1/3$  exactly, and CGW’s  $3/2$  is sharp there. (The upper bound  $\mathbb{P}_X[A] \leq 1$  is the trivial direction  $|J| \leq 2|X|$  of our identity; CGW’s constant  $1/2$  is its switching-count shadow.) The conjecture asks for  $\mathbb{P}_X[A] = \frac{1}{2} + o(1/\log n)$ .

**Remark 19.10** (a gap in the published proof of CGW’s Lemma 3.12). The lower bound in (27) is CGW’s upper bound  $3/2$ , and its proof requires every splitting pair of a  $\lambda$ -square to carry exactly two edges. Case 3 of their splitting procedure (“ $\{j, j'\}$  and  $\{\omega j, \omega j'\}$  both  $B$ -pairs: cross-switch with respect to  $\{\omega j, \omega j'\}$ , then backflip with respect to  $\{j, j'\}$ ”) does not produce an element of  $S(\mu, F)$  when  $\min(\alpha, \beta) \geq 3$ . Indeed the rows-1, 2 part of the cross-switch at a  $B$ -pair  $(J, J')$  with  $\omega^\alpha J = J'$  is forced by the proof of their Lemma 3.10 (row 1 keeps the symbol set  $\{1 \circ \omega^k J : 1 \leq k \leq \alpha\}$  on the cells  $\{\omega^k J : 1 \leq k < \alpha\} \cup \{J'\}$ , and in every interior column the two rows are swapped), which gives  $1' \circ \omega^k J = 1 \circ \omega^{k+1} J$  for  $1 \leq k < \alpha$ ,  $1' \circ J' = 1 \circ \omega J$ ,  $2' \circ \omega^k J = 1 \circ \omega^k J$ ,  $2' \circ J = 1 \circ J'$ ; hence  $\omega'(J) = \omega^{\alpha-1} J$ ,  $\omega'(\omega^k J) = \omega^{k-1} J$  for  $2 \leq k < \alpha$ ,  $\omega'(\omega J) = J'$ : the cross-switch *reverses the  $\omega$ -arc from  $J$  to  $J'$* . With  $(J, J') = (\omega j, \omega j')$  the reversed arc has  $j' = \omega^{\alpha-1} J$  in its interior, so afterwards  $\omega'(j) = \omega j$  and  $\omega'(\omega j) = j'$ : the pair  $\{j, j'\}$  is at distance 2, and the backflip at it splits the  $(\alpha + \beta)$ -cycle into parts 2 and  $\alpha + \beta - 2$ . (For  $\alpha = 2$  the arc has a single interior column and nothing changes, so the proof is complete for  $\min(\alpha, \beta) = 2$  — hence at  $n \leq 5$ , where it is sharp.) The joining rule 4 fails symmetrically, and the two multigraphs are not equal. Independent check: `s13_cgwcase3.py` implements the cross-switch (rows 1, 2 as above, columns  $j, j'$  swapped on the  $\delta_{j, j'}$ -arc from row 2 to row 1, which is what balances the corner cells), verifies that it is a Latin square of the same type in every instance, and finds that case 3 lands in the wrong type in 2676 of 2676 instances at  $n = 9, 12$  with  $\min(\alpha, \beta) \geq 3$  (a referee subagent found the gap first, with its own implementation). We found no published correction (the text we use is the journal version, DOI 10.1002/rsa.20216, and the example on its p. 293 exhibits the reversal:  $\omega = (1234567)$ ,  $j = 2$ ,  $j' = 5$ ,  $\alpha = 3$ , and after the cross-switch  $\omega' = (1243567)$ ).

*A partial repair.* A cross-switch at a  $B$ -pair  $(J, J')$  at any distance  $a \geq 2$  is a type-preserving trade (the proof of Lemma 3.10 does not use  $a = \alpha$ ), it reverses the  $\omega$ -arc from  $J$  to  $J'$ , and it toggles  $\{j, j'\}$  iff exactly one of  $j, j'$  lies in the interior of that arc, say at position  $i$ ; the distance

from  $j$  to  $j'$  then changes by  $a - 2i$ . So the distance-preserving toggles of  $\{j, j'\}$  are exactly the cross-switches at pairs *symmetric about  $j$  or about  $j'$*  ( $a = 2i$ ), and they exist as  $B$ -pairs or not. When  $\alpha$  is even and  $\alpha < 2\beta$ , the pair  $R = (\omega^{\alpha/2}j, \omega^{3\alpha/2}j)$  — symmetric about  $j'$ , at distance  $\alpha$ , with  $j$  outside its arc — always gives an edge: if  $R$  is  $B$ , cross-switch at  $R$  and backflip at  $\{j, j'\}$  (now an  $A$ -pair at distance  $\alpha$ ); if  $R$  is  $A$ , backflip at  $R$  itself. Symmetrically for  $\beta$  even and  $\beta < 2\alpha$ . (`s13_cgwnfix.py`: type  $\mu$  in all 3928 covered instances at  $n = 10, 13$ .) The  $\mu$ -side multiplicities change (an  $A$ -pair's backflip may serve two splitting pairs), so the constant  $3/2$  would have to be recomputed. When  $\alpha, \beta$  are both odd, every distance-preserving toggle has even arc length  $a \neq \alpha, \beta$ , so if all the symmetric pairs are  $A$ -pairs no one-cross-switch repair exists and we see no canonical repair; the same holds for  $\alpha \geq 2\beta$  with  $\beta$  odd. The gap is therefore open in those classes, which the reduction chains do use.

*Our own machinery bears on the missing direction.* By Theorem 19.2,  $\mathbb{P}_X[B] - \mathbb{P}_X[A] = \mathbb{P}_X[B, \text{Flip} = \emptyset] - \mathbb{P}_X[A, \text{Flip} = \emptyset] \leq \mathbb{P}_X[\text{Flip} = \emptyset]$ , so  $\mathbb{P}_X[B] \leq \frac{1}{2} + \frac{1}{2}\mathbb{P}_X[\text{Flip} = \emptyset]$ : CGW's  $3/2$ , i.e.  $\mathbb{P}_X[B] \leq 2/3$ , follows from  $\mathbb{P}_X[\text{Flip} \neq \emptyset] \geq 1/3$  — a constant-probability version of (L), e.g. (CP) with  $c \geq 1/3$  for a single ladder pair. Conversely nothing in the present notes uses CGW(3/2) on the main line.

*What survives:* the direction  $|S(\lambda)|N_\lambda \geq \frac{1}{2}|S(\mu)|N_\mu$ , which by Theorem 11.4 is the trivial  $|X| \geq |X_A|$ ; CGW's Lemma 4.1 upper bound  $(|S(\lambda, F)|(m-1)!/(|S((m), F)|\gamma(\lambda))) \leq 2^{\kappa-1}$ , Lemma 4.4's lower bound  $\mathbb{P}[(m)] \geq 2/m^2$ , and the upper bounds of Proposition 19.12 (the factor 2, hence  $\mathbb{E}[N_\ell] \leq (4 + o(1))/\ell$ ). *What is now conditional* on the truth of CGW's  $3/2$  — call it CGW(3/2), equivalently  $\mathbb{P}_X[B] \leq 2/3$  for all  $(\lambda, \alpha, \beta)$ , a weak form of our own target  $\mathbb{P}_X[B] = \frac{1}{2} + o(1/\log n)$  — is: the lower bound  $\frac{1}{3}$  in (27); the lower bounds (factor  $\frac{2}{3}$ ) of Proposition 19.12 and the witness bound  $(\frac{2}{3} - o(1))(\ell' - 1)/n$ ; the unconditional (CP) constant  $2/5$  of §19.4; the “ $\geq \frac{1}{3}$ ” of (32); CGW's  $\mathbb{P}[(n)] \leq 2n^{-2/3}$  and Corollary 4.5, hence the tail bound  $P_n(\kappa > A \log n) = O(n^{-2/3})$  used in the proof of Theorem 12.1 — now re-derived from the surviving direction in Proposition 19.11. The data ( $\mathbb{P}_X[B] \approx \frac{1}{2}$  throughout) leave no doubt about the statement, and the exact tables at  $n \leq 11$  verify it there.

**Proposition 19.11** (the number of cycles, from the surviving direction only). *Let  $L$  be a uniform Latin square of order  $n$ ,  $\sigma$  the column permutation of rows  $1, 2$ ,  $N_\ell$  its number of  $\ell$ -cycles and  $\kappa = \sum_\ell N_\ell$ . Then, using only the identity  $|J| = 2|X_A|$  of Theorem 11.4 and  $|X_A| \leq |X|$ :*

(i)  $\mathbb{E}[\ell N_\ell] \leq 2n/(n - \ell)$  for every  $\ell$ ; in particular  $\mathbb{E}[N_\ell] \leq 4/\ell$  for  $\ell \leq n/2$ .

(ii) For every  $k \geq 1$ , with  $(\kappa)_k = \kappa(\kappa - 1) \cdots (\kappa - k + 1)$ ,

$$\mathbb{E}[(\kappa)_k] \leq 2^{k-1} \sum_{\ell_1, \dots, \ell_k \geq 2} \frac{\mathbb{E}[m N_m]}{\ell_1 \cdots \ell_k} \Big|_{m=\ell_1+\dots+\ell_k} \leq 2^{k-1} \left(4 + \frac{k^2}{\log n}\right) (\log n)^k.$$

(iii) For  $A \geq 3$ ,  $\mathbb{P}[\kappa \geq A \log n] \leq (7 + \log n) n^{-\log((A-1)/2)}$ , which is  $O(n^{-2} \log n)$  for  $A \geq 16$ ; and  $\mathbb{P}[\text{all cycles of } \sigma \text{ have the same length } \ell < n] \leq 4d(n)/n$ ,  $d(n)$  the number of divisors.

*Proof.* (i) For  $\beta \neq \ell$  the joining instances  $(L, \{j, j'\})$  with  $j$  in an  $\ell$ -cycle and  $j'$  in a  $\beta$ -cycle number  $\ell N_\ell \beta N_\beta$  per square, the splitting instances (pairs at distance  $\ell$  on an  $(\ell + \beta)$ -cycle) number  $(\ell + \beta) N_{\ell+\beta}$ , and  $|J| = 2|X_A| \leq 2|X|$  gives  $\mathbb{E}[\ell N_\ell \beta N_\beta] \leq 2\mathbb{E}[(\ell + \beta) N_{\ell+\beta}]$ ; for  $\beta = \ell$  the same count with  $\ell^2 \binom{N_\ell}{2}$  and  $\ell N_{2\ell}$  gives  $\mathbb{E}[\ell^2 N_\ell (N_\ell - 1)] \leq 4\mathbb{E}[N_{2\ell}]$ . Summing the first over  $\beta \neq \ell$  ( $\sum_\beta \beta N_\beta = n$ ) and adding the second (its right side is the missing summand

$m = 2\ell$ ):  $\mathbb{E}[\ell N_\ell(n - \ell N_\ell)] + \mathbb{E}[\ell^2 N_\ell(N_\ell - 1)] \leq 2\mathbb{E}[\sum_{m \geq \ell+2} m N_m] \leq 2n$ . Hence  $n\mathbb{E}[\ell N_\ell] = \mathbb{E}[\ell N_\ell(n - \ell N_\ell)] + \mathbb{E}[\ell^2 N_\ell(N_\ell - 1)] + \ell\mathbb{E}[\ell N_\ell] \leq 2n + \ell\mathbb{E}[\ell N_\ell]$ .

(ii) Count the instances  $(L, (C_1, j_1), \dots, (C_k, j_k))$ :  $C_i$  pairwise distinct  $\sigma$ -cycles of  $L$  of lengths  $\ell_i$  and  $j_i \in C_i$ ; there are  $\ell_1 \cdots \ell_k$  of them per ordered  $k$ -tuple of distinct cycles with these lengths. Join  $C_1$  and  $C_2$  at the pair  $\{j_1, j_2\}$  by a joining step: flip if the pair is an  $A$ -pair; if it is a  $B$ -pair, first trade the row cycle  $C_2$  of  $j_2$  (a switch in the sense of Lemma 11.3, whose proof applies to the row cycle through either column of the pair; it inverts  $C_2$  in place and does not touch the cycle of  $j_1$ ), then flip. The flip replaces  $\sigma$  by  $\sigma \circ (j_1 j_2)$  on rows 1, 2, so the result has  $j_1, j_2$  on one cycle of length  $\ell_1 + \ell_2$  with  $j_2 = \sigma^{\ell_2}(j_1)$  whether or not the trade was applied (CGW Lemma 3.3), every other cycle of  $\sigma$  (in particular  $C_3, \dots, C_k$ ) untouched, and the map is at most two-to-one (the backflip at  $\{j_1, j_2\}$  returns  $L$  or the traded square). Repeat with  $\{j_1, j_3\}, \dots, \{j_1, j_k\}$ , always trading the cycle of the new column when needed, so that the cycle containing  $j_1$  is never inverted: after step  $i$  every earlier  $j_r$  is shifted by exactly  $\ell_{i+1}$ . After  $k-1$  steps all  $j_i$  lie on one cycle of length  $m = \sum \ell_i$  at the positions  $j_i = \sigma^{\ell_i + \dots + \ell_k}(j_1)$ , determined by  $j_1$  and  $(\ell_1, \dots, \ell_k)$ , and the composite is at most  $2^{k-1}$ -to-one. (With CGW's min-symbol rule the trade may invert the merged cycle and the positions are not determined; the count would then be  $2^{2k-3}$ -to-one.) Hence the number of instances with lengths  $(\ell_i)$  is at most  $2^{k-1} \sum_{L'} m N_m(L')$ , i.e.  $\mathbb{E}[\ell_1 \cdots \ell_k T(\ell_1, \dots, \ell_k)] \leq 2^{k-1} \mathbb{E}[m N_m]$  with  $T$  the number of ordered  $k$ -tuples of distinct cycles with these lengths; summing over  $(\ell_i)$  gives the first bound, since  $\sum_{(\ell_i)} T = (\kappa)_k$ . For the second, split at  $m \leq n/2$ , where  $\mathbb{E}[m N_m] \leq 4$  by (i) and  $\sum_{m \leq n/2} \sum_{\sum \ell_i = m} \prod \ell_i^{-1} \leq (\sum_{\ell \leq n/2} \ell^{-1})^k \leq (\log n)^k$ , and  $m > n/2$ , where some  $\ell_i \geq m/k$ , so  $\sum_{\sum \ell_i = m} \prod \ell_i^{-1} \leq k \cdot (k/m)(\log n)^{k-1}$ , and  $\sum_{m > n/2} m N_m \cdot k^2(\log n)^{k-1}/m \leq k^2(\log n)^{k-1}$  because at most one cycle is longer than  $n/2$ .

(iii) Markov for  $(\kappa)_k$  with  $k = \lceil \log n \rceil \leq \log n + 1$ : on  $\{\kappa \geq K\}$ ,  $(\kappa)_k \geq (K - k + 1)^k \geq ((A - 1) \log n)^k$  for  $K = A \log n$ , and  $k^2 / \log n \leq \log n + 3$ , so  $\mathbb{P}[\kappa \geq K] \leq 2^{k-1}(4 + k^2 / \log n)(\log n)^k / ((A - 1) \log n)^k \leq (7 + \log n)(2/(A - 1))^k \leq (7 + \log n)(2/(A - 1))^{\log n}$  for  $A \geq 3$ , and  $(A - 1)/2 \geq e^2$  for  $A \geq 16$ . For the equal-length types,  $\ell \mid n$  and  $\ell < n$  force  $\ell \leq n/2$ , so  $\mathbb{P}[N_\ell = n/\ell] \leq \mathbb{E}[N_\ell] \ell / n \leq 4/n$ ;  $\ell = n$  is the  $n$ -cycle.  $\square$

With this, the tail estimates in the proof of Theorem 12.1 no longer need [3, Cor. 4.5]:  $P_n(\kappa > K) = O(n^{-2} \log n)$  for  $K = \lceil 16 \log n \rceil$  and  $P_n(\text{non-plain}) = O(n^{-1+o(1)})$ , which is all the proof uses (the exponent of  $n$  in the tail only has to be negative).

**Proposition 19.12** (short cycles in two rows; lower bounds under CGW(3/2), Remark 19.10). *Let  $L$  be a uniform Latin square of order  $n$ ,  $\sigma$  the column permutation of two fixed rows,  $N_\ell$  the number of  $\ell$ -cycles of  $\sigma$ , and  $Q_q$  the  $\sigma$ -cycle through a fixed column  $q$ . For all  $\ell \neq \beta$  with  $\ell, \beta \geq 2$ ,*

$$\frac{2}{3} \mathbb{E}[(\ell + \beta) N_{\ell+\beta}] \leq \mathbb{E}[\ell N_\ell \cdot \beta N_\beta] \leq 2 \mathbb{E}[(\ell + \beta) N_{\ell+\beta}]. \quad (28)$$

Consequently, with  $\tau_n = \mathbb{P}[\sigma \text{ has at least } 10\sqrt{n} \text{ cycles}]$ , which is  $e^{-\Omega(\sqrt{n})}$  for  $n \geq n_0$  ([3], Theorem 4.9; the theorem is asymptotic, so no explicit  $n_0$ ), for every  $2 \leq \ell \leq \sqrt{n}/40$ ,

$$\frac{2}{3\ell} \left( 1 - \frac{(\ell + 1)(4 + n\tau_n)}{n} \right) \leq \mathbb{E}[N_\ell] \leq \frac{4 + n\tau_n}{\ell}, \quad \mathbb{P}[|Q_q| = \ell] = \frac{\ell \mathbb{E} N_\ell}{n} \in \left[ \frac{2/3 - o(1)}{n}, \frac{4 + o(1)}{n} \right].$$

In particular  $\mathbb{P}[|Q_q| \leq \ell', q' \notin Q_q] \geq (\frac{2}{3} - o(1))(\ell' - 1)/n$  for any second column  $q'$  and  $\ell' \leq \sqrt{n}/40$ : a fixed pair of rows carries a witness of length  $\leq \ell'$  with probability at least  $(2/3 - o(1))(\ell' - 1)/n$ .

*Proof.* For  $\ell \neq \beta$  let  $X(\ell, \beta)$  be the set of  $(L, \{q, q'\})$  with  $q, q'$  in a common  $\sigma$ -cycle of length  $\ell + \beta$  and  $q' = \sigma^\ell(q)$ , and  $J(\ell, \beta)$  the set of  $(L, \{q, q'\})$  with  $q$  in an  $\ell$ -cycle and  $q'$  in a  $\beta$ -cycle. Each  $(L, \{q, q'\})$  in  $X(\ell, \beta)$  has a type  $\lambda$  with an  $(\ell + \beta)$ -part and lies in  $X(n, \lambda; \ell, \beta)$ ; splitting that part gives the type  $\mu$  of  $J(n, \mu; \ell, \beta) \subseteq J(\ell, \beta)$ , and this is a bijection between the types occurring on the two sides. Summing the CGW inequality over types,  $\frac{2}{3}|X(\ell, \beta)| \leq |J(\ell, \beta)| \leq 2|X(\ell, \beta)|$ . Now  $|X(\ell, \beta)| = \sum_L (\ell + \beta) N_{\ell+\beta}(L)$ : every  $q$  in an  $(\ell + \beta)$ -cycle gives the pair  $\{q, \sigma^\ell(q)\}$ , and distinct  $q$  give distinct pairs because  $\{q, \sigma^\ell(q)\} = \{q_2, \sigma^\ell(q_2)\}$  with  $q_2 \neq q$  forces  $\sigma^{2\ell}q = q$ , i.e.  $\ell = \beta$ . And  $|J(\ell, \beta)| = \sum_L \ell N_\ell(L) \beta N_\beta(L)$  ( $q$  in an  $\ell$ -cycle,  $q'$  in a  $\beta$ -cycle; the unordered pair determines which is which since  $\ell \neq \beta$ ). Dividing by the number of squares gives (28).

Sum (28) over  $\beta \geq 2$ ,  $\beta \neq \ell$ . Since  $\sigma$  is a derangement,  $\sum_{\beta \geq 2, \beta \neq \ell} \beta N_\beta = n - \ell N_\ell$ , and  $\sum_{\beta} (\ell + \beta) N_{\ell+\beta} = n - W_\ell$  with  $W_\ell = \sum_{m \leq \ell+1} m N_m + 2\ell N_{2\ell}$  the number of columns in cycles of length  $\leq \ell + 1$  or exactly  $2\ell$ . Hence

$$\frac{2}{3} \mathbb{E}[n - W_\ell] \leq \mathbb{E}[\ell N_\ell (n - \ell N_\ell)] \leq 2 \mathbb{E}[n - W_\ell] \leq 2n. \quad (29)$$

Upper bound: on  $\{\ell N_\ell \leq n/2\}$  the middle term is at least  $\ell N_\ell \cdot n/2$ , so  $\mathbb{E}[\ell N_\ell; \ell N_\ell \leq n/2] \leq 4$  and  $\mathbb{E}[m N_m] \leq 4 + n \mathbb{P}[N_m > n/(2m)] \leq 4 + n \tau_n$  for every  $m \leq \sqrt{n}/20$ , because  $N_m > n/(2m) \geq 10\sqrt{n}$  means at least  $10\sqrt{n}$  cycles, and CGW's Theorem 4.9,  $\mathbb{P}[\geq k \text{ cycles}] = o((3/4)^k e^{\pi\sqrt{2n/3}})$ , is  $e^{-\Omega(\sqrt{n})}$  for  $k = 10\sqrt{n}$  ( $10 \log \frac{4}{3} = 2.88 > \pi\sqrt{2/3} = 2.57$ ). By column symmetry  $\mathbb{P}[|Q_q| = m] = \mathbb{E}[m N_m]/n \leq (4 + n \tau_n)/n$  for each  $m$ . Lower bound:  $\mathbb{E}[\ell N_\ell] \cdot n \geq \mathbb{E}[\ell N_\ell (n - \ell N_\ell)] \geq \frac{2}{3}(n - \mathbb{E}W_\ell)$  and  $\mathbb{E}W_\ell/n = \sum_{m \leq \ell+1} \mathbb{P}[|Q_q| = m] + \mathbb{P}[|Q_q| = 2\ell] \leq (\ell+1)(4 + n \tau_n)/n$  ( $\ell+1$  terms, all with  $m \leq 2\ell \leq \sqrt{n}/20$ , where the upper bound applies). Finally  $\mathbb{P}[q' \in Q_q \mid |Q_q| = m] = (m-1)/(n-1)$  by column symmetry, which gives the last claim.  $\square$

**Remark 19.13.** (i) For a uniform derangement the two sides of (28) are *exactly* equal (both count  $n! D_{n-\ell-\beta}/((n-\ell-\beta)! D_n)$ : ordered pairs of an  $\ell$ - and a  $\beta$ -cycle against marked  $(\ell + \beta)$ -cycles), so (28) is a constant-factor form of the cycle-count part of the CGW conjecture; type by type the exact ratio is  $2\mathbb{P}_X[A]$ , by Theorem 11.4. At  $n = 5$  the ratio  $|J(2, 3)|/|X(2, 3)|$  is  $2/3$  exactly. (ii) Until now the literature had  $\mathbb{E}N_\ell \geq c_\ell/\ell$  only with  $c_\ell$  decaying exponentially in  $\ell$  (from the universal bounds of Allsop–Morris [10]), and  $\mathbb{E}N_\ell \rightarrow 1/\ell$  only for  $\ell = 2$  (Kwan–Sudakov, Kwan–Sah–Sawhney). We have not found the constant-factor bound for all  $\ell \leq \sqrt{n}/40$  in the literature, although it is an immediate corollary of [3]; the point is that the switching theorem compares a cycle of length  $\ell + \beta$  with the *pair* of cycles  $(\ell, \beta)$  in one step, so no loss accumulates along  $\ell$ . (iii) The data (Remark 19.4(iv)):  $\mathbb{P}[|Q_q| = \ell] \approx 1/n$  for all  $\ell$  at  $n = 50, 100$ , inside  $[0.66, 4]/n$ .

**What is still missing for the witness lemma.** The proposition is unconditional. The witness lemma needs the same lower bound (a) for pairs of rows selected by the frame, i.e. conditionally on columns  $j, j'$ , and (b) inside  $X$ , i.e. conditionally on the type of  $\sigma$  and the mark, together with (c) the lower-tail concentration. For (a), CGW's framework already allows an arbitrary prescription of full columns, so conditioning on columns  $j, j'$  is of the kind their argument tolerates; the one obstacle is their standing assumption that the two distinguished rows carry the same symbols on the prescribed columns (so that their row-pair permutation restricted to the free columns is a permutation). For a ladder pair  $(x, y)$  this fails — the four symbols  $L(x, j), L(x, j'), L(y, j), L(y, j')$  are distinct, since an equality would force  $\pi(x) = y$  or  $\pi(y) = x$  — and the free part of rows  $x, y$  consists of cycles together with two paths; the witness is precisely “the path starting at the column where row  $y$  holds  $L(x, j)$  is short and ends at

the column where row  $x$  holds  $L(y, j)$ ". Whether CGW's three switchings and their counting extend to this path structure is examined in §19.4: they do not, and the reason is instructive. (c) is the second-moment version of the same question.

**A duality (heuristic).** Transposing the square exchanges the roles. With  $T(c, r) = L(r, c)$  and  $\rho$  in the convention  $L(y, \rho_{x,y}(q)) = L(x, q)$ , the frame  $\pi$  of  $(j, j')$  in  $L$  is the row-pair permutation  $\sigma_T := \rho_{j,j'}^T$  of rows  $j, j'$  of  $T$ , and  $\rho_{x,y}$  is the frame of the column pair  $(x, y)$  of  $T$ . So the witness event for the ladder pair  $(x, y)$  — a short  $\rho_{x,y}$ -cycle through column  $j$  avoiding  $j'$  — becomes the event that the column pair  $(x, y)$  of  $T$  has rows  $j, j'$  in different frame cycles with the cycle of row  $j$  short; the ladder pairs become the column pairs  $(\sigma_T^{-k}(c_1), \sigma_T^{-k}(c_2))$  of  $T$ , where  $c_1, c_2$  are the columns of  $T$  that were rows 1, 2 of  $L$ ; the bit of  $(L, P)$  becomes " $c_1, c_2$  in a common  $\sigma_T$ -cycle"; and the conditioning becomes: on  $\sigma_T$  (over which the hypothesis is annealed) and on the frame type of the column pair  $(c_1, c_2)$  of  $T$  (the type  $\lambda$  of  $L$ ). So the conditional witness lemma is equivalent to a *lower* bound on short frame cycles of the  $\sigma_T$ -shifts of a column pair, given  $\sigma_T$  — the opposite direction to Proposition 17.40, which bounds them from above by the splice-in switching. The splice-in move lengthens  $C_1$  using a free cycle and is always available; the cut that shortens it needs a flippable pair inside  $C_1$ . This is the same flippability obstruction in yet another guise; heuristically it suggests what a proof of the witness lemma must contain — a lower bound on flippable pairs inside a frame cycle — which is what CGW's cross-switch supplies for the pair (1, 2) at the level of constants (this is a reading of the structure, not a demonstrated reduction).

## 19.4 The path version of CGW's lemma: where it breaks, and what survives

Work in  $\Omega(F)$ ,  $F$  = the two full columns  $p = j$ ,  $p' = j'$ , for a pair of rows  $x, y$  carrying four distinct symbols on  $\{p, p'\}$  (every ladder pair); this extends CGW's notation, whose  $F$  is required to be suitable for the distinguished rows. Write  $Q_p, Q_{p'}$  for the  $\rho_{x,y}$ -cycles through  $p, p'$  (here  $\rho_{x,y}$  is the column permutation with  $L(y, \rho_{x,y}(q)) = L(x, q)$ , so that  $\sigma = \rho_{1,2}$  up to inversion). *Crossed*:  $p' \in Q_p$ , with arcs  $P_1$  (from  $p$  to  $p'$ ,  $d_1 - 1$  interior columns) and  $P_2$  (from  $p'$  to  $p$ ,  $d_2 - 1$ ); *parallel*:  $Q_p \neq Q_{p'}$ , interior sizes  $\ell_p - 1, \ell_{p'} - 1$  (these two sizes are unrelated to the split  $(\alpha, \beta)$  of the mark). Parallel is flippability. All other  $\rho_{x,y}$ -cycles are *free*, and a *free column* is a column other than  $p, p'$ . "Flip" below means the turn of a column cycle of a free column pair  $(q, c)$  through  $x$  or through  $y$  (CGW's flip and backflip alike); "flippable" remains the property of the row pair.

**Which of CGW's moves are legal.** A flip at a free column pair  $(q, c)$  (turn of the  $(q, c)$ -column cycle through  $y$ ) never touches columns  $p, p'$  and is legal in  $\Omega(F)$ ; it replaces  $\rho_{x,y}$  by  $(q, c) \circ \rho_{x,y}$  when  $(q, c)$  is an  $A$ -pair (rows  $x, y$  in different  $(q, c)$ -column cycles). A switch (trade of rows  $x, y$  along a  $\rho$ -cycle containing exactly one of  $q, c$ ) is legal iff that cycle is free. A cross-switch at a  $B$ -pair  $(q, c)$  in one cycle is legal iff the  $\rho$ -arc from  $q$  to  $c$  avoids  $p, p'$ . Hence: pairs  $(q, c)$  inside one free cycle, or  $q$  in a free cycle and  $c$  anywhere, or both on the same path with the arc inside the path, are fully repairable as in CGW (cases 1–3 of their splitting procedure), and the within-path/free-cycle statistics of Proposition 19.12 go through in  $\Omega(F)$ . But a *cross-path* pair —  $q \in P_1$ ,  $c \in P_2$  in the crossed state, or  $q \in Q_p \setminus p$ ,  $c \in Q_{p'} \setminus p'$  in the parallel state — cannot be repaired when it is a  $B$ -pair: in the parallel state the switch would trade along  $Q_p$  or  $Q_{p'}$  (through  $p$  or  $p'$ ), and in the crossed state both arcs between  $q$  and  $c$  contain  $p$  or  $p'$ , so the cross-switch is illegal too. And the cross-path pairs are exactly the ones whose flip

(at an  $A$ -pair) exchanges crossed and parallel. CGW’s condition that the distinguished rows carry the same symbols on  $F$  is therefore not cosmetic: it is what makes every pair repairable, by excluding paths. (This paragraph settles legality with respect to the prescribed columns only. Preservation of rows 1, 2, needed in  $X$ , is automatic for the switch, which trades rows  $x, y \notin \{1, 2\}$ , and requires for a flip or a cross-switch that the rows it changes — those of the traded column cycle or column arc — avoid  $\{1, 2\}$  or contain both of them; the latter conjugates  $\sigma$  by  $(q\ c)$  and keeps type and mark. Only the within-free-cycle statistics of Proposition 19.12 go through in  $\Omega(F)$  without further work; the within-path statistics meet boundary cases, when a neighbour pair  $\{\omega(j), \omega(j')\}$  hits  $p$  or  $p'$ , which we have not checked.)

**What survives: two exact identities in the cross space.** Identity (I) holds conditionally on the frame of  $(p, p')$  and on rows 1, 2 (it preserves both); identity (F) holds in  $X$  conditionally on the frame, and conditionally also on rows 1, 2 if legality is restricted to column cycles avoiding rows 1, 2.

- (F) The flip at a cross-path  $A$ -pair is an involution between crossed and parallel squares preserving the pair and its  $A$ -status (Lemma 11.2), legal when the  $(q, c)$ -column cycle through  $y$  avoids rows 1, 2 (or contains both — then  $\sigma$  is conjugated and the mark is kept, but rows 1, 2 change). Summing,  $\mathbb{E}[\#\text{legal } A \text{ cross-path pairs; crossed}] = \mathbb{E}[\#\text{legal } A \text{ cross-path pairs; parallel}]$ .
- (I) An intercalate on rows  $x, z$  ( $z \notin \{1, 2, x\}$ ;  $z \neq y$  is automatic, since  $\{q, c\}$  is never a 2-cycle of  $\rho_{x,y}$  for the pairs counted) and free columns  $q, c$  is a Latin trade whose switch replaces  $\rho_{x,y}$  by  $\rho_{x,y} \circ (q\ c)$  and touches nothing else that matters; it is always legal. It separates  $p, p'$  iff crossed with  $q \in P_1, c \in P_2$ , and merges iff parallel with  $q \in Q_p \setminus p, c \in Q_{p'} \setminus p'$ . With  $I_{\text{sep}}, I_{\text{merge}}$  the numbers of such intercalates,

$$\mathbb{E}[I_{\text{sep}}; \text{crossed}] = \mathbb{E}[I_{\text{merge}}; \text{parallel}] \quad \text{exactly, in every fibre (rows 1, 2 and columns } p, p' \text{ fixed).} \quad (30)$$

Identity (I) has the merit that its moves are local and automatically legal, and its weights are intercalate counts of a single row with columns constrained to two sets. On real squares (`s13_icsw.py`, `runs/s13/f1/icsw50.log`: 9000 ladder pairs at  $n = 50$ )  $\mathbb{E}[I_{\text{sep}} \mid \text{crossed}] = 0.98 \cdot \mathbb{E}[(d_1 - 1)(d_2 - 1)/n]$  and  $\mathbb{E}[I_{\text{merge}} \mid \text{parallel}] = 0.98 \cdot \mathbb{E}[(\ell_p - 1)(\ell_{p'} - 1)/n]$ , with ratio 0.97–0.99 in every bin of the product. But (I) is a *fair* identity and yields no inequality by itself: the only deterministic bound is  $I_{\text{merge}} \leq (\ell_p - 1)(\ell_{p'} - 1)$  (at most one row  $z$  per column pair), and this is attained up to constants in structured squares — in the Cayley table of  $\mathbb{Z}_8 \times \mathbb{Z}_2$  a parallel pair with  $\ell_p = \ell_{p'} = 8$  has  $I_{\text{merge}} = 12 > \min(\ell_p, \ell_{p'}) - 1$ . A constant lower bound on flippability would follow from (i)  $\mathbb{E}[I_{\text{sep}} \mid \text{crossed}, d_1, d_2] \geq \kappa(d_1 - 1)(d_2 - 1)/n$ , (ii) an arc-trapping bound (crossed squares have macroscopic arcs except with probability  $\leq \mathbb{P}[\text{crossed}] - c$ ), and (iii) the matching upper bound  $\mathbb{E}[I_{\text{merge}} \mid \text{parallel}, \ell_p, \ell_{p'}] \leq C(\ell_p - 1)(\ell_{p'} - 1)/n$ : two-sided control of the two-cell pattern “ $L(z, q_0) = L(x, c_0)$  in the row  $z$  holding  $L(x, q_0)$  at  $c_0$ ” conditionally on rows 1, 2, columns  $p, p'$  and the structure of rows  $x, y$ . (An earlier version of this paragraph claimed  $I_{\text{merge}} \leq \min(\ell_p, \ell_{p'}) - 1$  and deduced  $\mathbb{P}[\text{parallel}] \geq \frac{2}{n} \mathbb{E}[I_{\text{sep}}; \text{crossed}]$ ; both are wrong, as the example shows.)

**Why neither input is available.** The universal bounds of Allsop–Morris require the conditioning to be a partial square on  $\leq \alpha n$  rows and  $\leq \beta n$  columns; two full rows or two full columns violate this. CGW’s framework tolerates full columns but not full rows, and its repair

moves fail across the paths. For (ii), the splice-in switching that proves Proposition 17.40 needs turns of column cycles through  $y$  avoiding rows 1, 2 — legality is again a cycle-avoidance event — while the intercalate version of the splice has availability  $1/n$  per position and yields only  $\mathbb{P}[\text{crossed}, d_1 = d] \lesssim 1/d$ . In every formulation the obstruction is one and the same: in the space with two rows and two columns prescribed there is no exact move that is both automatically legal and able to change a cross-path status, and the probability that such moves are available is a cycle-avoidance event of the kind we are trying to bound.

**The core problem, stated once.** Let  $(L, P)$  be uniform on  $X$ , condition on rows 1, 2 and columns  $p, p'$ , and let  $(x, y)$  be a pair of rows carrying four distinct symbols on  $\{p, p'\}$ .

$$(CP) \quad \mathbb{P}[p, p' \text{ lie in different cycles of } \rho_{x,y} \mid \text{rows } 1, 2, \text{ columns } p, p'] \geq c > 0, \quad (31)$$

uniformly over the conditioning (or: for the ladder, not all pairs crossed). Which half of the conditioning is the obstacle is settled by transposition (§19.5 below): with only the two *columns* prescribed, (CP) is CGW’s Theorem 3.13 for the transposed square and holds with  $c = \frac{1}{3}$  under CGW(3/2) (Remark 19.10) ( $c = \frac{1}{2}$  exactly, unconditionally, when rows 1, 2 are apart in the frame); with only rows 1, 2 prescribed (their type and the mark — the conditioning that defines  $X$ ) it is the path version, which breaks for the reason above; with both it is what (L) needs, and the frame conditioning is then free. Everything in §19–§19.4 is exact except this one constant.

**The unconditional core problem is settled with an explicit constant (under CGW(3/2),**

**Remark 19.10).** For a uniform Latin square and fixed rows 1, 2 and columns  $p, p'$  let  $W = \{p, p' \text{ in a common } \sigma\text{-cycle}\}$  ( $\sigma$  the column permutation of rows 1, 2) and  $B = \{1, 2 \text{ in a common frame cycle of } (p, p')\}$ . Three exact facts determine the joint law of  $(W, B)$  up to one number. (i)  $\mathbb{P}[B] = \mathbb{P}[W]$ : transposition preserves the uniform measure and carries  $W$  to the event  $B$  for rows  $p, p'$  and columns 1, 2, which has the same probability as  $B$  by row and column relabelling. (ii)  $\mathbb{P}[B, \neg W] = \mathbb{P}[A, \neg W]$ : on  $\neg W$  the trade of rows 1, 2 along the  $\sigma$ -cycle through  $p$  inverts that cycle in place, so it is an involution preserving  $\neg W$ , and it changes column  $p$  at rows 1, 2 only, so it toggles  $B$  (Lemma 11.3, whose proof uses only  $p \in Q \not\equiv p'$ ). (iii)  $\mathbb{P}[A \mid W, \lambda, \alpha] \geq \frac{1}{3}$  for every type and every split with  $\alpha, \beta \geq 2$ : conditionally on the type of  $\sigma$  and on the mark data,  $L$  is uniform on a fibre of  $X(n, \lambda; \alpha, \beta)$  over the ordered mark, all such fibres have the same  $A$ -probability by column relabelling, so this is (27); the pairs at  $\sigma$ -distance 1 have  $B$  surely and total probability  $\delta \leq 2/(n-1)$  by column symmetry. Writing  $w = \mathbb{P}[W]$ :  $\mathbb{P}[A, \neg W] = \mathbb{P}[B, \neg W] = \frac{1-w}{2}$ ,  $\mathbb{P}[B, W] = w - \frac{1-w}{2} = \frac{3w-1}{2}$ ,  $\mathbb{P}[A, W] = \frac{1-w}{2}$ , hence  $\frac{1-w}{2} \geq \frac{1}{3}(w - \delta)$  and  $\mathbb{P}[B, W] \geq 0$  give

$$\frac{1}{3} \leq \mathbb{P}[W] \leq \frac{3}{5} + \frac{2}{5}\delta \leq \frac{3}{5} + \frac{4}{5(n-1)}, \quad \text{i.e.} \quad \mathbb{P}[p, p' \text{ in different cycles of } \rho_{1,2}] \geq \frac{2}{5} - \frac{4}{5(n-1)},$$

and  $\mathbb{E}|Q_p| = 1 + (n-1)\mathbb{P}[W] \leq (\frac{3}{5} + o(1))n$  by column symmetry: the cycle of two given rows through a given column covers at most three fifths of the columns on average. (The fairness conjecture is  $w = \frac{1}{2}$ ; the data give 0.50. Exhaustively at  $n = 5$ :  $w = \mathbb{P}[B] = 11/14$ ,  $\mathbb{P}[B, \neg W] = \mathbb{P}[A, \neg W] = 3/28$ ,  $\mathbb{P}[B, W] = 19/28$ , and  $w = \frac{3}{5} + \frac{2}{5}\delta$  exactly with  $\delta = 13/28$ , matching  $\mathbb{P}_X[A] = \frac{1}{3}$  at  $n = 5$ .) So (CP) holds without any conditioning, with  $c = 2/5$ , by combining the two exact involutions with CGW’s constant. What is missing is the same statement after conditioning on two rows and two columns. For a ladder pair  $(x, y)$  in place of (1, 2): (i) fails (transposition does not preserve the conditioning); (ii) survives the conditioning on rows 1, 2 but not on columns  $p, p'$ , which the trade of rows  $x, y$  alters — and with both prescribed

the apart/together status of  $(x, y)$  is fixed by the frame, so the  $2 \times 2$  table degenerates; (iii) is CGW's lemma in the path version, which breaks as described.

## 19.5 The orbit method: commuting column turns in $X$

This subsection records what the search for a switching proof of (CP) produced: (a) a transposition remark that locates the obstacle precisely; (b) an exact fairness statement for generic row pairs in  $X$ ; (c) a product bound for the ladder bits, which replaces the witness lemma by a *genericity* hypothesis; (d) data. Throughout  $X = X(n, \lambda; \alpha, \beta)$ , and “legal” means: preserves the type of rows 1, 2 and the mark, i.e. maps  $X$  to itself.

**(a) Transposition: which half of the conditioning is the obstacle.** Let  $L$  be uniform on *all* Latin squares of order  $n$  (no condition on rows 1, 2 in this paragraph), let  $x, y$  carry four distinct symbols on  $\{p, p'\}$ , and condition on the two full columns  $p, p'$  (the frame  $\pi$  of  $(p, p')$ ). In the transposed square  $L^\top$  the columns  $p, p'$  are the distinguished rows, the frame  $\pi$  is their column permutation, and the event “ $p, p'$  lie in different cycles of  $\rho_{x,y}$ ” is the event “ $\{x, y\}$  is an  $A$ -pair for the distinguished rows”. Conditioning on the two full rows  $p, p'$  of  $L^\top$  is conditioning on their type and on the position of the pair  $\{x, y\}$  in their cycle structure, which is free by symbol and column relabelling. Hence, by Lemma 11.3 ( $|J_A| = |J_B|$ , which needs no condition on the cycle lengths) and (27):

$$\mathbb{P}[p, p' \text{ in different cycles of } \rho_{x,y} \mid \text{columns } p, p'] = \frac{1}{2} \text{ if } x, y \text{ are apart in the frame,} \quad \geq \frac{1}{3} \text{ if together at dist } \geq 2 \quad (32)$$

So (CP) with the *columns* prescribed is CGW's theorem; it is the conditioning on rows 1, 2 — on their type and the mark, which is what defines  $X$  and cannot be removed — that is the path version. In the same transposed picture the ladder trades  $T_k$  of Theorem 19.2 at flippable pairs are CGW's flips and backflips at the column pairs  $\{x_k, y_k\}$  of  $L^\top$ , and Theorem 19.2 is exactly the legal half of CGW's Lemma 3.12 for the distinguished rows  $p, p'$ : the other half — the switch and the cross-switch at  $\{x_k, y_k\}$  — trades the columns  $p, p'$  of  $L$  along a frame cycle or frame arc through exactly one of rows 1, 2 (apart:  $C_1 \ni 1, C_1 \not\ni 2$ ; together: the arc from  $x_k$  to  $y_k$  contains 1 but not 2, the other arc 2 but not 1), so it changes the type of rows 1, 2 and leaves  $X$ . The switch is, in the original orientation, the unflip of rows 1, 2 at  $(p, p')$  (which maps  $X_A \rightarrow J_A$ ); it toggles the crossed/parallel status of *every* pair  $(x, y)$  with  $x$  on the side of row 1 and  $y$  on the side of row 2, and reverses the orientation of one side, so it carries the ladder (a diagonal of the side-by-side array of pairs) to a different family (an anti-diagonal). The cross-switch likewise maps  $X_B$  to  $J_B$  (it splits the  $\sigma$ -cycle through  $p, p'$  while keeping rows 1, 2 in one frame cycle, CGW Lemma 3.10) and toggles the interior pairs. Both are exact but not usable for the ladder bits in  $X$ .

**(b) Row trades are free in  $X$ ; column turns are conditionally free.** For rows  $z, w \notin \{1, 2\}$  every trade of rows  $z, w$  along a  $\rho_{z,w}$ -cycle or  $\rho_{z,w}$ -arc changes rows  $z, w$  only, hence is legal in  $X$ . In particular:

**Lemma 19.14** (exact fairness of column-pair status for generic rows). *Let  $x, y \notin \{1, 2\}$  and let  $(q, c)$  be a column pair lying in two different cycles of  $\rho_{x,y}$ . The CGW switch at  $(q, c)$  for the rows  $x, y$  (trade of rows  $x, y$  along the one of the two cycles that contains the smaller minimum symbol) is a legal involution of  $X$  that toggles whether  $x, y$  lie in one column cycle of  $(q, c)$ , and*

preserves the cycle partition of  $\rho_{x,y}$  and the separation status of every column pair with respect to  $\{p, p'\}$ . Hence, conditionally on everything determined by the other rows and by the cycle partition of  $\rho_{x,y}$ ,

$$\mathbb{P}_X[x, y \text{ in different column cycles of } (q, c)] = \frac{1}{2}.$$

*Proof.* As in Lemma 11.3 with  $(1, 2)$  replaced by  $(x, y)$ : the trade swaps the entries of rows  $x, y$  on the columns of the chosen cycle  $Q \ni q$ ,  $Q \not\ni c$ , so  $\delta_{q,c} \mapsto \delta_{q,c} \circ (xy)$ , which splits or joins the column cycles through  $x, y$ . It inverts  $Q$  in place, so the symbol sets of the two cycles are unchanged and the min-symbol rule re-selects  $Q$ : an involution. Rows 1, 2 are untouched, so type and mark are preserved. Inverting  $Q$  exchanges the two arcs between  $p$  and  $p'$  when  $Q \ni p, p'$ , so separation of any pair by  $\{p, p'\}$  is preserved. (If  $Q$  contains  $p$  or  $p'$  the frame changes at rows  $x, y$ ; so frame-determined data are part of the conditioning only when  $Q$  avoids  $p, p'$ .)  $\square$

A column turn, by contrast, changes the rows of a column cycle. Let  $(q, c)$  be a free column pair and  $Z$  one of its column cycles.

**Lemma 19.15** (legal column turns). *The turn of  $Z$  (swapping the entries of columns  $q, c$  in the rows of  $Z$ ) is a measure-preserving involution of  $X$  if and only if  $|Z \cap \{1, 2\}| \in \{0, 2\}$ ; when  $|Z \cap \{1, 2\}| = 2$  it conjugates  $\sigma$  by  $(qc)$  and keeps the type and the mark. It preserves the columns  $p, p'$  (hence the frame and every frame-determined family of row pairs) and the column-cycle partition of  $(q, c)$  and of every column pair disjoint from  $\{q, c\}$  (in particular of every pair of a matching containing  $(q, c)$ ). For a row pair  $(x, y)$  with  $|Z \cap \{x, y\}| = 1$  it replaces  $\rho_{x,y}$  by  $\rho_{x,y} \circ (qc)$  (if  $x \in Z$ ) or  $(qc) \circ \rho_{x,y}$  (if  $y \in Z$ ), which toggles the crossed/parallel status of  $(x, y)$  exactly when  $(q, c)$  is separated by  $\{p, p'\}$  in  $\rho_{x,y}$  (different arcs when crossed, different cycles  $Q_p, Q_{p'}$  when parallel), and leaves  $(q, c)$  separated afterwards; with  $|Z \cap \{x, y\}| \in \{0, 2\}$  the status is unchanged. Turns of cycles of column pairs from a fixed matching  $\mathcal{M}$  of the free columns pairwise commute.*

*Proof.* A column cycle is a Latin trade (Lemma 11.2), so the turn is a bijection of Latin squares and an involution, and it preserves the cycle partition of  $(q, c)$ ; column pairs disjoint from  $\{q, c\}$  are untouched, and so are columns  $p, p'$ . Rows 1, 2 change iff  $Z$  meets them; if  $Z \ni 1, 2$  both rows swap their entries at  $q, c$ , so  $\sigma' = (qc)\sigma(qc)$ , whose type is that of  $\sigma$  and in which  $p, p'$  (fixed by  $(qc)$ ) keep their distance. If  $Z$  meets exactly one of rows 1, 2,  $\sigma' = \sigma \circ (qc)$  or  $(qc) \circ \sigma$ , which changes the type. The effect on  $\rho_{x,y}$  is the computation of §19.4; separation is preserved by the arc computation there. Turns of cycles of disjoint column pairs act on disjoint sets of cells; two cycles of the same pair have disjoint row sets.  $\square$

**(c) The product bound.** Fix a matching  $\mathcal{M}$  of the free columns, independent of the square, and a frame-determined family  $\mathcal{G}$  of row pairs  $(x, y)$ ,  $x, y \notin \{1, 2\}$ , whose  $x$ -rows are pairwise distinct and disjoint from its  $y$ -rows (a block of the ladder, where the pairs are even disjoint; or a union of diagonals  $D_c$  with distinct  $x$ -rows, which may share  $y$ -rows). For  $(x, y) \in \mathcal{G}$  call a column pair  $(q, c) \in \mathcal{M}$  a *clean coordinate* of  $(x, y)$  if the column cycle  $Z$  of  $(q, c)$  through  $x$  has length at most  $\ell_1$  (a fixed constant), satisfies  $|Z \cap \{1, 2\}| \in \{0, 2\}$ , and contains no row of  $\mathcal{G}$  other than  $x$  (no other  $x$ -row and no  $y$ -row, in particular not  $y$ ). Let  $\mathcal{T}_{x,y}$  be the set of clean coordinates of  $(x, y)$  and  $\eta_{x,y} \in \{0, 1\}^{\mathcal{T}_{x,y}}$ . By Lemma 19.15 the turns of all clean coordinates of all pairs of  $\mathcal{G}$  commute, generate a group acting freely on  $X$ , preserve  $\mathcal{G}$  and the sets  $\mathcal{T}_{x,y}$ , and

$$\rho_{x,y}(\eta) = \rho_{x,y}(0) \circ \prod_{(q,c) \in \mathcal{T}_{x,y}: \eta_{x,y}(q,c)=1} (qc),$$

the bit  $b_{x,y}(\eta) = [(x, y) \text{ parallel}]$  depending on  $\eta_{x,y}$  only. Let the *orbit weight* of  $(x, y)$  be

$$\bar{y}_{x,y} = \mathbb{E}_{\eta_{x,y}} \left[ \frac{\#\{(q, c) \in \mathcal{T}_{x,y} : (q, c) \text{ separated by } \{p, p'\} \text{ in } \rho_{x,y}(\eta)\}}{|\mathcal{T}_{x,y}|} \right]$$

(0 if  $\mathcal{T}_{x,y} = \emptyset$ ), a function of the orbit, and  $Y_{\mathcal{G}} = \sum_{(x,y) \in \mathcal{G}} \bar{y}_{x,y}$ .

**Proposition 19.16** (product bound). *Conditionally on the orbit, the bits  $b_{x,y}$ ,  $(x, y) \in \mathcal{G}$ , are independent, and*

$$\mathbb{P}[(x, y) \text{ crossed} \mid \text{orbit}] \leq 1 - \frac{1}{2} \bar{y}_{x,y}, \quad \mathbb{P}[\text{no pair of } \mathcal{G} \text{ is flippable} \mid \text{orbit}] \leq \prod_{(x,y) \in \mathcal{G}} (1 - \frac{1}{2} \bar{y}_{x,y}) \leq e^{-Y_{\mathcal{G}}/2}.$$

Consequently  $\mathbb{P}_X[\text{Flip}_{\mathcal{G}} = \emptyset] \leq \mathbb{E}_X[e^{-Y_{\mathcal{G}}/2}]$ .

*Proof.* The action is free: distinct column pairs act on disjoint cells, and two clean coordinates of different pairs at the same  $(q, c)$  are the cycles through two distinct  $x$ -rows, each avoiding the other, hence different cycles; the sets  $\mathcal{T}_{x,y}$  are orbit-invariant by Lemma 19.15 (row sets of the cycles, their meeting  $\{1, 2\}$ , and the frame-determined family are preserved). The uniform measure on  $X$  restricted to an orbit is therefore uniform on the orbit, which is the product of the cubes  $\{0, 1\}^{\mathcal{T}_{x,y}}$ . A clean coordinate of  $(x, y)$  contains  $x$  and not  $y$ , and a clean coordinate of another pair contains neither  $x$  nor  $y$  (it avoids all  $x$ -rows but its own and all  $y$ -rows), so  $b_{x,y}$  depends on  $\eta_{x,y}$  alone, whence the independence. Fix  $(x, y)$  and  $Z = (q, c) \in \mathcal{T}_{x,y}$ . Whether  $Z$  is separated in  $\rho_{x,y}(\eta)$  does not depend on  $\eta_Z$  (Lemma 19.15: the turn of  $Z$  leaves  $Z$  separated), and when it is, the two values of  $\eta_Z$  give opposite bits. Hence  $\mathbb{P}[b_{x,y} = 0] \leq 1 - \frac{1}{2} \mathbb{P}_{\eta}[Z \text{ separated}]$  for every  $Z$ , and averaging the right side over  $Z \in \mathcal{T}_{x,y}$  gives the first bound; the second is the product of the first over the independent bits.  $\square$

**Hypothesis 19.17** (orbit genericity). *With  $\ell_0 = n/\log^2 n$ : (i) for the sub-ladder  $I$  of the first  $\min(K_0 - 1, \lceil n/8 \rceil)$  ladder pairs,  $\mathbb{P}[K_0 \geq \ell_0, Y_I < \kappa|I|] = o(1/\log n)$ ; (ii) for a  $B$ -instance with  $d_{12} = \ell \leq n/4$  and  $|C_1| \geq n/2 + \ell$ , with  $S = \{2\ell t : 0 \leq t \leq (n/2 - \ell)/(2\ell)\}$  (offsets at gaps  $2\ell$ , so that the  $x$ -rows  $\pi^{-(c+i)}(1)$ ,  $c \in S$ ,  $1 \leq i < \ell$ , are pairwise distinct,  $|S| \geq n/(4\ell)$ , and the diagonals  $D_c$ ,  $c \in S$ , involve at most  $n/4 + \ell$  rows), and cleanliness taken with respect to all rows of  $\bigcup_{c \in S} D_c$ : the event that fewer than half of the  $D_c$ ,  $c \in S$ , have  $Y_{D_c} \geq \kappa(\ell - 1)$  has probability  $o(1/\log n)$  (as a single event over all  $\ell$ ).*

**Proposition 19.18.** *Hypothesis 19.17 implies (L), hence the weak CGW conjecture.*

*Proof.* Follow the proof of Proposition 19.8. Long ladders: by Proposition 19.16, on  $\{K_0 \geq \ell_0\}$  and off the exceptional event of (i),  $\mathbb{P}[\text{Flip} = \emptyset \mid \text{orbit}] \leq e^{-\kappa|I|/2} \leq e^{-\kappa\ell_0/4} = o(1/\log n)$ . Short arcs: fix  $\ell$  and apply Proposition 19.16 to the family  $\bigcup_{c \in S} D_c$ : its  $x$ -rows  $\pi^{-(c+i)}(1)$  lie in the disjoint intervals of depths  $[c + 1, c + \ell - 1]$ ,  $c \in S$ , so they are pairwise distinct, and they lie on the frame arc from 2 to 1 while the  $y$ -rows  $\pi^{-i}(2)$  lie on the arc from 1 to 2, so the two sets are disjoint. Given the orbit the bits of all pairs are independent, hence so are the diagonals' flippabilities, and off the exceptional event at least  $|S|/2$  of the diagonals are flippable with conditional probability  $\geq 1 - e^{-\kappa(\ell-1)/2} \geq 4\theta_0$ ,  $\theta_0 := \frac{1}{4}(1 - e^{-\kappa/2})$ . The number of flippable offsets in  $S$  then has conditional mean  $\geq 2\theta_0|S|$ , and by Chernoff the fraction  $\theta_S$  is  $\geq \theta_0$  except with conditional probability  $\leq e^{-\theta_0|S|/4} \leq e^{-\theta_0 \log^2 n/16}$  (as  $|S| \geq n/(4\ell_0) = \log^2 n/4$ ), summable over  $\ell < \ell_0$  to  $o(1/n)$ . The identity behind (26) holds for the average over  $c \in S$  as well as over all  $c$ , so  $\sum_{\ell < \ell_0} \mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, \theta_S \geq \theta_0] \leq 36\ell_0/((n-2)\theta_0) = O(1/\log^2 n)$ ; the parts  $|C_1| < n/2 + \ell$  and  $K_0 < \ell_0$  on  $A$  are as before.  $\square$

**What the hypothesis says.** It replaces the witness lemma (a lower-tail statement about rare short cycles, probability  $\ell'/n$  per pair) by a statement about a quantity that is typically of constant size:  $\bar{y}_{x,y}$  is the average, over the cube of a few column turns that fix rows  $y$  and 1, 2 and the whole family, of the fraction of those turns' column pairs that are separated by  $\{p, p'\}$  in the row pair  $(x, y)$ . It is a *genericity* statement — the short column cycles of the matching pairs through  $x$  are not aligned with the arcs of  $\rho_{x,y}$  at  $p, p'$  — together with concentration of the sum over the family; concentration of the *flippabilities* themselves is no longer required, since Proposition 19.16 makes them conditionally independent. It still contains (CP)-type inputs: a positive orbit weight needs the cycles of  $\rho_{x,y}$  through  $p, p'$  to be macroscopic (the arc-trapping input (ii) of §19.4), and the genericity itself. The pure permutation core is: for a permutation  $\rho_0$  and a matching  $\mathcal{T}$  of transpositions on points other than  $p, p'$ , with  $\tau(\eta)$  the product of a uniformly random subset of  $\mathcal{T}$ , bound  $\mathbb{P}_\eta[p, p' \text{ separated in } \rho_0\tau(\eta)]$  from below. For an adversarial  $\mathcal{T}$  (pairs at  $\rho_0$ -distance 2) the probability is 0, so the content is that the clean coordinates of a random square are not adversarial; and the moves that re-randomise the clean coordinates without touching  $\rho_{x,y}$  — the free row trades of Lemma 19.14 — have availabilities that are again same-cycle events of generic row pairs. This is the recursion noted in §19.4: every availability is a cycle-avoidance event, and the exact identities in  $X$  (Lemma 19.14, (F), (I), Proposition 19.16) are all fair. What Proposition 19.16 adds is that the fairness can be *compounded* over a family: a product of conditionally independent coins, so that a constant per-pair genericity gives exponential decay in the family size.

**(d) Data.** `s13_orbit.py (runs/s13/orbit/orbit{30,50,100}.log; 600, 600 and 360 ladder pairs at  $n = 30, 50, 100$ , true marks,  $\ell_1 \leq 4, 5, 8$ ; here the matching is chosen greedily among all clean short pairs of the instance, which over-counts the coordinates of a fixed matching by a factor  $\approx n/\ell_1$  — observed 10.8 and 14.3 — but tests the same genericity):`

- the length of the column cycle of  $(q, c)$  through  $x$  is spread like a uniform variable on  $\{2, \dots, n\}$  (about  $1/n$  per length); the turn is legal for 98% of the 2-cycles and 64% of the long cycles, clean for 97% resp. 27% ( $n = 100$ );
- genericity:  $\mathbb{P}[(q, c) \text{ separated} \mid \text{clean, length } \ell] = 0.199, 0.198, 0.193, 0.201, 0.201, 0.198, 0.197$  for  $\ell = 2, \dots, 8$ , against  $\mathbb{P}[\text{separated}] = 0.195$  over all column pairs ( $n = 100$ ; at  $n = 50$ : 0.186–0.194 against 0.184); the dependence on the arc lengths of  $\rho_{x,y}$  at  $p, p'$  was not binned;
- the orbit coin: with  $|\mathcal{T}| \approx 42$  coordinates ( $n = 100$ ) the orbit average of the parallel bit is 0.478 starting from a crossed square and 0.523 from a parallel one (at  $n = 30$ : 0.398/0.594 with  $|\mathcal{T}| \approx 10$ ; at  $n = 50$ : 0.447/0.561 with  $|\mathcal{T}| \approx 18$ ); at  $n = 100$ , 243 of the 360 per-instance orbit averages lie in  $[0.4, 0.7]$ .

So  $\bar{y} \approx 0.2$  with no detectable dependence on the cycle length, and the orbit coin is fair to within the mixing allowed by  $\approx 0.2|\mathcal{T}|$  separated coordinates. With the *fixed* matching of consecutive free columns, as in Proposition 19.16 (`--fixed; orbit{50,100}_fixed.log`):  $|\mathcal{T}| = 1.7$  resp. 3.0 clean coordinates per pair on average ( $\approx 0.9(\ell_1 - 1)/2$ , as the uniform spread of the lengths predicts),  $\mathcal{T} = \emptyset$  for 15% resp. 4% of the pairs, and  $\mathbb{E}[\bar{y}] = 0.159$  resp. 0.189. So the per-pair factor of Proposition 19.16 is  $1 - \frac{1}{2}\bar{y} \approx 0.9$  on average, and Hypothesis 19.17(i) is the statement that the average over the sub-ladder does not collapse:  $\kappa \approx 0.15$  is what the data suggest.

(d') **The concentration input ( $\gamma$ ): data.** `s13_cov.py` (`runs/s13/orbit/cov{50,100}.log`; 543 and 286 instances) computes, with the fixed matching and cleanliness with respect to all rows of the sub-ladder  $I$  of the first  $\min(K_0 - 1, n/8)$  pairs, every  $\bar{y}_k$ ,  $k \in I$ , and  $Y_I = \sum_{k \in I} \bar{y}_k$ : mean  $|I| = 5.2$  resp. 9.9,  $\mathbb{E}[\bar{y}] = 0.130$  resp. 0.135,  $\text{Var}(\bar{y}) = 0.084$  resp. 0.080, and  $\text{Var}(Y_I) = 0.43$  resp. 0.90 against  $|I|\text{Var}(\bar{y}) = 0.44$  resp. 0.79: the ratio 0.99 resp. 1.14 (the latter including the variation of  $|I|$ ) shows no positive correlation between the orbit weights of different ladder pairs. So ( $\gamma$ ) is, numerically, an independence statement; the lower tail  $\mathbb{P}[Y_I < \kappa|I|]$  at these small  $|I|$  (0.18 at  $\kappa = 0.05$ ,  $n = 100$ ) is what  $|I| \approx 10$  independent summands with mean 0.13 and a 50% atom at 0 give.

(e) **The arc-trapping input ( $\alpha$ ): CGW's lemma for a generic row pair in  $X$ , and the blocked fraction.** The orbit weight  $\bar{y}_{x,y}$  vanishes when the  $\rho_{x,y}$ -cycles through  $p, p'$  are short, so Hypothesis 19.17 needs  $\mathbb{P}_X[|Q_p(\rho_{x,y})| \leq \varepsilon n] \leq C\varepsilon$  for the ladder pairs — CGW's short-cycle bound for the row pair  $(x, y)$  inside  $X$ . Run CGW's splitting/joining multigraph for the distinguished rows  $x, y \notin \{1, 2\}$  inside  $X$ . Switches and cross-switches of rows  $x, y$  are free (they change rows  $x, y$  only), so the repairs of their splitting and joining procedures are always available; the flip and backflip at a column pair  $(q, c)$  turn the  $(q, c)$ -column cycle through  $x$  or through  $y$  and are legal iff that cycle meets  $\{1, 2\}$  in 0 or 2 rows (Lemma 19.15). Since either cycle may be used, the flip at  $(q, c)$  is *blocked* exactly when rows 1, 2 and rows  $x, y$  are *interleaved* in the column-cycle structure of  $(q, c)$ : in the  $A$ -state ( $x, y$  in different cycles  $Z_x, Z_y$ ) one of 1, 2 lies in  $Z_x$  and the other in  $Z_y$ ; in the  $B$ -state the four rows alternate  $x, 1, y, 2$  on the common cycle. Interleaving is invariant under the switch of rows  $x, y$  (which maps  $A$ -crosswise to  $B$ -cross and back, Lemma 19.14) and under the switch of rows 1, 2 at a column pair in different  $\sigma$ -cycles (same computation with the roles exchanged; this move preserves  $X$  up to  $\alpha \leftrightarrow \beta$ ). Legality is a property of the bitrade, so CGW's edge correspondence survives the removal of the blocked edges, and the  $\mu$ -side flip edges alone give, for the cycle through the mark column ( $N_\lambda(p) = \ell$  splitting pairs leave  $p$  in the  $\ell$ -piece):

$$\mathbb{E}_X[\#\{(q, c) : q \in Q_p, c \notin Q_p, \text{ not interleaved}\}; |Q_p| = \ell] \leq 2\ell \mathbb{P}_X[|Q_p| \geq \ell + 2] \leq 2\ell. \quad (33)$$

Hence, with  $b_\ell$  the conditional expectation given  $|Q_p| = \ell$  of the fraction of interleaved pairs among the  $\ell(n - \ell)$  joining pairs,

$$\mathbb{P}_X[|Q_p(\rho_{x,y})| = \ell] \leq \frac{2}{(1 - b_\ell)(n - \ell)},$$

i.e. CGW's bound  $\leq 4/n$  holds in  $X$  up to the factor  $1/(1 - b_\ell)$ : ( $\alpha$ ) *follows from* (BL):  $b_\ell \leq 1 - c$ . Data (`s13_interleave.py`, `runs/s13/orbit/interleave50.log`, 300 ladder pairs at  $n = 50$ ,  $1.8 \cdot 10^5$  column pairs):  $\mathbb{P}[\text{interleaved}] = 0.186$  over all pairs in different  $\rho_{x,y}$ -cycles, 0.1867 in the  $A$ -state and 0.1861 in the  $B$ -state (the switch invariance), 0.188 on joining pairs of  $Q_p$  and 0.183 off them, 0.193/0.181 according as  $q, c$  lie in different/the same  $\sigma$ -cycles; and the lengths  $|Q_p|$  are spread uniformly (3, 7, 5, 7, 5, 9, 5, 5, 3 instances at  $\ell = 2, \dots, 10$  against 6 expected per length). So the blocked fraction is  $\approx 0.19$  and the loss is a factor 1.23.

What (BL) needs is again a same-cycle statement for generic row pairs: the interleaving at  $(q, c)$  is changed only by trades of third rows  $z, w \notin \{1, 2, x, y\}$  along  $\rho_{z,w}$ -cycles through exactly one of  $q, c$  ( $\delta_{q,c} \mapsto \delta_{q,c} \circ (zw)$ ), which de-interleave when  $z, w$  lie on adjacent arcs between the special rows (in the  $B$ -cross state) or separate  $x$  from the row of  $\{1, 2\}$  in its cycle (in the  $A$ -crosswise state); the cross-switch of  $z, w$  reverses a  $\delta_{q,c}$ -arc and is either illegal (the arc contains

exactly one of 1, 2), harmful (it contains exactly one of  $x, y$ ), or useless (it contains no special row). A double count of these involutions gives  $\mathbb{P}[\text{interleaved}] \leq 1/(1 + \kappa)$  as soon as, for an interleaved square, a  $\kappa$ -fraction of the row pairs  $(z, w)$  in the de-interleaving positions have  $q, c$  in different  $\rho_{z,w}$ -cycles — the generic same-cycle event of §19.4 once more, now for the third rows at the prescribed column pair. This is the explicit form of the recursion: each level's availability is the next level's genericity, and no level is settled by an exact identity. For the ladder pairs there is the further point that  $(x, y)$  is selected by the frame, so  $(\alpha)$  is needed conditionally on the frame, while (33) is stated for fixed rows  $x, y$  with the frame free. The same blocked-edge multigraph can be run with columns  $p, p'$  fixed if every move avoids them: the flips at free column pairs do; the switch of rows  $x, y$  used to repair a  $B$ -pair  $(q, c)$  trades along the  $\rho_{x,y}$ -cycle selected by the min-symbol rule, which is unavailable when that cycle is  $Q_p$  or  $Q_{p'}$  (one may then trade along the other cycle, if it avoids  $p, p'$ , at the cost of the involution property); and CGW's cross-switch repairs trade along  $\rho_{x,y}$ -arcs that may pass through  $p$  or  $p'$ . This enlarges the blocked set by a second cycle-avoidance event; we have not carried the bookkeeping through.

**(f) The permutation problem ( $\Pi$ ), and the one statement everything reduces to.**

Fix a permutation  $\rho_0$  of the columns, two columns  $p \neq p'$ , and a set  $\mathcal{T}$  of pairwise disjoint transpositions avoiding  $p, p'$ ; for  $\eta \in \{0, 1\}^{\mathcal{T}}$  let  $\tau(\eta)$  be the product of the transpositions with  $\eta_Z = 1$ . Call  $Z = (q, c)$  *crossing* in a permutation  $\rho$  if  $\{q, c\}$  and  $\{p, p'\}$  are interleaved in the cycle structure of  $\rho$ :  $q, c$  on different arcs of a common cycle through  $p, p'$ , or  $q$  in the cycle of  $p$  and  $c$  in the (different) cycle of  $p'$ , or vice versa. By Lemma 19.15,  $\rho \mapsto \rho \circ Z$  toggles the separation of  $p, p'$  exactly when  $Z$  is crossing in  $\rho$ , and  $Z$  is crossing in  $\rho \circ Z$  iff it is crossing in  $\rho$ . Hence, for  $\eta$  uniform and  $b(\eta) = [p, p' \text{ separated in } \rho_0 \tau(\eta)]$ ,

$$\mathbb{P}_\eta[b = 1] \geq \frac{1}{2} \max_{Z \in \mathcal{T}} \mathbb{P}_\eta[Z \text{ crossing in } \rho_0 \tau(\eta)] \geq \frac{1}{2} \mathbb{E}_\eta \left[ \frac{\#\{Z \text{ crossing in } \rho_0 \tau(\eta)\}}{|\mathcal{T}|} \right], \quad (34)$$

and  $b \equiv 0$  on the cube iff no  $Z$  is crossing in any  $\rho_0 \tau(\eta)$ . This happens iff the two arcs  $P_1, P_2$  of the cycle through  $p, p'$  (crossed state) lie in different components of the graph whose vertices are  $P_1, P_2$  and the other cycles of  $\rho_0$  and whose edges are the transpositions of  $\mathcal{T}$  joining different vertices: a transposition inside a vertex splits off a piece without changing the separation, one joining a third cycle to  $P_1$  merges it into the arc, and once a path of such merges connects  $P_1$  to  $P_2$  some transposition has become crossing. In particular a single crossing  $Z$  at  $\eta = 0$  gives  $\mathbb{P}_\eta[b = 1] \geq \frac{1}{2}$  on the orbit of  $Z$  alone, and the examples of §(c) (chords at  $\rho_0$ -distance 2) are precisely the disconnected case. There is no loss in using one coordinate per pair: with  $Z_k$  the first clean coordinate of the  $k$ -th pair in a fixed order of the matching (an orbit-invariant choice), Proposition 19.16 reads

$$\mathbb{P}_X[\text{Flip}_I = \emptyset] \leq \mathbb{E}_X \left[ \prod_{k \in I} (1 - \frac{1}{2} X_k) \right], \quad X_k = [Z_k \text{ crossing in } \rho_{x_k, y_k}], \quad (35)$$

and  $(\beta)$  becomes:  $\mathbb{P}_X[X_k = 1] \geq c$  for most  $k$ , i.e. the column pair selected by the column-cycle structure through  $x_k$  is interleaved with  $\{p, p'\}$  in the row-cycle structure of  $(x_k, y_k)$  with probability bounded below.

Every input of Hypothesis 19.17 is now of one form. Call two pairs of parallel lines *interleaved* in a third pair of lines if they are interleaved in its cycle structure, as above. Then  $(\beta)$  asks that the column pairs  $\{p, p'\}$  (the mark) and  $\{q, c\}$  (a clean coordinate) be interleaved in the row pair  $(x_k, y_k)$  with probability  $\geq c$ ; (BL), hence  $(\alpha)$ , asks that the row pairs  $\{1, 2\}$  and  $\{x, y\}$  be

not interleaved in a joining column pair  $(q, c)$  with probability  $\geq c$ ; and the availability of every third-row repair of §(e) is the event that  $\{p, p'\}$  and  $\{q, c\}$  are not interleaved in  $(z, w)$ . So the whole residual is the two-sided statement

$$(U) \quad c \leq \mathbb{P}[\text{two given pairs of lines are interleaved in a third pair}] \leq 1 - c$$

in  $X$  (type and mark of rows 1, 2 prescribed; the other lines generic or selected by cycle structure), and the exact tools of  $X$  — the free row trades of Lemma 19.14, which preserve interleaving of  $\{1, 2\}$  with the traded pair (§(e)), and the column turns of Lemma 19.15, which are legal exactly off the interleaved configurations — are either blind to (U) or need (U). Without any conditioning, (U) for four generic columns in a generic row pair follows from  $\mathbb{P}[W] \leq 3/5$  under CGW(3/2) (§19.4); from the surviving direction alone one does not even get  $\mathbb{P}[\sigma \text{ is an } n\text{-cycle}] \leq 1 - c$  (that is CGW’s Lemma 4.4, upper half), although it follows from the intercalate counts of Kwan–Sudakov [11] by a second moment. The data give 0.19 for the interleaving probability in every version we measured (§(d), §(e)).

## 20 Open problems

**Problem 20.1.** Prove Hypothesis 10.4 in any nontrivial regime, e.g.  $\delta(n) = o(1)$  ( $\delta(n) = o(1/\log n)$  gives Conjecture 1.2 by Theorem 12.1). By Theorem 11.4 this is exactly the statement that a marked pair  $\{j, \sigma^\alpha(j)\}$  of a uniform  $L \in S(\lambda)$  is a  $B$ -pair with probability  $\frac{1}{2} + O(\delta)$ ; the programme at the end of §14, as sharpened in §19.5, reduces it further to the annealed bias bound of Problem 15.8 — a genericity statement for the rank event of Theorem 15.6 under the orbit measure, for which generic position gives  $\mathbb{E}|\text{bias}| \asymp k^{-1/2}$  against a budget of  $k^{-1/11}$ . After §16 the outstanding pieces are: the  $L^2$  mixing bound of Problem 16.4 (which closes the iid model via the factorization Theorem 16.1; after §17 this is reduced to the parity-decay statement Problem 17.9 alone with any exponent  $\alpha > 2/9$  — the gap law Problem 17.7 is not needed over the short horizon  $j \leq 8d$  of Theorem 16.1 (Corollary 17.11) — and the rate  $1/d$  is proved sharp; after §17.9 the environment is controlled over the horizon by the Harer–Zagier block count, Corollary 17.16, and Problem 17.9 is reduced to the conditions (R1)–(R2) of Proposition 17.21), the transfer from iid to orbit positions, and the supply–mark decoupling of Problem 16.7.

**Problem 20.2.** Determine the layer profile  $\gamma$  of §5 and the constant  $c_2 = \int_0^1 \gamma$  ( $\gamma(0) = 1$  is Corollary 4.3; Theorems 7.15 and 8.5 pin the fourth and fifth layers, and Remark 8.7 every fixed layer, so  $\gamma$  is flat at the origin; see Remark 5.1 for why the  $n \leq 11$  value 1.30 of  $n^3 h_n(2)$  does not yet determine the limit). Even a heuristic derivation via permanents of complements of random rectangles with a planted rows-(1, 2) intercalate would be valuable; the remaining unknown is the bulk  $k \asymp tn$ ,  $t > 0$ .

**Problem 20.3.** Prove the third-order intercalate expectation  $\mathbb{E}[\mathbf{N}] = n^2/4 - n/4 + \Theta(1/n)$  (§3.1) — a statement plausibly within reach of the switching methods of [15].

**Problem 20.4.** Prove the sharp parity asymptotics  $\mathbb{P}(\sigma_{1,2} \text{ even}) - \frac{1}{2} \sim (-1)^{n-1}(n-1)/(2D_n)$ , in the spirit of [9].

**Problem 20.5.** Give a corrected exact joining bookkeeping for  $\alpha = \beta \geq 3$  (Remark 10.2), restoring the identity for all adjacent pairs.

**Problem 20.6.** Extend the switch calculus of §7 to the fifth layer (Theorem 7.15 settles the fourth), where the analogue of Proposition 6.1 is an inclusion–exclusion over the three pairwise coincidence patterns of a triple  $(\tau, \eta, \theta)$ . Beyond that lies the bulk of the profile  $\gamma$ . (Classical context: Riordan’s aggregate  $K(3, n) = \sum_k \binom{n}{k} D_{n-k} D_k U_{n-2k}$  [19, Ch. 8, eq. (30a)]; Moser’s very reduced  $4 \times n$  rectangles [17]; the  $k$ -discordant literature [21]. Theorem 9.2 resolves the 2-cycle-level joint statistics for three rows; the full cycle-type-resolved joint law remains open.)

## References

- [1] P. J. Cameron, *A niggling problem*, blog post, 24 January 2015, <https://cameroncounts.wordpress.com/2015/01/24/a-niggling-problem/>; see also *Some challenges on Latin squares*, 6 October 2023.
- [2] P. J. Cameron, Almost all quasigroups have rank 2, *Discrete Math.* **106/107** (1992), 111–115.
- [3] N. J. Cavenagh, C. Greenhill and I. M. Wanless, The cycle structure of two rows in a random Latin square, *Random Structures Algorithms* **33** (2008), 286–309.
- [4] M. Cohn and A. Lempel, Cycle decomposition by disjoint transpositions, *J. Combin. Theory Ser. A* **13** (1972), 83–89.
- [5] P. Flajolet and M. Soria, Gaussian limiting distributions for the number of components in combinatorial structures, *J. Combin. Theory Ser. A* **53** (1990), 165–182.
- [6] R. Häggkvist and J. C. M. Janssen, All-even Latin squares, *Discrete Math.* **157** (1996), 199–206.
- [7] J. Harer and D. Zagier, The Euler characteristic of the moduli space of curves, *Invent. Math.* **85** (1986), 457–485.
- [8] M. T. Jacobson and P. Matthews, Generating uniformly distributed random Latin squares, *J. Combin. Des.* **4** (1996), 405–437.
- [9] M. Kwan, K. Petrova and M. Sawhney, Parities in random Latin squares, arXiv:2509.13125 (2025).
- [10] J. Allsop and P. Morris, Universal probability bounds for partial Latin squares, arXiv:2606.18174 (2026).
- [11] M. Kwan and B. Sudakov, Intercalates and discrepancy in random Latin squares, *Random Structures Algorithms* **52** (2018), 181–196.
- [12] M. Kwan, A. Sah, M. Sawhney and M. Simkin, Substructures in Latin squares, *Israel J. Math.* (2023); arXiv:2202.05088. See also *Large deviations in random Latin squares*, *Bull. Lond. Math. Soc.* **54** (2022).
- [13] O. Schramm, Compositions of random transpositions, *Israel J. Math.* **147** (2005), 221–243.
- [14] T. Łuczak and L. Pyber, On random generation of the symmetric group, *Combin. Probab. Comput.* **2** (1993), 505–512.

- [15] B. D. McKay and I. M. Wanless, Most Latin squares have many subsquares, *J. Combin. Theory Ser. A* **86** (1999), 323–347.
- [16] B. D. McKay and I. M. Wanless, On the number of Latin squares, *Ann. Comb.* **9** (2005), 335–344.
- [17] W. O. J. Moser, The number of very reduced  $4 \times n$  Latin rectangles, *Canad. J. Math.* **19** (1967), 1011–1017.
- [18] J. Riordan, Discordant permutations, *Scripta Math.* **20** (1954), 14–23.
- [19] J. Riordan, *An Introduction to Combinatorial Analysis*, Wiley, New York, 1958; Chapter 8 (*Permutations with restricted position II*), esp. §3, pp. 201–204.
- [20] J. Touchard, Permutations discordant with two given permutations, *Scripta Math.* **19** (1953), 109–119.
- [21] E. G. Whitehead, Jr., Four discordant permutations, *J. Austral. Math. Soc. Ser. A* **28** (1979), 369–377.

## A Data and computation manifest

All computations were performed in this working session; code and exact data are available alongside these notes.

1. `cgw_data.py`: the exact  $C_n(\lambda)$ ,  $4 \leq n \leq 11$ , from [3, Tables 1–2], verified against the published  $P_n(\lambda)$  fractions and against  $L_n$  for all  $n \leq 11$ .
2. `fit_additive.py`: additive fits of  $\log C_n(\lambda)$ , chain increments,  $d_{TV}$ , spreads.
3. `third_row.py`, `third_row_asymp.py`: exact  $N^{(3)}$  via matching polynomials; fusion identity verified for all types,  $n \leq 25$ ; exact ratios to  $n = 400$  with Richardson extrapolation.
4. `layer4.c`: exact  $N^{(4)}$  for  $n \leq 11$  by summation of Ryser–Gray permanents over all third rows ( $\approx 5 \times 10^6$  permanents of order 11).
5. `verify_identity.py`: complete enumeration of Latin squares of orders 5 and 6 (via first-row normalisation) implementing the CGW switch/flip operations; exact rational match of Theorem 10.1 in the cases reported, and the exact values  $\mathbb{P}(B_1) = \mathbb{P}(B_2)$ .
6. `jm.c`, `jm_stats.py`: Jacobson–Matthews sampler ( $O(1)$  per move) with fixed-grid proper-state sampling, and the measurement pipeline.
7. `puncture.py`: the punctured-board calculus ((4)), verified against exhaustive enumeration at  $n = 6, 7$ .
8. `pair_ie.py`, `layer4_production.py`, `layer4_big.txt`: the pair inclusion–exclusion identity (verified exactly at  $n = 6, 7$ ), exact  $T_j$  sums for  $j \leq 3$ , truncated layer-4 effects to  $n = 18$  with validation against the exact  $n \leq 11$  values.
9. `three_row_moments.py`: exact joint intercalate moments of  $3 \times n$  rectangles (certified against brute force at  $n = 7$ ),  $n \leq 13$  tables, the tilt identity and covariance data.

10. `switch.py`: the switch calculus of §7: board  $B_0$ , punctured counts on arbitrary path/cycle boards, the empty bracket  $= M_{n-4}$  for  $n \leq 14$ , brute-force verification of Lemma 7.1 at  $n = 6, 7$ , and per- $M$  bracket ratio scans.
11. `fusion_lemmas.py`: symbolic proof-checks of Lemma 7.3 and exact integer verification of the master formula (12) on random punctures at  $n = 9, 11, 13$ .
12. `valuation_scan.py`: exhaustive verification of Lemma 7.5 (valuation  $\geq 2$ ; the  $[x^2]$  classification) over all  $|M| \leq 2$  at  $n = 9, 10$ .
13. `switch_j1.py`: stratified sums  $S_j$  (13) against direct enumeration ( $n = 8, 9$ ), class-resolved contributions, and the worry-class (rows  $1 \wedge 2$ ) resummation scans.
14. `loop_reduction_check.py`: complete-enumeration proof of Remark 1.3 at  $n = 5, 6$  (the loop second-row type law is exactly  $P_n$ ).
15. `anatomy.c`: exact  $\text{Br}_G$ ,  $N_0^{(4)}$ , masses  $m^\pm$  and conditional means  $E^\pm[A]$  by full enumeration of  $A_0$  with per- $\tau$  Ryser permanents,  $n \leq 11$ .
16. `triple_ie.py`: the triple inclusion–exclusion identity (Proposition 8.1), both forms, verified against brute-force triple enumeration at  $n = 5, 6, 7$  (both cycle types).
17. `n5_direct.c`, `n5_direct2.c`: exact  $N^{(5)}$  by direct enumeration with 64-/128-bit cell masks,  $n \leq 9$ , both types;  $N^{(3)}, N^{(4)}$  reproduced as certification.
18. `switch5.py`, `anatomy5.c`: the fifth-layer switch decomposition (Lemma 8.3) verified by complete enumeration at  $n = 6, 7$  and against the direct  $\Delta N^{(5)}$  at  $n = 8$ ; exact  $\text{Br}_G^{(5)}$ ,  $N_0^{(5)}$ ,  $G_2^{(5)}$ -,  $H^{(5)}$ -pieces on  $A_0$ .
19. `layer5_production.py`: the truncated triple-IE series  $S(Q)$ ,  $Q \leq 3$ , both types,  $n \leq 11+$ , validated against the exact  $N^{(5)}$  anchors.
20. `pf_orbit.py`: the CGW cross-switch ([3, Def. 3.9]) implemented and verified exhaustively at  $n = 5$  (Latin, type-preserving, involutive, marked status preserved, shifted-pair toggle profile), and the splitting-instance statistics  $\mathbb{P}(B_i \mid \text{marked } A/B)$ .
21. `pf_bridge.py`: instance-by-instance verification of the bijections behind Theorem 11.4 at  $n = 5, 6$  (unflip statuses, roundtrips, hit multiplicities, orphan structure) for  $(2, 3) \rightarrow (5)$ ,  $(2, 4) \rightarrow (6)$ ,  $(2, 2, 2) \rightarrow (2, 4)$ ,  $(3, 3) \rightarrow (6)$ .
22. `xside.c`, `xside2.c`: the  $X$ -side counts  $|X_A|, |X_B|$  for all types and merges — by complete enumeration of first-row-normalised squares ( $n \leq 6$ ), and per fixed representative  $\sigma_0$  of each type ( $n = 7$ ; conjugation symmetry); exact match with  $2C_n(\lambda)/C_n(\mu) - 1$  in all cases.
23. `pf_cl.py`: instance-by-instance verification of Theorem 15.1, Theorem 15.3, the passenger construction, and randomized two-sided products with shared rows and conjugating coordinates (all activation subsets, exhaustively).
24. `pf_master.py`: end-to-end verification of the master formula on real squares —  $n = 5$  complete (7920 subset checks),  $n = 6$  sampled (1,706,949 checks, all four column-incidence classes): 0 failures.

25. `pf_bias.py`, `bias_scale.c`: the exactly solvable bias families of §19.5 and the  $k$ -scaling of the generic-position bias (rank route).
26. `s7_bias_lab.py`: pendant-star family (Proposition 16.5), exhaustive predicate scan over all chord configurations  $k \leq 6$ , per-coordinate toggle statistics.
27. `s7_twopoint.c`: cycle-route measurements of  $4\mathbb{E}[\text{bias}^2]$ , the one-point function, and  $\psi(j, d)$  (Theorem 16.1); independent of the rank route.
28. `s7_real_bias.py` (with `jm.c`): the real-ensemble study of §16.4 at  $n = 30$ –100, including ground-truth re-verification of the master formula.
29. `s8_splitmerge.c`: the continuum marked split–merge chain of §17 (modes: `psi/psibar` gap-binned  $\psi$ , `one`, `iota` symmetry test, `gapdist` stationary gap law, `traj` invariants); validated against `s7_twopoint.c` at matched  $(j, d)$  and against the PD(1) invariants.
30. `s8_real_bias.py`: `s7_real_bias.py` plus the  $\sigma$ -cycle length (class size) column and  $k = 0$  rows, for the supply–mark decoupling measurements; `s8_inst200.csv` extends the real-ensemble study to  $n = 200$ .

## B Exact $\bar{q}$ values from the identity

All  $\alpha \neq \beta$  joins at  $n = 11$ , with  $\bar{q} = 2C_{11}(\lambda)/C_{11}(\mu) - 1$  (exact values, displayed to six decimals) and the normalised deficit  $\frac{1}{2}(1 - \bar{q})n^3$ :

| $\mu$           | join | $\lambda$    | $\bar{q}$ | $\frac{1}{2}(1 - \bar{q})n^3$ |
|-----------------|------|--------------|-----------|-------------------------------|
| (2, 9)          | 2+9  | (11)         | 0.998080  | +1.278                        |
| (2, 2, 7)       | 2+7  | (2, 9)       | 0.998078  | +1.279                        |
| (2, 3, 6)       | 2+6  | (3, 8)       | 0.998084  | +1.275                        |
| (2, 4, 5)       | 2+5  | (4, 7)       | 0.998084  | +1.275                        |
| (2, 2, 2, 5)    | 2+5  | (2, 2, 7)    | 0.998080  | +1.278                        |
| (2, 4, 5)       | 2+4  | (5, 6)       | 0.998077  | +1.280                        |
| (2, 2, 3, 4)    | 2+4  | (2, 3, 6)    | 0.998079  | +1.278                        |
| (2, 3, 6)       | 2+3  | (5, 6)       | 0.998046  | +1.301                        |
| (2, 2, 3, 4)    | 2+3  | (2, 4, 5)    | 0.998048  | +1.299                        |
| (2, 3, 3, 3)    | 2+3  | (3, 3, 5)    | 0.998054  | +1.295                        |
| (2, 2, 2, 2, 3) | 2+3  | (2, 2, 2, 5) | 0.998052  | +1.297                        |
| (3, 8)          | 3+8  | (11)         | 0.999966  | +0.023                        |
| (2, 3, 6)       | 3+6  | (2, 9)       | 0.999970  | +0.020                        |
| (3, 3, 5)       | 3+5  | (3, 8)       | 0.999970  | +0.020                        |
| (3, 4, 4)       | 3+4  | (4, 7)       | 0.999963  | +0.024                        |
| (2, 2, 3, 4)    | 3+4  | (2, 2, 7)    | 0.999971  | +0.019                        |
| (4, 7)          | 4+7  | (11)         | 0.999997  | +0.002                        |
| (2, 4, 5)       | 4+5  | (2, 9)       | 1.000001  | −0.001                        |
| (5, 6)          | 5+6  | (11)         | 1.000004  | −0.003                        |

The deficits stratify precisely by the smallest part destroyed by the join —  $\approx c_2 n^{-3}$  when a 2-cycle is destroyed,  $\approx h_n(3)$  when a 3-cycle is, and at noise level below — and are essentially independent of the ambient type  $\mu$ : the identity turns the refined law of Conjecture 3.1 into an equidistribution statement, join by join.